

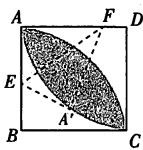
参考答案

全国重点高中提前招生 考试全真试卷(一)

1. B 【解析】 $\therefore a = \frac{\sqrt{3}}{\sqrt{2}+\sqrt{3}+\sqrt{5}}, b = 2+\sqrt{6}-\sqrt{10} = \sqrt{2}(\sqrt{2}+\sqrt{3}-\sqrt{5})$,

$\therefore \frac{a}{b} = \frac{\sqrt{3}}{\sqrt{2}+\sqrt{3}+\sqrt{5}} \cdot \frac{1}{\sqrt{2}(\sqrt{2}+\sqrt{3}-\sqrt{5})} = \frac{1}{4}$. 故选 B.

2. D 【解析】如图,以 EF 为折线翻折, A 点落在正方形 $ABCD$ 所在的 A' 点的位置,那么 A' 所有可能位置形成的图形是图中阴影部分. $\therefore S_{\text{阴}} = 2 \cdot S_{\text{扇形}BAC} - S_{\text{正方形}ABCD} = \frac{\pi}{2} - 1$. 故选 D.



第 2 题图

3. C 【解析】由题意知只需考虑 a, b, c 均为正数.

解法一:由题意可知 $\begin{cases} b^2 - 4ac \geq 0, \\ a + b + c = 1, \end{cases}$ 由对称性不妨假设

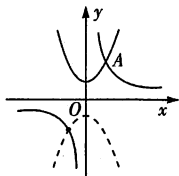
$a \leq c$, 所以 $b^2 \geq 4ac \geq 4a^2$, 所以 $b \geq 2a$, 把 a, b, c 中最小者记为 $\min\{a, b, c\}$,

此时 $\min\{a, b, c\} = a, \begin{cases} b \geq 2a, \\ c \geq a, \end{cases}$ 所以 $1 = a + b + c \geq a + 2a + a = 4a$, 从而 $a \leq \frac{1}{4}$, 故 a 的最大值记为 $\max\{a, b, c\} = \frac{1}{4}$, 此时 $y = \frac{1}{4}x^2 + \frac{1}{2}x + \frac{1}{4}$, 满足题意.

解法二:由题意可知 $\begin{cases} b^2 - 4ac \geq 0, \\ a + b + c = 1, \end{cases}$ 所以 $b \geq 2\sqrt{ac}$. 结合基本不等式知 $1 = b + a + c \geq 2\sqrt{ac} + 2\sqrt{ac}$, 解得 $\sqrt{ac} \leq \frac{1}{4}$, 设 $m = \min\{a, b, c\}$, 则 $\begin{cases} m \leq a, \\ m \leq c, \end{cases}$ 两式相乘可得 $m^2 \leq ac \leq \frac{1}{16}$, 所以 $m \leq \frac{1}{4}$, $\max\{m\} = \frac{1}{4}$, 此时 $y = \frac{1}{4}x^2 + \frac{1}{2}x + \frac{1}{4}$ 满足题意. 故选 C.

4. B 【解析】如图,点 A 关于原点的对称点的横坐标为 -2 , 所以不等式 $\frac{k}{x} + ax^2 + 1 < 0$, 即不等式

$\frac{k}{x} < -ax^2 - 1$ 的解集是 $-2 < x < 0$. 故选 B.



第 4 题图

5. B 【解析】过点 A 作直线 $PQ \parallel BC$, 分别延长 BD, CD 交 PQ 于点 P , 交 PQ 于点 Q . 则可证 $\triangle PQD \cong \triangle BCD$, $\triangle BCE \sim \triangle PAE$, $\therefore \frac{AE}{CE} = \frac{AP}{BC}$ ①, 同理:

$\triangle CBF \sim \triangle QAF$, $\therefore \frac{AF}{BF} = \frac{AQ}{BC}$ ②, ①+②得 $\frac{AC-CE}{CE} + \frac{AB-BF}{BF} = \frac{AP+AQ}{BC}$,

$\therefore \frac{AC}{CE} + \frac{AB}{BF} = 3$. 又 $\frac{1}{CE} + \frac{1}{BF} = \frac{1}{3}$, $AC = AB$, $\therefore \triangle ABC$ 的边长 $= 9$. 故选 B.

6. B 【解析】令 $x = 0$, 则 $y = 1$, $\therefore OP = 1$, 设点 A 横坐标为 m , 则 $AD = -m^2 + 1$, $\therefore AB \perp y$ 轴, $AD \perp x$ 轴, $\therefore AF = OD = m$, $OF = -m^2 + 1$, $PF = 1 - (-m^2 + 1) = m^2$. 在 $\text{Rt} \triangle PAF$ 中, $PA^2 = PF^2 + AF^2 = m^4 + m^2$. 在 $\text{Rt} \triangle POD$ 中, $PD = \sqrt{OP^2 + OD^2} = \sqrt{1 + m^2}$. 由 $AB \parallel x$ 轴, $\therefore \triangle PEF \sim \triangle PDO$, $\therefore \frac{PF}{OP} = \frac{PE}{PD}$, 即 $\frac{m^2}{1} = \frac{PE}{\sqrt{1 + m^2}}$, $PE = m^2 \sqrt{1 + m^2}$, $\therefore PA^2 = PD \cdot PE = m^4 + m^2$, $\therefore \frac{PA}{PD} = \frac{PE}{PD}$, $\therefore \angle APE = \angle DPA$, $\therefore \triangle PAD \sim \triangle PEA$, 即两三角形始终相似.

7. 两 【解析】把 $y\sqrt{x} = 1$ 两边同时平方得 $y^2x = 1$, 则 $x = y^{-2}$, 把 $x = y^{-2}$ 代入 $x^{x-y} = y^{x+y}$ 得 $y^{-2(x-y)} = y^{x+y}$, 当 $y = 1$ 时, $x = 1$; 当 $y \neq 1$ 时, 则 $-2(x-y) = x+y$, $y = 3x$, $x = \frac{y}{3}$, $\therefore \frac{y}{3} = y^{-2}$, 解得 $y^3 = 3$, $y = \sqrt[3]{3}$, 此时 $x = \frac{\sqrt[3]{3}}{3}$. 经验方

程组的解为 $\begin{cases} x=1, \\ y=1 \end{cases}$ 或 $\begin{cases} x=\frac{\sqrt[3]{3}}{3}, \\ y=\sqrt[3]{3}. \end{cases}$

8. 4 【解析】设 $m = x - 2, n = y - 2$, 则有 $m^3 + 2013m = -2, n^3 + 2013n = 2$, 两式相加得 $m^3 + 2013m + n^3 + 2013n = 0$, 整理得 $(m+n)(m^2 - mn + n^2 + 2013) = 0$, $\therefore m+n=0$ 或 $m^2 - mn + n^2 + 2013 = 0$, 而 $m^2 - mn + n^2 + 2013 = (m - \frac{1}{2}n)^2 + \frac{3}{4}n^2 + 2013 > 0$, $\therefore m+n=0$, 即 $x+y=4$.

9. $\frac{65}{9}$ 【解析】若 4 为斜边, 方程 $x^2 - 6x + m = 0$ 两根为 x_1, x_2 , 则 x_1, x_2 为直角边, $x_1^2 + x_2^2 = 4^2$, $(x_1 + x_2)^2 - 2x_1x_2 = 4^2$, $36 - 2m = 16$, 解得 $m = 10$, 此时 $x^2 - 6x + 10 = 0$ 无解, 舍去; 若 4 为直角边, $x^2 - 6x + m = 0$ 两根中不妨设 4 为斜边, 则 $x_2^2 + 4^2 = x_1^2$, $(x_1 + x_2)(x_1 - x_2) = 16$, $x_1 - x_2 = \frac{16}{6} = \frac{8}{3}$, 由

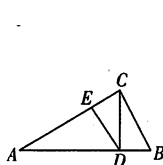
$\begin{cases} x_1 + x_2 = 6 \\ x_1 - x_2 = \frac{8}{3} \end{cases}$ ①, ②, 得 $\begin{cases} x_1^2 + x_2^2 + 2x_1x_2 = 36 \\ x_1^2 + x_2^2 - 2x_1x_2 = \frac{64}{9} \end{cases}$ ③, ④, ③-④得 $4x_1x_2 = 36 - \frac{64}{9}$

$\frac{64}{9}, 4m = 36 - \frac{64}{9}, m = \frac{65}{9}$.

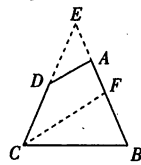
10. 2 【解析】 $\because CD \perp AB, DE \perp AC, \therefore \text{Rt} \triangle ADE \sim \text{Rt} \triangle ABC, \text{Rt} \triangle CBD \sim \text{Rt} \triangle ABC$. 令 $BC = a, AB = c$, 则 $DB = \frac{a^2}{c}, AD = AB - DB = c - \frac{a^2}{c} = \frac{c^2 - a^2}{c}$,

因为相似三角形的周长之比等于相似比, 于是 $\frac{p_1 + p_2}{p} = \frac{p_1}{p} + \frac{p_2}{p} = \frac{AD}{AB} + \frac{BC}{AB} = \frac{c^2 - a^2}{c^2} + \frac{a}{c} = -(\frac{a}{c})^2 + \frac{a}{c} + 1$, 由二次函数性质知, 当 $\frac{a}{c} =$

$-\frac{1}{2 \times (-1)} = \frac{1}{2}$ 即 $\frac{BC}{AB} = \frac{1}{2}$ 时, $\frac{p_1 + p_2}{p}$ 取得最大值, 而 $BC = 1$, 所以此时 $AB = 2$.



第 10 题图



第 11 题图

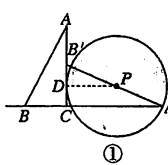
11. $0 < AD < 2$ 【解析】如图, 延长 CD, BA , 交于点 E , 过点 C 作 $CF \parallel DA$, 交 BA 于点 F . $\because$ 在平面四边形 $ABCD$ 中, $\angle A = \angle B = \angle C = 75^\circ, \therefore \angle CFB = \angle A = \angle B, \therefore CF = BC = 2. \therefore 0 < AD < CF, \therefore 0 < AD < 2$.

12. $\frac{156}{25}$ 或 $\frac{102}{13}$ 【解析】如图①, $\odot P$ 与 AC 相切于点 D , 设 $PD = PA' = r$,

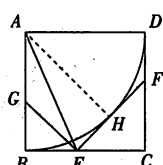
$\therefore PD \parallel A'C, \therefore \frac{r}{12} = \frac{13-r}{13}, r = \frac{156}{25}$. 如图②, $B'C = 5, AB' = 7$,

易证 $\triangle AB'D \sim \triangle A'B'C, \therefore \frac{B'D}{AB'} = \frac{B'C}{A'B'}, \frac{B'D}{7} = \frac{5}{13}, B'D = \frac{35}{13}$, 直径 $A'D =$

$\frac{35}{13} + 13$, 半径 $r = \frac{1}{2}(\frac{35}{13} + 13) = \frac{102}{13}$.



第 12 题图



第 15 题图

13. 解: 依题意有 $\Delta = [-(2k+1)]^2 - 4(k^2+2) \geq 0$, 解得 $k \geq \frac{7}{4}$. $\therefore x_1 + x_2 = 2k+1, x_1 \cdot x_2 = k^2+2, \therefore (x_1-2)^2 + (x_2-2)^2 = x_1^2 + x_2^2 - 4(x_1+x_2) + 8 = (x_1+x_2)^2 - 4(x_1+x_2) - 2x_1x_2 + 8 = (2k+1)^2 - 4(2k+1) - 2(k^2+2) + 8 = 2k^2 - 4k + 1 = 2(k-1)^2 - 1$, 当 $k = \frac{7}{4}$ 时, $(x_1-2)^2 + (x_2-2)^2$

的最小值为 $\frac{1}{8}$.

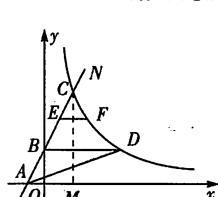
14. 解: (1) 作 $CM \perp x$ 轴于 M , 如图①.

$\because$ 直线 $y = 3x + b$ 经过点 $A(-1, 0), \therefore -3 + b = 0$, 解得 $b = 3, \therefore$ 直线解析式为 $y = 3x + 3, \therefore B(0, 3). \therefore \angle BOA = \angle CMA, \angle BAO = \angle CAM, \therefore \triangle BOA \sim \triangle CMA, \therefore AB:AC = AO:AM$. 又 $AC = 3AB, \therefore CM = 3BO = 9, AM = 3OA = 3, \therefore C$ 点坐标为 $(2, 9), \therefore$ 将 C 点坐标代入 $y = \frac{k}{x}$, 得 $k = 18$.

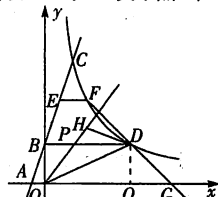
(2) $\because BD \parallel x$ 轴, $\therefore D$ 点的纵坐标为 3, 代入 $y = \frac{18}{x}$, 得 $x = 6, \therefore D$ 点坐标为 $(6, 3)$. 将 E 点横坐标代入 $y = 3x + 3$, 得 $y = 3m + 3, \therefore EF \parallel BD, \therefore F$ 点纵坐标为 $3m + 3$. 代入 $y = \frac{18}{x}$, 得 $x = \frac{6}{m+1}, \therefore F$ 点坐标为 $(\frac{6}{m+1}, 3m+3)$. $\therefore EF = \frac{1}{3}BD, \therefore \frac{6}{m+1} - m = \frac{1}{3} \times 6$, 解方程得 $m = 1$ 或 -4 (舍), $\therefore m = 1$.

(3) 存在. 理由如下: 如图②, 过点 D 作 $DQ \perp x$ 轴于 Q , 由 (2) 知 $D(6, 3), F(3, 6), \therefore$ 直线 FD 的解析式为 $y = -x + 9, OQ = 6, DQ = 3, \therefore OG = 9, \therefore GQ = OG - OQ = 9 - 6 = 3, \therefore DQ = GQ, \therefore \angle QGD = \angle QDG = 45^\circ. \therefore OD = 3\sqrt{5}, DG = 3\sqrt{2}$.

I. 当 $\angle HOD = \angle DOG$ 时, 如图②所示, 设 BD 与 OH 交于点 P ,



第 14 题图①

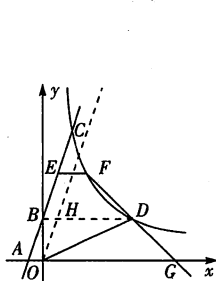


第 14 题图②

由 (2) 知, $BD \parallel x$ 轴, $\therefore \angle BDO = \angle DOG, \therefore \angle BDO = \angle HOD$,

$\therefore OP=PD$. 设 $OP=m$, 则 $BP=6-m$, 在 $Rt\triangle OBP$ 中, 由勾股定理可得 $m^2-3^2=(6-m)^2$, 解得 $m=\frac{15}{4}$, $\therefore BP=\frac{9}{4}$, $\therefore P(\frac{9}{4}, 3)$, $\therefore$ 直线 OP 的解析式为 $y=\frac{4}{3}x$. ①若 $\triangle ODG \sim \triangle ODH$, 则 $OD:OD=OG:OH=1$, 不符合题意, 舍去; ②若 $\triangle ODG \sim \triangle OHD$, 如图③所示, $\therefore OD:OH=OG:OD$, 即 $3\sqrt{5}:OH=9:3\sqrt{5}$, 解得 $OH=5$, 设 $H(3t, 4t)$, $\therefore (3t)^2+(4t)^2=5^2$, 解得 $t=1$ (负值舍去), $\therefore H(3, 4)$;

Ⅱ. 当 $\angle HOD = \angle DGO$ 时,

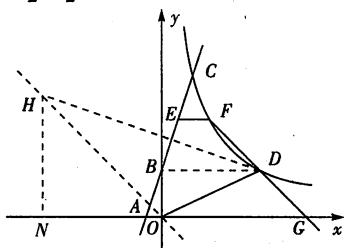


第 14 题图③

①若 $\triangle ODG \sim \triangle DHO$, 如图③, $\therefore \angle DOG = \angle ODH$, $DG:OH=OG:DO$, $\therefore DH \parallel OG$, 即点 H 在 BD 上, $3\sqrt{2}:OH=9:3\sqrt{5}$, $\therefore OH=\sqrt{10}$, $\therefore BH=1$, $\therefore H(1, 3)$, 直线 OH 的解析式为 $y=3x$;

②若 $\triangle ODG \sim \triangle HDO$, 如图④, $DG:OD=OG:OH$, 即 $3\sqrt{2}:3\sqrt{5}=9:OH$, 解得 $OH=\frac{9\sqrt{10}}{2}$, 设 $H(t, 3t)$, $\therefore t^2+(3t)^2=(\frac{9\sqrt{10}}{2})^2$, 解得 $t=\frac{9}{2}$ (负值舍去), $\therefore H(\frac{9}{2}, \frac{27}{2})$;

Ⅲ. 当 $\angle HOD = \angle ODG$ 时, $OH \parallel FG$, $\therefore$ 直线 OH 的解析式为 $y=-x$; ①若 $\triangle ODG \sim \triangle DOH$, 则 $OD:OD=OG:DH=1$, 不符合题意, 舍去; ②若 $\triangle ODG \sim \triangle HOD$, 如图⑤, $\therefore OD:OH=DG:OD$, 即 $3\sqrt{5}:OH=3\sqrt{2}:3\sqrt{5}$, 解得 $OH=\frac{15\sqrt{2}}{2}$. 设 $H(t, -t)$, $\therefore t^2+(-t)^2=(\frac{15\sqrt{2}}{2})^2$, 解得 $t=-\frac{15}{2}$ (正值舍去), $\therefore H(-\frac{15}{2}, \frac{15}{2})$.



第 14 题图⑤

综上, 符合题意的点 H 的坐标为 $(3, 4)$ 或 $(1, 3)$ 或 $(\frac{9}{2}, \frac{27}{2})$ 或 $(-\frac{15}{2}, \frac{15}{2})$.

15. 解: (1) 如图, 连接 AH , $\therefore AH \perp EF$, $GE \perp EF$, $\therefore GE \parallel AH$, $\therefore \angle GEA = \angle EAH$. $\therefore AH=AB$, $AE=AE$, $\angle ABE = \angle AHE$, $\therefore \triangle AHE \cong \triangle ABE$, $\therefore \angle BAE = \angle EAH$, $\therefore \angle BAE = \angle GEA$. $\therefore AG=EG$, 即 $\triangle AGE$ 是等腰三角形.

(2) $\therefore EH=EB=x$, $\therefore EC=1-x$, $CF=1-FD$. $\therefore FD=EH$, $\therefore EF=EH+HF=x+FD$. 在 $Rt\triangle ECF$ 中, $EF^2=EC^2+CF^2$, $\therefore (1-x)^2+(1-FD)^2=(x+FD)^2$. 整理得 $(1+x)FD=1-x$, $\therefore FD=\frac{1-x}{1+x}$, $CF=1-FD=1-\frac{1-x}{1+x}=\frac{2x}{1+x}$, 又易证 $\triangle GEB \sim \triangle EFC$,

$\therefore \frac{BE}{CF} = \frac{BG}{EC}$, $\therefore \frac{\frac{1}{2}BE \cdot BG}{\frac{1}{2}EC \cdot CF} = \frac{BE}{CF} \cdot \frac{BG}{EC} = (\frac{BE}{CF})^2 = (\frac{x}{1+x})^2 = \frac{1}{4}(1+x)^2$, $\therefore y = \frac{1}{4}x^2 + \frac{1}{2}x + \frac{1}{4}$ ($0 < x < 1$).

(3) 假设 BC 上存在一点 E , 能使 $GE=EF$, 则 $\frac{GE}{EF} = \frac{BE}{CF} = 1$, $\therefore x = \frac{2x}{1+x}$, 解得 $x=0$ 或 $x=1$, 经检验 $x=0$ 或 $x=1$ 是原方程的解, 但动点 E 不能与 B, C 点重合, 故 $x \neq 0$ 且 $x \neq 1$, $\therefore BC$ 边上符合条件的点 E 不存在.

16. 解: (1) $\therefore BC = \sqrt{10}$, $\therefore AB = \sqrt{5}$. ①当点 D 落在线段 AB 上, $BD=AB-AD=\sqrt{5}-\sqrt{2}$; 当点 D 落在线段 BA 的延长线上时, $BD=AB+AD=\sqrt{5}+\sqrt{2}$, $\therefore BD$ 的长为 $\sqrt{5}-\sqrt{2}$ 或 $\sqrt{5}+\sqrt{2}$. ②显然 $\angle ABD$ 不能为直角, 当 $\angle ADB$ 为直角时, $AD^2+BD^2=AB^2$, $\therefore BD=\sqrt{3}$; 当 $\angle BAD$ 为直角时, $AB^2+AD^2=BD^2$, $\therefore BD=\sqrt{7}$, $\therefore BD$ 长为 $\sqrt{3}$ 或 $\sqrt{7}$.

(2) 连接 D_1D_2, D_1C , 则 $\triangle AD_1D_2$ 为等腰直角三角形, $\therefore D_1D_2=\sqrt{2}AD_1=2$, $\therefore AD_1=AD_2, AB=AC$. $\therefore \angle BAC = \angle D_1AD_1$, $\therefore \angle BAD_2 = \angle CAD_1$, $\therefore \triangle BAD_2 \cong \triangle CAD_1$ (SAS), $\therefore BD_2=CD_1$. 又 $\therefore \angle AD_2C = 135^\circ$, $\therefore \angle D_1D_2C =$

$\angle AD_2C - \angle AD_2D_1 = 135^\circ - 45^\circ = 90^\circ$, $\therefore CD_1 = \sqrt{CD_2^2 + D_1D_2^2} = \sqrt{5}$, $\therefore BD_2 = \sqrt{5}$.

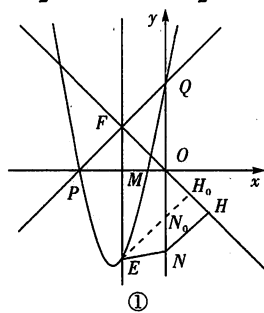
(3) 连接 CD_1 , $\therefore$ 点 P, M 分别是 CD_2, D_2D_1 的中点, $\therefore PM = \frac{1}{2}CD_1$, $PM \parallel CD_1$. $\therefore$ 点 N, M 分别是 BC, D_1D_2 的中点, $\therefore PN = \frac{1}{2}BD_2$, $PN \parallel BD_2$. $\therefore BD_2 = CD_1$, $\therefore PM = PN$, $\therefore \triangle PMN$ 是等腰三角形. $\therefore PM \parallel CD_1$, $\therefore \angle D_2PM = \angle D_2CD_1$. $\therefore PN \parallel BD_2$, $\therefore \angle PNC = \angle D_2BC$. $\therefore \angle D_2PN = \angle D_2CB + \angle PNC = \angle D_2CB + \angle D_2BC$, $\therefore \angle MPN = \angle D_2PM + \angle D_2PN = \angle D_2CD_1 + \angle D_2CB + \angle D_2BC = \angle BCD_1 + \angle D_2BC = \angle ACB + \angle ACD_1 + \angle D_2BC = \angle ACB + \angle ABD_2 + \angle D_2BC = \angle ACB + \angle ABC$. $\therefore \angle BAC = 90^\circ$, $\therefore \angle ACB + \angle ABC = 90^\circ$, $\therefore \angle MPN = 90^\circ$. $\therefore \triangle PMN$ 为等腰直角三角形.

$\therefore S_{\triangle PMN} = \frac{1}{2}(PM)^2 = \frac{1}{2}(\frac{1}{2}CD_1)^2 = \frac{1}{2}(\frac{1}{2}BD_2)^2$, $\therefore$ 当 BD_2 取最大值时, $\triangle PMN$ 的面积最大, 此时最大面积 $S = \frac{1}{2}[\frac{1}{2}(\sqrt{5}+\sqrt{2})]^2 = \frac{1}{8}(7+2\sqrt{10})$; 当 BD_2 取最小时, $\triangle PMN$ 面积最小, 此时最小面积 $S = \frac{1}{2}[\frac{1}{2}(\sqrt{5}-\sqrt{2})]^2 = \frac{1}{8}(7-2\sqrt{10})$.

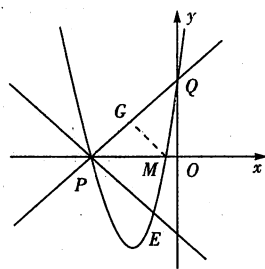
17. 解: (1) 由题可知, P, Q 两点坐标分别为 $P(-6, 0), Q(0, 6)$, 抛物线经过 P, Q 两点, 则 $\begin{cases} 36-6m+n=0 \\ n=6 \end{cases}$, 解得 $\begin{cases} m=7 \\ n=6 \end{cases}$, 即抛物线的解析式为 $y = x^2 + 7x + 6$.

(2) ①设 $E(t, t^2 + 7t + 6)$, 则 $F(t, t+6)$, $EF = t+6 - (t^2 + 7t + 6) = -t^2 - 6t = -(t+3)^2 + 9$, 故当 $t = -3$ 时, EF 取得最大值, 最大为 9, 此时 E 的坐标为 $(-3, 6)$, F 的坐标为 $(-3, 3)$, 如图①, 过 y 轴上的点 N 作直线 $y = -x$ 的垂线, 垂足为 H , 过 E 作 EH_0 垂直于直线 $y = -x$ 于点 H_0 , 交 y 轴于 N_0 点, 则 $\frac{\sqrt{2}}{2}ON = NH$, 故 $EN + \frac{\sqrt{2}}{2}ON = EN + NH \geq EH_0$, 即当 $E, N,$

H 三点共线时, $EN + \frac{\sqrt{2}}{2}ON$ 取得最小值, 又 $F(-3, 3)$, 则点 F 在直线 $y = -x$ 上, 故此时 $\triangle EFH_0$ 为等腰直角三角形, 即 $EH_0 = \frac{\sqrt{2}}{2}EF = \frac{9\sqrt{2}}{2}$, 则 $EN + \frac{\sqrt{2}}{2}ON$ 的最小值为 $\frac{9\sqrt{2}}{2}$.



①



②

第 17 题图

②如图②, 连接 MQ , 过点 M 作 $MG \perp PQ$ 于点 G . 易得 $\triangle MGP$ 和 $\triangle POQ$ 均为等腰直角三角形, 又 $OP = OQ = 6, MP = 5$, 故 $PQ = 6\sqrt{2}, MG = PG = \frac{5\sqrt{2}}{2}$, 即 $QG = \frac{7\sqrt{2}}{2}$, 故 $\tan \angle MQP = \frac{5}{7}$, $\therefore \tan \angle EPM = 1$. 设 $E(m, m^2 + 7m + 6)$, 则 $\tan \angle EPM = \frac{|m^2 + 7m + 6|}{|m + 6|} = |m + 1| = 1$, $\therefore m = 0$ 或 -2 . 又 $-6 < m < 0$, $\therefore m = -2$, 点 E 的坐标为 $(-2, -4)$.

全国重点高中提前招生 考试全真试卷(二)

1. A 【解析】 $\therefore$ 点 $P(-1, \frac{\sqrt{3}}{3})$ 在“勾股一次函数” $y = \frac{a}{c}x + \frac{b}{c}$ 的图象上, $\therefore \frac{\sqrt{3}}{3} = -\frac{a}{c} + \frac{b}{c}$, 即 $a - b = -\frac{\sqrt{3}}{3}c$. 又 a, b, c 分别是 $Rt\triangle ABC$ 的三条边长, $\angle C = 90^\circ$, $Rt\triangle ABC$ 的面积是 4, $\therefore \frac{1}{2}ab = 4$, 即 $ab = 8$. 又 $a^2 + b^2 = c^2$, $\therefore (a-b)^2 + 2ab = c^2$, 即 $(-\frac{\sqrt{3}}{3}c)^2 + 2 \times 8 = c^2$, 解得 $c = 2\sqrt{6}$. 故选 A.

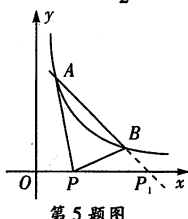
2. C 【解析】方程 $\frac{x}{x^2+1} = \frac{c}{c^2+1}$ 去分母后整理得 $cx^2 - (c^2+1)x + c = 0$, 当 $c \neq 0$ 时解得 $x_1 = c, x_2 = \frac{1}{c}$, 当 $c = 0$ 时解得 $x = 0$, ①不正确; 方程组 $\begin{cases} 2x+y=2, \\ ax+2y=1, \end{cases}$ 消去 y 得 $(4-a)x = 3$, $\therefore a = 4$ 时, 方程 $(4-a)x = 3$ 无解, 从而方程组无解, ②正确; $\therefore x^2 - xy + y^2 - x - y + 1 = \frac{1}{2}(2x^2 - 2xy + 2y^2 - 2x - 2y + 2) = \frac{1}{2}[(x-y)^2 + (x-1)^2 + (y-1)^2] \geq 0$, $\therefore$ ③正确; 对于方程 $\sqrt{2}x + \sqrt{3}y =$

$\sqrt{5}$, 当 $x=0, y=\frac{\sqrt{5}}{3}$ 时使得方程成立, $\therefore$ ④错误 故选 C.

3. B 【解析】由题意可得, 当各班人数除以 10 的余数不大于 6 时, 应舍去, 当各班人数除以 10 的余数大于等于 7 时, 就增加一名代表, 故 y 与 x 的函数关系式是 $y = \left[\frac{x+3}{10} \right]$ 故选 B.

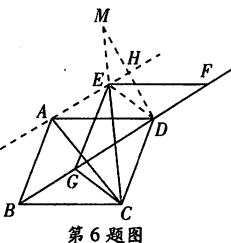
4. B 【解析】 $\because \angle AOB = 90^\circ, AO = 4, BO = 8, \therefore AB = \sqrt{OA^2 + OB^2} = \sqrt{4^2 + 8^2} = 4\sqrt{5}$. $\because \triangle AOB$ 绕顶点 O 逆时针旋转到 $\triangle A'O'B'$ 处, $\therefore AO = A'O' = 4, A'B' = AB = 4\sqrt{5}$. $\because$ 点 E 为 BO 的中点, $\therefore OE = \frac{1}{2}BO = \frac{1}{2} \times 8 = 4, \therefore OE = A'O' = 4$, 过点 O 作 $OF \perp A'B'$ 于点 F , 则 $S_{\triangle A'O'B'} = \frac{1}{2} \times 4\sqrt{5} \cdot OF = \frac{1}{2} \times 4 \times 8$, 解得 $OF = \frac{8\sqrt{5}}{5}$, 在 $Rt\triangle EOF$ 中, $EF = \sqrt{OE^2 - OF^2} = \sqrt{4^2 - \left(\frac{8\sqrt{5}}{5}\right)^2} = \frac{4\sqrt{5}}{5}$, $\therefore OE = A'O', OF \perp A'B', \therefore A'E = 2EF = 2 \times \frac{4\sqrt{5}}{5} = \frac{8\sqrt{5}}{5}$. $\therefore B'E = A'B' - A'E = 4\sqrt{5} - \frac{8\sqrt{5}}{5} = \frac{12\sqrt{5}}{5}$. 故选 B.

5. D 【解析】如图, 设 AB 的延长线与 x 轴的交点为 P_1 , 把 $A\left(\frac{1}{2}, y_1\right), B\left(2, y_2\right)$ 分别代入反比例函数 $y = \frac{1}{x}$, 得 $y_1 = 2, y_2 = \frac{1}{2}$. $\therefore A\left(\frac{1}{2}, 2\right), B\left(2, \frac{1}{2}\right)$. 在 $\triangle ABP$ 中, 由三角形的三边关系定理, 得 $AP - BP < AB$, 当 P, A, B 三点共线时, $AP - BP = AB$, 所以对 x 轴正半轴上的点, 总有 $AP - BP \leq AB$, 所以线段 AP 与 BP 差的最大值是线段 AB 的长, 此时点 P 即为直线 AB 与 x 轴的交点 P_1 , 设直线 AB 的解析式为 $y = kx + b$, 把点 A, B 的坐标代入, 得 $\begin{cases} \frac{1}{2}k + b = 2, \\ 2k + b = \frac{1}{2}. \end{cases}$ 解得 $\begin{cases} k = -1, \\ b = \frac{5}{2}. \end{cases}$ $\therefore$ 直线 AB 的解析式是 $y = -x + \frac{5}{2}$, $\therefore$ 当 $y = 0$ 时, $x = \frac{5}{2}$, 即 $P\left(\frac{5}{2}, 0\right)$. 故选 D.



第 5 题图

6. B 【解析】 $\because$ 在边长为 4 的菱形 $ABCD$ 中, $\angle ABC = 60^\circ$, $\therefore AB = CD = 4, \angle ABD = 30^\circ$. $\because \triangle ABD$ 将沿射线 BD 的方向平移得到 $\triangle EGF$, $\therefore EG = AB = 4, EG \parallel AB$. $\because$ 四边形 $ABCD$ 是菱形, $\therefore AB = CD, AB \parallel CD$, $\therefore \angle BAD = 120^\circ, \therefore EG = CD, EG \parallel CD$. 连接 ED , $\therefore$ 四边形 $EGCD$ 是平行四边形, $\therefore ED = GC$, $\therefore EC + GC$ 的最小值 $= EC + ED$ 的最小值. $\because$ 点 E 在过点 A 且平行于 BD 的定直线上, $\therefore$ 作点 D 关于定直线的对称点 M , 连接 CM 交定直线于 E , 则 CM 的长度即为 $EC + ED$ 的最小值. $\because \angle EAD = \angle ADB = 30^\circ, AD = 4, \therefore \angle ADM = 60^\circ, DH = MH = \frac{1}{2}AD = 2, \therefore DM = 4, \therefore DM = CD$. $\because \angle CDM = \angle MDG + \angle CDB = 90^\circ + 30^\circ = 120^\circ, \therefore \angle M = \angle DCM = 30^\circ, CM = 2 \times \frac{\sqrt{3}}{2}CD = 4\sqrt{3}$. 故选 B.



第 6 题图

7. 2 020 【解析】 $\because a + b + ab + 1 = (a+1)(b+1)$, 及后来的数加 1 是被擦去的 2 个数加 1 的乘积. 设最终黑板上剩下的数为 x , 则 $x+1 = (1+1) \times \left(\frac{1}{2}+1\right) \times \left(\frac{1}{3}+1\right) \times \left(\frac{1}{4}+1\right) \times \cdots \times \left(\frac{1}{2020}+1\right) = 2 \times \frac{3}{2} \times \frac{4}{3} \times \cdots \times \frac{2021}{2020} = 2021$. 最终黑板上剩下的数是 2 020.

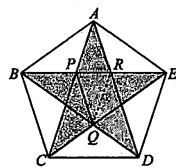
8. -2 022 【解析】 $x^2 + y^2 - xy + 2x - y + 1 = 0$, 两边乘 2 得 $2x^2 + 2y^2 - 2xy + 4x - 2y + 2 = 0, x^2 + y^2 - 2xy + 2x - 2y + 1 + x^2 + 2x + 1 + y^2 = 0$, 配方得 $(x-y)^2 + 2(x-y) + 1 + (x+1)^2 + y^2 = 0, (x-y+1)^2 + (x+1)^2 + y^2 = 0, x-y+1=0, x+1=0, y=0$, 解得 $x = -1, y = 0, \therefore 2 022x + 2 023y = -2 022 + 0 = -2 022$.

9. $\frac{1}{5}$ 【解析】六个人选取 2 人到社区 1 有 $\frac{6 \times 5}{2} = 15$ 种可能, 余下 4 人选取 2 人到社区 2 有 $\frac{4 \times 3}{2} = 6$ 种可能, 最后 2 人只能到社区 3, 所以, 所有可能的情况有 $15 \times 6 \times 1 = 90$ 种. 再考虑甲乙在一起的情况, 把甲乙看成一个整体, 则分到 3 个社区有 3 种可能, 余下 4 人选取 2 人到社区 2 有 $\frac{4 \times 3}{2} = 6$ 种可能, 最后 2 人只能到社区 3, 所以, 甲乙在一起所有可能的情况有 $3 \times 6 \times 1 = 18$ 种, 则甲、乙两名志愿者分到同一个社区的概率为 $\frac{18}{90} = \frac{1}{5}$.

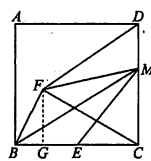
10. $\sqrt{2} - \sqrt{3}$

【解析】 $\frac{1+a}{1-a} = \frac{1+\frac{\sqrt{3}+1-\sqrt{2}}{\sqrt{3}-1-\sqrt{2}}}{1-\frac{\sqrt{3}+1-\sqrt{2}}{\sqrt{3}-1-\sqrt{2}}} = \frac{\sqrt{3}-1-\sqrt{2}}{1-\sqrt{3}-1+\sqrt{2}} = \frac{\sqrt{3}-1-\sqrt{2}}{-\sqrt{3}+\sqrt{2}} = \frac{\sqrt{3}-1-\sqrt{2}}{\sqrt{2}-\sqrt{3}} = \sqrt{2} - \sqrt{3}$.

11. $\frac{1}{2}$ 【解析】如图, 连接 RQ , $\because$ 由五角星性质可知四边形 $APQR$ 为平行四边形, $\therefore \triangle PRA$ 与 $\triangle PQR$ 面积相等, $\therefore 6S_1 + 2S_2 = 1, \therefore S_{\text{四边形}APQD} = 3S_1 + S_2 = \frac{1}{2}$.



第 11 题图



第 12 题图

12. $\frac{5}{2}$ 【解析】如图, 取 $BC = \frac{1}{2}$, 连接 FG , $\because BC = 2, E$ 是 BC 中点, $\therefore BE = 1$, 由翻折性质可知 $BF = BE = 1, \therefore BF = 1, BC = 2, GB = \frac{1}{2}, \therefore BF^2 = BC \cdot GB, \therefore \frac{BF}{BC} = \frac{GB}{FB}$, 又 $\because \angle FBG = \angle FBC, \therefore \triangle BCF \sim \triangle BFC, \therefore \frac{FG}{FC} = \frac{BF}{BC} = \frac{1}{2}, \therefore FG = \frac{1}{2}FC, \therefore DF + \frac{1}{2}FC = DF + FG \geq DG = \sqrt{DC^2 + CG^2} = \sqrt{2^2 + \left(\frac{3}{2}\right)^2} = \frac{5}{2}$.

13. 解: (1) 设两位数的十位数字为 a , 个位数字为 b , 则 $\frac{n}{S(n)} = \frac{10a+b}{a+b} = \frac{a+b+9a}{a+b} = 1 + \frac{9a}{a+b} = 1 + \frac{9}{1+\frac{b}{a}}$, 要使 $\frac{10a+b}{a+b}$ 值最小, 则 $\frac{b}{a}$ 最大, $\therefore a = 1, b = 9, \therefore \frac{19}{1+9} = 1.9$.

(2) 设百位数字为 a , 十位数字为 b , 个位数字为 c , 要使 $\frac{n}{S(n)} = \frac{100a+10b+c}{a+b+c}$ 的值最小, $\frac{n}{S(n)} = 1 + \frac{99a+9b}{a+b+c}$, 其值最小, 则 $a=1, c=9, \therefore \frac{n}{S(n)} = \frac{99+9b}{10+b} + 1 = 10 + \frac{9}{10+b}, \therefore b=9, \therefore \frac{n}{S(n)} = \frac{199}{19} = 10\frac{9}{19}$. (3) 设千位数字为 a , 百位数字为 b , 十位数字为 c , 个位数字为 d , 则 $\frac{n}{S(n)} = \frac{1000a+100b+10c+d}{a+b+c+d} = 1 + \frac{999a+99b+9c}{a+b+c+d}$, 其值最小, 则 $a=1, d=9, \therefore \frac{n}{S(n)} = \frac{999+99b+9c}{b+c+10} + 1$, 类似分析, $b=0, c=9$ 时符合题意, $\frac{n}{S(n)}$ 最小值为 $\frac{1099}{19}$.

14. 解: (1) 第 10 个 1 是 $1+2+3+\cdots+9+1=46$ 项. (2) $1+2+3+\cdots+62+63 = \frac{(1+63) \times 63}{2} = 2016$, 第 2 018 项应为第 64 组的第 2 项, 该数列的第 2 018 项为 2. (3) 由题意, 这列数分为 $(n+1)$ 组: 前 n 组含有的项数分别为 $1, 2, 3, \dots, n$, 最后一组 x 项, 前 n 组每组的和为 $2^1 - 1, 2^2 - 1, 2^3 - 1, \dots, 2^n - 1$, 前 n 组所有项数的和 $S_n = 1 + (1+2) + (1+2+3) + \cdots + (1+2+\cdots+n-1) = 2^1 - 1 + 2^2 - 1 + 2^3 - 1 + \cdots + 2^n - 1 = 2^{n+1} - n - 2$, 前 n 组共有的项数为 $N = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$. 由题意得, 2^{n+1} 为 2 的整数次幂, 只需将最后一组 x 项的和与 $-2-n$ 消去即可. ① $1+2+(-2-n) = 0, n=1$, 总项数 $N = \frac{1 \times (1+1)}{2} + 2 = 3, N < 100$, 舍去. ② $1+2+4+(-2-n) = 0, n=5$, 总项数 $N = \frac{(1+5) \times 5}{2} + 3 = 18, N < 100$, 舍去. ③ $1+2+4+8+(-2-n) = 0, n=13$, 总项数 $N = \frac{(1+13) \times 13}{2} + 4 = 95, N < 100$, 舍去. ④ $1+2+4+8+16+(-2-n) = 0, n=29$, 总项数 $N = \frac{(1+29) \times 29}{2} + 5 = 440, N$ 符合, $\therefore$ 最小整数 $N = 440$.

15. (1) 证明: 连接 OC , $\because AB$ 是 $\odot O$ 的直径, C 是弧 AB 的中点, $\therefore OC \perp AB, \therefore CD = AC, OA = OB, \therefore OC$ 为 $\triangle ABD$ 的中位线. $\therefore OC \parallel BD, \therefore BD \perp AB$. 又点 B 在 $\odot O$ 上, $\therefore BD$ 是 $\odot O$ 的切线. (2) 解: $\because E$ 是 OB 的中点, $\therefore OE = BE, \therefore OC \parallel BD, \therefore \triangle OCE \sim \triangle BFE$. $\therefore \frac{OC}{BF} = \frac{OE}{BE}$. 又 $\because BF = 1, \therefore OC = 1$. 在 $Rt\triangle ABF$ 中, $AF = \sqrt{AB^2 + BF^2} = \sqrt{2^2 + 1^2} = \sqrt{5}$. $\because AB$ 是 $\odot O$ 的直径, $\therefore \angle AHB = 90^\circ, \therefore \frac{1}{2}AF \cdot BH = \frac{1}{2}AB \cdot BF, \therefore BH = \frac{2 \times 1}{\sqrt{5}} = \frac{2\sqrt{5}}{5}$. $\therefore BH$ 的长为 $\frac{2\sqrt{5}}{5}$.

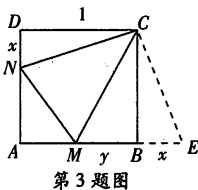
16. 解: (1) 设建造 A 型处理点 x 个, 则建造 B 型处理点 $(20-x)$ 个. 依题意有 $\begin{cases} 15x + 20(20-x) \leq 370, \\ 18x + 30(20-x) \geq 490, \end{cases}$ 解得 $6 \leq x \leq 9.7$. $\because x$ 为整数, $\therefore x = 6, 7, 8, 9$ 有四种方案. 设建造 A 型处理点 x 个时, 总费用为 y 万元. 则 $y = 1.5x + 2.1(20-x) = -0.6x + 42$. $\because -0.6 < 0, \therefore y$ 随 x 增大而减小, 当 $x = 9$ 时, y 的值最小, 此时 $y = 36.6$ (万元), $\therefore$ 当建造 A 型处理点 9 个, 建造 B 型处理点 11 个时最省钱.

(2)由题意得,每吨垃圾的处理成本为 $\frac{y}{x}$ (元/吨),当 $0 \leq x < 144$ 时, $\frac{y}{x} = \frac{1}{3}x^2 - 80x + 5040$, $\therefore \frac{1}{3} > 0$,故 $\frac{y}{x}$ 有最小值,当 $x = -\frac{b}{2a} = -\frac{-80}{2 \times \frac{1}{3}} = 120$ (吨)时, $\frac{y}{x}$ 的最小值为240(元/吨);当 $144 \leq x < 300$ 时, $\frac{y}{x} = \frac{1}{x}(10x + 72000) = 10 + \frac{72000}{x}$,当 $x = 300$ (吨)时, $\frac{y}{x} = 250$,即 $\frac{y}{x} > 250$ (元/吨). $\therefore 240 < 250$,故当 $x = 120$ 吨时, $\frac{y}{x}$ 的最小值为240元/吨. $\therefore$ 每个B型处理点的垃圾月处理量是A型处理点的1.2倍且A型处理点9个,建造B型处理点11个, $\therefore$ 每个A型处理点每月处理量 $= \frac{9 \times 1}{9 \times 1 + 11 \times 1.2} \times 120 \times \frac{1}{9} \approx 5.4$ (吨),故每个A型处理点每月处理量为5.4吨时,才能使该街道每月垃圾的月处理成本最低.

17. 解:(1) $\because x + \frac{4}{x} \geq 2\sqrt{x \times \frac{4}{x}} = 4$, $\therefore y_{\text{最大}} = 2 - 4 = -2$.
 (2) $y = 9x + \frac{1}{x-1} = 9(x-1) + \frac{1}{x-1} + 9 \geq 2\sqrt{9(x-1) \times \frac{1}{x-1}} + 9 = 15$,
 当 $9(x-1) = \frac{1}{x-1}$ 时,即当 $x = \frac{4}{3}$ 时, $y_{\text{最小}} = 15$.
 (3)把 $x = -2, y = 3$ 代入 $y = \frac{k}{x}$ 得 $3 = \frac{k}{-2}$, $\therefore k = -6$, $\therefore y = -\frac{6}{x}$,设点A(a,0),B(0,b),(a>0,b<0), $\therefore$ 直线AB的解析式为 $y = -\frac{b}{a}x + b$,
 由 $-\frac{6}{x} = -\frac{b}{a}x + b$ 得 $bx^2 - abx - 6a = 0$,
 $\therefore$ 直线AB与双曲线 $y = \frac{k}{x}$ 只有一个公共点, $\therefore \Delta = (-ab)^2 + 24ab = 0$,
 $\therefore b = -\frac{24}{a}$.由 $y = \frac{3}{2}x + 6$ 得D(0,6),C(-4,0),
 $\therefore AC = a + 4, BD = 6 - b = 6 + \frac{24}{a}$, $\therefore S_{\text{四边形ABCD}} = \frac{1}{2}AC \cdot BD = \frac{1}{2}(a + 4) \cdot (6 + \frac{24}{a}) = 3(a + \frac{16}{a}) + 24 \geq 3 \times 2\sqrt{a \times \frac{16}{a}} + 24 = 48$, $\therefore$ 当 $a = \frac{16}{a}$,即 $a = 4$ 时,四边形ABCD的面积最小值为48.

全国重点高中提前招生 考试全真试卷(三)

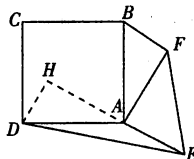
1. C 【解析】原式 $= \left| \frac{(a-b)^2 - (a-b)}{a-b} \right| = \left| \frac{(a-b)(a-b-1)}{a-b} \right| = |a-b-1|$,由数轴可知 $a-b < 0$,原式 $= -(a-b-1) = -a+b+1$.故选C.
 2. C 【解析】 $\because$ 函数 $y = x^2 + x - 1$ 的对称轴为直线 $x = -\frac{1}{2}$, $\therefore$ 当 $x = -\frac{1}{2}$ 时,y有最小值,此时 $y = \frac{1}{4} - \frac{1}{2} - 1 = -\frac{5}{4}$, $\therefore$ 函数 $y = x^2 + x - 1$ 在 $m \leq x \leq 1$ 上的最小值是 $-\frac{5}{4}$, $\therefore m \leq -\frac{5}{4}$;当 $x = 1$ 时, $y = 1 + 1 - 1 = 1$,对称轴为直线 $x = -\frac{1}{2}$, $\therefore$ 当 $x = -\frac{1}{2} - [1 - (-\frac{1}{2})] = -2$ 时, $y = 1$, $\therefore$ 函数 $y = x^2 + x - 1$ 在 $m \leq x \leq 1$ 上的最大值是1,且 $m \leq -\frac{5}{4}$, $\therefore -2 \leq m \leq -\frac{5}{4}$.故选C.
 3. D 【解析】如图所示,把 $\triangle CDN$ 绕点C逆时针旋转 90° 至 $\triangle CBE$ 的位置,显然 $\triangle CDN \cong \triangle CBE$, $\therefore DN = BE, CN = CE, \angle DCN = \angle BCE$.不妨设 $DN = x, BM = y$,则 $AN = 1 - x, AM = 1 - y, ME = x + y$. $\therefore \triangle AMN$ 周长为2, $\therefore MN = 2 - (1 - x) - (1 - y) = x + y$, $\therefore MN = ME$, $\therefore \triangle CNM \cong \triangle CEM$, $\therefore \angle NCM = \angle ECM$.由旋转的性质知 $\angle NCE = 90^\circ$,即 $\angle NCM + \angle ECM = 90^\circ$,而 $\angle NCM = \angle ECM$, $\therefore \angle MCN = 45^\circ$,②正确; $\because S_{\triangle CDN} + S_{\triangle CEM} = \frac{1}{2}x + \frac{1}{2}y, S_{\triangle CMN} = S_{\text{正方形ABCD}} - S_{\triangle CDN} - S_{\triangle CEM} - S_{\triangle AMN} = 1 - \frac{1}{2}x - \frac{1}{2}y - \frac{1}{2}(1-x)(1-y) = \frac{1}{2} - \frac{1}{2}xy$,在Rt $\triangle AMN$ 中,由勾股定理得 $(1-x)^2 + (1-y)^2 = (x+y)^2$,整理得 $2 - 2x - 2y = 2xy$, $\therefore x + y = 1 - xy$, $\therefore \frac{1}{2}x + \frac{1}{2}y = \frac{1}{2} - \frac{1}{2}xy$, $\therefore S_{\triangle CMN} = S_{\triangle CDN} + S_{\triangle CEM}$,①正确;不妨设C点到MN的距离为d, $\therefore S_{\triangle CMN} = \frac{1}{2}x + \frac{1}{2}y = \frac{1}{2}(x+y)$,又 $\because S_{\triangle CMN} = \frac{1}{2} \times MN \times d$, $\therefore \frac{1}{2}(x+y) = \frac{1}{2} \times MN \times d$, $\therefore MN = x+y$, $\therefore d = 1$,③正确.故选D.
 4. C 【解析】 $\because$ 点B(1,5)在双曲线 $y = \frac{k}{x}(k \neq 0)$ 的图象上, $\therefore k = 5$. $\because A(0,1)$,曲线AB与BC组成图形G.由点C开始不断重复图形G形成一条“波浪线”, $\therefore$ 点C的纵坐标为1. $\therefore$ 点C在 $y = \frac{5}{x}(k \neq 0)$ 的图象上, $\therefore$ 点C的



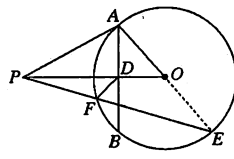
第3题图

横坐标是5, $\therefore$ 点C的坐标为(5,1). $\because 2020 + 5 = 404$, $\therefore P(2020, m)$ 中 $m = 1$, $\therefore$ 点Q(x,n)在该“波浪线”上, $\therefore n$ 的最大值是5.综上所述, $m = 1, n = 5$.故选C.

5. A 【解析】连接OE,则可得 $\angle FOE = \angle DEH$, $\therefore AF = 3, FB = \frac{4}{3}$,
 $\therefore AB = AF + BF = \frac{13}{3}$, $\therefore OB = \frac{1}{2}AB = \frac{13}{6}$, $\therefore OF = OB - FB = \frac{5}{6}$.
 在Rt $\triangle OEF$ 中, $EF = \sqrt{OE^2 - OF^2} = \sqrt{\left(\frac{13}{6}\right)^2 - \left(\frac{5}{6}\right)^2} = 2$, $\therefore \tan \angle DEH = \tan \angle EOF = \frac{EF}{OF} = \frac{2}{\frac{5}{6}} = \frac{12}{5}$.故选A.
 6. A 【解析】依题意得 $(2a + b + 3)^2 - 4(a^2 + ab + 6) = 0$,即 $(b + 3)^2 = 12(2 - a)$
 ①, $(4a - 2b - 2)^2 - 8a(2a - 2b - 1) = 0$,即 $(b + 1)^2 = 2a$ ②,①+② $\times 6$ 得
 $7b^2 + 18b - 9 = 0$, $(b + 3)(7b - 3) = 0$, $b_1 = -\frac{3}{7}$ (舍), $b_2 = -3$, $\therefore b = -3, a = 2$,
 $\therefore$ 两个方程分别为 $x^2 + 4x + 4 = 0$ 和 $4x^2 + 12x + 9 = 0$, $\therefore \alpha = -2, \beta = -\frac{3}{2}$,以
 α, β 为根的整系数一元二次方程是 $2x^2 + 7x + 6 = 0$.故选A.
 7. 6 【解析】 $\because 0 < m + \frac{1}{2022} < m + \frac{2}{2022} < \dots < m + \frac{2021}{2022} < 2$,
 $\therefore [m + \frac{1}{2022}], [m + \frac{2}{2022}], [m + \frac{3}{2022}], \dots, [m + \frac{2021}{2022}]$ 等于0或1,由
 题设知其中有21个等于1,有2000个等于0,
 $\therefore \begin{cases} 0 < m + \frac{2001}{2022} < 1 \\ 1 \leq m + \frac{2002}{2022} < 2 \end{cases}$,解得 $\frac{20}{2022} \leq m < \frac{21}{2022}$, $\therefore \frac{20}{3} \leq 674m < 7$,
 $\therefore [674m] = 6$.
 8. (2,1) 【解析】方程 $x + my - m - 2 = 0$ 可化为 $(x - 2) + m(y - 1) = 0$,
 $\therefore$ 对于任意实数m,直线 $x + my - m - 2 = 0$ 恒过定点,
 $\therefore \begin{cases} x - 2 = 0 \\ y - 1 = 0 \end{cases}$,故定点坐标是(2,1).
 9. 2 【解析】设直角三角形三边为a, b, c,且 $a < b < c$,则
 $\begin{cases} a^2 + b^2 = c^2 \\ ab = 2(a + b + c) \end{cases}$,有 $c^2 = a^2 + b^2 < 2b^2$, $\therefore c < \sqrt{2}b, ab < 2(a + b + \sqrt{2}b) < 2(b + b + \sqrt{2}b) = (2 + 2\sqrt{2})b < 7b$, $\therefore a < 7$,而 $ab = 2(a + b + c) > 2(a + b + b)$, $\therefore a > 4$,又 $\because a$ 为整数, $\therefore a$ 可取5或6.当 $a = 5$ 时, $c^2 - b^2 = 25$, $(c + b)(c - b) = 25 \times 1, c = 13, b = 12$ 符合;当 $a = 6$ 时, $36 = (c + b)(c - b) = 18 \times 2, c = 10, b = 8$ 符合题意.综上 $(a, b, c) = (6, 8, 10)$ 或 $(5, 12, 13)$.
 10. 9 【解析】由三个半圆依次与直线 $y = \frac{\sqrt{3}}{3}x$ 相切并且圆心都在x轴上,
 $\therefore$ 直线 $y = \frac{\sqrt{3}}{3}x$ 的倾斜角是 30° , $\therefore OO_1 = 2r_1, OO_2 = 2r_2 = OO_1 + r_1 + r_2 = 3r_1 + r_2, OO_3 = 2r_3, 2r_2 = 3r_1 + r_2, \therefore r_2 = 3r_1, \therefore r_1 = 1, \therefore OO_1 = 2, OO_2 = 2r_2 = 6r_1 = 6, OO_3 = 18, \therefore r_3 = 9$.
 11. 24 【解析】作 $DH \perp AE$ 于H,如图, $\because AF = 8$,当 $\triangle AEF$ 绕点A旋转时,点F在以A为圆心,8为半径的圆上, $\therefore$ 当BF为此圆的切线时, $\angle ABF$ 最大,即 $BF \perp AF$.在Rt $\triangle ABF$ 中, $BF = \sqrt{AB^2 - AF^2} = \sqrt{100 - 64} = 6$,
 $\therefore \angle EAF = 90^\circ, \therefore \angle BAF + \angle BAH = 90^\circ. \therefore \angle DAH + \angle BAH = 90^\circ$,
 $\therefore \angle AHD = \angle AFB$,
 $\therefore \angle DAH = \angle BAF$.在 $\triangle ADH$ 和 $\triangle ABF$ 中, $\begin{cases} \angle DAH = \angle BAF \\ \angle AHD = \angle AFB \end{cases}, \therefore \triangle ADH \cong \triangle ABF$ (AAS), $\therefore DH = BF = 6, \therefore S_{\triangle ADE} = \frac{1}{2}AE \cdot DH = \frac{1}{2} \times 6 \times 8 = 24$.



第11题图



第12题图

12. $\frac{3}{10}$ 【解析】如图,连接OE,设 $OA = 3a, OD = 2a$,则 $AO^2 = OD \cdot PO$,
 $PO = \frac{9}{2}a, PD = \frac{5}{2}a$.由切割线定理得 $PF \cdot PE = PA^2$,由射影定理,得
 $PA^2 = PD \cdot PO, \therefore PF \cdot PE = PD \cdot PO, \therefore \frac{PF}{PD} = \frac{PO}{PE}$,且 $\angle DPF = \angle OPE$,
 $\therefore \triangle PFD \sim \triangle POE, \therefore \frac{DF}{OE} = \frac{PD}{PE} = \frac{PF}{PO}$,代入数据解得 $PE = \frac{5\sqrt{6}}{2}a, PF = \frac{3\sqrt{6}}{4}a, \therefore \frac{PF}{PE} = \frac{3}{10}$.
 13. 解:(1)依题意可得 $S(351) = \frac{531 + 153 + 315}{111} = 9$,
 $S(985) = \frac{895 + 589 + 958}{111} = 22$.
 (2)为方便起见,不妨设三位数a百位上数字为p,十位上数字为q,个位上数字为r,那么这个三位数可表示为 $100p + 10q + r$,依题意易知三个新“互异好数”分别可表示为 $100q + 10p + r, 100r + 10q + p, 100p + 10r + q$,则 $S(a) =$

$$\frac{100q+10p+r+100r+10q+p+100p+10r+q}{111} = \frac{111p+111q+111r}{111} = p+q+r$$

$r, \therefore S(m) = S(200x+15) = 2x+1+5 = 2x+6, S(n) = S(230+y) = 2+3+y = 5+y$. 当 $S(m) + 2S(n) = 26$ 时, 有 $2x+6+2(5+y) = 26, \therefore x+y=5$. 又 $\because n=230+y$ 是“互异好数”, $1 \leq x \leq 4, 1 \leq y \leq 9, x, y$ 是正整数, $\therefore \begin{cases} x=1, \\ y=4 \end{cases}$ 或 $\begin{cases} x=4, \\ y=1 \end{cases}$. 当 $\begin{cases} x=1, \\ y=4 \end{cases}$ 时, $\lambda = \frac{S(m)}{S(n)} = \frac{2x+6}{5+y} = \frac{2 \times 1 + 6}{5+4} = \frac{8}{9}$; 当 $\begin{cases} x=4, \\ y=1 \end{cases}$ 时, $\lambda = \frac{S(m)}{S(n)} = \frac{2x+6}{5+y} = \frac{2 \times 4 + 6}{5+1} = \frac{7}{3}$. 综上所述, λ 的值为 $\frac{8}{9}$ 或 $\frac{7}{3}$.

14. 证明: (1) $\because \triangle ABE$ 是等边三角形, $\therefore BA = BE, \angle ABE = 60^\circ, \therefore \angle MBN = 60^\circ, \therefore \angle MBN - \angle ABN = \angle ABE - \angle ABN$ 即 $\angle ABM = \angle NBE$, 又 $\because MB = NB, \therefore \triangle ABM \cong \triangle ENB$ (SAS).

(2) 解: ① 当点 M 落在 BD 的中点时, A, M, C 三点共线, $AM + CM$ 的值最小. ② 连接 CE , 当 M 点位于 BD 与 CE 的交点处时, $AM + BM + CM$ 的值最小, 理由如下: 连接 MN , 由 (1) 知 $\triangle ABM \cong \triangle ENB, \therefore AM = EN, \therefore \angle MBN = 60^\circ, MB = NB, \therefore \triangle MBN$ 是等边三角形, $\therefore BM = MN, \therefore AM + BM + CM = EN + MN + CM$, 根据两点之间线段最短, 得 $EN + MN + CM = EC$ 最短, 可证 $\triangle ABM \cong \triangle CBM$, 故 $\angle BAM = \angle BCM = \angle BEN, \therefore EB = CB, \therefore \angle BEC = \angle BCE, \therefore \angle BCM = \angle BCE, \angle BEN = \angle BEC, \therefore M, N$ 可同时在 EC 上, $\therefore$ 当 M 点位于 BD 与 CE 的交点处时, $AM + BM + CM$ 的值最小, 即等于 EC 的长.

(3) 解: 过 E 点作 $EF \perp BC$ 交 CB 的延长线于点 $F, \therefore \angle EBF = 90^\circ - 60^\circ = 30^\circ$, 设正方形的边长为 x , 则 $BF = \frac{\sqrt{3}}{2}x, EF = \frac{x}{2}$. 在 $Rt\triangle EFC$ 中, $\therefore EF^2 + FC^2 = EC^2, \therefore \left(\frac{x}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}x + x\right)^2 = (\sqrt{3}+1)^2$, 解得 $x = \pm\sqrt{2}$ (舍去负值), $\therefore$ 正方形的边长为 $\sqrt{2}$.

15. (1) 证明: $\because BA = BC, \angle ABC = 90^\circ, \therefore \angle BAC = 45^\circ, \therefore AB$ 是 $\odot O$ 的直径, $\therefore \angle ADB = \angle AEB = 90^\circ, \therefore \angle ADF = \angle BDG = 90^\circ, \therefore \angle DAF + \angle BGD = \angle DBG + \angle BCD = 90^\circ, \therefore \angle DAF = \angle DBG, \therefore \angle ABD + \angle BAC = 90^\circ, \therefore \angle ABD = \angle BAC = 45^\circ, \therefore AD = BD, \therefore \triangle ADF \cong \triangle BDG$ (ASA).

(2) 解: 连接 $OE, \therefore$ 四边形 $OBEH$ 为菱形, $\therefore OB = BE$. 又 $\because OE = OB, \therefore \triangle OBE$ 是等边三角形, $\therefore \angle EBO = 60^\circ$. 又 $\because \angle AEB = 90^\circ, \therefore \angle EAB = 30^\circ$.

(3) 解: 连接 $OD, OE, \therefore$ 点 E 是弧 BD 上靠近点 B 的三等分点, $\therefore \angle DOE = \frac{2}{3} \angle DOB$, 由 (1) 知 $\triangle ADB$ 是等腰直角三角形, $\therefore \angle DOB = 90^\circ, \therefore \angle DOE = 60^\circ, \therefore \angle DBE = 30^\circ, DG = BD \cdot \tan 30^\circ$. 又 $\because AB = 4, \therefore BD = 2\sqrt{2}, \therefore DG = 2\sqrt{2} \times \frac{\sqrt{3}}{3} = \frac{2\sqrt{6}}{3}$.

16. (1) ① 证明: $\because$ 四边形 $ABCD$ 和 $EFCD$ 均为正方形, $\therefore \frac{AC}{BC} = \frac{CE}{CF} = \sqrt{2}, \therefore \angle ACB = \angle ECF = 45^\circ, \therefore \angle ACE = \angle BCF, \therefore \triangle ACE \sim \triangle BCF$.

② 解: $\because \triangle ACE \sim \triangle BCF, \therefore \angle CAE = \angle CBF, \frac{AE}{BF} = \frac{AC}{BC}$. 又 $\because \angle CAE + \angle CBE = 90^\circ, \therefore \angle CBF + \angle CBE = 90^\circ, \therefore \angle EBF = 90^\circ, \therefore \frac{AE}{BF} = \frac{AC}{BC} = \sqrt{2}, AE = 2, \therefore BF = \sqrt{2}, \therefore EC = \sqrt{6}, \therefore EF = CF = \sqrt{3}, \therefore BE = \sqrt{EF^2 - BF^2} = \sqrt{3 - 2} = 1$.

(2) 解: 如图①, 连接 $BF, \therefore \angle ACB = \angle ECF = \alpha, \therefore \tan \angle ACB = \tan \angle ECF, \therefore \frac{AB}{BC} = \frac{EF}{FC}$.

设 $\tan \alpha = \frac{AB}{BC} = \frac{EF}{FC} = k, BC = a, FC = b$, 则 $AB = ka, EF = kb$, $\therefore AC = \sqrt{AB^2 + BC^2} = a \cdot \sqrt{k^2 + 1}, CE = \sqrt{EF^2 + CF^2} = b \cdot \sqrt{k^2 + 1}, \therefore \frac{AC}{BC} = \frac{EC}{FC} = \sqrt{k^2 + 1}, \angle ACE = \angle BCF, \therefore \triangle ACE \sim \triangle BCF$,

$\therefore \frac{AE}{BF} = \frac{AC}{BC} = \sqrt{k^2 + 1}, \angle CAE = \angle CBF$. 又 $\because AE = 6, \therefore \frac{6}{BF} = \sqrt{k^2 + 1}, \therefore BF = \frac{6}{\sqrt{k^2 + 1}}$.

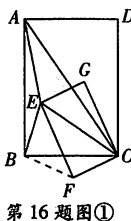
$\because \angle CAE = \angle CBF, \angle CAE + \angle CBE = 90^\circ, \therefore \angle CBE + \angle CBF = 90^\circ, \therefore \angle EBF = 90^\circ, \therefore EF^2 = BE^2 + BF^2 = 9 + \frac{36}{k^2 + 1}$.

$\therefore \frac{EC}{CF} = \sqrt{k^2 + 1}, \therefore \frac{EC}{EF} = \frac{\sqrt{k^2 + 1}}{k}, \therefore CE = 5, \therefore EF = \frac{5k}{\sqrt{k^2 + 1}}$, $\therefore 9 + \frac{36}{k^2 + 1} = \frac{25k^2}{k^2 + 1}, \therefore k^2 = \frac{45}{16}$.

$\therefore k > 0, \therefore k = \frac{3\sqrt{5}}{4}, \therefore \tan \alpha = \frac{3\sqrt{5}}{4}$.

(3) 解: 连接 BF , 同理可得 $\angle EBF = 90^\circ$, 过 C 点作 $CH \perp AB$ 交 AB 的延长线于 H , 如图②.

$\because$ 四边形 $ABCD$ 为菱形, $\therefore AB = BC$. 设 $AB = BC = x$, $\therefore \angle CBH = \angle DAB = 60^\circ$, $\therefore BH = \frac{1}{2}x, CH = \frac{\sqrt{3}}{2}x$,



第 16 题图①

$$\therefore AC^2 = AH^2 + CH^2 = \left(x + \frac{1}{2}x\right)^2 + \left(\frac{\sqrt{3}}{2}x\right)^2 = 3x^2, \therefore AB^2 : BC^2 : AC^2 = 1 : 1 : 3,$$

$$\text{同理可得 } EF^2 : FC^2 : EC^2 = 1 : 1 : 3, \therefore EC = p, \therefore EF^2 = \frac{p^2}{3}.$$

$$\therefore \frac{AC}{BC} = \frac{EC}{FC} = \sqrt{3}, \angle ACE = \angle BCF, \therefore \triangle ACE \sim \triangle BCF,$$

$$\therefore \frac{AE^2}{BF^2} = \frac{AC^2}{CB^2} = 3, \angle CAE = \angle CBF.$$

$$\text{又 } \because AE = n, \therefore BF^2 = \frac{n^2}{3}.$$

$$\therefore \angle EBF = 90^\circ, \therefore EF^2 = BE^2 + BF^2, \therefore \frac{p^2}{3} = m^2 + \frac{n^2}{3}, \therefore 3m^2 + n^2 = p^2.$$

$$\text{即 } m, n, p \text{ 三者之间满足的等量关系是 } 3m^2 + n^2 = p^2.$$

17. 解: (1) 对于抛物线 $y = -\frac{1}{3}x^2 + \frac{1}{3}x + 4$, 令 $y = 0$ 得 $y = -\frac{1}{3}x^2 + \frac{1}{3}x + 4 = 0$, 解得 $x = -3$ 或 4 . 令 $x = 0$ 得 $y = 4, \therefore A(-3, 0), B(4, 0), C(0, 4)$.

$\therefore x = 5$ 时, $y = -\frac{8}{3}, \therefore E\left(5, -\frac{8}{3}\right)$. 设直线 AD 的解析式为 $y = kx + b$, 则有 $\begin{cases} -3k + b = 0, \\ 5k + b = -\frac{8}{3}, \end{cases}$ 解得 $\begin{cases} k = -\frac{1}{3}, \\ b = -1, \end{cases}$ $\therefore$ 直线 AD 的解析式为 $y = -\frac{1}{3}x - 1$.

(2) 如图①, 设 $F\left(m, -\frac{1}{3}m^2 + \frac{1}{3}m + 4\right)$, 则 $G\left(m, -\frac{1}{3}m - 1\right)$, $FG = -\frac{1}{3}m^2 + \frac{2}{3}m + 5, \therefore FG \parallel CD, FH \parallel AC, \therefore \angle FGH = \angle CAD,$

$\angle FGH = \angle CDA, \therefore \angle FGH, \angle FHG$ 是定值. $\therefore$ 当 FG 最大时, $\triangle FGH$ 的周长最短. $\therefore FG = -\frac{1}{3}m^2 + \frac{2}{3}m + 5 = -\frac{1}{3}(m-1)^2 + \frac{16}{3}, \therefore -\frac{1}{3} < 0, \therefore m = 1$ 时, FG 有最大值, 此时 $F(1, 4)$, 作点 F 关于 y 轴的对称点 F' , 连接 AF' , 延长 AF' 交 y 轴于点 T , 此时 $|TA - TF|$ 最大. $\therefore A(-3, 0), F'(-1, 4), \therefore$ 直线 AF' 的解析式为 $y = 2x + 6, \therefore$ 点 T 坐标为 $(0, 6)$.

(3) ① 如图②, 当 $\angle MNP = 90^\circ$, 重叠部分是 $\triangle MNP$ 是直角三角形. $\therefore \triangle FMN \sim \triangle COA, \therefore \frac{FN}{CO} = \frac{FM}{AC}, \therefore FN = -\frac{1}{3}m^2 + \frac{1}{3}m + 4, \therefore FM = PM = PG = \frac{5}{4}FN = \frac{5}{4}\left(-\frac{1}{3}m^2 + \frac{1}{3}m + 4\right), \therefore 2FN + PG = FG,$

$\therefore 2\left(-\frac{1}{3}m^2 + \frac{1}{3}m + 4\right) + \frac{5}{4}\left(-\frac{1}{3}m^2 + \frac{1}{3}m + 4\right) = -\frac{1}{3}m^2 + \frac{2}{3}m + 5$, 整理得 $9m^2 - 5m - 96 = 0$, 解得 $m = \frac{32}{9}$ 或 -3 (舍去), $\therefore FN = -\frac{1}{3} \times \left(\frac{32}{9}\right)^2 + \frac{1}{3} \times \frac{32}{9} + 4 = \frac{236}{243}$.

② 如图③中, 当 $PM \perp GF$ 时, 重叠部分是 $\triangle MNK$ 是直角三角形. $\therefore KM : FK : FM = 3 : 4 : 5, PM = PG, \therefore PK : PG = 2 : 5, \therefore KG : PK = \sqrt{21} : 2, \therefore \left(\frac{1}{3}m + 1\right) : \left(-\frac{1}{3}m^2 + \frac{1}{3}m + 4\right) = \sqrt{21} : 4,$

$\therefore (m+3) : (m+3)(-m+4) = \sqrt{21} : 4$, 解得 $m = 4 - \frac{4\sqrt{21}}{21}$ 或 -3 (舍去), $\therefore FK = \frac{28\sqrt{21}}{63} - \frac{16}{63}$.

$\therefore FN : NK = FM : MK = 5 : 3, \therefore FN = \frac{5}{8}FK = \frac{35\sqrt{21}}{126} - \frac{10}{63}$.

综上所述, 当 $\triangle PMN$ 与 $\triangle FGH$ 的重叠部分图形为直角三角形, 且 $PM \perp PG$ 时, 线段 FN 的长为 $\frac{236}{243}$ 或 $\frac{35\sqrt{21}}{126} - \frac{10}{63}$.

③ 如图④中, 当 $PM \perp GF$ 时, 重叠部分是 $\triangle MNK$ 是直角三角形. $\therefore KM : FK : FM = 3 : 4 : 5, PM = PG, \therefore PK : PG = 2 : 5, \therefore KG : PK = \sqrt{21} : 2, \therefore \left(\frac{1}{3}m + 1\right) : \left(-\frac{1}{3}m^2 + \frac{1}{3}m + 4\right) = \sqrt{21} : 4,$

$\therefore (m+3) : (m+3)(-m+4) = \sqrt{21} : 4$, 解得 $m = 4 - \frac{4\sqrt{21}}{21}$ 或 -3 (舍去), $\therefore FK = \frac{28\sqrt{21}}{63} - \frac{16}{63}$.

$\therefore FN : NK = FM : MK = 5 : 3, \therefore FN = \frac{5}{8}FK = \frac{35\sqrt{21}}{126} - \frac{10}{63}$.

综上所述, 当 $\triangle PMN$ 与 $\triangle FGH$ 的重叠部分图形为直角三角形, 且 $PM \perp PG$ 时, 线段 FN 的长为 $\frac{236}{243}$ 或 $\frac{35\sqrt{21}}{126} - \frac{10}{63}$.

④ 如图⑤中, 当 $PM \perp GF$ 时, 重叠部分是 $\triangle MNK$ 是直角三角形. $\therefore KM : FK : FM = 3 : 4 : 5, PM = PG, \therefore PK : PG = 2 : 5, \therefore KG : PK = \sqrt{21} : 2, \therefore \left(\frac{1}{3}m + 1\right) : \left(-\frac{1}{3}m^2 + \frac{1}{3}m + 4\right) = \sqrt{21} : 4,$

$\therefore (m+3) : (m+3)(-m+4) = \sqrt{21} : 4$, 解得 $m = 4 - \frac{4\sqrt{21}}{21}$ 或 -3 (舍去), $\therefore FK = \frac{28\sqrt{21}}{63} - \frac{16}{63}$.

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$\therefore FN : NK = FM : MK = 5 : 3, \therefore FN = \frac{5}{8}FK = \frac{35\sqrt{21}}{126} - \frac{10}{63}$.

第 16 题图②

第 17 题图

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1. C 【解析】 m, n 是方程 $x^2 + 2022x + 7 = 0$ 的两个根, $\therefore m + n = -2022, mn = 7, \therefore m^2 + 2022m + 7 = 0, n^2 + 2022n + 7 = 0, \therefore (m^2 + 2022m + 6) \cdot (n^2 + 2022n + 8) = (m^2 + 2022m + 7 - m - 1)(n^2 + 2022n + 7 + n + 1) = -(m+1)(n+1) = -(mn + m + n + 1) = -(7 - 2022 + 1) = 2014$. 故选 C.

2. C 【解析】当 $k = 0$ 时, 不符合题意; 当 $k \neq 0$ 时, 二次函数 $y = kx^2 + (k-6)x + 2 > 0$, 即函数图象与 x 轴无交点, $\therefore \Delta = (k-6)^2 - 4 \times k \times 2 < 0$, 解得 $(k-2)(k-18) < 0$, 即 $2 < k < 18$. 故选 C.

3. A 【解析】由题意得 $a^2 - 4 = 0$, 且 $a + 2 \neq 0$, 解得 $a = 2, \therefore 2a + 1 = 2 \times 2 + 1 = 5$. 故选 A.

4. A 【解析】因为 $AB+BC=9$, 设 $AB=m$, 则 $BC=9-m$. 当点 M 在 BC 上时, 则 $AB=m$, $BM=x-m$, $MC=9-x$, $NC=y$, $\therefore MN \perp AM$, 则 $\angle MAB = \angle NMC$, $\tan \angle MAB = \tan \angle NMC$, 即 $\frac{BM}{AB} = \frac{CN}{CM}$, 即 $\frac{x-m}{m} = \frac{y}{9-x}$, 化简得 $y = -\frac{1}{m}x^2 + \frac{9+m}{m}x - 9$, 当 $x = -\frac{b}{2a} = \frac{9+m}{2}$ 时, $y = -9 + \frac{(9+m)^2}{4m} = \frac{4}{5}$, 解得 $m=5$, 则 $AB=5$, $BC=4$, $S_{\text{矩形}ABCD} = 20$. 故选 A.

5. A 【解析】连接 OE, OF, ON, OG , 则易证四边形 $AFOE, FBGO$ 是正方形, $\therefore AF=BF=AE=BG=2$, $DE=3$, 又 $\therefore DM$ 是 $\odot O$ 的切线, $\therefore DN=DE=3$, $MN=MG$, $\therefore CM=5-2-MN=3-MN$. 在 $\text{Rt} \triangle DMC$ 中, $DM^2 = CD^2 + CM^2$, $\therefore (3+NM)^2 = (3-NM)^2 + 4^2$, $\therefore NM = \frac{4}{3}$, $\therefore DM = 3 + \frac{4}{3} = \frac{13}{3}$. 故选 A.

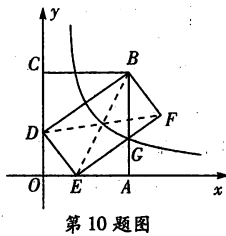
6. A

7. 7 【解析】设 $\frac{1}{x} = a, y^4 = b$, 则 $a - \sqrt{a} = 3$, 即 $a - 3 = \sqrt{a} > 0, b + \sqrt{b} = 3$, 即 $3 - b = \sqrt{b} > 0$; 由以上所得式子对比看, a, b 的值为方程 $x = (x-3)^2$ 的两个根且 a 大 b 小, 即 $a = \frac{7+\sqrt{13}}{2}, b = \frac{7-\sqrt{13}}{2}$, $\therefore a+b=7$.

8. $-2 \leq P < -\frac{1}{3}$ 【解析】 $\therefore T(1, -1) = -2, T(4, 2) = 1, \therefore \frac{a-b}{2+(-1)} = -2, \frac{4a+2b}{2 \times 4+2} = 1$, 解得 $a=1, b=3, T(2m, 5-4m) = \frac{2m+3(5-4m)}{4m+5-4m} \leq 4$, 解得 $m \geq -\frac{1}{2}, T(m, 3-2m) = \frac{m+3(3-2m)}{2m+3-2m} > P$, 解得 $m < \frac{9-3P}{5}$, $\therefore$ 关于 m 的不等式组 $\begin{cases} T(2m, 5-4m) \leq 4 \\ T(m, 3-2m) > P \end{cases}$ 恰好有 3 个整数解, $\therefore 2 < \frac{9-3P}{5} \leq 3, \therefore -2 \leq P < -\frac{1}{3}$, $\therefore$ 实数 P 的取值范围是 $-2 \leq P < -\frac{1}{3}$.

9. $2\sqrt{13}$ 【解析】延长 CG 交 AB 于点 H , $\therefore BE$ 和 AD 分别是边 AC, BC 上的中线, $\therefore$ 点 G 是 $\triangle ABC$ 的重心, $\therefore AG=2GD=4, BG=2GE=6, CG=2GH$. $\therefore AD \perp BE$, $\therefore \angle AGB=90^\circ, \therefore AB = \sqrt{AG^2 + BG^2} = \sqrt{4^2 + 6^2} = 2\sqrt{13}$. 又 $\therefore H$ 是 AB 的中点, $\therefore CH = \frac{1}{2}AB = \sqrt{13}, \therefore CG = 2GH = 2\sqrt{13}$.

10. $\frac{24}{5}$ 【解析】如图, 连接 DF, BE , $\therefore$ 四边形 $OABC, BDEF$ 是矩形, $\therefore OC=AB, BE=DF, \angle BAO = \angle BDE = \angle DEF = 90^\circ, \therefore BD = OC, \therefore BD=AB$. 又 $\therefore BE=BE$, $\therefore \text{Rt} \triangle BDE \cong \text{Rt} \triangle BAE (\text{HL}), \therefore AE=DE=2$, $\therefore EG = \sqrt{AE^2 + AG^2} = \sqrt{4 + \frac{9}{4}} = \frac{5}{2}$. $\therefore \angle DEO + \angle AEG = 90^\circ, \angle EDO + \angle DEO = 90^\circ, \therefore \angle AEG = \angle EDO$. 又 $\therefore \angle EOD =$



第 10 题图

$\angle EAG = 90^\circ, \therefore \triangle DEO \sim \triangle EGA, \therefore \frac{AG}{OE} = \frac{EG}{DE}, \therefore \frac{3}{\frac{2}{5}} = \frac{5}{2}, \therefore OE = \frac{6}{5}$, $\therefore OA = 2 + \frac{6}{5} = \frac{16}{5}, \therefore$ 点 $G(\frac{16}{5}, \frac{3}{2})$. $\therefore$ 反比例函数 $y = \frac{k}{x} (k \neq 0)$ 的图象经过点 $G, \therefore k = \frac{16}{5} \times \frac{3}{2} = \frac{24}{5}$.

11. $\frac{75}{16}$ 【解析】 $AB = \sqrt{6^2 + 8^2} = 10, \therefore D$ 点为 AB 的中点, $\therefore AD=5$. $\therefore \triangle ABC \cong \triangle FDE$, 易证 $\triangle AHD \sim \triangle ABC, \therefore \frac{HD}{BC} = \frac{AD}{AC}$, 即 $\frac{HD}{6} = \frac{5}{8}$, 解得 $HD = \frac{15}{4}, \therefore S_{\triangle ADH} = \frac{1}{2} \times 5 \times \frac{15}{4} = \frac{75}{8}, \therefore S_{\triangle HGD} = \frac{1}{2} S_{\triangle ADH} = \frac{75}{16}$.

12. -2018 【解析】 $(a^{2018})^2 - (c^{2018} + d^{2018}) \cdot a^{2018} + (cd)^{2018} - 2018 = 0$, $(b^{2018})^2 - (c^{2018} + d^{2018}) \cdot b^{2018} + (cd)^{2018} - 2018 = 0$, 所以 a^{2018}, b^{2018} 可以看作是 x 的一元二次方程 $x^2 - (c^{2018} + d^{2018}) \cdot x + (cd)^{2018} - 2018 = 0$ 的两个不相等的实数根, 所以 $a^{2018} \cdot b^{2018} = (cd)^{2018} - 2018$. 原式 $= -2018$.

13. 解: $\therefore ab=1, b = \frac{1}{a}, \therefore S = \frac{1}{1+a} + \frac{1}{1+2b} = \frac{1}{1+a} + \frac{1}{1+\frac{2}{a}} = 1 - \frac{a}{a^2+3a+2} = 1 - \frac{1}{\frac{a^2+3a+2}{a}}$. 设 $N = a + \frac{2}{a} + 3 = (\sqrt{a} - \sqrt{\frac{2}{a}})^2 + 3 + 2\sqrt{2} \geq 3 + 2\sqrt{2}$ (当 $a = \sqrt{2}$ 时, 取等号), $\therefore 0 < \frac{1}{N} \leq \frac{1}{3+2\sqrt{2}}, S = 1 - \frac{1}{N} \geq 1 - \frac{1}{3+2\sqrt{2}} = 2\sqrt{2} - 2$. 故当 $a = \sqrt{2}, b = \frac{\sqrt{2}}{2}$ 时, S 取得最小值为 $2\sqrt{2} - 2$.

14. 解: (1) 过点 A 作 $AH \perp x$ 轴于点 H . 在 $\text{Rt} \triangle AOH$ 中, $\therefore OA = 2\sqrt{5}, \tan \angle AOH = \frac{1}{2}, \therefore AH=2, OH=4, \therefore A(-4, 2)$. $\therefore A(-4, 2)$ 在 $y = \frac{k}{x}$ 的图象上, $\therefore k = -8, \therefore B(m, -4)$ 在 $y = -\frac{8}{x}$ 的图象上, $\therefore m=2$, 则 $\begin{cases} -4k+b=2 \\ 2k+b=-4 \end{cases}$, 解得 $\begin{cases} k=-1 \\ b=-2 \end{cases}$, $\therefore$ 反比例函数的解析

式为 $y = -\frac{8}{x}$, 一次函数的解析式为 $y = -x - 2$.

(2) 由 $y = -x - 2$, 令 $x=0$, 则 $y = -2$; 令 $y=0$, 则 $x = -2, \therefore D(0, -2), C(-2, 0), \therefore S_{\triangle AOB} = S_{\triangle AOD} + S_{\triangle BOD} = \frac{1}{2} \times 2 \times (4+2) = 6$, 若点 E 在 y 轴

上, 设 $E(0, y)$, 则 $DE = |y - (-2)|$. 由 $S_{\triangle AED} = 3S_{\triangle AOB}$, 可得 $\frac{1}{2} \times |y - (-2)| \times 4 = 3 \times 6$. 解得 $y = 7$ 或 $-11, \therefore$ 点 E 的坐标为 $(0, 7)$ 或 $(0, -11)$; 若点 E 在 x 轴上, 设 $E(x, 0)$, 则 $CE = |x - (-2)|$.

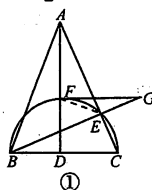
由 $S_{\triangle AED} = 3S_{\triangle AOB}$, 可得 $\frac{1}{2} \times |x - (-2)| \times 4 = 3 \times 6$. 解得 $x = 7$ 或 $-11, \therefore$ 点 E 的坐标为 $(7, 0)$ 或 $(-11, 0)$. 故点 E 的坐标为 $(0, 7)$ 或 $(0, -11)$ 或 $(7, 0)$ 或 $(-11, 0)$.

15. 解: (1) $\therefore 4+5=9, \therefore 459$ 是“桃园数”; $\therefore 6+1 \neq 5, \therefore 615$ 不是“桃园数”.

(2) 设这个数的百位为 a , 十位是 b , 个位是 $c(a, b, c$ 均为正整数, 且 b 可以为 0), $\therefore m=10a+b, n=10c+b, m+n=10a+2b+10c$, 又 $\therefore m+n=24k (k \geq 1, \text{且 } k \text{ 为整数}), c=a+b, \therefore 10a+2b+10c=24k, 5a+b+5c=12k, 5a+b+5a+5b=12k, 10a+6b=12k, 5a+3b=6k$. 当 $a=1, a=2$ 时, 没有满足条件的 b 的值; 当 $a=3$ 时, b 可以是 1, 3, 5, 此时“桃园数” p 为 314, 336, 358; 当 $a=4, a=5$ 时, 没有满足条件的 b 的值; 当 $a=6$ 时, b 可以为 0, 2, 此时“桃园数” p 为 606, 628; 当 $a=7, a=8, a=9$ 时, 没有满足条件的 b 的值. 综上所述, p 可以为 314, 336, 358, 606, 628.

16. 解: (1) 如图①, 连接 $EF, \therefore AB=AC, AD$ 为 BC 边上的中线, $\therefore AD \perp BC, \therefore \angle BAD = \angle CAD, \therefore \angle ADB = 90^\circ$, 点 E 在半圆 D 上, $\therefore \angle BEF = \frac{1}{2} \angle BDA = 45^\circ, \therefore BC$ 是直径, $\therefore \angle BEC = 90^\circ, \therefore \angle AEB = 90^\circ$.

$\therefore \angle BEF = \frac{1}{2} \angle BEA, \therefore EF$ 平分 $\angle BEA, \therefore F$ 点是 $\triangle ABE$ 的内心.



第 16 题图

(2) 方法一: 如图②, 过 F 作 $FH \perp AE, FN \perp BE$, 垂足分别为 $H, N, \therefore EF$ 平分 $\angle AEB, \therefore FH=FN$, 由题意得 $\angle FAH = \angle FGN, \therefore \text{Rt} \triangle AHF \cong \text{Rt} \triangle GNF, \therefore AF=FG$.

方法二: 如图②, 连接 $AG, \therefore \angle AFG = \angle AEG = 90^\circ, \therefore A, F, E, G$ 四点共圆, $\therefore \angle FEB = \angle FAG = 45^\circ, \therefore AF=FG$.

(3) 如图③, 在 AE 上截取 $AM=EG$, 连接 FM, EF, ED , 过 E 作 $EK \perp AD$ 于 K , 由题意 $\triangle AMF \cong \triangle GEF, \angle AFM = \angle GFE, \angle MFE = 90^\circ, \therefore \triangle MFE$ 为等腰直角三角形, $ME = AE - EG = \frac{2\sqrt{34}}{17}, EF = \frac{2\sqrt{17}}{17}, \therefore DE = DF = 1$,

$\therefore$ 解 $\triangle DEF$ 得 $EK = \frac{8}{17}, FK = \frac{2}{17}, \triangle AKE \sim \triangle ADC, \frac{AK}{AD} = \frac{EK}{DC}$, 得 $AF = \frac{2}{3}$, 因此 $FG = \frac{2}{3}$.

17. 解: (1) 将 $A(-1, 0), C(0, \sqrt{3})$ 代入抛物线 $y = ax^2 + \frac{2\sqrt{3}}{3}x + c (a \neq 0)$, 则

$$\begin{cases} a - \frac{2\sqrt{3}}{3} + c = 0 \\ c = \sqrt{3} \end{cases}, \text{解得} \begin{cases} a = -\frac{\sqrt{3}}{3} \\ c = \sqrt{3} \end{cases}.$$

(2) 由 (1) 得抛物线解析式 $y = -\frac{\sqrt{3}}{3}x^2 + \frac{2\sqrt{3}}{3}x + \sqrt{3}, \therefore$ 点 D 是点 C 关于抛物线对称轴的对称点, $C(0, \sqrt{3}), \therefore D(2, \sqrt{3}), \therefore DH = \sqrt{3}$, 令 $y=0$, 即 $-\frac{\sqrt{3}}{3}x^2 + \frac{2\sqrt{3}}{3}x + \sqrt{3} = 0$, 解得 $x_1 = -1, x_2 = 3, \therefore A(-1, 0), B(3, 0)$.

又 $\therefore AE \perp AC, EH \perp AH, \therefore \triangle ACO \sim \triangle EAH, \therefore \frac{OC}{AH} = \frac{OA}{EH}$, 即 $\frac{\sqrt{3}}{3} = \frac{1}{EH}$, 解得 $EH = \sqrt{3}$, 则 $DE = 2\sqrt{3}$.

(3) 找点 C 关于 DE 的对称点 $N(4, \sqrt{3})$, 找点 C 关于 AE 的对称点 $G(-2, -\sqrt{3})$, 连接 GN , 交 AE 于点 F , 交 DE 于点 P , 即 G, F, P, N 四点共线时, $\triangle CPF$ 的周长 $= CF + PF + CP = GF + PF + PN$ 最小, $\therefore$ 直线 GN 的解析式 $y = \frac{\sqrt{3}}{3}x - \frac{\sqrt{3}}{3}$ ①, 由 (2) 得 $E(2, -\sqrt{3}), A(-1, 0), \therefore$ 直线 AE 的

解析式 $y = -\frac{\sqrt{3}}{3}x - \frac{\sqrt{3}}{3}$ ②, 联立①, ②解得 $\begin{cases} x=0 \\ y=-\frac{\sqrt{3}}{3} \end{cases}, \therefore F(0, -\frac{\sqrt{3}}{3})$,

$\therefore DH \perp x$ 轴, $\therefore$ 将 $x=2$ 代入 $y = \frac{\sqrt{3}}{3}x - \frac{\sqrt{3}}{3}$ 中, 得 $P(2, \frac{\sqrt{3}}{3})$,

$\therefore F(0, -\frac{\sqrt{3}}{3})$ 与 $P(2, \frac{\sqrt{3}}{3})$ 的水平距离为 2, 过点 M 作 y 轴的平行线交 FP 于点 Q , 设点 $M(m, -\frac{\sqrt{3}}{3}m^2 + \frac{2\sqrt{3}}{3}m + \sqrt{3})$, 则 $Q(m, \frac{\sqrt{3}}{3}m - \frac{\sqrt{3}}{3})$

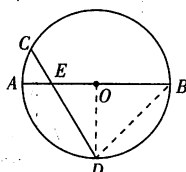
$$\left(\frac{1-\sqrt{17}}{2} < m < \frac{1+\sqrt{17}}{2}\right), \therefore S_{\triangle MPF} = S_{\triangle MQF} + S_{\triangle NQP} = \frac{1}{2} \times MQ \times 2 = MQ = \left(-\frac{\sqrt{3}}{3}m^2 + \frac{2\sqrt{3}}{3}m + \sqrt{3}\right) - \left(\frac{\sqrt{3}}{3}m - \frac{\sqrt{3}}{3}\right) = -\frac{\sqrt{3}}{3}m^2 + \frac{\sqrt{3}}{3}m + \frac{4\sqrt{3}}{3} = -\frac{\sqrt{3}}{3}\left(m - \frac{1}{2}\right)^2 + \frac{17\sqrt{3}}{12}, \therefore -\frac{\sqrt{3}}{3} < 0, \therefore \text{当 } m = \frac{1}{2} \text{ 时, } \triangle MPF \text{ 面积有最大值 } \frac{17\sqrt{3}}{12}.$$

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1. B 【解析】设 $x = \sqrt{3} + \sqrt{5} + \sqrt{3 - \sqrt{5}}$, 则 $x^2 = (\sqrt{3} + \sqrt{5} + \sqrt{3 - \sqrt{5}})^2 = 10$, $\therefore x = \pm\sqrt{10}$. $\because x > 0, \therefore x = \sqrt{10}$. 故选 B.

2. D 【解析】设 $y = ax^2 + bx + c$ 交 y 轴于点 $C(0, c)$, $c \neq 0$, 交 x 轴于点 $A(x_1, 0)$, $B(x_2, 0)$, 且 $x_1 < 0 < x_2$, 由 $\triangle ABC$ 是直角三角形知, 点 C 必为直角顶点, 且 $c^2 = (-x_1)x_2 = -x_1x_2$ (射影定理的逆定理), 由根与系数的关系得 $x_1 + x_2 = -\frac{b}{a}$, $x_1x_2 = \frac{c}{a}$, $\therefore c^2 = -\frac{c}{a}$, $c = -\frac{1}{a}$, 又 $\frac{4ac - b^2}{4a} = -2$, 即 $8a = 4 + b^2 \geq 4$, $\therefore a \geq \frac{1}{2}$. $\therefore S_{\triangle ABC} = \frac{1}{2} |c| \cdot |x_1 - x_2| = \frac{1}{2a} \sqrt{(x_1 + x_2)^2 - 4x_1x_2} = \frac{1}{2a} \sqrt{\frac{b^2}{a^2} + \frac{4}{a^2}} = \frac{1}{2a} \cdot \frac{\sqrt{b^2 + 4}}{a} = \frac{\sqrt{b^2 + 4}}{2a^2} \leq 4$, 当且仅当 $a = \frac{1}{2}$, $b = 0$, $c = -2$ 时等号成立, 因此, $\text{Rt} \triangle ABC$ 的最大面积是 4. 故选 D.

3. B 【解析】如图, 连接 OD, AC, BD , $\therefore D$ 是半圆中点, $\therefore \widehat{AD} = \widehat{BD}$, $\therefore \angle AOD = \angle BOD = \frac{1}{2} \times 180^\circ = 90^\circ$. 设 $OA = OB = OD = r$, $AE = r - OE$, $BE = r + OE$, $CE = 2$, $DE = 4$, $OE^2 = DE^2 - OD^2$, $\therefore OE^2 = 16 - r^2$, $\therefore \angle C = \angle B$, $\angle A = \angle BDE$, $\therefore \triangle AEC \sim \triangle DEB$, $\therefore \frac{AE}{DE} = \frac{CE}{BE}$, $\therefore AE \cdot BE = CE \cdot DE = 2 \times 4 = 8$, $\therefore (r - OE)(r + OE) = 8$. 整理得 $OE^2 = r^2 - 8$, $\therefore 16 - r^2 = r^2 - 8$, 解得 $r = 2\sqrt{3}$ 或 $r = -2\sqrt{3}$ (不符合题意, 舍去), $\therefore \odot O$ 的半径为 $2\sqrt{3}$. 故选 B.



第3题图

4. A 【解析】根据图形和题意有 $(a+b)^2 = b(a+2b)$, $a=1$, 则有 $(1+b)^2 = b(1+2b)$. 解得 $b = \frac{\sqrt{5}+1}{2}$, 故正方形的面积为 $(1+\frac{\sqrt{5}+1}{2})^2 = \frac{7+3\sqrt{5}}{2}$. 故选 A.

5. B 【解析】过点 A 作 $AG \perp x$ 轴, 过点 D 作 $DE \perp x$ 轴, 作 $DF \perp AG$ 交 y 轴于点 H , 易证 $\triangle DHM \cong \triangle DEN$ (ASA), $\therefore S_{\triangle DHM} = S_{\triangle DEN}$, $\therefore S_{\text{阴影}} = \frac{2}{3} = S_{\text{四边形 } DHOF}$, $DH \times DE$, $\therefore DH = DE = \frac{\sqrt{6}}{3}$. 同理可证 $\triangle AFD \cong \triangle BGA \cong \triangle COB \cong \triangle DHC$, $\therefore OC = HD = \frac{\sqrt{6}}{3} = AF = BG$, $\therefore CH = \frac{2\sqrt{6}}{3}$, $\therefore AG = \frac{2\sqrt{6}}{3} = BO$, $\therefore GO = \frac{\sqrt{6}}{3}$, $\therefore$ 点 A 的坐标为 $(-\frac{\sqrt{6}}{3}, \frac{2\sqrt{6}}{3})$, $\therefore k = -\frac{\sqrt{6}}{3} \times \frac{2\sqrt{6}}{3} = -\frac{4}{3}$. 故选 B.

6. C
7. 1, 2, 4 【解析】方程 $(2-a)x = 2$, 当 $a=2$ 时无解; 当 $a=1$ 时, 当 $x=-1$ 时, $a=4$, $x=2$ 或 -1 是增根, 原方程无解.

8. 2 【解析】 $\sqrt{5-2\sqrt{6}} + \sqrt{7-4\sqrt{3}} + \frac{4\sqrt{2}-4}{4-2\sqrt{2}} = \sqrt{\frac{10-4\sqrt{6}}{2}} + \sqrt{\frac{14-8\sqrt{3}}{2}} + \frac{4\sqrt{2}-4}{4-2\sqrt{2}} = \frac{\sqrt{6}-2}{2} + \frac{\sqrt{2}-1}{2} + \frac{4\sqrt{2}-4}{4-2\sqrt{2}} = \frac{\sqrt{2}}{2}(\sqrt{6}-2) + \frac{\sqrt{2}}{2}(2\sqrt{2}-\sqrt{6}) + \sqrt{2} = 2$.

9. 14 【解析】由题意, 设 $OA = 3a$, 则 $OB = 4a$, 则 $A(3a, 0)$, $B(0, 4a)$, 求得直线 AB 的解析式为 $y = -\frac{4}{3}x + 4a$, $\therefore$ 直线 CD 是 $\angle AOB$ 的平分线, 则

$$\text{直线 } OD \text{ 的解析式是 } y = x, \text{ 联立 } \begin{cases} y = -\frac{4}{3}x + 4a, \\ y = x, \end{cases} \text{ 解得 } \begin{cases} x = \frac{12}{7}a, \\ y = \frac{12}{7}a, \end{cases} \text{ 则 } D$$

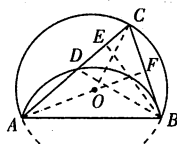
点的坐标是 $(\frac{12}{7}a, \frac{12}{7}a)$, 又 $\therefore OA$ 的中垂线的解析式是 $x = \frac{3}{2}a$, 则 C 点的坐标是 $(\frac{3}{2}a, \frac{3}{2}a)$, 则 $k = \frac{9}{4}a^2$, $\therefore$ 以 CD 为边的正方形的面积为 $\frac{4}{7}$, 且

$$CD^2 = (x_D - x_C)^2 + (y_D - y_C)^2 = 2\left(\frac{12}{7}a - \frac{3}{2}a\right)^2,$$

$$\text{则 } 2\left(\frac{12}{7}a - \frac{3}{2}a\right)^2 = \frac{4}{7}, \text{ 解得 } a^2 = \frac{56}{9}, \therefore k = \frac{9}{4} \times \frac{56}{9} = 14.$$

10. $4\sqrt{2}$ 【解析】如图, 连接 BD , 作 $BE \perp AC$ 于 E , 连接 AO 并延长交 BC 于 F , 连接 OC , $\therefore \angle DAB = \angle CAB$, $\therefore \widehat{BD} = \widehat{BC}$, $\therefore BC = BD$, $\therefore DE = CE = \frac{1}{2}CD = 2$, 在 $\text{Rt} \triangle ABE$ 中, $AB = 8$, $AE = AD + DE = 6$, $\therefore BE = \sqrt{8^2 - 6^2} = 4\sqrt{2}$.

$2\sqrt{7}$, 在 $\text{Rt} \triangle BDE$ 中, $DE = 2$, $BE = 2\sqrt{7}$, $\therefore BC = BD = \sqrt{(2\sqrt{7})^2 + 2^2} = 4\sqrt{2}$, 根据对称性可得 $AF \perp BC$, $CF = BF = \frac{1}{2}BC = 2\sqrt{2}$, 在 $\text{Rt} \triangle ACF$ 中, $CF = 2\sqrt{2}$, $AC = 8$, $AF = 2\sqrt{14}$, 设 $OA = OC = R$, 在 $\text{Rt} \triangle COF$ 中, 由勾股定理得 $R^2 - (2\sqrt{14} - R)^2 = (2\sqrt{2})^2$, $\therefore R = \frac{8\sqrt{14}}{7}$.



第10题图

11. $\frac{3}{4}$ 【解析】 $\because x + y + z = 1, \therefore z = 1 - x - y, \therefore M = xy + 2yz + 3zx = xy + 2y(1 - x - y) + 3x(1 - x - y) = xy + 2y - 2xy - 2y^2 + 3x - 3x^2 - 3xy = -3x^2 - 4xy - 2y^2 + 2y + 3x = -2(x + y)^2 + 2(x + y) - x^2 + x = -2\left[(x + y) + \frac{1}{4} - \frac{1}{4}\right] - \left(x^2 - x + \frac{1}{4} - \frac{1}{4}\right) = -2\left(x + y - \frac{1}{2}\right)^2 - \left(x - \frac{1}{2}\right)^2 + \frac{3}{4} \leq \frac{3}{4}, \therefore M = xy + 2yz + 3zx$ 的最大值为 $\frac{3}{4}$.

12. 10 【解析】过点 C 作 $CE \perp AB$ 于点 E , 交 BD 于点 M , 过点 M 作 $MN \perp BC$ 于 N , $\because BD$ 平分 $\angle ABC$, $ME \perp AB$ 于点 E , $MN \perp BC$ 于 N , $\therefore MN = ME$. $\therefore CE = CM + ME = CM + MN$ 的最小值.

$\therefore S_{\triangle ABC} = 40$, $AB = 8$, $\therefore \frac{1}{2} \times 8 \times CE = 40, \therefore CE = 10$.

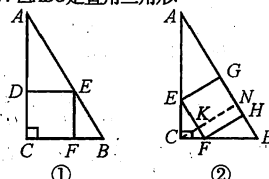
13. 160 【解析】由题意可设 $x_1, x_2, x_3, \dots, x_{100}$ 中有 a 个 -1 , b 个 0 , c 个 1 , d 个 2 , 则 $a + b + c + d = 100$, $-a + c + 2d = 20$, $a + c + 4d = 100$, 可得 $a = 40 - d$, $b = 3d$, $c = 60 - 3d$, $\therefore x_1^3 + x_2^3 + x_3^3 + \dots + x_{100}^3 = -a + c + 8d =$

$$\begin{cases} d \geq 0, \\ 40 - d \geq 0, \\ 3d \geq 0, \\ 60 - 3d \geq 0, \end{cases} \text{ 解得 } 0 \leq d \leq 20, \therefore \text{当 } d = 0 \text{ 时, } x_1^3 + x_2^3 + x_3^3 + \dots + x_{100}^3 \text{ 的最小值为 } 20; \text{当 } d = 20 \text{ 时, } x_1^3 + x_2^3 + x_3^3 + \dots + x_{100}^3 \text{ 的最大值为 } 140. \therefore x_1^3 + x_2^3 + x_3^3 + \dots + x_{100}^3 \text{ 的最小值和最大值的和为 } 160.$$

14. 解: (1) $y = x^2 + 2px - 2 = (x + p)^2 - p^2 - 2$, 当 $-2 \leq x \leq 0$ 时, ①当 $-p < -2$, 即 $p > 2$ 时, 函数最小值 $M = 2 - 4p$; ②当 $0 \leq p \leq 2$ 时, 函数最小值 $M = -p^2 - 2$; ③当 $p < 0$ 时, 函数最小值 $M = -2$. $\therefore M$ 关于 p 的函数解析式为 $M = \begin{cases} 2 - 4p, & (p > 2), \\ -p^2 - 2, & (0 \leq p \leq 2), \\ -2, & (p < 0), \end{cases}$

(2) 由 $M = -3$ 知, $\begin{cases} p > 2, \\ 2 - 4p = -3 \end{cases}$ 或 $\begin{cases} 0 \leq p \leq 2, \\ -p^2 - 2 = -3 \end{cases}$ $\therefore p = 1, \therefore y = x^2 + 2x - 2 = (x + 1)^2 - 3$, 当 $-2 \leq x \leq 0$ 时, 函数最大值为 -2 .

15. (1) 证明: 由题意, 得 $\frac{2(a+b+c)(a+b-c) - (2\sqrt{ab})^2}{4(a+b+c)} = 0$, 化简, 得 $(a+b)^2 - c^2 = 2ab$, $a^2 + b^2 = c^2$, $\therefore \triangle ABC$ 是直角三角形.



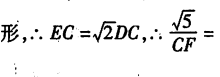
第15题图

(2) 解: $\because \cos^2 A + \cos^2 B = 1, \therefore (\cos A + \cos B)^2 - 2\cos A \cdot \cos B = 1$. $\therefore \cos A + \cos B = \frac{2m-5}{m+5}$, $\cos A \cdot \cos B = \frac{m-8}{m+5}$. $\therefore (2m-5)^2 - 2(m-8)(m+5) = (m+5)^2$, $\therefore m_1 = 4$ (舍), $m_2 = 20$, $\therefore m = 20$.

(3) 解: 设斜边为 c , 则 $\pi \cdot \left(\frac{c}{2}\right)^2 = \frac{25}{4}\pi$, $c = 5$, 由 (2) 得 $m = 20$, $25x^2 - 35x + 12 = 0$, $x_1 = \frac{3}{5}$, $x_2 = \frac{4}{5}$. 如图①, 设 $DE = x$, 则 $AD = 4 - x$, $\therefore DE \parallel BC$, $\therefore \frac{DE}{BC} = \frac{AD}{AC}$, $\frac{x}{3} = \frac{4-x}{4}$, $x = \frac{12}{7}$. 如图②, $S_{\triangle ABC} = \frac{3 \times 4}{2} = \frac{5 \times CN}{2}$, $CN = 2.4$, $\therefore EF \parallel AB$, $\therefore \triangle CEF \sim \triangle CAB$, $\therefore \frac{EF}{AB} = \frac{CK}{CN}$, 设 $EF = y$, $\therefore \frac{y}{5} = \frac{2.4 - y}{2.4}$, $12 - 5y = 2.4y$, $y = \frac{60}{37}$. $\therefore \triangle ABC$ 内接正方形的边长为 $\frac{12}{7}$ 或 $\frac{60}{37}$.

16. 解: (1) 把 $x = 0$ 代入 $y = 2x + 2$ 得 $y = 2$, $\therefore E$ 点坐标为 $(0, 2)$. 把 $y = 0$ 代入 $y = 2x + 2$ 得 $2x + 2 = 0$, 解得 $x = -1$, $\therefore A$ 点坐标为 $(-1, 0)$, $\therefore AE = \sqrt{AO^2 + OE^2} = \sqrt{1^2 + 2^2} = \sqrt{5}$.

(2) CF 的长度不变. 理由如下: 连接 AC , 如图. $\because$ 四边形 $BEDC$ 为正方形, $\therefore \angle DCE = \angle DEC = 45^\circ$, $\therefore \angle AEC = 135^\circ$. $\therefore EC \perp CF$, $\therefore \angle DCF = 45^\circ + 90^\circ = 135^\circ$, $\therefore \angle AEC = \angle DCF$. $\therefore \angle DAC = \angle DFC$, $\therefore \triangle ACE \sim \triangle FDC$, $\therefore \frac{AE}{CF} = \frac{EC}{DC}$. $\therefore \triangle DEC$ 为等腰直角三角形, $\therefore EC = \sqrt{2}DC$, $\therefore \frac{\sqrt{5}}{CF} =$



第16题图

$$\sqrt{2}, \therefore CF = \frac{\sqrt{10}}{2}.$$

17. 解: (1) $y = (x-1)^2 - 4$ 向左平移 1 个单位长度, 再向上平移 4 个单位长度即可得到 $y = x^2$.

(2) 如题图①, ①设抛物线 C_1 与 y 轴交于 C 点, 直线 AB 与 y 轴交于 D 点, $\therefore C_1: y = (x-1)^2 - 4, \therefore A(3, 0), C(0, -3)$.

$\therefore$ 直线 $y = -\frac{4}{3}x + b$ 经过点 $A, \therefore b = 4, \therefore D(0, 4), \therefore AP = AQ, PQ \parallel y$ 轴, $\therefore P, Q$ 两点关于 x 轴对称, 设 $D(0, 4)$ 关于 x 轴的对称点为 D' , 则 $D'(0, -4), \therefore$ 直线 AD' 的解析式为 $y = \frac{4}{3}x - 4$, 由 $\begin{cases} y = x^2 - 2x - 3, \\ y = \frac{4}{3}x - 4, \end{cases}$ 得

$x_1 = 3, x_2 = \frac{1}{3}, \therefore x_0 = \frac{1}{3}, \therefore x_P = x_Q = \frac{1}{3}, \therefore P$ 点横坐标为 $\frac{1}{3}$; ②设 $P(m, 4 - \frac{4}{3}m), Q(m, m^2 - 2m - 3), \therefore PA = PQ, \therefore (m^2 - \frac{2}{3}m - 7)^2 = (m-3)^2 + (4 - \frac{4}{3}m)^2, \therefore |m^2 - \frac{2}{3}m - 7| = \frac{5}{3}|m-3|, \therefore -1 < m < 3, \therefore -m^2 + \frac{2}{3}m + 7 = \frac{5}{3}(3-m), \therefore m = -\frac{2}{3}$ 或 $m = 3$ (舍), $\therefore P$ 点横坐标为 $-\frac{2}{3}$.

(3) 设经过 M 与 E 的直线解析式为 $y = k(x-m) + m^2$, $\therefore \begin{cases} y = x^2, \\ y = k(x-m) + m^2, \end{cases}$ 则有 $x^2 - kx + km - m^2 = 0, \Delta = k^2 - 4km + 4m^2 = (k-2m)^2 = 0, \therefore k = 2m, \therefore$ 直线 ME 的解析式为 $y = 2mx - m^2$; 同理, 直线 NE 的解析式为 $y = 2nx - n^2, \therefore E(\frac{m+n}{2}, mn), \therefore \frac{1}{2}[(n^2 - mn) + (m^2 - mn)] \times (m-n) - \frac{1}{2}(n^2 - mn) \times (\frac{m+n}{2} - n) - \frac{1}{2}(m^2 - mn) \times (\frac{m+n}{2} - m) = 2, \therefore (m-n)^3 - \frac{(m-n)^3}{2} = 4, \therefore (m-n)^3 = 8, \therefore m-n = 2$.

全国重点高中提前招生 考试全真试卷(六)

1. A 【解析】 $\therefore \sqrt{x^2 - 4x + 13} + \sqrt{x^2 + 2x + 2} = \sqrt{(x-2)^2 + (0-3)^2} + \sqrt{(x+1)^2 + (0-1)^2}, \therefore \sqrt{x^2 - 4x + 13} + \sqrt{x^2 + 2x + 2}$ 的最小值可看成是平面直角坐标系中 x 轴上一点 $P(x, 0)$ 到两点 $M(2, 3)$ 与 $N(-1, 1)$ 之间的距离之和最短. $N(-1, 1)$ 关于 x 轴对称点为 $N'(-1, -1)$, 则 MN' 即为所求最小值. 此时, $MN' = \sqrt{[3 - (-1)]^2 + [2 - (-1)]^2} = 5$. 故选 A.

2. A 【解析】 $M = 5x^2 - 12xy + 10y^2 - 6x - 4y + 13 = 4x^2 - 12xy + 9y^2 + x^2 - 4y + 4 + x^2 - 6x + 9 = (2x - 3y)^2 + (y - 2)^2 + (x - 3)^2 \geq 0$, 则 M 一定是非负数. 故选 A.

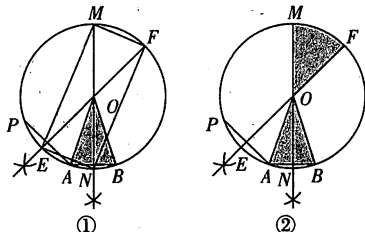
3. A 【解析】当 $CP \perp AN, CQ \perp PM$ 时, CQ 最小.

4. B 【解析】易得正方形的边长为 $4\sqrt{15}, \therefore BD = 4\sqrt{30}, \therefore \frac{BE}{DC} = \frac{BC}{GD} = \frac{1}{2},$

$$\therefore BG = \frac{4\sqrt{30}}{3}, \therefore S_{\triangle BEG} = \frac{1}{2}BE \times BG \sin \angle EBG = 20. \therefore \triangle CFH \sim \triangle CEB,$$

$$\therefore \frac{S_{\triangle CFH}}{S_{\triangle CEB}} = \frac{1}{5}, \therefore S_{\triangle CFH} = 12, \therefore S_{\text{四边形BFHC}} = S_{\triangle CEB} - S_{\triangle BEG} - S_{\triangle CFH} = 28. \text{ 故选 B.}$$

5. C 【解析】如图①, $\therefore MN$ 是 AB 的垂直平分线, EF 是 AP 的垂直平分线, 且 M, N, E, F 都在 $\odot O$ 上, $\therefore MN$ 和 EF 都经过圆心 O , 线段 MN 和 EF 是 $\odot O$ 的直径, $\therefore OM = ON, OE = OF, \therefore$ 四边形 $MENF$ 是平行四边形. $\therefore$ 线段 MN 是 $\odot O$ 的直径, $\therefore \angle MEN = 90^\circ, \therefore$ 平行四边形 $MENF$ 是矩形. $\therefore$ 结论 II 正确.



第 5 题图

如图②, 当点 P 在直线 MN 左侧且 $AP = AB$ 时, $\therefore AP = AB, \therefore \widehat{AB} = \widehat{AP}$.

$$\therefore MN \perp AB, EF \perp AP, \therefore \widehat{AN} = \frac{1}{2}\widehat{AB}, \widehat{AE} = \frac{1}{2}\widehat{AP},$$

$$\therefore \angle AOE = \angle AON = \frac{1}{2}\angle AOB = 20^\circ, \therefore \angle EON = 40^\circ, \therefore \angle MOF = \angle EON = 40^\circ. \therefore$$

扇形 FOM 与扇形 AOB 的半径、圆心角度数都分别相等, $\therefore S_{\text{扇形FOM}} = S_{\text{扇形AOB}}$. 同理, 当点 P 在直线 MN 右侧时, 也存在一点 P , 使得 $S_{\text{扇形FOM}} = S_{\text{扇形AOB}}$. $\therefore$ 结论 I 错误. 故选 C.

6. A 【解析】 $a = 3 + 2\sqrt{2} - [3 + 2\sqrt{2}] = 3 + 2\sqrt{2} - 5 = 2\sqrt{2} - 2,$

$$b = -3 - 2\sqrt{2} + 6 = 3 - 2\sqrt{2}, \frac{1}{4b} - \frac{1}{a} = \frac{1}{4}. \text{ 故选 A.}$$

7. $\frac{11}{25}$ 【解析】 $\therefore abc = 1, a, b, c$ 都为正数, $\therefore ab = \frac{1}{c}, \therefore a + \frac{1}{b} = 3, \therefore ab +$

$$1 = 3b \text{ ①. } \therefore b + \frac{1}{c} = 17, \therefore b + ab = 17 \text{ ②. 由 ①② 解得 } ab = \frac{25}{2}, b = \frac{9}{2},$$

$$\therefore c = \frac{2}{25}, a = \frac{25}{9}, \therefore c + \frac{1}{a} = \frac{2}{25} + \frac{9}{25} = \frac{11}{25}.$$

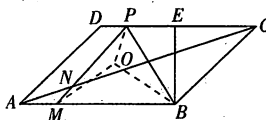
8. $2 - \sqrt{3}$ 【解析】过点 A 作 $AH \perp CD$ 于点 $H, \therefore$ 四边形 $ABCD$ 是菱形, $\angle BAD = 120^\circ, \therefore AB \parallel CD, \therefore \angle D + \angle BAD = 180^\circ, \therefore \angle D = 60^\circ, \therefore AD = AB = 2, \therefore AH = AD \cdot \sin 60^\circ = \sqrt{3}, \therefore B, B'$ 关于 EF 对称, $\therefore BE = EB',$ 当 BE 的值最小时, AE 的值最大, 根据垂线段最短可知, 当 $EB' = AH = \sqrt{3}$ 时, BE 的值最小, $\therefore AE$ 的最大值 $= 2 - \sqrt{3}$.

9. 10 【解析】设 $C(a, a), D(b, b)$, 则 $B(a, \frac{1}{a}), E(b, \frac{1}{b}), 9OB^2 - 4OE^2 = 9(a^2 + \frac{1}{a^2}) - 4(b^2 + \frac{1}{b^2}) = 9[(a - \frac{1}{a})^2 + 2] - 4[(b - \frac{1}{b})^2 + 2],$ 由已知得 $(a - \frac{1}{a}) + (b - \frac{1}{b}) = 2k, 3k$, 所以原式 $= 9 \times 4k^2 + 18 - 4 \times 9k^2 - 8 = 10$.

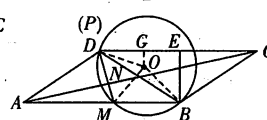
10. $\frac{7}{4}$ 【解析】设 $x^2 = y$, 原方程变为 $y^2 - 5y + (4 - k) = 0$, 设此方程有实根 $\alpha, \beta (0 < \alpha < \beta)$, 则原方程的四个实根为 $\pm\sqrt{\alpha}, \pm\sqrt{\beta}$, 由于它们对应的点在数轴上等距排列, $\sqrt{\beta} - \sqrt{\alpha} = \sqrt{\alpha} - (-\sqrt{\alpha}),$ 即 $\beta = 9\alpha$. 又 $\alpha + \beta = 5, \alpha\beta = 4 - k$, 则 $k = \frac{7}{4}$ 且满足 $\Delta = 25 + 4k - 16 > 0, \therefore k = \frac{7}{4}$.

11. -3 【解析】 $\therefore |1 + x| + |2 - x| = |x - (-1)| + |x - 2|$, 根据绝对值的几何意义可知 $|x - (-1)| + |x - 2|$ 表示数轴上一点 x 到 -1 与 2 的距离之和, 显然 $|x - (-1)| + |x - 2| \geq 3$, 同理 $|y + 2| + |y - 1| \geq 3$, 又 $\therefore (|1 + x| + |2 - x|)(|y + 2| + |y - 1|) = 9, \therefore |1 + x| + |2 - x| = 3, |y + 2| + |y - 1| = 3, \therefore -1 \leq x \leq 2, -2 \leq y \leq 1, \therefore x = -1, y = 1$ 时, $x - 2y$ 的最小值为 -3.

12. $2\sqrt{2} - 2$ 或 $1 < x \leq \frac{5}{3}$ 【解析】 $\therefore \square ABCD$ 的面积为 4, $AB = 4, BE \perp CD, \therefore BE = 1. \therefore$ 点 P 是唯一使得 $\angle MPB = 45^\circ$ 的点, 则可看成弦 MB 所对的圆周角 $\angle MPB = 45^\circ$ (如图①). 设 $\triangle MBP$ 外接圆的圆心为 O , 则 $\angle MOB = 90^\circ, \triangle OMB$ 为等腰直角三角形, $\therefore OB = \frac{\sqrt{2}}{2}x,$



第 12 题图①



第 12 题图②

(1) 当 $\odot O$ 经过点 D 时, 即点 P 在点 D 处时, 如图②所示,

$$\therefore CD \text{ 与 } AB \text{ 之间的距离为 } 1, \therefore OG = 1 - \frac{1}{2}x, \odot O \text{ 半径 } r = OB = \frac{\sqrt{2}}{2}x,$$

$$DG = 2 - \frac{1}{2}x, \therefore (\frac{\sqrt{2}}{2}x)^2 = (2 - \frac{1}{2}x)^2 + (1 - \frac{1}{2}x)^2, \text{ 解得 } x = \frac{5}{3}; \text{ 显然 } \odot O \text{ 与线段 } DE \text{ 只有一个交点, 符合题意;}$$

(2) 要让 $\odot O$ 与线段 DE 有且只有一个点, D 点应该在外接圆外, E 点应该在外接圆内. 于是 $OD = \sqrt{OG^2 + DG^2} = \sqrt{(1 - \frac{1}{2}x)^2 + (2 - \frac{1}{2}x)^2} > r = \frac{\sqrt{2}}{2}x, OE = \sqrt{OG^2 + GE^2} = \sqrt{(1 - \frac{1}{2}x)^2 + (\frac{1}{2}x)^2} < r = \frac{\sqrt{2}}{2}x,$

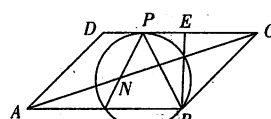
$$\text{即 } \begin{cases} \sqrt{(1 - \frac{1}{2}x)^2 + (2 - \frac{1}{2}x)^2} > \frac{\sqrt{2}}{2}x, \\ \sqrt{(1 - \frac{1}{2}x)^2 + (\frac{1}{2}x)^2} < \frac{\sqrt{2}}{2}x, \end{cases} \text{ 解得 } 1 < x < \frac{5}{3};$$

(3) 当圆与 DE 相切时, 如图③,

$$\frac{1}{2}x + \frac{\sqrt{2}}{2}x = 1, \text{ 解得 } x = 2\sqrt{2} - 2.$$

综上所述, 线段 BM 长 x 的取值范围

$$\text{是 } x = 2\sqrt{2} - 2 \text{ 或 } 1 < x \leq \frac{5}{3}.$$



第 12 题图③

13. 解: (1) 由 $\Delta = (2k - 1)^2 - 4k^2 \geq 0$, 解得 $k \leq \frac{1}{4}, \therefore x_1 + x_2 = 2k - 1, x_1 x_2 = k^2,$

$$\therefore k \leq \frac{1}{4}, \therefore x_1 + x_2 = 2k - 1 < 0, \text{ 而 } x_1 x_2 = k^2 \geq 0, \therefore x_1 \leq 0, x_2 \leq 0.$$

$$\therefore x_1 x_2 - |x_1| - |x_2| = 2, \therefore x_1 \cdot x_2 + x_1 + x_2 = 2, \text{ 即 } k^2 + (2k - 1) = 2, \text{ 整理得 } k^2 + 2k - 3 = 0, \text{ 解得 } k_1 = -3, k_2 = 1, \text{ 而 } k \leq \frac{1}{4}, \therefore k = -3.$$

$$(2) \therefore \frac{a}{b} + \frac{b}{a} = 6, \therefore a^2 + b^2 = 6ab, \therefore (a + b)^2 = 8ab, \therefore (b - a)^2 = (a + b)^2 - 4ab = 4ab, \therefore \left(\frac{a + b}{b - a}\right)^2 = \frac{(a + b)^2}{(b - a)^2} = 2, \therefore \frac{a + b}{b - a} = \pm\sqrt{2}.$$

$$\therefore a < b < 0, \therefore a + b < 0, b - a > 0, \therefore \frac{a + b}{b - a} < 0,$$

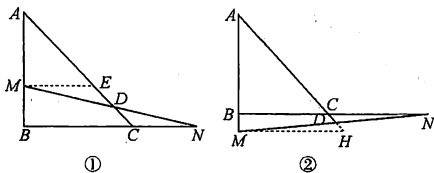
$$\therefore \frac{a+b}{b-a} = -\sqrt{2}, \therefore \left(\frac{a+b}{b-a}\right)^3 = -2\sqrt{2}.$$

14. 解: (1) $\because AB-4=0$ 且 $AB-BC=0$, $\therefore AB=BC=4$,

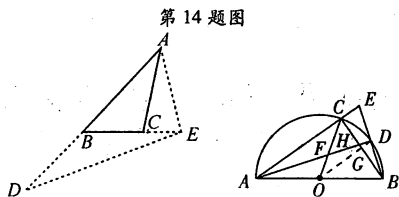
$$\therefore \frac{m}{AB-m} = \frac{AB}{BC} + 2, \therefore \frac{m}{4-m} = 3, \therefore m=3, \text{经检验得 } m=3.$$

(2) $DM=DN$. 理由如下: 如图①, 过 M 作 $ME \perp AB$ 交 AC 于点 E , $\because AB=BC$, $\angle B=90^\circ$, $\therefore \angle A = \angle ACB = 45^\circ$, $\therefore \angle AEM = 90^\circ - \angle A = 45^\circ$, $\therefore AM=ME$, $\therefore AM=CN$, $\therefore ME=CN$, $\therefore \angle AME = \angle B = 90^\circ$, $\therefore ME \parallel BC$, $\therefore \angle EMD = \angle N$, $\therefore \triangle MED \cong \triangle NCD$ (AAS), $\therefore DM=DN$.

(3) $DM=DN$. 理由如下: 如图②, 过 M 作 $MH \perp AM$ 交 AD 的延长线于点 H , 同(2)可证: $\triangle MHD \cong \triangle NCD$, $\therefore DM=DN$.



第 14 题图



第 15 题图

第 16 题图

15. 证明: 延长 BC 至点 E , 使 $AE=AC$, 延长 AB 至点 D , 使 $BD=AC$, 连接 DE . $\because \angle A: \angle B: \angle C = 1: 2: 4$, $\therefore \angle A = \frac{180^\circ}{7}$, $\angle B = \frac{360^\circ}{7}$, $\angle C = \frac{720^\circ}{7}$.

$$\therefore \angle EAC = 180^\circ - \left(\frac{180^\circ}{7} + \frac{360^\circ}{7}\right) \times 2 = \frac{180^\circ}{7}, \therefore EA=EB=BD, \therefore \angle D = \frac{180^\circ}{7}. \therefore \triangle ABC \sim \triangle DAE, \therefore \frac{AD}{AB} = \frac{AE}{BC}, \frac{c+b}{c} = \frac{b}{a}, ac+ab=bc, \therefore \frac{1}{b} + \frac{1}{c} = \frac{1}{a}.$$

16. (1) 证明: 如图①, $\because CF=CH$, $\therefore \angle CFH = \angle AHC$.

$$\therefore \angle AFO = \angle CFH, \therefore \angle AFO = \angle AHC.$$

$$\therefore \angle BDC = \angle CAD, \therefore \triangle AFO \sim \triangle AHC.$$

$$\therefore \frac{AF}{AH} = \frac{OF}{CH}, \therefore \frac{AF}{AH} = \frac{OF}{CF}, \therefore CF \cdot AF = OF \cdot AH.$$

(2) 解: 如图, 连接 OD 交 BC 于点 G . 设 $OG=x$, 则 $DG=2-x$.

$\because \angle COD = \angle BOD$, $\therefore OC=OB$, $\therefore OD \perp BC$, $CG=BG$. 在 $Rt\triangle OCG$ 和 $Rt\triangle BGD$ 中, 则有 $2^2 - x^2 = 1^2 - (2-x)^2$, $\therefore x = \frac{7}{4}$, 即 $OG = \frac{7}{4}$. $\because OA=OB$, $\therefore OG$ 是 $\triangle ABC$ 的中位线, $\therefore OG = \frac{1}{2}AC$, 即 $AC = \frac{7}{2}$.

17. (1) 证明: 如图①, 分别连接 OE, OF . $\because$ 四边形 $ABCD$ 是菱形, $\therefore AC \perp BD$, BD 平分 $\angle ADC$, $AD=DC=BC$, $\therefore \angle COD = \angle COB = \angle AOD = 90^\circ$, $\angle ADO = \frac{1}{2} \angle ADC = \frac{1}{2} \times 60^\circ = 30^\circ$, 又 $\because E, F$ 分别为 DC, CB 中点, $\therefore OE = \frac{1}{2}CD$, $OF = \frac{1}{2}BC$, $AO = \frac{1}{2}AD$, $\therefore OE=OF=OA$, $\therefore$ 点 O 即为 $\triangle AEF$ 的外心.

(2) 证明: 猜想: 外心 P 一定落在直线 DB 上. 理由如下: 如图②, 分别连接 PE, PA , 过点 P 分别作 $PI \perp CD$ 于点 I , $PJ \perp AD$ 于点 J , $\therefore \angle PIE = \angle PJD = 90^\circ$, $\therefore \angle ADC = 60^\circ$, $\therefore \angle IPJ = 360^\circ - \angle PIE - \angle PJD - \angle JDI = 120^\circ$, $\therefore$ 点 P 是等边三角形 AEF 的外心, $\therefore \angle EPA = 120^\circ$, $PE=PA$, $\therefore \angle IPJ = \angle EPA$, $\therefore \angle IPE = \angle JPA$, $\therefore \triangle PIE \cong \triangle PJA$, $\therefore PI=PJ$, $\therefore$ 点 P 在 $\angle ADC$ 的平分线上, 即点 P 落在直线 DB 上.

(3) $\frac{1}{DM} + \frac{1}{DN}$ 为定值 2. 理由如下: 当 $AE \perp DC$ 时, $\triangle AEF$ 面积最小, 此时点 E, F 分别为 DC, CB 的中点, 连接 BD, AC 交于点 P , 由(2)可得点 P 在 BD 上, 即为 $\triangle AEF$ 的外心. 如图③, 设 MN 交 BC 于点 G , $DM=x$, $DN=y$ ($x \neq 0, y \neq 0$), 则 $CN=y-1$. $\because BC \parallel DA$, $\therefore \triangle GBC \cong \triangle MDP$. $\therefore BG=DM=x$, $\therefore CG=1-x$, $\because BC \parallel DA$, $\therefore \triangle NCG \sim \triangle NDM$, $\therefore \frac{CG}{DM} = \frac{CN}{DN}$, $\therefore \frac{1-x}{y} = \frac{1-x}{x}$, $\therefore x+y=2xy$, $\therefore \frac{1}{x} + \frac{1}{y} = 2$, 即 $\frac{1}{DM} + \frac{1}{DN} = 2$.

全国重点高中提前招生 考试全真试卷(七)

1. C 【解析】关于 x 的分式方程 $\frac{ax-3}{x+1} = 1 - \frac{2}{x+1}$, 可化为 $ax-3=x+1-2$, 即 $(a-1)x=2$, 解得 $x = \frac{2}{a-1}$ ($a \neq 1$), 由于 $x=-1$ 是分式方程的增根,

$\therefore x=-1$ 是方程 $(a-1)x=2$ 的根, 即 $a=-1$, $\therefore a \neq -1$,

又 $\because$ 分式方程的解为负数, 即 $\frac{2}{a-1} < 0$, $\therefore a-1 < 0$, 即 $a < 1$, 且 $a \neq -1$,

由于 $\begin{cases} \frac{3y-1}{2} \leq -5, \\ y-a > 1 \end{cases}$ 无解, 即 $\begin{cases} y \leq -3, \\ y > a+1 \end{cases}$ 无解, $\therefore a+1 \geq -3$, $\therefore a \geq -4$,

$\therefore -4 \leq a < 1$, 且 $a \neq -1$, $\therefore$ 满足条件的整数 a 的值之和是 $-4-3-2+0=-9$. 故选 C.

2. D 【解析】 $\because x_1, x_2$ 是方程 $ax^2+bx+c=0$ 的两根, $\therefore x_1+x_2 = -\frac{b}{a}$,

$x_1x_2 = \frac{c}{a}$. 又 $\because$ 以 $-\frac{1}{x_1}, -\frac{1}{x_2}$ 为两根的一元二次方程为 $x^2 - \left(-\frac{1}{x_1} - \frac{1}{x_2}\right)x + \frac{1}{x_1x_2} = 0$, $\therefore x^2 + \left(\frac{1}{x_1} + \frac{1}{x_2}\right)x + \frac{1}{x_1x_2} = 0$, $\therefore x^2 - \frac{b}{c}x + \frac{a}{c} = 0$, 化简得 $cx^2 - bx + a = 0$. 不妨设方程 $cx^2 - bx + a = 0$ 的两根为 x_3, x_4 ($x_3 < x_4$), $\therefore x_3 = -\frac{1}{x_1}, x_4 = -\frac{1}{x_2}$, 又 $\because 1 < x_1 < 2 < x_2 < 4$, $\therefore -1 < x_3 < -\frac{1}{2}$. 故选 D.

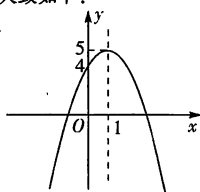
3. D 【解析】三个函数的判别式中至少一个非负: $1-4m \geq 0$, 或 $4(m^2-4) \geq 0$, 或 $m(4-3m) \geq 0$, 解得 $m \leq \frac{1}{4}$; $m \leq -2$ 或 $m \geq 2$; $0 \leq m \leq \frac{4}{3}$. 取其并集 $m \leq \frac{4}{3}$ 且 $m \neq 0$ 或 $m \geq 2$. 故选 D.

4. B 【解析】过点 D 作 $DH \perp AC$ 于 H , 设 $DH=x$, 则 $AD=2x$, $AH=\sqrt{3}x$, $CH=\sqrt{3}-\sqrt{3}x$. 由 $\triangle CDH \sim \triangle CBA$ 可得 $x = \frac{4}{5}$, $\therefore AD = \frac{8}{5}$. 故选 B.

5. D 【解析】过点 E 作 $EF \perp AD$ 于点 F , 则 $\triangle EFH \sim \triangle CDH$, 又 $\because EF:CD=EH:AD=1:2$, $\therefore S_{\triangle EFH}:S_{\triangle CDH}=1:4$, $\therefore \triangle GHE$ 是等腰三角形, 且 $S_{\triangle GHE}=2$, $\therefore S_{\triangle EFH}=1$, $\therefore S_{\triangle CDH}=4$. 故选 D.

6. C $7. n=1$ 或 $-3 \leq n < 0$ 【解析】当 $\Delta=4-4n=0$ 时, $n=1$, 此时抛物线的顶点 $(-1, 0)$ 即为函数的图象与 x 轴的唯一一个公共点, 此时 $x=-1$, 显然在 $-2 \leq x \leq 1$ 的范围内, 符合题意; 当 $\Delta=4-4n > 0$ 时, 若抛物线与 x 轴在 $-2 \leq x \leq 1$ 的范围内只有一个交点, 则还应满足: $x=-2$ 时, 函数值小于 0, 且 $x=1$ 时, 函数值大于或等于 0, 于是有 $\begin{cases} 4-4n > 0, \\ 4+2 \times (-2) + n < 0, \\ 1+2+n \geq 0, \end{cases}$ 得 $-3 \leq n < 0$. 综上所述, n 的取值范围是 $n=1$ 或 $-3 \leq n < 0$.

8. 0.5 【解析】二次函数 $y=-(x-1)^2+5$ 的图象大致如下:
① 当 $m < 0 \leq x \leq n < 1$ 时, 当 $x=m$ 时, y 取得最小值, 即 $2m=-(m-1)^2+5$, 解得 $m=-2$, 当 $x=n$ 时, y 取得最大值, 即 $2n=-(n-1)^2+5$, 解得 $n=2$ 或 -2 (均不符合题意, 舍去);
② 当 $m < 0 \leq x \leq 1 \leq n$ 时, 当 $x=m$ 时, y 取得最小值, 即 $2m=-(m-1)^2+5$, 解得 $m=-2$, 当 $x=1$ 时, y 取得最大值, 即 $2n=-(1-1)^2+5$, 解得 $n=2.5$, 或当 $x=n$ 时, y 取得最小值, 当 $x=1$ 时 y 取得最大值, 即 $2m=-(n-1)^2+5$, $n=2.5$, $\therefore m = \frac{11}{8}$. $\therefore m < 0$, 此种情况不符合题意, $\therefore m+n=-2+2.5=0.5$.



第 8 题图

9. $\frac{4041}{2}$ 【解析】由题意知 $P_{2021}(P_{2021}, 4041)$, 又 $\because P_{2021}$ 在 $y = \frac{6}{x}$ 上, $\therefore P_{2021} = \frac{6}{4041}$, 而 Q_{2021} (即 P_{2021}) 在 $y = \frac{3}{x}$ 上, $\therefore Q_{2021} = \frac{3}{\frac{6}{4041}} = \frac{4041}{2}$.

$$\therefore |P_{2021}Q_{2021}| = |P_{2021} - Q_{2021}| = \left| \frac{6}{4041} - \frac{4041}{2} \right| = \frac{4041}{2}.$$

10. $\frac{45\sqrt{34}}{17}$ 【解析】连接 AE , 由旋转变换的性质可知, $\angle ADE = \angle CDG$, $AD=BC \neq DE=17$, $AB=CD=DG=15$, 由勾股定理得, $CE = \sqrt{DE^2 - CD^2} = \sqrt{17^2 - 15^2} = 8$, $\therefore BE = BC - CE = 17 - 8 = 9$, 则 $AE = \sqrt{AB^2 + BE^2} = \sqrt{15^2 + 9^2} = 3\sqrt{34}$, $\therefore \frac{AD}{DC} = \frac{DE}{DG}$, $\angle ADE = \angle CDG$, $\therefore \triangle ADE \sim \triangle CDG$, $\therefore \frac{CG}{AE} = \frac{DC}{AD} = \frac{15}{17}$, 解得 $CG = \frac{45\sqrt{34}}{17}$.

11. 677 【解析】已知 C_1 方程为 $y = -x(x-3)$, C_2 方程为 $y = (x-3)(x-6)$, C_3 方程为 $y = -(x-6)(x-9)$, $\dots$, C_{2n-1} 的方程为 $y = -[x-3(2n-2)][x-3(2n-1)]$, C_{2n} 方程为 $y = [x-3(2n-1)] \cdot (x-3 \times 2n)$, 则 $P(2023, m)$ 在抛物线 C_{675} 上, 当 $x=2023$ 时, $m = -1 \times (-2) = 2$, 此时 $n=675$.

12. $\frac{\sqrt{3}}{337}$ 【解析】 $\because AC = \frac{BC}{\tan 30^\circ} = \sqrt{3}BC = 6$, $\therefore S_{\triangle ABC} = \frac{1}{2}AC \cdot BC = 6\sqrt{3}$, $\therefore D_1E_1 \perp AC$, $\therefore D_1E_1 \parallel BC$. $\therefore \triangle BD_1E_1$ 与 $\triangle CD_1E_1$ 同底同高, 面积相等. $\therefore D_1$ 是斜边 AB 的中点, $\therefore D_1E_1 = \frac{1}{2}BC$, $CE_1 = \frac{1}{2}AC$.

$$\therefore S_1 = \frac{1}{2} BC \cdot CE_1 = \frac{1}{2} BC \times \frac{1}{2} AC = \frac{1}{2} \times \frac{1}{2} AC \cdot BC = \frac{1}{2} S_{\triangle ABC},$$

$$\therefore \text{在} \triangle ACB \text{中}, D_2 \text{ 为其重心}, \therefore D_2 E_1 = \frac{1}{3} BE_1, \therefore D_2 E_2 = \frac{1}{3} BC, CE_2 = \frac{1}{3} AC, S_2 = \frac{1}{3} \times \frac{1}{2} \times AC \cdot BC = \frac{1}{3} S_{\triangle ABC}, \therefore D_3 E_3 = \frac{1}{4} BC, CE_3 = \frac{1}{4} AC,$$

$$S_3 = \frac{1}{4} S_{\triangle ABC}, \dots, S_n = \frac{1}{n+1} S_{\triangle ABC}, \therefore S_{2021} = \frac{1}{2021+1} \times 6\sqrt{3} = \frac{3\sqrt{3}}{1011} = \frac{\sqrt{3}}{337}.$$

$$13. (1) \text{解: 由等式 } \frac{(x^2-1)(y^2-1)}{xy} + \frac{(y^2-1)(z^2-1)}{yz} + \frac{(z^2-1)(x^2-1)}{zx} = 4 \text{ 去}$$

$$\text{分母得 } x(z^2-1)(y^2-1) + x(y^2-1)(z^2-1) + y(z^2-1)(x^2-1) = 4xyz, \\ \therefore x^2 y^2 z^2 + x y^2 z^2 + x^2 y z^2 - [x(y^2+z^2) + y(z^2+x^2) + z(x^2+y^2) + 3xyz] + \\ (x+y+z) - xyz = 0, \therefore xyz(xy+yz+zx) - (x+y+z)(xy+yz+zx) + (x+y+z) - xyz = 0, \\ \therefore [xyz - (x+y+z)](xy+yz+zx-1) = 0, \therefore xy+yz+zx \neq 1, \therefore xy+yz+zx-1 \neq 0, \therefore xyz - (x+y+z) = 0, \therefore xyz = x+y+z,$$

$$\therefore \text{原式} = \frac{x+y+z}{xyz} = 1.$$

$$(2) \text{证明: 由(1)得 } xyz = x+y+z, \text{ 又 } x, y, z \text{ 为正实数}, \therefore 9(x+y)(y+z)(z+x) - 8xyz(xy+yz+zx) = 9(x+y)(y+z)(z+x) - 8(x+y+z)(xy+yz+zx) \\ = x(y^2+z^2) + y(z^2+x^2) + z(x^2+y^2) - 6xyz = x(y-z)^2 + y(z-x)^2 + z(x-y)^2 \geq 0. \therefore 9(x+y)(y+z)(z+x) \geq 8xyz(xy+yz+zx).$$

$$14. \text{解: (1) 如图①中, 过点 } C \text{ 作 } CT \parallel AB \text{ 交 } EF \text{ 于点 } T.$$

$$\therefore BE \parallel CT, \therefore \angle B = \angle DCT,$$

$$\therefore BD = DC, \angle BDE = \angle CDT,$$

$$\therefore \triangle BDE \cong \triangle CDT (\text{ASA}), \therefore BE = CT.$$

$$\therefore CT \parallel AE, \therefore \frac{CF}{AE} = \frac{CT}{AE} = \frac{BE}{AE}, \therefore \frac{AB}{AE} + \frac{AC}{AF} =$$

$$\frac{AE+EB}{AE} + \frac{AF+CF}{AF} = 1 + \frac{EB}{AE} + 1 + \frac{CF}{AF} = 2.$$

$$(2) \text{如图②, 过点 } B \text{ 作 } BT \perp AD \text{ 交 } AD \text{ 的延长线于点 } T, \text{ 设 } DG = x, AG = y.$$

$$\therefore CG \perp AD, \therefore \angle CGD = \angle CGA = 90^\circ.$$

$$\therefore \angle ACD = 90^\circ,$$

$$\therefore \angle DCG + \angle ACG = 90^\circ, \angle ACG + \angle CAG = 90^\circ,$$

$$\therefore \angle DCG = \angle CAG, \therefore \triangle CGD \sim \triangle AGC,$$

$$\therefore \frac{GD}{CG} = \frac{CG}{AG}, \therefore CG = \sqrt{xy}.$$

$$\therefore BT \perp AT, \therefore \angle T = \angle CGD = 90^\circ.$$

$$\therefore \angle BDT = \angle CDG, BD = DC,$$

$$\therefore \triangle BDT \cong \triangle CDG (\text{AAS}),$$

$$\therefore BT = CG = \sqrt{xy}, DT = DG = x.$$

$$\therefore \angle CDG = \angle CDA, \angle ACD = \angle DGC = 90^\circ, \therefore \triangle DCG \sim \triangle DAC, \therefore CD^2 = DG \cdot DA, \therefore BD^2 = DG \cdot DA.$$

$$\therefore \frac{BD}{DG} = \frac{DA}{BD}, \text{ 又 } \angle BDG = \angle ADB, \therefore \triangle BDG \sim \triangle ADB,$$

$$\therefore \angle DBG = \angle BAD, \therefore \tan \angle BAT = \tan \angle CBF = \frac{BT}{AT} = \frac{\sqrt{6}}{7},$$

$$\therefore \frac{\sqrt{xy}}{2x+y} = \frac{\sqrt{6}}{7}, \text{ 整理得 } 24x^2 - 25xy + 6y^2 = 0, \therefore (3x-2y)(8x-3y) = 0,$$

$$\therefore 3x-2y=0 \text{ 或 } 8x-3y=0,$$

$$\therefore \frac{x}{y} = \frac{2}{3} \text{ 或 } \frac{3}{8}, \therefore \frac{DG}{AG} \text{ 的值为 } \frac{2}{3} \text{ 或 } \frac{3}{8}.$$

$$15. \text{解: (1) } a^2 = 2a+2, b^2 = 2b+2,$$

$$\therefore \frac{a^2}{b} + \frac{b^2}{a} = \frac{2(a+b)^2 - 4ab + 2(a+b)}{ab} = -10, \therefore m = -2.$$

$$(2) \text{由 } m = -2, y = x^2 - 2x + n. \text{ 令 } y = 0, \text{ 故 } x_1 + x_2 = 2, x_1 x_2 = n.$$

$$\therefore \frac{|x_1 - x_2|}{2} \cdot \sqrt{3} = |n-1|, \frac{3}{4} [(x_1 + x_2)^2 - 4x_1 x_2] = (n-1)^2,$$

$$3(1-n) = (n-1)^2, \therefore n = 1, \text{ 不合题意, 舍去}, \therefore n = -2.$$

$$(3) \text{由 } m = -2 \text{ 得 } y = -2x^2 - tx + 2, \text{ 对称轴 } x = -\frac{t}{4}.$$

$$(i) \text{ 当 } -\frac{t}{4} \geq 1, \text{ 即 } t \leq -4 \text{ 时, 若要 } y \leq 3 \text{ 成立, 则 } -2-t+2 \leq 3 \text{ 恒成立, 即 } t \geq -3, \text{ 与 } t \leq -4 \text{ 相矛盾, 故不可能};$$

$$(ii) \text{ 当 } -\frac{t}{4} \leq -1, \text{ 即 } t \geq 4 \text{ 时, 若要 } y \leq 3 \text{ 成立, 则 } -2+t+2 \leq 3 \text{ 恒成立, 即 } t \leq 3, \text{ 与 } t \geq 4 \text{ 相矛盾, 故不可能};$$

$$(iii) \text{ 当 } -1 < -\frac{t}{4} < 1, \text{ 即 } -4 < t < 4 \text{ 时, 若要 } y \leq 3 \text{ 恒成立, 则 } -2 \times \left(-\frac{t}{4}\right)^2 + t \times \left(-\frac{t}{4}\right) + 2 \leq 3 \text{ 恒成立, 即 } \frac{t^2}{4} \leq 1 \text{ 恒成立, 所以 } -2 \leq t \leq 2.$$

综上所述 t 的取值范围是 $-2 \leq t \leq 2$.

$$16. (1) \text{证明: 如图①, 连接 } OM, \therefore M \text{ 是 } \widehat{AC} \text{ 的中点}, \therefore OM \perp AC. \therefore MN \parallel AC, \therefore OM \perp MN. \therefore OM \text{ 为 } \odot O \text{ 的半径},$$

$$\therefore MN \text{ 为 } \odot O \text{ 的切线}.$$

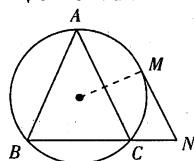
$$(2) \text{解: 如图②, 连接 } OM \text{ 交 } AC \text{ 于 } K, \text{ 连接 } AM,$$

$$\therefore M \text{ 是 } \widehat{AC} \text{ 的中点}, \therefore \widehat{AM} = \widehat{CM}, \therefore AM = CM = c.$$

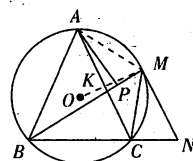
$$\therefore AP \perp BM, \therefore \angle APM = \angle APB = 90^\circ,$$

$$\therefore AP^2 = AM^2 - PM^2 = c^2 - b^2, \therefore AB^2 = AP^2 + BP^2 = c^2 - b^2 + a^2,$$

$$\therefore AC = AB = \sqrt{c^2 - b^2 + a^2}.$$



第 16 题图①



第 16 题图②

$$\therefore M \text{ 是 } \widehat{AC} \text{ 的中点}, \therefore OM \perp AC,$$

$$\therefore AK = CK = \frac{1}{2} AC = \frac{1}{2} \sqrt{c^2 - b^2 + a^2}.$$

$$\therefore \angle APB = \angle CKM = 90^\circ, \angle ABP = \angle MCK,$$

$$\therefore \triangle ABP \sim \triangle MCK, \therefore \frac{BP}{AB} = \frac{CK}{CM}, \therefore BP \cdot CM = CK \cdot AB,$$

$$\therefore ac = \frac{1}{2} \sqrt{c^2 - b^2 + a^2} \cdot \sqrt{c^2 - b^2 + a^2}, \therefore 2ac = c^2 - b^2 + a^2,$$

$$\therefore (a-c)^2 - b^2 = 0, \therefore (a+b-c)(a-b-c) = 0.$$

$$\therefore a+b-c > 0, \therefore a-b-c = 0, \therefore a = b+c.$$

$$(3) \text{解: 过点 } B \text{ 作 } BH \parallel AC, \text{ 过点 } D \text{ 作 } DH \parallel BC, BH \text{ 与 } DH \text{ 交于点 } H, \text{ 连接 } CH, \text{ 如图③, 则 } \angle BDH = \angle ABC = 60^\circ, \angle DBH = \angle ACB = 60^\circ,$$

$$\therefore \triangle BDH \text{ 是等边三角形}, \therefore BH = BD, \angle DBH = 60^\circ.$$

$$\therefore BH = CE, \angle CBH = \angle ABC + \angle DBH = 60^\circ + 60^\circ = 120^\circ.$$

$$\therefore \angle ACE = 180^\circ - \angle ACB = 120^\circ = \angle CBH, AC = BC,$$

$$\therefore \triangle ACE \cong \triangle CBH (\text{SAS}),$$

$$\therefore \angle CAE = \angle BCH, AE = CH.$$

$$\therefore DH \parallel BC, DH = CE,$$

$$\therefore \text{四边形 } CEDH \text{ 是平行四边形},$$

$$\therefore CE \parallel ED, CH = ED,$$

$$\therefore \angle BCH = \angle BED, CH = AE,$$

$$\therefore \angle BED = \angle CAE, AE = ED,$$

$$\text{过点 } E \text{ 作 } ET \perp AB \text{ 于点 } T, \text{ 交 } AC \text{ 于点 } L, \text{ 连接 } DL,$$

$$\text{则 } AT = TD = \frac{1}{2} AD, AL = DL. \therefore \angle BAC = 60^\circ, \therefore \triangle ADL \text{ 是等边三角形},$$

$$\therefore \angle ALD = 60^\circ = \angle ACB, \therefore DL \parallel BC, \text{ 即 } HD \text{ 与 } DL \text{ 在同一直线上},$$

$$\therefore \text{四边形 } BCLH \text{ 是平行四边形},$$

$$\therefore CL = BH = BD = CE, LH = BC. \text{ 设 } CE = x, \text{ 则 } CL = x, BC = AC = 5 - x,$$

$$AD = DL = AL = AC - CL = 5 - 2x, AT = \frac{5-2x}{2}. \therefore DF \parallel CH,$$

$$\therefore \frac{LF}{CL} = \frac{LD}{LH}, \text{ 即 } \frac{LF}{x} = \frac{5-2x}{5-x}, \therefore LF = \frac{(5-2x)x}{5-x},$$

$$\therefore AF = AL + LF = 5 - 2x + \frac{(5-2x)x}{5-x} = \frac{5(5-2x)}{5-x},$$

$$\text{在 Rt} \triangle BET \text{ 中}, ET = BE \cdot \sin 60^\circ = \frac{5\sqrt{3}}{2}. \therefore AE^2 = AT^2 + ET^2,$$

$$\therefore AE^2 = \left(\frac{5-2x}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2 = x^2 - 5x + 25. \text{ 延长 } BH, ED \text{ 交于点 } R, \text{ 则}$$

$$\angle RHD = \angle FCE, \angle R = \angle CFE, DH = CE,$$

$$\therefore \triangle HDR \cong \triangle CEF (\text{AAS}), \therefore DR = EF,$$

$$\therefore ER = ED + DR = AE + EF = 9 - AF = 9 - \frac{5(5-2x)}{5-x} = \frac{x+20}{5-x}.$$

$$\therefore CH \parallel ED, \therefore \frac{CH}{ER} = \frac{BC}{BE},$$

$$\therefore CH = \frac{BC}{BE} \cdot ER = \frac{5-x}{5} \times \frac{x+20}{5-x} = \frac{x+20}{5}, \therefore AE = \frac{x+20}{5},$$

$$\therefore x^2 - 5x + 25 = \left(\frac{x+20}{5}\right)^2, \text{ 解得 } x_1 = 5 (\text{舍去}), x_2 = \frac{15}{8},$$

$$\therefore AD = 5 - 2 \times \frac{15}{8} = \frac{5}{4}, AF = \frac{5(5-2x)}{5-x} = 10 - \frac{25}{5-\frac{15}{8}} = 2.$$

$$\text{作 } DM \perp AL \text{ 于点 } M, \text{ 则 } DM = AD \cdot \sin 60^\circ = \frac{5}{4} \times \frac{\sqrt{3}}{2} = \frac{5\sqrt{3}}{8},$$

$$\therefore S_{\triangle AEF} = S_{\triangle ADE} - S_{\triangle ADF} = \frac{1}{2} AD \cdot ET - \frac{1}{2} AF \cdot DM = \frac{1}{2} \times \frac{5}{4} \times \frac{5\sqrt{3}}{2} -$$

$$\frac{1}{2} \times 2 \times \frac{5\sqrt{3}}{8} = \frac{15\sqrt{3}}{16}.$$

$$17. \text{解: (1) } \therefore f(1) = f(2) = f(3) = \dots = f(9) = f(10) = 1, \therefore g(10) = \frac{f(10)}{81 + f(1) + f(2) + \dots + f(9)} = \frac{1}{81 + 1 + 1 + \dots + 1} = \frac{1}{90}.$$

$$(2) \text{当 } 1 \leq x \leq 20 \text{ 时}, f(1) = f(2) = \dots = f(19) = f(20) = 1,$$

$$\therefore g(x) = \frac{f(x)}{80 + f(1) + f(2) + \dots + f(x-1)} = \frac{1}{81 + 1 + 1 + \dots + 1} = \frac{1}{81 + x - 1} =$$

$$\frac{1}{x+80}; \text{ 当 } 21 \leq x \leq 60 \text{ 时}, f(x) = \frac{x}{10}, f(21) = \frac{1}{10} \times 21, f(22) = \frac{1}{10} \times 22,$$

$$\therefore g(x) = \frac{f(x)}{81 + f(1) + f(2) + \cdots + f(20) + f(21) + f(22) + \cdots + f(x-1)} = \frac{\frac{1}{10}x}{81 + \frac{1}{10} \times 21 + \frac{1}{10} \times 22 + \cdots + \frac{1}{10} \times (x-1)} = \frac{\frac{1}{10}x}{\frac{1}{10}(1010 + 21 + 22 + \cdots + x-1)} = \frac{x}{1010 + (21 + 22 + \cdots + x-1)} = \frac{2x}{x^2 - x + 1600} \therefore \text{当第 } x \text{ 个月的当月利润率为}$$

$$g(x) = \begin{cases} \frac{1}{x+80} (1 \leq x \leq 20), \\ \frac{2x}{x^2 - x + 1600} (21 \leq x \leq 60) \end{cases} (x \text{ 为正整数}).$$

(3) 当 $1 \leq x \leq 20$ 时, $g(x)$ 的最大值为 $g(1) = \frac{1}{81}$; 当 $21 \leq x \leq 60$ 时,

$$g(x) = \frac{2x}{x^2 - x + 1600} = \frac{2}{x + \frac{1600}{x} - 1} \leq \frac{2}{79}, \text{ 当且仅当 } \sqrt{x} = \frac{40}{\sqrt{79}}, \text{ 即 } x = \frac{1600}{79} \text{ 时, } g(x) \text{ 有最大值 } \frac{2}{79}.$$

$$\frac{2}{79} > \frac{1}{81}, \therefore \text{当 } x = 40 \text{ 时, } g(x) \text{ 有最大值 } \frac{2}{79}.$$

全国重点高中提前招生考试全真试卷(八)

1. D 【解析】 $\sqrt{6-3\sqrt{3}} + \sqrt{6+3\sqrt{3}} = \sqrt{\frac{(3-\sqrt{3})^2}{2}} + \sqrt{\frac{(3+\sqrt{3})^2}{2}} = \frac{\sqrt{2}}{2}(3-\sqrt{3}+3+\sqrt{3}) = 3\sqrt{2}$. 故选 D.

2. D 【解析】 $\because x^2 - x - a^2 + a + 1 > 0$ 对任意实数 x 成立, 即函数 $y = x^2 - x - a^2 + a + 1$ 与 x 轴没有交点, $\therefore \Delta = 1 - 4(-a^2 + a + 1) < 0$. 即 $4a^2 - 4a - 3 < 0$. $\therefore (2a+1)(2a-3) < 0$. $\therefore -\frac{1}{2} < a < \frac{3}{2}$. 故选 D.

3. B 【解析】 $(x-2)(x-3) - p^2 = 0, x^2 - 5x + 6 - p^2 = 0, a=1, b=-5, c=6-p^2, \Delta = b^2 - 4ac = (-5)^2 - 4 \times 1 \times (6-p^2) = 1+4p^2 > 0, x = \frac{5 \pm \sqrt{1+4p^2}}{2}, \therefore 1+4p^2 \geq 1, \therefore x_1 = \frac{5+\sqrt{1+4p^2}}{2} \geq \frac{5+1}{2} = 3, x_2 = \frac{5-\sqrt{1+4p^2}}{2} \leq \frac{5-1}{2} = 2, \therefore \alpha \leq 2 \text{ 且 } \beta \geq 3$. 故选 B.

4. C 【解析】延长 AD, BC 相交于点 E , $\therefore \angle A = 90^\circ, \angle ABC = 30^\circ, \therefore \angle E = 90^\circ - 30^\circ = 60^\circ, \therefore \angle EDC = 90^\circ - 60^\circ = 30^\circ$.

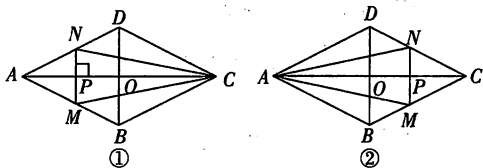
在 $\text{Rt}\triangle DCE$ 中, $CD = 2\sqrt{3}$,

$$\therefore CE = CD \tan 30^\circ = 2\sqrt{3} \times \frac{\sqrt{3}}{3} = 2,$$

$$\therefore DE = 2CE = 4, \therefore AE = DE + AD = 4 + 2 = 6,$$

在 $\text{Rt}\triangle ABE$ 中, $\angle ABC = 30^\circ, AE = 6, \therefore AB = \frac{AE}{\tan 30^\circ} = 6\sqrt{3}$. 在 $\text{Rt}\triangle ABD$ 中, $BD = \sqrt{AD^2 + AB^2} = \sqrt{4 + 108} = 4\sqrt{7}$. 故选 C.

5. A 【解析】(1) 当 $0 < x \leq 1$ 时, 如图①, 在菱形四边形 $ABCD$ 中, $AC = 2, BD = 1, AO = 1$, 且 $AC \perp BD$. $\therefore MN \perp AC, \therefore MN \parallel BD$,



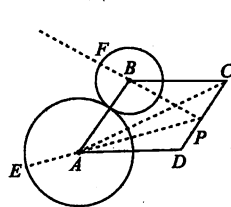
第 5 题图

$$\therefore \triangle AMN \sim \triangle ABD, \therefore \frac{AP}{AO} = \frac{MN}{BD}, \text{ 即 } \frac{x}{1} = \frac{MN}{1} \therefore MN = x, \therefore y = \frac{1}{2} CP \times$$

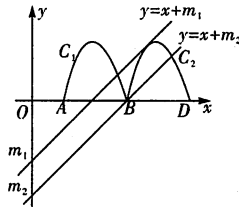
$$MN = \frac{1}{2} (2-x)x = -\frac{1}{2}x^2 + x (0 < x \leq 1). \therefore -\frac{1}{2} < 0, \therefore \text{函数图象开口向下}; (2) \text{ 当 } 1 < x < 2 \text{ 时, 如图②, 同理可证 } \triangle CDB \sim \triangle CNM, \frac{CP}{OC} = \frac{MN}{BD}, \text{ 即}$$

$$\frac{2-x}{1} = \frac{MN}{1}, \therefore MN = 2-x, \therefore y = \frac{1}{2} CP \times MN = \frac{1}{2} (2-x) \times (2-x) = \frac{1}{2} (x-2)^2. \therefore \frac{1}{2} > 0, \therefore \text{函数图象开口向上. 综上, 答案 A 的图象大致符合. 故选 A.}$$

6. A 【解析】如图, 连接 PA , 延长 PA 交 $\odot A$ 于点 E , 连接 PB , 延长 PB 交 $\odot B$ 于点 F , 观察图象可知, 当点 P 与点 C 重合时, $PE + PF$ 的值最大, $\therefore AC = 2 \times 6 \times \frac{\sqrt{3}}{2} = 6\sqrt{3}, \therefore PE + PF$ 的最大值 $= AC + AE + BC + BF = 6\sqrt{3} + 12$. 故选 A.



第 6 题图



第 8 题图

7. -1 【解析】由 $a - b + 4 = 0$ 可得 $b = a + 4, \therefore ab = a(a+4) < 0$.

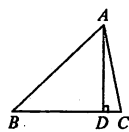
$$\therefore -4 < a < 0. \therefore \text{整数 } a \text{ 的最大值为 } -1.$$

8. $-3 < m < -\frac{15}{8}$ 【解析】令 $y = -2x^2 + 8x - 6 = 0$, 即 $x^2 - 4x + 3 = 0$, 解得 $x = 1$ 或 $x = 3$, 则点 $A(1, 0), B(3, 0)$. 由于将 C_1 向右平移 2 个单位长度得 C_2 , 则 C_2 对应的函数解析式为 $y = -2(x-4)^2 + 2 (3 \leq x \leq 5)$, 如图所示, 当 $y = x + m_1$ 与 C_2 相切时, 令 $x + m_1 = -2(x-4)^2 + 2$, 即 $2x^2 - 15x + 30 + m_1 = 0$, 由 $\Delta = 15^2 - 8(30 + m_1) = -8m_1 - 15 = 0$, 解得 $m_1 = -\frac{15}{8}$; 当 $y = x + m_2$ 过点 $B(3, 0)$ 时, $0 = 3 + m_2$, 解得 $m_2 = -3$. 结合图象知当 $-3 < m < -\frac{15}{8}$ 时, 直线 $y = x + m$ 与 C_1, C_2 共有 3 个不同的交点.

9. $\frac{1}{132}$ 【解析】 $\because a_{n,n} = a_{n,1} = \frac{1}{n}, \therefore a_{11,11} = a_{11,1} = \frac{1}{11}, a_{12,12} = a_{12,1} = \frac{1}{12}$. 又 $\because a_{n,m} = a_{n+1,m} + a_{n+1,m+1}, \therefore a_{12,11} = a_{11,11} - a_{12,12} = \frac{1}{11} - \frac{1}{12} = \frac{1}{132}$.

10. $\frac{1+\sqrt{5}}{2}$ 【解析】面积相等, 得 $(a+b)^2 = (a+b+b) \cdot b$.

11. 4 【解析】如图, 设 AB, BC, AC 分别为 $n+1, n, n-1$, 则 $BD - CD = \frac{BD^2 - CD^2}{BD + CD} = \frac{(AB^2 - AD^2) - (AC^2 - AD^2)}{BC} = \frac{AB^2 - AC^2}{BC} = \frac{(n+1)^2 - (n-1)^2}{n} = \frac{4n}{n} = 4$.

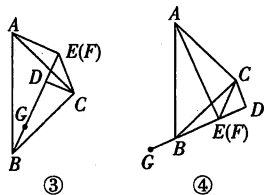


第 11 题图

12. $\frac{2\sqrt{5}}{5}$ 或 $\frac{\sqrt{2}}{2}$ 【解析】由题意可得点 B 坐标为 $(\frac{2}{\sqrt{k}}, 2\sqrt{k})$, 点 A 的坐标为 $(\frac{1}{\sqrt{k}}, \sqrt{k})$, $\therefore BD \perp x$ 轴, $\therefore$ 点 C 横坐标为 $\frac{2}{\sqrt{k}}$, 纵坐标为 $\frac{1}{2}\sqrt{k}$, $\therefore BA = \sqrt{\frac{1}{k} + k}, AC = \sqrt{\frac{1}{k} + \frac{k}{4}}, BC = \frac{3}{2}\sqrt{k}$, $\therefore BA^2 - AC^2 = \frac{3}{4}k > 0, \therefore BA \neq AC$, ①当 $AB = BC$ 时, 则 $\sqrt{\frac{1}{k} + k} = \frac{3}{2}\sqrt{k}$, 解得 $k = \pm \frac{2\sqrt{5}}{5}$ (舍负); ②当 $AC = BC$ 时, 同理可得 $k = \frac{\sqrt{2}}{2}$.

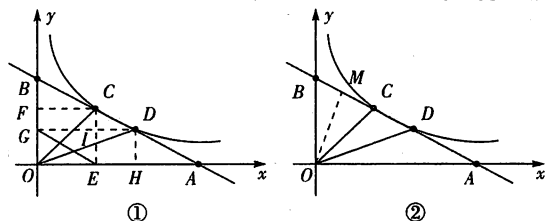
13. 解: 依题意有 $\Delta = 4(m-2)^2 - 4(m^2 - 3m + 3) = -4m + 4 > 0, \therefore m < 1$, 结合题意知 $-1 \leq m < 1$. (1) $\because x_1^2 + x_2^2 = (x_1 + x_2)^2 - 2x_1x_2 = 4(m-2)^2 - 2(m^2 - 3m + 3) = 2m^2 - 10m + 10 = 6, \therefore m = \frac{5 \pm \sqrt{17}}{2}, \therefore -1 \leq m < 1, \therefore m = \frac{5 - \sqrt{17}}{2}$. (2) $\frac{mx_1^2}{1-x_1} + \frac{mx_2^2}{1-x_2} = \frac{m[x_1^2 + x_2^2 - x_1x_2(x_1 + x_2)]}{(1-x_1)(1-x_2)} = \frac{m(2m^2 - 8m^2 + 8m - 2)}{m^2 - m} = \frac{2m(m-1)(m^2 - 3m + 1)}{m(m-1)} = 2(m^2 - 3m + 1) = 2(m - \frac{3}{2})^2 - \frac{5}{2} (-1 \leq m < 1), \therefore$ 对称轴 $m = \frac{3}{2}, 2 > 0, \therefore$ 当 $m = -1$ 时, 原代数式取最大值为 10.

14. 解: (1) $CD \perp BG, CD = BG$. 理由如下: 如图①, 延长 CD 交 AB 于点 P , $\because \triangle EDC$ 是等腰直角三角形, $\therefore \angle ECD = 45^\circ, \therefore \angle ACB = 90^\circ, \therefore \angle ACP = \angle BCP = 45^\circ, \therefore AC = BC, \therefore PC \perp AB$, 即 $CD \perp BG, \therefore \triangle PCB$ 是等腰直角三角形, $\therefore PC = PB, \therefore DF \perp AC, AC \perp BC, \therefore FG \parallel BC, \therefore \angle DGP = \angle B = 45^\circ, \therefore \triangle PDG$ 是等腰直角三角形. $\therefore PD = PG, \therefore PC - PD = PB - PG$, 即 $CD = BG$. (2) 如图②, 延长 ED 至点 H , 使 $DH = DE$, 连接 CH, BH , 延长 BH 交 AE 于点 K , 设 AC 与 BK 交于点 $O, \therefore \triangle CDE$ 是等腰直角三角形, $\therefore CD \perp DE, \therefore CE = CH, \angle ECD = \angle HCD = 45^\circ, \therefore \angle ECH = 90^\circ, \therefore \triangle CEH$ 为等腰直角三角形. $\therefore \angle ECH = \angle ACB = 90^\circ, \therefore \angle ACE = \angle BCH$. 又 $\because AC = BC, EC = HC, \therefore \triangle ACE \cong \triangle BCH (SAS), \therefore AE = BH, \angle EAC = \angle HBC$. 又 $\because \angle AOK = \angle BOC, \therefore \angle AKB = \angle ACB = 90^\circ$. 又 $\because DF \perp AE, \therefore BH \parallel GE$. 又 $\because BG \parallel EH, \therefore$ 四边形 $BHDG$ 为平行四边形, $\therefore DH = BG$. 又 $\because CD = DE = DH, CD \perp DH, \therefore CD \perp BG, CD = BG$. (3) 存在两种情况: ①如图③, 由 (2) 知 $CD = DE = BG = 2$, 当 E, D, G 三点共线时, $\text{Rt}\triangle BCD$ 中, $\angle CDB = 90^\circ, \therefore CD = 2, BC = 4\sqrt{2}, \therefore BD = \sqrt{(4\sqrt{2})^2 - 2^2} = 2\sqrt{7}, \therefore DG = BD - BG = 2\sqrt{7} - 2$; ②如图④, 当 E, D, G 三点共线, 同理可得 $BD = 2\sqrt{7}, \therefore DG = BD + BG = 2\sqrt{7} + 2$. 综上所述, DG 的长为 $2\sqrt{7} - 2$ 或 $2\sqrt{7} + 2$.



第 14 题图

15. 解: 如图①, 作 $CE \perp OA$ 于 E , $CF \perp OB$ 于 F , $DG \perp OB$ 于 G , $DH \perp OA$ 于 H , CE 与 DG 相交于 I , 连接 GE . 设 $C(m, \frac{k}{m})$, $D(n, \frac{k}{n})$, $\therefore GI = OE = m$, $DG = OH = n$, $IE = OG = \frac{k}{n}$, $CE = OF = \frac{k}{m}$, $\therefore \frac{GI}{DG} = \frac{IE}{CE} = \frac{m}{n}$,



第 15 题图

$\therefore GE \parallel CD$, $\therefore DG \parallel OA$, $\therefore$ 四边形 $GEAD$ 是平行四边形, $\therefore AD = GE$, 同理可得 $BC = GE$, $\therefore BC = AD$, 如图②, 作 $OM \perp AB$ 于 M ,

$\therefore OB = OC$, $\therefore BM = CM = \frac{1}{2}BC$, $\angle COM = \frac{1}{2}\angle BOC$.

$\therefore \angle COD = \frac{1}{2}\angle AOC$,

$\therefore \angle DOM = \angle COM + \angle COD = \frac{1}{2}(\angle BOC + \angle AOC) = 45^\circ$.

设 $OB = a$, $OA = b$, $\therefore AB = \sqrt{OA^2 + OB^2} = \sqrt{a^2 + b^2}$.

$\therefore \cos \angle ABO = \frac{BM}{OB} = \frac{OB}{AB}$, $\therefore BM = \frac{OB^2}{AB} = \frac{a^2}{\sqrt{a^2 + b^2}}$,

$\therefore AD = BC = 2 \cdot \frac{a^2}{\sqrt{a^2 + b^2}}$, $\therefore CD = AB - 2AD = \sqrt{a^2 + b^2} - 4 \cdot \frac{a^2}{\sqrt{a^2 + b^2}}$,

$\therefore DM = CM + CD = \sqrt{a^2 + b^2} - 3 \cdot \frac{a^2}{\sqrt{a^2 + b^2}}$

$\therefore S_{\triangle AOB} = \frac{1}{2}AB \cdot OM = \frac{1}{2}OA \cdot OB$, $\therefore OM = \frac{ab}{\sqrt{a^2 + b^2}}$

$\therefore \triangle DOM$ 是等腰直角三角形, $\therefore OM = DM$, $\therefore \frac{ab}{\sqrt{a^2 + b^2}} = \sqrt{a^2 + b^2} - 3 \cdot \frac{a^2}{\sqrt{a^2 + b^2}}$,

$3 \cdot \frac{a^2}{\sqrt{a^2 + b^2}}$, $\therefore 2a^2 + ab - b^2 = 0$, $\therefore 2(\frac{a}{b})^2 + \frac{a}{b} - 1 = 0$, $\therefore (\frac{2a}{b} - 1) \cdot (\frac{a}{b} + 1) = 0$, $\therefore \frac{a}{b} = \frac{1}{2}$ 或 $\frac{a}{b} = -1$ (舍去), $\therefore \frac{a}{b} = \frac{1}{2}$.

16. 解: (1) 由射影定理, 得 $OB^2 = OC \cdot OA$, $OA = 8$, $OB = 4$, $\therefore OC = 2$, $C(-2, 0)$.

(2) $\therefore MN \parallel x$ 轴, $\therefore \triangle BPM \cong \triangle BCA$, $\therefore \frac{PM}{10} = \frac{BP}{BC}$ ①.

同理, $\frac{PN}{10} = \frac{PD}{CD} = \frac{PC + BC}{BC}$ ②.

① + ②, 得 $\frac{PM + PN}{10} = \frac{PC + BP + BC}{BC} = \frac{BC + BC}{BC} = 2$,

$\therefore PM + PN = 20$.

17. 解: (1) 由抛物线 $y = ax^2 - 2ax + 4$ 知: 对称轴 $x = 1$, $C(0, 4)$.

$\therefore S_{\text{四边形AFBC}} = S_{\triangle ABC} + S_{\triangle ABF} = \frac{1}{2}AB \cdot (OC + OF) = \frac{1}{2}AB \cdot (4 + 1) = 15$,

$\therefore AB = 6$. 又 A, B 两点关于 $x = 1$ 对称, 且 $AB = 6$, $\therefore A(-2, 0)$, $B(4, 0)$, 将 $B(4, 0)$ 代入 $y = ax^2 - 2ax + 4$ 中, 得 $16a - 8a + 4 = 0$, 解得 $a = -\frac{1}{2}$,

$\therefore$ 抛物线的解析式为 $y = -\frac{1}{2}x^2 + x + 4$.

(2) 在 $\triangle ACF$ 中, $OA = 2$, $OF = 1$, $OC = 4$, 即 $\frac{OA}{OF} = \frac{OC}{OA}$.

又 $\therefore \angle COA = \angle AOF$, $\therefore \triangle AOC \sim \triangle FOA$. $\therefore \angle CAO = \angle AFO$, $\angle CAF = \angle CAO + \angle FAO = \angle AFO + \angle FAO = 90^\circ$, 延长 AF 交直线 CP 于点 D , 在 $\text{Rt}\triangle ADC$ 中, $AC = \sqrt{OC \cdot CF} = 2\sqrt{5}$, $\tan \angle DCA = \frac{3}{2}$, 则 $AD = 3\sqrt{5}$.

又 $\therefore \tan \angle OAF = \frac{OF}{OA} = \frac{1}{2}$,

$\therefore \sin \angle OAF = \frac{\sqrt{5}}{5}$, $\cos \angle OAF = \frac{2\sqrt{5}}{5}$. 由 $AD = 3\sqrt{5}$ 解得 $D(4, -3)$, 设直线

$y_{CD} = kx + 4$, 代入 D 点的坐标可得 $k = -\frac{7}{4}$.

$$\text{联立} \begin{cases} y = -\frac{7}{4}x + 4, \\ y = -\frac{1}{2}x^2 + x + 4, \end{cases} \text{解得} \begin{cases} x_1 = 0, \\ x_2 = \frac{11}{2}, \end{cases} \begin{cases} y_1 = 4, \\ y_2 = -\frac{45}{8}. \end{cases} \therefore P(\frac{11}{2}, -\frac{45}{8}).$$

(3) 设圆心 O' 的坐标为 $(1, y)$, 则 $O'A^2 = 9 + y^2$, $O'C^2 = 1 + (y - 4)^2 = y^2 - 8y + 17$, $\therefore O'A = O'C$, $\therefore 9 + y^2 = y^2 - 8y + 17$, 解得 $y = 1$, $\therefore \odot O'$ 的半径 $R = \sqrt{10}$. 延长 AE 交 $\odot O'$ 于点 G ,

$\therefore \angle EAB = \frac{1}{2}\angle MNB$, $\therefore G$ 是 $\widehat{MB}$ 的中点, 即 $\widehat{MG} = \widehat{BG}$, 过 G 作 $\odot O'$ 的直径 GH , 连接 GH, HM, MG , 则 $\triangle HMG$ 是直角三角形, 且 $\angle HMG = 90^\circ$.

$\therefore \angle MAG = \angle EAS$ ($\widehat{MG} = \widehat{BG}$), $\angle HMG = \angle ESA = 90^\circ$, $\therefore \triangle HMG \sim \triangle ASE$, 得 $\frac{HG}{AE} = \frac{MG}{ES}$, 即 $\frac{AE \cdot MG}{ES} = HG = 2R$ ①. 连接 AM, AN .

$\therefore \widehat{MG} = \widehat{BG}$, $\widehat{AN} = \widehat{BN}$, $\therefore \angle GAB = \angle MAE$, $\angle AME = \angle BAN$. $\therefore \angle GEM = \angle MAE + \angle AME$, $\angle GMN = \angle GAB + \angle BAN$, $\therefore \angle GEM = \angle GMN$, 即 $MG = GE$, 代入 ① 式, 得 $\frac{AE \cdot GE}{ES} = 2R = 2\sqrt{10}$. 由相交弦定理得 $ME \cdot NE = AE \cdot GE$, $\therefore \frac{ME \cdot NE}{ES} = \frac{AE \cdot GE}{ES} = 2\sqrt{10}$. 综上所述, $\frac{ME \cdot NE}{ES}$ 的值不会发生变化, 且值为 $2\sqrt{10}$.

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1. C 【解析】依题意, 得 $\begin{cases} m - 2 \neq 0 \text{ ①,} \\ \Delta = 4^2 - 4(m - 2) \times (-1) \geq 0 \text{ ②,} \end{cases}$ 解得 $m \geq -2$ 且 $m \neq 2$. 故选 C.

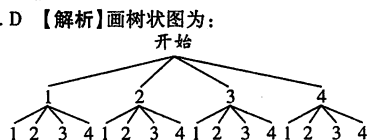
2. A 【解析】过点 B 作 $BM \perp DE$ 于点 M , 设 $A(a, \frac{k}{a})$, 则 $B(\frac{a}{2}, \frac{2k}{a})$. 由题意得 $OE = -\frac{1}{3}a$, $DE = -\frac{2}{3}a$, $ME = -\frac{1}{6}a$, $BM = \frac{2k}{a}$, $AD = \frac{k}{a}$, $DM = -\frac{1}{2}a$, $\therefore S_{\triangle ABE} = S_{\text{梯形ADMB}} + S_{\triangle BEM} - S_{\triangle ADE} = 7$, $\therefore \frac{1}{2}(\frac{k}{a} + \frac{2k}{a}) \times (-\frac{1}{2}a) + \frac{1}{2} \times (-\frac{1}{6}a) \times (\frac{2k}{a}) - \frac{1}{2} \times \frac{k}{a} \times (-\frac{2}{3}a) = 7$, 解得 $k = -12$. 故选 A.

3. C

4. B 【解析】点 A, B 在二次函数 $y_1 = x^2$ 第一象限的图象上, 则 $k = a^2$ 且 $k + 1 = b^2$, 即 $b^2 = a^2 + 1$, 对于函数 y_2 , $\Delta = (\frac{b}{a})^2 - 4 \times \frac{m}{a} = \frac{b^2 - 4am}{a^2}$,

当 $m = \frac{1}{4}$ 时, $\Delta = \frac{b^2 - 4am}{a^2} = \frac{(a - \frac{1}{2})^2 + \frac{3}{4}}{a^2} > 0$, 故 $m = \frac{1}{4}$, 则 y_2 与 x 轴必有 2 个交点, 故 D 正确, 不符合题意; 当 $m = -1$ 时, 同理可得 $\Delta = \frac{a^2 + 4a + 1}{a^2} > 0$, 故 C 正确, 不符合题意; 当 $m = \frac{1}{2}$ 时, 同理可得 $\Delta = \frac{(a - 1)^2}{a^2} \geq 0$, 故 B 错误, 符合题意; 同理可得 A 正确, 不符合题意. 故选 B.

5. D 【解析】画树状图如下:



共有 16 种等可能的结果数, 其中两次抽取的卡片上数字之积为偶数的结果数为 12, $\therefore$ 所取两卡片上数字之积为偶数的概率是 $\frac{12}{16} = \frac{3}{4}$. 故选 D.

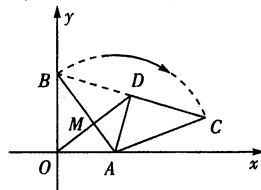
6. B 【解析】连接 PP' 交 BC 于点 O , 若四边形 $QPCP'$ 为菱形, 则 $PP' \perp QC$, $\therefore \angle POQ = 90^\circ$. $\therefore \angle ACB = 90^\circ$, $\therefore PO \parallel AC$, $\therefore \frac{AP}{AB} = \frac{CO}{CB}$. 设点 Q 运动的时间为 t s, 则 $AP = \sqrt{2}t$, $QB = t$, $\therefore QC = 6 - t$, $CO = 3 - \frac{1}{2}t$, $AB = 6\sqrt{2}$.

$$\therefore \frac{\sqrt{2}t}{6\sqrt{2}} = \frac{3 - \frac{1}{2}t}{6}, \text{解得 } t = 2. \text{ 故选 B.}$$

7. 4 【解析】由隐含条件 $(m - 2)n^2 \geq 0$ 可得 $m - 2 \geq 0$. 即 $m \geq 2$, 原式化简得 $-(4 - 2m) + (n - 2) + \sqrt{(m - 2)n^2} = 2m - 4$, $\therefore (n - 2)^2 + \sqrt{(m - 2)n^2} = 0$, $\therefore n - 2 = 0$, $n = 2$, $\sqrt{(m - 2)n^2} = 0$, $m - 2 = 0$, $m = 2$, $m + n = 4$.

8. $y = -\frac{7}{24}x + 4$ 【解析】 $\therefore \triangle BOA$ 绕点 A

按顺时针方向旋转得 $\triangle CDA$, $\therefore \triangle BOA \cong \triangle CDA$. $\therefore \angle DOA = \angle OBA$, $\angle OAM = \angle BAO$, $\therefore \triangle AOM \sim \triangle ABO$, $\therefore \angle AMO = \angle AOB = 90^\circ$, $\therefore OD \perp AB$.



第 8 题图

$\therefore AO = AD$,

$\therefore \angle OAM = \angle DAM$. 在 $\triangle AOB$ 和 $\triangle ABD$ 中, $\begin{cases} OA = DA, \\ \angle BAO = \angle BAD, \\ AB = AB, \end{cases} \therefore \triangle AOB \cong \triangle ABD$ (SAS), $\therefore OM = DM$,

$\therefore \triangle ABD \cong \triangle ACD$, $\therefore \angle ADB = \angle ADC = 90^\circ$,

$\therefore B, D, C$ 三点共线. 设直线 AB 解析式为 $y = kx + b$, 把 A 与 B 坐标代入

$$\begin{cases} 3k + b = 0, \\ b = 4, \end{cases} \text{解得} \begin{cases} k = -\frac{4}{3}, \\ b = 4, \end{cases}$$

$\therefore$ 直线 AB 解析式为 $y = -\frac{4}{3}x + 4$, $\therefore$ 直线 OD 解析式为 $y = \frac{3}{4}x$, 联立

$$\begin{cases} y = \frac{3}{4}x, \\ y = -\frac{4}{3}x + 4, \end{cases} \text{解得} \begin{cases} x = \frac{36}{25}, \\ y = \frac{48}{25}, \end{cases} \text{即 } M(\frac{36}{25}, \frac{48}{25}).$$

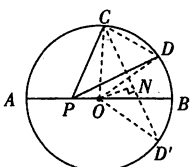
$\therefore M$ 为线段 OD 的中点, $\therefore D(\frac{72}{25}, \frac{96}{25})$. 设直线 CD 解析式为 $y = mx + n$,

$$\begin{cases} n = 4, \\ \frac{96}{25}m + n = \frac{72}{25}, \end{cases}$$

解得 $m = -\frac{7}{24}$, $n = 4$, 则直线 CD 解析式为 $y = -\frac{7}{24}x + 4$.

9. -1 或 8 【解析】当 $x + y + z = 0$ 时, 原式 = -1; 当 $x + y + z \neq 0$ 时, $x + y = 2z$, $x + z = 2y$, $y + z = 2x$. 原式 = 8.

10. $\sqrt{3}$ 【解析】将半圆补成整圆 (圆心为 O), 作 D 点关于直径 AB 的对称点 D' , 连接 CD' , 作 $ON \perp CD'$, 则 $\angle DOB = 36^\circ$, $\angle AOC = 96^\circ$, $\therefore \angle COD = 48^\circ$, $\therefore \angle BOD' = 36^\circ$, $\therefore \angle COD' = 36^\circ + 36^\circ + 48^\circ = 120^\circ$. 又 $ON = \frac{1}{2}$,



第 10 题图

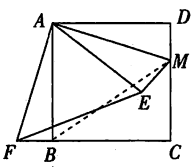
$$\therefore CN = \sqrt{1 - (\frac{1}{2})^2} = \frac{\sqrt{3}}{2}.$$

$$\therefore CD' = \sqrt{3}.$$

$$\therefore CD' = PC + PD', \therefore PC + PD' = \sqrt{3}.$$

即 $CP + PD$ 的最小值为 $\sqrt{3}$.

11. 5 【解析】如图, 连接 BM , 依题意有 $AE = AD$, $\angle MAD = \angle MAE$, $AF = AM$, $\angle FAB = \angle MAD$, $\therefore \angle FAB = \angle MAE$. $\therefore \angle FAB + \angle BAE = \angle BAE + \angle MAE$. $\therefore \angle FAE = \angle MAB$. $\therefore \triangle FAE \cong \triangle MAB$. $\therefore EF = BM$. 又 $BC = CD = AB = 4$, $DM = 1$, $\therefore CM = 3$. $\therefore$ 在 $Rt\triangle BCM$ 中, $BM = \sqrt{3^2 + 4^2} = 5$, $\therefore EF = 5$.



第 11 题图

12. 4 $2 < x \leq 4$ 【解析】由题意知 $x < 5$, 当 $x = 4$ 时, 第四次值为 244, 当 $x = 2$ 时, 第五次值为 244.

13. 解: (1) 原式 = $(\sqrt{5xy} + 3\sqrt{5xy} - \frac{1}{2}\sqrt{5xy}) \div \frac{\sqrt{xy}}{y} = \frac{7\sqrt{xy}}{2} \div \frac{\sqrt{xy}}{y} = \frac{7}{2}y$.

(2) 原方程变形为 $\frac{1}{x-1} + \frac{k}{x+2} = \frac{2k+3}{(x+2)(x-1)}$, 两边乘以 $(x-1)(x+2)$ 得 $x+2+k(x-1) = 2k+3$, 整理得 $(k+1)x = 3k+1$, 当 $k = -1$ 时, $(k+1)x = 3k+1$ 无解, 原方程也无解, 当 $k \neq -1$ 时, $(k+1)x = 3k+1$ 的解是 $x = \frac{3k+1}{k+1}$. 若 $\frac{3k+1}{k+1} = 1$ 或 $\frac{3k+1}{k+1} = -2$, 原方程无解, $\therefore k = 0$ 或 $k = -\frac{3}{5}$. 综上所述, $k = -1$ 或 $k = 0$ 或 $k = -\frac{3}{5}$, 原方程无实数解.

14. 解: $\therefore a = \frac{1}{2}\sqrt{\sqrt{2} + \frac{1}{8} - \frac{\sqrt{2}}{8}}$, $\therefore (a + \frac{\sqrt{2}}{8})^2 = \frac{1}{4} \cdot (\sqrt{2} + \frac{1}{8})$, 整理得 $4a^2 + \sqrt{2}a = \sqrt{2}$, 则 $2\sqrt{2}a^2 + a = 1$, $\therefore a = 1 - 2\sqrt{2}a^2$, $\therefore a^4 + a + 1 = a^4 + 2 - 2\sqrt{2}a^2 = (a^2 - \sqrt{2})^2$, $\therefore a^2 + \sqrt{a^4 + a + 1} = a^2 + \sqrt{a^2 - \sqrt{2}} = a^2 + \sqrt{2} - a^2 = \sqrt{2}$, $\therefore a^2 + \sqrt{a^4 + a + 1} = a^2 + |a^2 - \sqrt{2}| = a^2 + \sqrt{2} - a^2 = \sqrt{2}$.

15. 解: (1) $BD = MF$, $BD \perp MF$. 理由如下: 如图①, 延长 FM 交 BD 于点 N , 由题意得 $\triangle BAD \cong \triangle MAF$, $\therefore BD = MF$, $\angle ADB = \angle AFM$. 又 $\therefore \angle DMN = \angle AMF$, $\therefore \angle ADB + \angle DMN = \angle AFM + \angle AMF = 90^\circ$. $\therefore \angle DNM = 90^\circ$, $\therefore BD \perp MF$.

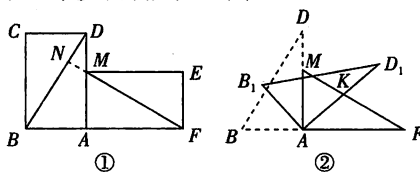
(2) 如图②, ①当 $AK = FK$ 时, $\angle KAF = \angle F = 30^\circ$, 则 $\angle BAB_1 = 180^\circ - \angle B_1AD_1 - \angle KAF = 180^\circ - 90^\circ - 30^\circ = 60^\circ$, 即 $\beta = 60^\circ$; ②当 $AF = FK$ 时, $\angle FAK = \frac{1}{2}(180^\circ - \angle F) = 75^\circ$, $\therefore \angle BAB_1 = 90^\circ - \angle FAK = 15^\circ$, 即 $\beta = 15^\circ$.

综上所述, β 的度数为 60° 或 15° .

(3) 如图③, 由题意得矩形 PNA_2A . 设 $A_2A = x$, 则 $PN = x$, 在 $Rt\triangle A_2M_2F_2$ 中, $\therefore F_2M_2 = FM = 16$, $\angle F = \angle A_2F_2P = \angle ADB = 30^\circ$, $\therefore A_2M_2 = 8$, $A_2F_2 = 8\sqrt{3}$, $\therefore AF_2 = 8\sqrt{3} - x$. $\therefore AP = AF_2 \cdot \tan 30^\circ = 8 - \frac{\sqrt{3}x}{3}$, $\therefore PD = AD - AP = 8\sqrt{3} - 8 + \frac{\sqrt{3}x}{3}$. $\therefore NP \parallel AB$, $\therefore \angle DNP = \angle B$. $\therefore \angle D = \angle D$,

$\therefore \triangle DPN \sim \triangle DAB$, $\therefore \frac{PN}{AB} = \frac{DP}{DA}$, $\therefore \frac{x}{8} = \frac{8\sqrt{3} - 8 + \frac{\sqrt{3}x}{3}}{8\sqrt{3}}$, 解得 $x = 12 - 4\sqrt{3}$, 即

$A_2A = 12 - 4\sqrt{3}$. $\therefore$ 平移的距离是 $(12 - 4\sqrt{3})$ cm.



第 15 题图

16. (1) 证明: 如图①所示, 连接 PA, PB, AG .

$\therefore AT \perp AB$, $\therefore \angle TAB = 90^\circ$, 即 $\angle TAP + \angle PAB = 90^\circ$.

又 $\therefore \widehat{PA} = \widehat{PA}$, $\therefore \angle PBA = \angle PCA$.

$\therefore AB$ 是 $\odot O$ 的直径, $\therefore \angle APB = 90^\circ$,

$\therefore \angle PBA + \angle PAB = 90^\circ$,

$\therefore \angle TAP = \angle PBA = \angle PCA$.

又 $\therefore \angle ATP = \angle GTA$, $\therefore \triangle ATP \sim \triangle GTA$, $\therefore \frac{AT}{GT} = \frac{PT}{AT}$,

$\therefore AT^2 = TP \cdot TG$.

(2) 解: 如图②所示, 设 $\triangle BDC$ 的外接圆为 $\odot O$, 过 O 作 $OP \perp DC$, 垂足为 P ,

显然, $\angle DBC = \frac{1}{2} \angle DOC$,

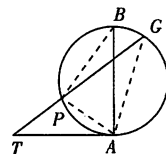
$\therefore OD = OC$, $OP \perp DC$,

$\therefore \angle DOP = \angle COP = \frac{1}{2} \angle DOC$,

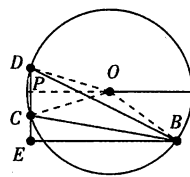
$\therefore \angle DBC = \angle DOP$.

故当 $\angle DOP$ 最大时, $\angle DBC$ 最大,

$\therefore$ 直径 $OP \perp$ 弦 DC , $DC = 4$, $\therefore DP = \frac{1}{2}DC = 2$.



第 16 题图①



第 16 题图②

又 $\therefore \sin \angle DOP = \frac{DP}{OD} = \frac{2}{OB}$, 故当 OB 最小时, $\angle DOP$ 最大,

$\therefore$ 圆心在直线 OP 上运动, 故半径 OB 最小时, $\angle DOP$ 最大, 根据垂线段最短知当 $OB \perp BE$ 时, 半径 OB 取得最小值 4, 此时 $BE = OP = \sqrt{4^2 - 2^2} = 2\sqrt{3}$.

$\therefore$ 当 $BE = 2\sqrt{3}$ 时, $\angle CBD$ 最大.

17. 解: (1) $\therefore$ 点 B 的横坐标为 4, 点 D 的纵坐标为 8, $AD \parallel x$ 轴, $AB \parallel y$ 轴, $\therefore$ 点 A 的坐标为 $(4, 8)$. 将 $A(4, 8)$, $C(8, 0)$ 两点坐标分别代入 $y = ax^2 + bx$, 得 $\begin{cases} 16a + 4b = 8, \\ 64a + 8b = 0, \end{cases}$ 解得 $\begin{cases} a = -\frac{1}{2}, \\ b = 4, \end{cases}$

故抛物线的解析式为 $y = -\frac{1}{2}x^2 + 4x$.

(2) ①易知 $\tan \angle CAB = \frac{EP}{AP} = \frac{1}{2}$, $\therefore AP = t$, $BP = 8 - t$, $\therefore EP = \frac{1}{2}t$.

$\therefore HG = t$. $\therefore H$ 坐标为 $[4 - \frac{1}{2}t, -\frac{1}{2}(4 - \frac{1}{2}t)^2 + 4(4 - \frac{1}{2}t)]$, 易求 $\tan \angle APH = \frac{4}{8-t}$, $\tan \angle CPB = \frac{4}{8-t}$, 即 $\angle APH = \angle CPB$, $\therefore H, P, C$ 三点在同一直线上, $\therefore S_{\triangle AHC} = \frac{1}{2} \times AP \times (4 + \frac{1}{2}t) = \frac{1}{4}t^2 + 2t = 16$, 解得 $t = -4 + 4\sqrt{5}$ 或 $-4 - 4\sqrt{5}$ (舍负), 即当 $t = -4 + 4\sqrt{5}$ 时, $\triangle HAC$ 的面积为 16;

②取 AC 中点 R , 连接 DR , 过 D 点作 $DH \perp AC$, 易求 $AC = 4\sqrt{5}$, $DR = 2\sqrt{5}$, $DH = \frac{8\sqrt{5}}{5}$, $\therefore \sin \angle DRA = \frac{DH}{DR} = \frac{4}{5}$,

$\therefore E(4 + \frac{1}{2}t, 8 - t)$, $Q(8, t)$, $\therefore MQ = CQ \times \sin \angle ACD = t \cdot \frac{\sqrt{5}}{5} = \frac{\sqrt{5}}{5}t$, $EQ = \sqrt{(\frac{1}{2}t - 4)^2 + (8 - 2t)^2}$.

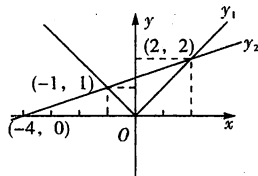
$\therefore \angle QEM = \angle QRA = 2\angle QCE$, $\therefore \frac{\frac{\sqrt{5}}{5}t}{\sqrt{(\frac{1}{2}t - 4)^2 + (8 - 2t)^2}} = \frac{4}{5}$, 整理

得 $63t^2 - 576t + 1280 = 0$, $\therefore (3t - 16)(21t - 80) = 0$, 解得 $t = \frac{16}{3}$ 或 $\frac{80}{21}$.

故 $t = \frac{16}{3}$ 或 $\frac{80}{21}$ 时, $\angle QEM = 2\angle QCE$.

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1. D 【解析】如图所示, 当 $y_1 > y_2$ 时, $x > 2$ 或 $x < -1$. 故选 D.



第1题图

2. A

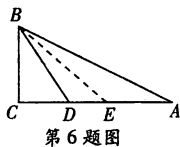
3. A 【解析】将点 $(0, 2)$, $(1, 2)$ 的坐标代入 $y = ax^2 + bx + c$, 得 $\begin{cases} 2 = c, \\ 2 = a + b + c, \end{cases}$ 解得 $\begin{cases} c = 2, \\ b = -a, \end{cases}$ $\therefore$ 二次函数的解析式为 $y = ax^2 - ax + 2$.

$\therefore$ 当 $x = \frac{3}{2}$ 时, 对应的函数值 $y < 0$, $\therefore \frac{9}{4}a - \frac{3}{2}a + 2 < 0$, $\therefore a < -\frac{8}{3}$, 即 $b > \frac{8}{3}$, $\therefore a < 0, b > 0, c > 0$, $\therefore abc < 0$, 故 A 项不正确; $\therefore$ 当 $x = -1$ 时, $y = m$, 当 $x = 2$ 时, $y = n$, $\therefore m = a + a + 2 = 2a + 2, n = 4a - 2a + 2 = 2a + 2$, $\therefore mn = (2a + 2)^2$, $\therefore a < -\frac{8}{3}$, $\therefore mn = (2a + 2)^2 > [2 \times (-\frac{8}{3}) + 2]^2$, $\therefore mn > \frac{100}{9}$, 故 B 项正确; 设方程 $ax^2 + bx + c = 0$ 两个实数根分别为 x_1, x_2 , 根据根与系数有 $x_1 + x_2 = -\frac{b}{a}, x_1 x_2 = \frac{c}{a}$, 又 $\begin{cases} 2 = c, \\ b = -a, \end{cases} a < -\frac{8}{3}$, $\therefore x_1 + x_2 = 1, -\frac{3}{4} < x_1 x_2 = \frac{2}{a} < 0$, $\therefore$ 关于 x 的方程 $ax^2 + bx + c = 0$ 一定有一正、一负两个实数根, $\therefore$ 抛物线过点 $(0, 2), (1, 2)$, $\therefore$ 抛物线的对称轴为直线 $x = \frac{1}{2}$, 又 $\therefore$ 当 $x = \frac{3}{2}$ 时, 对应的函数值 $y < 0$, 根据对称性, 当 $x = -\frac{1}{2}$ 时, 对应的函数值 $y < 0$, 而当 $x = 0$ 时, $y = 2 > 0$, $\therefore$ 抛物线与 x 轴负半轴交点横坐标在 $-\frac{1}{2}$ 和 0 之间, $\therefore$ 关于 x 的方程 $ax^2 + bx + c = 0$ 的负实数根在 $-\frac{1}{2}$ 和 0 之间, 故 C 项正确; $\therefore P_1(t+2, y_1)$ 和 $P_2(t-2, y_2)$ 在该二次函数的图象上, $\therefore y_1 = a(t+2)^2 - a(t+2) + 2, y_2 = a(t-2)^2 - a(t-2) + 2$. 若 $y_1 > y_2$, 则 $a(t+2)^2 - a(t+2) + 2 > a(t-2)^2 - a(t-2) + 2$, 即 $a(t+2)^2 - a(t+2) > a(t-2)^2 - a(t-2)$, $\therefore a < 0$, $\therefore (t+2)^2 - (t+2) < (t-2)^2 - (t-2)$, 解得 $t < \frac{1}{2}$, 故 D 项正确. 故选 A.

4. B 【解析】 $2^2 = 3 + 5$, 第一项为 $2^2 - 2 + 1$, 最后一项为 $3 + 2 \times 1$; $3^2 = 7 + 9 + 11$, 第一项为 $3^2 - 3 + 1$, 最后一项为 $7 + 2 \times 2$; $4^2 = 13 + 15 + 17 + 19$, 第一项为 $4^2 - 4 + 1$, 最后一项为 $13 + 2 \times 3$, $\therefore 45^2$ 的第一项为 $45^2 - 45 + 1 = 1981$, 最后一项为 $1981 + 2 \times 44 = 2069$, 1981 到 2069 之间有奇数 2019 , $\therefore m$ 的值为 45 . 故选 B.

5. B 【解析】延长 BC 交 CH 于点 E , 易证 $\triangle ABC \cong \triangle CDH$ (SSS), $\triangle ABC \cong \triangle BCE$ (ASA), $\therefore BE = AG = 8, CE = BC = 6, \angle BEC = \angle AGB = 90^\circ$, $\therefore GE = BE - BC = 8 - 6 = 2$, 同理可得 $HE = 2$, 在 $Rt\triangle GHE$ 中, $GH = \sqrt{2^2 + 2^2} = 2\sqrt{2}$. 故选 B.

6. C 【解析】设 $BC = x$, 则 $BD = \sqrt{x^2 + 1}, AB = \sqrt{x^2 + 25}$, 如图, 作 $\triangle ABD$ 角平分线 BE , 则 $\triangle BDE \sim \triangle ADB$, 因此 $BD^2 = DE \cdot DA = 4DE$. 由角平分线定理可知 $\frac{DE}{AE} = \frac{BD}{AB}$, $\therefore \frac{DE}{AE + DE} = \frac{BD}{AB + BD}$, 即 $\frac{DE}{4} = \frac{BD}{AB + BD}$, $\therefore DE = \frac{4BD}{AB + BD}$, $\therefore BD^2 = 4DE = \frac{16BD}{AB + BD}$, $\therefore x^2 + 1 = \frac{16\sqrt{x^2 + 1}}{\sqrt{x^2 + 25} + \sqrt{x^2 + 1}}$, 解得 $BC = x = \frac{5\sqrt{7}}{7}$. 故选 C.



第6题图

7. 1 【解析】将 p 代入方程①②, 得 $\begin{cases} p^2 + ap + b = 0, \\ p^2 + bp + c = 0, \end{cases}$ $\therefore (a-b)p = c-b$, 同理可得 $(b-c)q = a-c; (c-a)r = b-a$. 三式相乘, 得 $pqr = -1$. 考虑到若 p, q, r 互不相等, 则可应用韦达定理, 故对 p, q, r 能否相等进行分析.

(1) 若 $p = r$, 则方程①②③有公共根 p , 故 $\begin{cases} (a-b)p = c-b, \\ (b-c)p = a-c, \\ (c-a)p = b-a, \end{cases}$ $\therefore 1 - a + b = 0, 1 - b + c = 0, 1 - c + a = 0$, 三式相加, 得 $3 \neq$

0, 不成立. 由此可知, p, q, r 互不相等.

(2) p, q, r 互不相等时, 由韦达定理, 得 $\begin{cases} pr = b, \\ pq = c, \\ qr = a, \end{cases}$

$\therefore abc = (pqr)^2 = 1$. 综上所述, $abc = 1$.

8. 1, 3 或 5 【解析】显然 $a \neq 0$. 故原方程为关于 x 的二次方程, 则 $\Delta = [-(3a^2 - 8a)]^2 - 4a^2(2a^2 - 13a + 15) = [a(a+2)]^2$ 是完全平方, 故 $x = \frac{(3a^2 - 8a) \pm a(a+2)}{2a^2}$, 即 $x_1 = \frac{2a-3}{a} = 2 - \frac{3}{a}, x_2 = \frac{a-5}{a} = 1 - \frac{5}{a}$. 当 $2 - \frac{3}{a}$ 是整数时, $a = 1, 3$; 当 $1 - \frac{5}{a}$ 是整数时, $a = 1, 5$. 综上所述 $a = 1, 3$ 或 5 .

9. $y = -\frac{3}{x} (x > 0)$

10. $4 + 4\sqrt{2}$ 【解析】取 MN 的中点 E , 连接 OE, PE, OP , 则有 $OE = \frac{1}{2}MN =$

$4, PE = \sqrt{PN^2 + NE^2} = 4\sqrt{2}$, 又 $\therefore OP \leq PE + OE = 4 + 4\sqrt{2}$, $\therefore OP$ 的最大值为 $4 + 4\sqrt{2}$, 即点 P 到原点 O 距离的最大值是 $4 + 4\sqrt{2}$.

11. $2 - \sqrt{2} \leq d \leq 1$ 【解析】如图, 连接 AO , 过点 O 作 $OE \perp AB$ 于点 E , 根据题意, O 为正方形中心, $\therefore OE = AE = 1, \therefore AO = \sqrt{OE^2 + AE^2} = \sqrt{2}$. 当点 A 落在 OP 上时, 点 P 到正方形的最小距离为 $OP - AO = 2 - \sqrt{2}$. 当点 E 落在 OP 上时, 点 P 到正方形的最小距离为 $OP - OE = 2 - 1 = 1$, $\therefore 2 - \sqrt{2} \leq d \leq 1$.

12. -32

13. 解: (1) $\frac{1}{\frac{n(n+1)-1}{n(n+1)}} = \frac{1}{n^2 + n - 2 + \frac{1}{n(n+1)}} =$

$\frac{1}{n^2 + n - 2} = \frac{1}{(n-1)(n+2)}$, $\therefore S = \frac{1}{9 \times 12} + \frac{1}{10 \times 13} + \dots + \frac{1}{2021 \times 2024} = \frac{1}{3} \times \left[\left(\frac{1}{9} - \frac{1}{12} \right) + \left(\frac{1}{10} - \frac{1}{13} \right) + \dots + \left(\frac{1}{2021} - \frac{1}{2024} \right) \right] = \frac{1}{3} \times \left(\frac{1}{9} + \frac{1}{10} + \frac{1}{11} - \frac{1}{2022} - \frac{1}{2023} - \frac{1}{2024} \right)$,

$90S = \frac{30}{9} + 3 + \frac{30}{11} - \frac{30}{2022} - \frac{30}{2023} - \frac{30}{2024} = 9 + \frac{6}{99} - \frac{30}{2022} - \frac{30}{2023} - \frac{30}{2024}$, $\therefore [90S] = 9$

(2) 设 $\frac{x-1}{2} = k$, 则 $(2k)^2 - 4 = 12[k], k^2 - 1 = 3[k], 3k - 3 < k^2 - 1 \leq 3k$, 左边可得 $k < 1$ 或 $k > 2$, 右边有 $k^2 - 3k \leq 1 < 4, -1 < k < 4$, 也即 $-1 < k < 1$ 或 $2 < k < 4, [k] = -1, 0, 2$ 或 3 , 代入 $k^2 - 1 = 3[k]$ 验证得 $k = \sqrt{7}$ 或 $\sqrt{10}$, $\therefore x = 2\sqrt{7} + 1$ 或 $2\sqrt{10} + 1$.

14. 解: (1) 小玲摸到 C 棋的概率等于 $\frac{3}{10}$.

(2) 小玲在这一轮中胜小军的概率是 $\frac{4}{9}$.

(3) ①若小玲摸到 A 棋, 小玲胜小军的概率是 $\frac{5}{9}$; ②若小玲摸到 B 棋, 小玲胜小军的概率是 $\frac{7}{9}$; ③若小玲摸到 C 棋, 小玲胜小军的概率是 $\frac{4}{9}$; ④若小玲摸到 D 棋, 小玲胜小军的概率是 $\frac{1}{9}$. 综上所述, 小玲希望摸到 B 棋, 小玲胜小军的概率最大.

15. 解: (1) 设抛物线的解析式为 $y = ax^2 + c$, $\therefore$ 正方形 OABC 的边长为 8, 点 A(8, 0), C(0, 8),

$\therefore$ 将 A, C 点代入 $y = ax^2 + c$, 得 $\begin{cases} 8^2 + c = 0, \\ c = 8, \end{cases}$ 解得 $\begin{cases} a = -\frac{1}{8}, \\ c = 8, \end{cases}$ $\therefore$ 抛物线

的解析式为 $y = -\frac{1}{8}x^2 + 8$.

(2) 对于任意一点 P, PD 与 PF 的差为定值, 这个猜想是正确的. 理由如下: 设 $P(x, -\frac{1}{8}x^2 + 8)$, 则 $PF = 8 - (-\frac{1}{8}x^2 + 8) = \frac{1}{8}x^2$, 过点 P 作 $PM \perp y$ 轴于点 M, 则 $PD^2 = PM^2 + DM^2 = (-x)^2 + [6 - (-\frac{1}{8}x^2 + 8)]^2 = \frac{1}{64}x^4 + \frac{1}{2}x^2 + 4 = (\frac{1}{8}x^2 + 2)^2$, $\therefore PD = \frac{1}{8}x^2 + 2$.

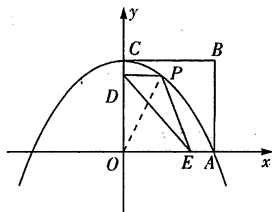
$\therefore PD - PF = \frac{1}{8}x^2 + 2 - \frac{1}{8}x^2 = 2$, 故该猜想正确.

(3) $\therefore$ 当点 P 运动时, DE 的大小不变, $\therefore PE$ 与 PD 的和最小时, $\triangle PDE$ 的周长最小. $\therefore PD - PF = 2, \therefore PD = PF + 2, \therefore PE + PD = PE + PF + 2$, $\therefore$ 当 P, E, F 三点共线时, PE + PF 最小, 此时, 点 P, E 的横坐标均为 4, 将 $x = 4$ 代入 $y = -\frac{1}{8}x^2 + 8$, 得 $y = 6, \therefore P$ 点坐标为 (4, 6), 此时 $\triangle PDE$ 周长最小, 且 $\triangle PDE$ 的面积为 $\frac{1}{2} \times 4 \times 6 = 12$, 点 P 恰为“好点”,

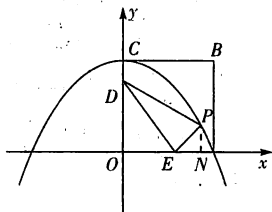
$\therefore \triangle PDE$ 周长最小时“好点”的坐标为 (4, 6). 由 (2) 得 $P(a, -\frac{1}{8}a^2 +$

8) 点D, E 的坐标分别为(0,6), (4,0).

如图①, 当 $0 < a \leq 4$ 时, $S_{\triangle PDE} = S_{\triangle PEO} + S_{\triangle POD} - S_{\triangle DOE} = \frac{1}{2} \times 4 \times (-\frac{1}{8}a^2 + 8) + \frac{1}{2} \times 6 \times a - \frac{1}{2} \times 4 \times 6 = -\frac{1}{4}a^2 + 3a + 4$,



①



②

第15题图

$\therefore 4 < S_{\triangle PDE} \leq 12$;

$a=0$ 时, $S_{\triangle PDE}=4$;

如图②, 当 $4 < a < 8$ 时, 过点P作 $PN \perp x$ 轴于点N.

$S_{\triangle PDE} = S_{\text{梯形}PNOD} - S_{\triangle PNE} - S_{\triangle DOE} = \frac{1}{2}a(-\frac{1}{8}a^2 + 8 + 6) - \frac{1}{2}(a-4) \cdot (-\frac{1}{8}a^2 + 8) - \frac{1}{2} \times 4 \times 6 = -\frac{1}{4}a^2 + 3a + 4, \therefore 12 < S_{\triangle PDE} \leq 13$;

当 $a=8$ 和 4 时, $S_{\triangle PDE}=12$. 故 $\triangle PDE$ 的面积可以等于 4 到 13 所有整数, 在面积为 12 时, a 的值有两个为 4 和 8 , 所以面积为整数时好点有 11 个, 经过验证周长最小的好点包含这 11 个之内, 所以好点共 11 个. 综上所述, 11 个好点, $\triangle PDE$ 周长最小时“好点”的坐标 $P(4, 6)$.

16. (1) 证明: 连接 AB , $\because CA$ 切 $\odot O'$ 于 A , $\therefore \angle CAB = \angle F$.

$\therefore \angle CAB = \angle E$, $\therefore \angle E = \angle F$. $\therefore AF \parallel CE$. $\therefore \frac{PE}{PF} = \frac{PC}{PA}$,

$\therefore PA \cdot PE = PC \cdot PF$.

(2) 证明: $\because \frac{PE}{PF} = \frac{PC}{PA}$, $\therefore \frac{PE^2}{PF^2} = \frac{PC^2}{PA^2}$,

$\therefore \frac{PE^2}{PC^2} = \frac{PF^2}{PA^2}$. 由切割线定理得 $PA^2 = PB \cdot PF$, $\therefore \frac{PE^2}{PC^2} = \frac{PF}{PB}$.

(3) 解: 连接 AE , 由(1)知 $\triangle PEC \sim \triangle PFA$, 而 $PC:CE:EP=3:4:5$, $\therefore PA:FA:PF=3:4:5$. 设 $PC=3x$, $CE=4x$, $EP=5x$, $PA=3y$, $FA=4y$, $PF=5y$, $\therefore EP^2 = PC^2 + CE^2$, $PF^2 = PA^2 + FA^2$,

$\therefore \angle C = \angle CAF = 90^\circ$, $\therefore AE$ 为 $\odot O$ 的直径, AF 为 $\odot O'$ 的直径.

$\therefore \odot O$ 与 $\odot O'$ 为等圆, $\therefore AE=AF=4y$. $\therefore AC^2 + CE^2 = AE^2$,

$\therefore (3x+3y)^2 + (4x)^2 = (4y)^2$, 即 $25x^2 + 18xy - 7y^2 = 0$,

$\therefore (25x-7y)(x+y) = 0$, $\therefore \frac{x}{y} = \frac{7}{25}$, $\therefore S_{\triangle ECP}:S_{\triangle FAP} = \frac{x^2}{y^2} = \frac{49}{625}$.

17. 解: (1) 将 $y = -\frac{4}{3}x - 4$ 化为 $4x + 3y + 12 = 0$, 由上述距离公式得 $d = \frac{|4 \times 3 + 3 \times 2 + 12|}{\sqrt{4^2 + 3^2}} = 6$, $\therefore$ 点M到直线AB的距离为6.

(2) 假设抛物线上存在点P, 使得 $\triangle PAB$ 的面积最小,

设 $P(x, x^2 - 4x + 5)$, 则点P到直线AB的距离为 $d = \frac{|4x + 3(x^2 - 4x + 5) + 12|}{\sqrt{4^2 + 3^2}}$, 由图象知, 点P到直线AB的距离最小时

$x > 0$, $x^2 - 4x + 5 > 0$,

$\therefore d = \frac{|4x + 3(x^2 - 4x + 5) + 12|}{\sqrt{4^2 + 3^2}} = \frac{3x^2 - 8x + 27}{5} = \frac{3}{5}(x - \frac{4}{3})^2 + \frac{13}{3}$,

$\therefore$ 当 $x = \frac{4}{3}$ 时, d 最小, 为 $\frac{13}{3}$; 当 $x = \frac{4}{3}$ 时, $x^2 - 4x + 5 = (\frac{4}{3})^2 - 4 \times \frac{4}{3} + 5 = \frac{13}{9}$, $\therefore P(\frac{4}{3}, \frac{13}{9})$;

在 $y = -\frac{4}{3}x - 4$ 中, 令 $x=0$, 则 $y=-4$, $\therefore B(0, -4)$,

令 $y=0$, 则 $x=-3$, $\therefore A(-3, 0)$, $\therefore AB = \sqrt{4^2 + 3^2} = 5$,

$\therefore \triangle PAB$ 面积的最小值为 $\frac{1}{2} \times 5 \times \frac{13}{3} = \frac{65}{6}$. 综上所述, 存在这样的点

P, 且P的坐标为 $(\frac{4}{3}, \frac{13}{9})$, $\triangle PAB$ 面积的最小值为 $\frac{65}{6}$.

全国重点高中提前招生 考试全真试卷(十一)

1. A 【解析】 $\begin{cases} -5-x \leq \frac{1}{11}(x-a) \text{ ①,} \\ \frac{3x+1}{2} > 2x+1 \text{ ②,} \end{cases}$ 解不等式①得 $x \geq \frac{a-55}{12}$, 解不等式

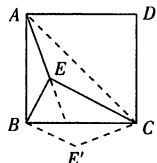
②得 $x < -1$, $\therefore$ 原不等式组的解集为 $\frac{a-55}{12} \leq x < -1$, $\therefore$ 不等式组的解集恰好有3个负整数解, $\therefore -5 < \frac{a-55}{12} \leq -4$, $\therefore -5 < a \leq 7$, 所以整数 $a =$

$-4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7$, 关于 y 的方程 $\frac{2y-a}{y-1} - \frac{3y-2}{1-y} = 1$, 化简得 $2y-a+3y-2=y-1$, 解得 $y = \frac{a+1}{4}$, $\therefore$ 关于 y 的方程有非负整数

解, $\therefore y \geq 0$, 且 $\frac{a+1}{4} \neq 1$, $\therefore$ 符合条件的所有整数 a 的值为 $-1, 7$, $\therefore$ 符合

条件的所有整数 a 的和为 6 . 故选 A.

2. C 【解析】如图, 将 BE 绕点 B 顺时针旋转 90° 至 BE' , 连接 CE' , $\therefore \angle EBE' = \angle CBE + \angle CBE' = 90^\circ$, $\angle ABC = \angle ABE + \angle CBE = 90^\circ$, $\therefore \angle ABE = \angle CBE'$, $\therefore \triangle ABE \cong \triangle CBE'$ (SAS), $\therefore AE = CE' = 2$, $BE = BE' = 1$. 连接 EE' , 则 $EE' = \sqrt{2}$, $EC = \sqrt{6}$, 则 $EE'^2 + E'C^2 = CE^2$, $\therefore \angle EE'C = 90^\circ$, $\angle BE'C = 135^\circ$, $\therefore \angle AEB = 135^\circ$. $\therefore \angle BEE' = 45^\circ$, $\therefore A, E, E'$ 三点共



第2题图

线. 连接 AC , $S_{\triangle ABC} = S_{\triangle ABE} + S_{\triangle ACE} + S_{\triangle BCE'}$, $\therefore \triangle ABE \cong \triangle CBE'$, $\therefore S_{\triangle ABE} = S_{\triangle CBE'}$, $\therefore S_{\triangle ABC} = S_{\triangle CBE'} + S_{\triangle ACE} + S_{\triangle BCE} = S_{\triangle ACE} + S_{\triangle BEE'} = \frac{1}{2} \times (2 + \sqrt{2}) \times 2 + \frac{1}{2} \times 1 \times 1 = \frac{5}{2} + \sqrt{2}$, $\therefore$ 四边形ABCD =

$S_{\triangle ACD} + S_{\triangle ACE} = S_{\triangle ABC} + S_{\triangle ACE} = \frac{5}{2} + \sqrt{2} + \frac{1}{2} \times 2 \times 2 = \sqrt{2} + \frac{9}{2}$. 故选 C.

3. D 【解析】依题意可得 $OB = 4$. ①当P点在OA上时, 即 $0 \leq x \leq 2$ 时, $PC = OC = x$, $S_{\triangle POC} = y = \frac{1}{2}PC \cdot OC = \frac{1}{2}x^2$, 是开口向上的抛物线, 当 $x=2$ 时, $y=2$; ②当P点在AB上时, 即 $2 < x \leq 4$ 时, $OC = x$, 则 $BC = 4 - x$, $PC = BC = 4 - x$, $S_{\triangle POC} = y = \frac{1}{2}PC \cdot OC = \frac{1}{2}x(4-x) = -\frac{1}{2}x^2 + 2x$, 是开口向下的抛物线, 当 $x=4$ 时, $y=0$. 综上所述, D 答案符合运动过程中 y 与 x 的函数关系式. 故选 D.

4. D 【解析】对于A, 由柱状图可得5月份出货量最高, 故A正确; 对于B, 根据曲线幅度可得下半年波动比上半年波动小, 故B正确; 对于C, 根据曲线上数据可得仅仅4月5月同比高, 其余各月均低于2018, 且明显出货量低于2018年, 故C正确; 对于D, 可计算得2018年12月出货量为 $3044.4 \div (1 - 14.7\%) = 3569.05$, 8月出货量为 $3087.5 \div (1 - 5.3\%) = 3260.3$, 而 $3260.3 < 3569.05$, 故12月更高, D错误. 故选D.

5. D 【解析】设 x_0 是它们的一个公共实数根, 则 $ax_0^2 + bx_0 + c = 0$, $bx_0^2 + cx_0 + a = 0$, $cx_0^2 + ax_0 + b = 0$. 把上面三个式子相加, 并整理得 $(a+b+c)(x_0^2 + x_0 + 1) = 0$. $\therefore x_0^2 + x_0 + 1 = (x_0 + \frac{1}{2})^2 + \frac{3}{4} > 0$, $\therefore a+b+c=0$. 于是 $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab} = \frac{a^3 + b^3 + c^3}{abc} = \frac{a^3 + b^3 - (a+b)^3}{abc} = \frac{-3ab(a+b)}{abc} = 3$. 故选 D.

6. A 【解析】结论②是错误的. 故选 A.

7. -3 【解析】因为 $\sqrt{7-4\sqrt{3}} = \sqrt{(2-\sqrt{3})^2} = 2-\sqrt{3}$, 代入整理, 得 $\sqrt{3}(a+4) - (2a+b+7) = 0$, 所以 $a+4=0$ 且 $2a+b+7=0$, 解得 $a=-4$, $b=1$. 故 $a+b=-3$.

8. -6 或 $-\frac{25}{4}$

9. $n+1$ 【解析】由 $a_1=2, a_2=3, a_3=4$, 知 $a_n=n+1$.

10. $\frac{27}{4}$ 【解析】过点D作 $DF \parallel x$

轴, 交BE于点F, 交y轴于点G,

延长BE交x轴于点H, 连接OD,

$\therefore E$ 为 OC 的四等分点 ($OE >$

EC), $\triangle DEC$ 的面积为3,

$\therefore \triangle DEO$ 的面积为9.

$\therefore BE \parallel y$ 轴, $\therefore$ 四边形BMOE是

平行四边形,

$\therefore BM = OE$, $\therefore AM = EC = \frac{1}{4}AB$.

$\therefore D$ 为 AB 中点, $\therefore DM = EC = \frac{1}{3}OE$, 由平行四边形得 $\angle OEH = \angle EBM =$

$\angle DMG$, $\angle OHE = \angle DGM = 90^\circ$, $\therefore \triangle OHE \sim \triangle DGM$, $\therefore \frac{DG}{OH} = \frac{DM}{OE} = \frac{1}{3}$. 设

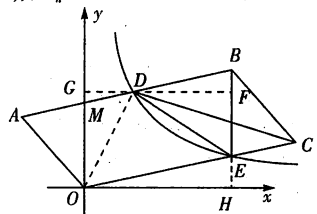
D点坐标为 $(a, \frac{k}{a})$, 则E点坐标为 $(3a, \frac{k}{3a})$, $S_{\triangle ODE} = 3a \times \frac{k}{a} - \frac{1}{2}a \times$

$\frac{k}{a} - \frac{1}{2} \times 3a \times \frac{k}{3a} - \frac{1}{2}(3a-a)(\frac{k}{a} - \frac{k}{3a}) = 3k - \frac{1}{2}k - \frac{1}{2}k - \frac{2}{3}k =$

9, 解得 $k = \frac{27}{4}$.

11. $3\sqrt{5}-3$ 【解析】 $\because$ 四边形ABCD是正方形, $\therefore CD=AB=6$. $\therefore E$ 是CD的中点, $\therefore EC = \frac{1}{2}CD = 3$. $\therefore \triangle ECF$ 沿EF所在的直线折叠得到 $\triangle ECF'$, $\therefore EC' = EC = 3$. $\therefore$ 当点A, C', E三点共线时, AC' 最小. $\therefore AC' = AE - EC' = 3\sqrt{5}-3$.

12. $\frac{\pi}{4} - \frac{1}{2}$ 【解析】连接CD, 作 $DM \perp BC$, $DN \perp AC$. 则 $DC = \frac{1}{2}AB = 1$, 四边形DMCN是正方形, $DM = \frac{\sqrt{2}}{2}$. (H, G分别是DE, DF与AC, BC的交点)



第10题图

$$\therefore S_{\text{扇形}FDE} = \frac{90\pi \times 1^2}{360} = \frac{\pi}{4}. \text{易证 } \triangle DMG \cong \triangle DNH (\text{ASA}), \therefore S_{\text{四边形}DGCH} = S_{\text{四边形}DMCN} = \frac{1}{2}, \text{则 } S_{\text{阴影}} = \frac{\pi}{4} - \frac{1}{2}.$$

13. 解: (1) 由 $\alpha\beta=1$, 可知 $\frac{1}{\alpha}=\beta$, $\therefore m=\pm 1$, 其中 $m=-1$ 不合题意, 舍去. 由 $x^2+5x+1=0$, 可知 $\alpha+\beta=-5, \alpha\beta=1. \therefore \alpha<0, \beta<0. \therefore \alpha^2-\beta^2=(\alpha+\beta) \cdot (\alpha-\beta)=(\alpha+\beta) [\pm \sqrt{(\alpha-\beta)^2}]=(\alpha+\beta) [\pm \sqrt{(\alpha+\beta)^2-4\alpha\beta}]=\pm 5\sqrt{21}.$

(2) $m=1$, 第二个方程化为 $x^2+2(a+1)x+2a+1=0$, 即 $(x+1)(x+2a+1)=0, x_1=-1, x_2=-(2a+1).$

由 $\begin{cases} -(2a+1)>0, \\ -(2a+1)<2, \end{cases}$ 解得 $-\frac{3}{2}<a<-\frac{1}{2}.$

14. 解: (1) $\therefore (\sqrt{2025}-\sqrt{2024})(\sqrt{2025}+\sqrt{2024})=1,$
 $\therefore a=\sqrt{2025}-\sqrt{2024}=\frac{1}{\sqrt{2025}+\sqrt{2024}},$ 同理 $b=\frac{1}{\sqrt{2024}+\sqrt{2023}}, c=\frac{1}{\sqrt{2023}+\sqrt{2022}}.$

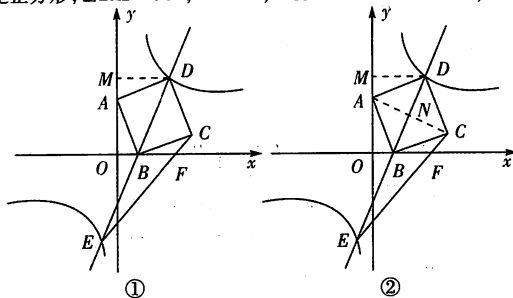
显然, a, b, c 均大于 0, 且 $\sqrt{2025}+\sqrt{2024}>\sqrt{2024}+\sqrt{2023}>\sqrt{2023}+\sqrt{2022}.$
 $\therefore \frac{1}{\sqrt{2025}+\sqrt{2024}}<\frac{1}{\sqrt{2024}+\sqrt{2023}}<\frac{1}{\sqrt{2023}+\sqrt{2022}},$ 即 $a<b<c.$

(2) $(\sqrt{5}+\sqrt{3})^6=(8+2\sqrt{15})^3.$ 令 $8+2\sqrt{15}=a, 8-2\sqrt{15}=b$, 则 $ab=4, a+b=16$, 所以 a, b 是方程 $x^2-16x+4=0$ 的两个实根, 从而有 $a^2=16a-4, b^2=16b-4$, 得 $a^3=16a^2-4a, b^3=16b^2-4b. \therefore a^3+b^3=16(a^2+b^2)-4(a+b)=16[(a+b)^2-2ab]-4(a+b)=16[16^2-2\times 4]-4\times 16=3904.$ 因为 $0<b<1$, 所以 $0<b^3<1$, 从而不超过 a^3 的最大整数为 3903.

故不超过 $(\sqrt{5}+\sqrt{3})^6$ 的最大整数为 3903.

15. 解: (1) 作 $DM \perp y$ 轴, 如图①.

$\therefore$ 点 A 的坐标 $(0, 4)$, 点 B 坐标 $(2, 0), \therefore OA=4, OB=2, \therefore$ 四边形 $ABCD$ 是正方形, $\angle BAD=90^\circ, AB=AD, \therefore \angle OAB+\angle DAM=90^\circ,$



第 15 题图

$\therefore \angle OAB+\angle ABO=90^\circ, \therefore \angle DAM=\angle ABO, \therefore \angle AOB=\angle DMA=90^\circ, \therefore \triangle AOB \cong \triangle DMA (\text{AAS}), \therefore AM=OB=2, DM=OA=4, \therefore$ 点 $D(4, 6),$ 将点 D 代入双曲线 $y=\frac{k}{x}$ 得 $k=4\times 6=24, \therefore$ 双曲线解析式为 $y=\frac{24}{x},$ 设直线 BD 的解析式为 $y=mx+n$, 把 $B(2, 0), D(4, 6)$ 代入得 $\begin{cases} 2m+n=0 \\ 4m+n=6 \end{cases}$, 解得 $\begin{cases} m=3 \\ n=-6 \end{cases}, \therefore$ 直线 BD 的解析式为 $y=3x-6.$

(2) 连接 AC , 交 BD 于 N , 如图②.

$\therefore$ 四边形 $ABCD$ 是正方形, $\therefore BD$ 垂直平分 $AC, AC=BD,$

$\begin{cases} y=3x-6, \\ y=\frac{24}{x} \end{cases}$ 得 $\begin{cases} x_1=4, \\ y_1=6 \end{cases}$ 或 $\begin{cases} x_2=-2, \\ y_2=-12, \end{cases} \therefore E(-2, -12),$

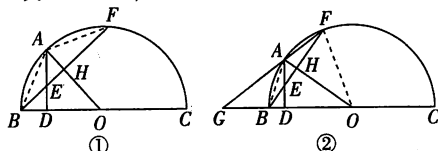
$\therefore B(2, 0), D(4, 6), \therefore DE=\sqrt{(4+2)^2+(6+12)^2}=6\sqrt{10}, DB=\sqrt{(4-2)^2+(6-0)^2}=2\sqrt{10},$

$\therefore DN=CN=\frac{1}{2}BD=\sqrt{10},$

$\therefore S_{\triangle BEC}=S_{\triangle DEC}-S_{\triangle DBC}=\frac{1}{2}DE \cdot CN-\frac{1}{2}BD \cdot CN=\frac{1}{2}\times 6\sqrt{10}\times \sqrt{10}-\frac{1}{2}\times 2\sqrt{10}\times \sqrt{10}=20.$

(3) 由图象可以得出不等式 $ax+b>\frac{k}{x}$ 的解集是 $-2<x<0$ 或 $x>4.$

16. (1) 证明: $\therefore AD \perp BC, BH \perp AO, \therefore \angle ADO=\angle BHO=90^\circ.$
 又 $\therefore \angle AOD=\angle BOH, OA=OB, \therefore \triangle ADO \cong \triangle BHO (\text{AAS}),$
 $\therefore OH=OD.$ 又 $\therefore OA=OB, \therefore AH=BD.$



第 16 题图

(2) 解: 如图①连接 $AB, AF, \therefore BE \cdot BF=y, BF=2BH, \therefore BE \cdot BH=\frac{1}{2}y. \therefore \triangle BED \sim \triangle BOH, \therefore \frac{BE}{OB}=\frac{BD}{BH}, \therefore OB \cdot BD=BE \cdot BH, \therefore 4x=\frac{1}{2}y, \therefore y=8x(0<x<4).$

(3) 解: 如图②连接 $AB, OF, \therefore \angle GFB$ 是公共角, $\angle FAE>\angle G, \therefore$ 当 $\triangle FAE \sim \triangle FBC$ 时, $\angle AEF=\angle G. \therefore \angle BHA=\angle ADO=90^\circ, \therefore \angle AEF+\angle DAO=90^\circ, \angle AOD+\angle DAO=90^\circ. \therefore \angle AEF=\angle AOD. \therefore \angle G=\angle AOD.$
 $\therefore AG=AO=4. \therefore \angle AOD=\angle AOF, \therefore \angle G=\angle AOF.$ 又 $\therefore \angle GFO$ 是公共角,
 $\therefore \triangle FAO \sim \triangle FOG.$

$\therefore \frac{AF}{OF}=\frac{OF}{FG}$ 由 (2) 知 $AB^2=8x, \widehat{AB}=\widehat{AF}, \therefore AF=2\sqrt{2x},$

$\therefore \frac{2\sqrt{2x}}{4}=\frac{4}{4+2\sqrt{2x}},$ 解得 $x=3\pm\sqrt{5}, \therefore 3+\sqrt{5}>4$, 舍去,

$\therefore BD=3-\sqrt{5}.$

17. 解: (1) $\therefore$ 抛物线 $y=ax^2-x+3(a\neq 0)$ 的对称轴为直线 $x=-2,$

$\therefore -\frac{1}{2a}=-2, \therefore a=-\frac{1}{4}, \therefore y=-\frac{1}{4}x^2-x+3, \therefore D(-2, 4).$

(2) 探究一: 当 $0<t<4$ 时, W 有最大值. $\therefore$ 抛物线 $y=-\frac{1}{4}x^2-x+3$ 交 x 轴于 A, B 两点, 交 y 轴于点 $C, \therefore A(-6, 0), B(2, 0), C(0, 3). \therefore OA=6, OC=3.$ 当 $0<t<4$ 时, 过点 D 作 $DM \perp y$ 轴于点 M , 则 $DM=2, OM=4.$

$\therefore P(0, t), \therefore OP=t, MP=OM-OP=4-t. \therefore S_{\triangle PAD}=S_{\text{梯形}OADM}-S_{\triangle AOP}-S_{\triangle DMP}=\frac{1}{2}\times(2+6)\times 4-\frac{1}{2}\times 6\times t-\frac{1}{2}\times 2\times(4-t)=12-2t, \therefore W=t(12-2t)=-2(t-3)^2+18, \therefore$ 当 $t=3$ 时, W 有最大值, $W_{\text{最大值}}=18.$

探究二: 存在. 理由如下: ①当 $\angle P_1DA=90^\circ$ 时, 过点 D 作 $DE \perp x$ 轴于点 E , 则 $OE=2, DE=4, \angle DEA=90^\circ, \therefore AE=OA-OE=6-2=4=DE, \therefore \angle DAE=\angle ADE=45^\circ, AD=\sqrt{2}DE=4\sqrt{2}, \therefore \angle P_1DE=\angle P_1DA-\angle ADE=90^\circ-45^\circ=45^\circ. \therefore DM \perp y$ 轴, $OA \perp y$ 轴, $\therefore DM \parallel OA, \therefore \angle MDE=\angle DEA=90^\circ. \therefore \angle MDP_1=\angle MDE-\angle P_1DE=90^\circ-45^\circ=45^\circ. \therefore P_1M=DM=2,$

$P_1D=\sqrt{2}DM=2\sqrt{2}, \frac{OC}{P_1D}=\frac{OA}{AD}=\frac{3\sqrt{2}}{4}.$ 又 $\text{Rt} \triangle ADP_1 \sim \text{Rt} \triangle AOC, \therefore OP_1=OM-P_1M=4-2=2, \therefore P_1(0, 2).$ $\therefore$ 当 $\angle P_1DA=90^\circ$ 时, 存在点 P_1 , 使 $\text{Rt} \triangle ADP_1 \sim \text{Rt} \triangle AOC$, 此时 $P_1(0, 2);$

②当 $\angle P_2AD=90^\circ$ 时, 则 $\angle P_2AO=45^\circ, \therefore P_2A=6\sqrt{2}, \therefore \frac{P_2A}{OA}=\frac{6\sqrt{2}}{6}=\sqrt{2}.$

$\therefore \frac{AD}{OC}=\frac{4\sqrt{2}}{3}, \frac{P_2A}{OA} \neq \frac{AD}{OC}, \therefore \triangle P_2AD$ 与 $\triangle AOC$ 不相似, 此时点 P_2 不存在;

③当 $\angle AP_3D=90^\circ$ 时, 以 AD 为直径作 $\odot O_1$, 则 $\odot O_1$ 的半径 $r=\frac{AD}{2}=2\sqrt{2},$ 圆心 O_1 到 y 轴的距离 $d=4. \therefore d>r, \therefore \odot O_1$ 与 y 轴相离, $\therefore$ 不存在点 P_3 , 使 $\angle AP_3D=90^\circ. \therefore$ 综上所述, 只存在一点 $P(0, 2)$ 使 $\text{Rt} \triangle ADP$ 与 $\text{Rt} \triangle AOC$ 相似.

全国重点高中提前招生 考试全真试卷(十二)

1. C 【解析】 $\therefore a>0, b=-(a+c), c=-(a+b), \therefore m=|x_1-x_2|=\sqrt{(x_1+x_2)^2-4x_1x_2}=\sqrt{\frac{b^2-4ac}{a^2}}=\sqrt{\frac{b^2-4ac}{a}}=\frac{a-c}{a}=1-\frac{c}{a}=1-\frac{-(a+b)}{a}=2+\frac{b}{a}, \therefore 2<m<3.$ 故选 C.

2. D 【解析】 $\therefore$ 方程有一个整数根, $\therefore \Delta=25m^2-8(2m^2-5)=9m^2+40>0, \therefore \Delta=p^2(p$ 为正整数), $\therefore (3m-p)(3m+p)=-40. \therefore 3m-p\leq 3m+p$ 且同奇偶, $\therefore \begin{cases} 3m-p=-4, \\ 3m+p=10 \end{cases}$ 或 $\begin{cases} 3m-p=-10, \\ 3m+p=4 \end{cases}$ 或 $\begin{cases} 3m-p=-2, \\ 3m+p=20 \end{cases}$ 解得 $m=\pm 3, \pm 1$, 经检验, 均有一根为整数, $\therefore$ 符合条件的整数 m 的值有 4 个. 故选 D.

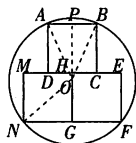
3. C 【解析】连接 BR , 设 $S_{\triangle CDR}=a, S_{\triangle BRF}=b, \therefore BD=2CD, AF=2BF,$
 $\therefore S_{\triangle BDR}=2a, S_{\triangle ARF}=2b, \therefore S_{\triangle ABC}=1, \therefore S_{\triangle CDR}+S_{\triangle BDR}+S_{\triangle BRF}=\frac{1}{3}, S_{\triangle BDR}+$

$S_{\triangle BRF}+S_{\triangle ARF}=\frac{2}{3}, \therefore \begin{cases} a+2a+b=\frac{1}{3}, \\ 2a+b+2b=\frac{2}{3}, \end{cases}$ 解得 $\begin{cases} a=\frac{1}{21}, \\ b=\frac{4}{21}, \end{cases} \therefore S_{\triangle CDR}=\frac{1}{21}.$ 同

理, $S_{\triangle AFE}=S_{\triangle BFQ}=\frac{1}{21}, S_{\triangle PQR}=S_{\triangle BCE}-(S_{\triangle BCF}-S_{\triangle BFQ})-(S_{\triangle ACD}-S_{\triangle AFE}-S_{\triangle CDR})=\frac{2}{3}-\frac{1}{3}+S_{\triangle BFQ}-\frac{1}{3}+S_{\triangle AFE}+S_{\triangle CDR}=S_{\triangle BFQ}+S_{\triangle AFE}+S_{\triangle CDR}=\frac{1}{21}\times 3=\frac{1}{7}.$

4. C 【解析】如图, 设圆心为 O , GH 与 AB 的交点为 P , 连接 $OA, OB, ON,$
 $\therefore PG$ 垂直平分 $NF, OA=OB=ON, \therefore$ 点 O 在 PG 上, $AP=PB=\frac{1}{2}AB=$

20. 设 $OG = x$, 则 $OP = PG - OG = 80 - x$. 在 $\text{Rt} \triangle APO$ 中, $OA^2 = AP^2 + OP^2$, 在 $\text{Rt} \triangle NGO$ 中, $ON^2 = NG^2 + OG^2$, $\therefore AP^2 + OP^2 = NG^2 + OG^2$. $\therefore 20^2 + (80 - x)^2 = 40^2 + x^2$, 解得 $x = \frac{65}{2}$, 则 $ON = \sqrt{NG^2 + OG^2} = \frac{25\sqrt{17}}{2}$, $\therefore$ 圆形硬纸板的最小直径为 $25\sqrt{17}$. 故选 C.



第4题图

5. D 【解析】记3款新疆棉纱制成的毛巾为1,2,3,2款普通棉纱制成的毛巾为4,5. 从5款中任选2款, 所有可能出现的情况为①1,2; ②1,3; ③1,4; ④1,5; ⑤2,3; ⑥2,4; ⑦2,5; ⑧3,4; ⑨3,5; ⑩4,5; 共10种. 至少有一款是新疆棉纱的情况为9种, 至少有一款是新疆棉纱的概率为 $\frac{9}{10}$. 故选 D.

6. D 【解析】如图, $\therefore AE = DF$, $\therefore DE + DF = DE + AE = AD = 4$. $\therefore EF < DE + DF$, $\therefore EF < 4$, 故①正确;

$\therefore$ 四边形 $ABCD$ 是正方形, $\therefore AB = AD$, $\angle BAE = \angle ADF = \angle BAD = 90^\circ$. $\therefore AE = DF$, $\therefore \triangle BAE \cong \triangle ADF (\text{SAS})$, $\therefore \angle ABE = \angle DAF$. $\therefore \angle DAF + \angle BAH = 90^\circ$, $\therefore \angle ABE + \angle BAH = 90^\circ$, $\therefore \angle AHB = 90^\circ$, 即 $BE \perp AF$, $\therefore \angle AHE = \angle EAB = 90^\circ$. $\therefore \angle AEH = \angle BEA$, $\therefore \triangle EAH \sim \triangle EBA$, $\therefore \frac{AE}{EB} = \frac{EH}{AE}$, $\therefore EA^2 = EH \cdot EB$, 故②正确;

连接 BF 交 EC 于点 K , 同法可证 $BF \perp CE$.

$\therefore EK, FH$ 都是 $\triangle BEF$ 的高,

$\therefore BG \perp EF$, 故③正确;

连接 DH .

$\therefore \angle EHF = \angle EDF = 90^\circ$,

$\therefore D, E, H, F$ 四点共圆,

$\therefore$ 当 $DE = DF$ 时, $\angle DHF = \angle DHE = 45^\circ$, 故④正确;

设 AB 的中点为 O , 过点 O 作 $OJ \perp DM$ 于点 J .

$\therefore AM = 1, AD = 4, \angle DAM = 90^\circ$,

$\therefore DM = \sqrt{AM^2 + AD^2} = \sqrt{1^2 + 4^2} = \sqrt{17}$.

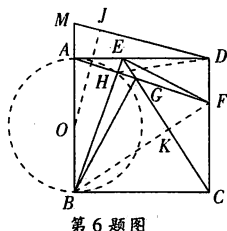
$\therefore OA = OB = 2, \therefore OM = 3$.

$\therefore \angle M = \angle M, \angle OJM = \angle DAM, \therefore \triangle OJM \sim \triangle DAM$,

$\therefore \frac{OJ}{AD} = \frac{OM}{DM}, \therefore \frac{OJ}{4} = \frac{3}{\sqrt{17}}, \therefore OJ = \frac{12\sqrt{17}}{17}$.

$\therefore \angle AHB = 90^\circ, \therefore$ 点 H 在以 AB 为直径的圆上运动,

当点 H 落在 OJ 上时, 点 H 到 DM 的距离最小, 最小距离 $= \frac{12\sqrt{17}}{17} - 2$, 故⑤正确. 故选 D.



第6题图

7. 2 9455 【解析】 $\therefore n^2 = \frac{n(n+1)(n+1)}{1 \times 2 \times 3} - \frac{(n-1)n[(n-1)+1]}{1 \times 2 \times 3} = \frac{(n^2+n)(n+1) - (n^2-n)(n+1)}{6} = \frac{3n^2 - n + 2n}{6}$,

$\therefore a = 2$ 时等式成立.

$\therefore 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1 \times 2 \times 3}{1 \times 2 \times 3} - \frac{0}{1 \times 2 \times 3} + \frac{2 \times 3 \times 5}{1 \times 2 \times 3} - \frac{1 \times 2 \times 3}{1 \times 2 \times 3} + \dots + \frac{n(n+1)(2n+1)}{1 \times 2 \times 3} - \frac{(n-1)n(2n-1)}{1 \times 2 \times 3} = \frac{n(n+1)(2n+1)}{6}$,

$\therefore 1^2 + 2^2 + 3^2 + \dots + 30^2 = \frac{30 \times 31 \times 61}{6} = 9455$.

8. 3 【解析】由 $y = x^2 - 2x + 3 = (x-1)^2 + 2$ 可知抛物线 $y = x^2 - 2x + 3$ 的对称轴为直线 $x = 1$. $\therefore$ 当 $x = a$ 与 $x = b (a \neq b)$ 时, 代数式 $x^2 - 2x + 3$ 的值相等, $\therefore$ 当 $x = a$ 或 $x = b (a \neq b)$ 时, 二次函数 $y = x^2 - 2x + 3$ 的函数值相等, $\therefore$ 以 a, b 为横坐标的点关于直线 $x = 1$ 对称, 则 $\frac{a+b}{2} = 1, \therefore a + b = 2$,

$\therefore x = a + b = 2$, 当 $x = 2$ 时, 代数式 $x^2 - 2x + 3 = 4 - 4 + 3 = 3$.

9. (4, 33) 【解析】由 $(-x+4)p = 0$ 知 $x = 4$, 从而 $y = 33$.

10. 4 $\frac{1}{3}$

11. $16 + 12\sqrt{2}$ 【解析】连接 OA, OD , 过点 O 作 $OP \perp AB$ 于点 $P, OM \perp AD$ 于点 $M, ON \perp CD$ 于点 N . $\therefore S_{\text{矩形}ABCD} = 4S_{\triangle AOD}$, 且 OA, OD 的长是定值, 当 $\angle AOD$ 的正弦值最大时, $S_{\triangle AOD}$ 最大, 即 $\angle AOD = 90^\circ$, 则 $AD = 6\sqrt{2}, OM = 4$, 即 $AB = 8$. 则矩形 $ABCD$ 的周长是 $16 + 12\sqrt{2}$.

12. $(-2)^{1010} \sqrt{3} \times 2^{1010}$

13. 解: $\therefore m, n$ 是方程 $x^2 + 3x + 1 = 0$ 的两根, $\therefore m + n = -3, mn = 1$.

(1) 原式 $= \frac{(3+m)(3-m)}{5-m} \cdot \frac{2(m-5)}{3-m} - \frac{2}{m} = -6 - 2 \left(m + \frac{1}{m} \right) =$

$$-6 - 2 \times \frac{m^2 + 1}{m} = -6 - 2 \times \frac{-3m}{m} = -6 + 6 = 0.$$

$$(2) \text{原式} = \sqrt{\frac{m^4}{mn}} + \sqrt{\frac{n^4}{mn}} = \frac{m^2}{\sqrt{mn}} + \frac{n^2}{\sqrt{mn}} = \frac{m^2 + n^2}{\sqrt{mn}} = \frac{(m+n)^2 - 2mn}{\sqrt{mn}} = \frac{(-3)^2 - 2 \times 1}{\sqrt{1}} = 7.$$

14. (1) 证明: $\therefore$ 在 $\triangle ABC$ 中, $\angle BAC = 90^\circ, AB = AC, \therefore \angle B = \angle C = 45^\circ, \therefore \angle B + \angle BOE + \angle BEO = 180^\circ, \therefore \angle BOE + \angle BEO = 135^\circ$.

$\therefore \angle EOF = 45^\circ, \angle BOE + \angle EOF + \angle COF = 180^\circ, \therefore \angle BOE + \angle COF = 135^\circ$.

$\therefore \angle BEO = \angle COF$. 又 $\angle B = \angle C, \therefore \triangle BOE \sim \triangle CFO$.

(2) 解: ① $\triangle BOE \sim \triangle CFO$; ② $\triangle BOE$ 与 $\triangle OFE$ 相似. 理由如下: 同(1),

可证 $\triangle BOE \sim \triangle CFO$, 则 $CO : BE = OF : OE$, 而 $CO = BO, \therefore OB : BE = OF : OE$.

$\therefore \angle EBO = \angle EOF, \therefore \triangle BOE \sim \triangle OFE$; ③ $\triangle ABC$ 为等腰直角三角形, 且 $AB = AC = 2, O$ 为 BC 的中点, $\therefore BO = \sqrt{2}$. 设 $EO = y, \therefore \triangle BOE \sim$

$\triangle OFE, \therefore \frac{EO}{BE} = \frac{FO}{BO} = \frac{FE}{EO}, \text{即} \frac{y}{BE} = \frac{FO}{\sqrt{2}} = \frac{FE}{y}, \text{解得} FO = \frac{\sqrt{2}x}{y}, \text{则} S_{\triangle EOF} =$

$$\frac{1}{2} \cdot \sin 45^\circ \cdot EO \cdot FO = \frac{\sqrt{2}}{4} \cdot EO \cdot FO = \frac{\sqrt{2}}{4} \times \sqrt{2}x = \frac{1}{2}x.$$

15. (1) ① 证明: $\Delta = (-m)^2 - 4 \left(\frac{1}{2}m^2 - m + 1 \right) = -(m-2)^2 \geq 0, \therefore (m-2)^2 \leq 0, \therefore m-2 = 0, \therefore \Delta = 0, \therefore$ 一元二次方程 $x^2 - mx + \frac{1}{2}m^2 - m + 1 =$

0 有两个相等实根, $\therefore AM = AN$.

② 证明: 由①可知 $AM = AN, \therefore \angle AMN = \angle ANM = \angle CND$.

$\therefore \angle BAC = 90^\circ, AD \perp BC$, 又 $\angle AMN + \angle ACN = 90^\circ, \angle CND + \angle NCD = 90^\circ$,

$\therefore \angle ACN = \angle MCB, \therefore \angle NAC + \angle BAD = \angle MBC + \angle BAD = 90^\circ$,

$\therefore \angle NAC = \angle MBC$,

$\therefore \triangle ACN \sim \triangle BCM, \therefore \frac{AC}{BC} = \frac{AN}{BM}$, 即 $\frac{AC}{BC} = \frac{AM}{BM}$

(2) 解: 由题意知 $\angle ADC = \angle ADB = 90^\circ, \angle DAC = \angle DBA$,

$\therefore \triangle ADC \sim \triangle BDA, \therefore \frac{AD}{BD} = \frac{DC}{AD}, \therefore AD^2 = BD \cdot DC$.

$\therefore CF \perp BE, \therefore \angle FCB + \angle EBD = 90^\circ$.

$\therefore \angle E + \angle EBD = 90^\circ, \therefore \angle E = \angle FCB$.

$\therefore \angle NDC = \angle EDB = 90^\circ, \therefore \triangle EBD \sim \triangle CND$,

$\therefore \frac{ED}{CD} = \frac{BD}{DN}, \therefore BD \cdot DC = ED \cdot DN, \therefore AD^2 = ED \cdot DN$.

$\therefore AN = 2, DN = 3, \therefore AD = DN + AN = 5, \therefore 5^2 = 3DE$,

$\therefore DE = \frac{25}{3}$.

16. 解: (1) GH 不变. 延长 HC 交 OP 于点 $E, \therefore$ 点 G 是 $\triangle OPH$ 的重心,

$\therefore GH = \frac{2}{3}EH, \therefore PO$ 是半径, 它是 $\text{Rt} \triangle OPH$ 的斜边, 它的中线等于它的

一半, $\therefore EH = \frac{1}{2}OP, \therefore GH = \frac{2}{3} \times \frac{1}{2}OP = \frac{1}{3}OP = 2$.

(2) 延长 PG 交 OA 于点 C , 则 $y = \frac{2}{3} \times PC$.

令 $OC = a = CH$, 在 $\text{Rt} \triangle PHC$ 中,

$PC = \sqrt{PH^2 + CH^2} = \sqrt{x^2 + a^2}$, 则 $y = \frac{2}{3} \sqrt{x^2 + a^2}$;

在 $\text{Rt} \triangle PHO$ 中, $OP^2 = x^2 + (2a)^2 = 36$, 则 $a^2 = 9 - \frac{x^2}{4}$, 将其代入

$y = \frac{2}{3} \sqrt{x^2 + a^2}$ 中得 $y = \frac{2}{3} \sqrt{\frac{3}{4}x^2 + 9} = \frac{\sqrt{3x^2 + 36}}{3} (0 < x < 6)$.

(3) 当 $PG = GH$ 时, 则 $y = GH = 2$, 解方程得 $x = 0$, 不合题意; 当 $PH =$

$GH = 2$ 时, 则可以解得 $x = 2$; 当 $PH = PG$ 时, 则 $x = y$ 代入可以求得 $x =$

$\sqrt{6}$. 综上所述, 线段 PH 的长是 $\sqrt{6}$ 或 2.

17. 解: (1) $\therefore$ 直线 $l: y = \frac{3}{4}x + m$ 经过点 $B(0, -1), \therefore m = -1, \therefore$ 直线 l 的解析

式为 $y = \frac{3}{4}x - 1$. $\therefore$ 直线 $l: y = \frac{3}{4}x - 1$ 经过点 $C(4, n), \therefore n = \frac{3}{4} \times 4 - 1 =$

2. $\therefore$ 抛物线 $y = \frac{1}{2}x^2 + bx + c$ 经过点 $C(4, 2)$ 和点 $B(0, -1)$,

$\therefore \begin{cases} \frac{1}{2} \times 4^2 + 4b + c = 2, \\ c = -1, \end{cases}$ 解得 $\begin{cases} b = -\frac{5}{4}, \\ c = -1, \end{cases} \therefore$ 抛物线的解析式为 $y = \frac{1}{2}x^2 -$

$\frac{5}{4}x - 1$.

(2) 令 $y = 0$, 则 $\frac{3}{4}x - 1 = 0$, 解得 $x = \frac{4}{3}, \therefore$ 点 $A \left(\frac{4}{3}, 0 \right)$,

$\therefore OA = \frac{4}{3}$, 在 $\text{Rt} \triangle OAB$ 中, $AB = \sqrt{\left(\frac{4}{3} \right)^2 + 1^2} = \frac{5}{3}$,

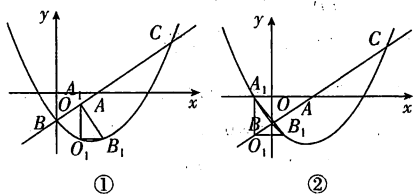
$\therefore DE \parallel y$ 轴,

$\therefore \angle ABO = \angle DEF$, 在矩形 $DFEG$ 中, $EF = \frac{3}{5}DE, DF = \frac{4}{5}DE$,

$\therefore p = 2(DF + EF) = 2 \times \left(\frac{4}{5} + \frac{3}{5} \right) DE = \frac{14}{5}DE$,

∵ 点D的横坐标为 $t(0 < t < 4)$,

$$\therefore D\left(t, \frac{1}{2}t^2 - \frac{5}{4}t - 1\right), E\left(t, \frac{3}{4}t - 1\right), \therefore DE = \left(\frac{3}{4}t - 1\right) - \left(\frac{1}{2}t^2 - \frac{5}{4}t - 1\right) = -\frac{1}{2}t^2 + 2t, \therefore p = \frac{14}{5} \times \left(-\frac{1}{2}t^2 + 2t\right) = -\frac{7}{5}t^2 + \frac{28}{5}t = -\frac{7}{5}(t-2)^2 + \frac{28}{5}, \text{且 } -\frac{7}{5} < 0, \therefore \text{当 } t=2 \text{ 时, } p \text{ 有最大值为 } \frac{28}{5}.$$



第17题图

(3) ∵ $\triangle AOB$ 绕点 M 沿逆时针方向旋转 90° , $\therefore A_1O_1 \parallel y$ 轴, $B_1O_1 \parallel x$ 轴, 设点 A_1 的横坐标为 x .

①如图①, 点 O_1, B_1 在抛物线上时, 点 O_1 的横坐标为 x , 点 B_1 的横坐标为 $x+1$, $\therefore \frac{1}{2}x^2 - \frac{5}{4}x - 1 = \frac{1}{2}(x+1)^2 - \frac{5}{4}(x+1) - 1$, 解得 $x = \frac{3}{4}$;

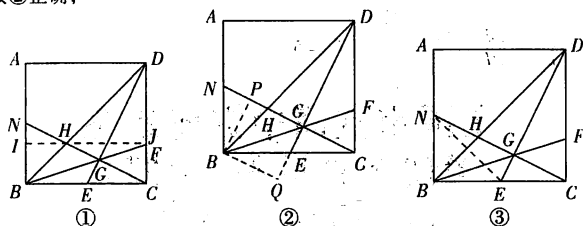
②如图②, 点 A_1, B_1 在抛物线上时, 点 B_1 的横坐标为 $x+1$, 点 A_1 的纵坐标比点 B_1 的纵坐标大 $\frac{4}{3}$, $\therefore \frac{1}{2}x^2 - \frac{5}{4}x - 1 = \frac{1}{2}(x+1)^2 - \frac{5}{4}(x+1) - 1 + \frac{4}{3}$, 解得 $x = -\frac{7}{12}$. 综上所述, 点 A_1 的横坐标为 $\frac{3}{4}$ 或 $-\frac{7}{12}$.

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1. B 【解析】 $\because a^2 - 2 \cdot 020a + 4 = 0, \therefore a^2 + 4 = 2 \cdot 020a$,

$$\therefore a^2 - 2 \cdot 019a + \frac{8 \cdot 080}{a^2 + 4} + 4 = a^2 - 2 \cdot 019a + \frac{8 \cdot 080}{2 \cdot 020a} + 4 = a^2 + 4 - 2 \cdot 019a + \frac{4}{a} = 2 \cdot 020a - 2 \cdot 019a + \frac{4}{a} + 4 = a + \frac{4}{a} = \frac{a^2 + 4}{a} = \frac{2 \cdot 020a}{a} = 2 \cdot 020. \text{ 故选 B.}$$

2. D 【解析】① ∵ 在正方形 $ABCD$ 中, $\angle NBC = \angle ECD = 90^\circ, BC = CD, \therefore \angle BCN + \angle GCD = 90^\circ, \angle CDE + \angle GCD = 90^\circ$, 即 $\angle BCN = \angle CDE, \therefore \triangle NBC \cong \triangle ECD (\text{ASA}), \therefore CN = DE$, 故①正确; ②在正方形 $ABCD$ 中, $AB \parallel CD, \therefore \triangle NBH \sim \triangle CDH, \therefore \frac{NB}{DC} = \frac{BH}{DH}, \therefore \triangle NBC \cong \triangle ECD, E$ 为 BC 的中点, 四边形 $ABCD$ 是正方形, $\therefore NB = \frac{1}{2}BC = \frac{1}{2}CD, \therefore \frac{NB}{DC} = \frac{BH}{DH} = \frac{1}{2}$, 故②正确;



第2题图

③如图①, 过 H 点作 $IJ \parallel AD$,

$\therefore$ 根据 $\triangle NBH \sim \triangle CDH$, 有 $HI = \frac{1}{2}HJ$, 则 $HI = \frac{1}{3}IJ = \frac{1}{3}DC, \therefore S_{\triangle DCH} = \frac{1}{2}EC \cdot DC, S_{\triangle BNH} = \frac{1}{2}BN \cdot HI = \frac{1}{2} \times EC \times \frac{1}{3}DC = \frac{1}{3} \times \left(\frac{1}{2} \times EC \times DC\right)$, 即 $S_{\triangle DEC} = 3S_{\triangle BNH}$, 故③正确;

④过 B 作 $BP \perp CN$ 于 $P, BQ \perp DG$, 交 DE 的延长线于 E , 如图②, $\therefore \angle BPC = \angle BQD = \angle PGQ = 90^\circ, \therefore$ 四边形 $PBQG$ 是矩形, $\therefore \angle PBQ = 90^\circ, \therefore \angle ABC = 90^\circ, \therefore \angle NBP = \angle QBE$, 由①得 $\triangle BNC \cong \triangle CED, \therefore EC = BN, \therefore E$ 是 BC 的中点, $\therefore BE = EC, \therefore BE = BN, \therefore \angle BPN = \angle BQE = 90^\circ, \therefore \triangle BPN \cong \triangle BQE, \therefore BP = BQ, \therefore$ 四边形 $PBQG$ 是正方形, $\therefore \angle BGE = 45^\circ$, 故④正确;

⑤如图③, 连接 NE .

设 $BN = x$, 则 $BE = EC = x, BC = 2x, \therefore CG \perp DE, \angle NBC = 90^\circ, \therefore CN = \sqrt{BC^2 + BN^2} = \sqrt{(2x)^2 + x^2} = \sqrt{5}x, EN = \sqrt{BE^2 + BN^2} = \sqrt{x^2 + x^2} = \sqrt{2}x$, 由 $\triangle ECN$ 的面积可得 $\frac{1}{2}CN \cdot GE = \frac{1}{2}EC \cdot BN$ 化简得 $GE = \frac{\sqrt{5}}{5}x$,

$$\therefore GN = \sqrt{EN^2 - GE^2} = \sqrt{(\sqrt{2}x)^2 - \left(\frac{\sqrt{5}}{5}x\right)^2} = \frac{3\sqrt{5}}{5}x,$$

$$\text{则有 } GN + GE = \frac{3\sqrt{5}}{5}x + \frac{\sqrt{5}}{5}x = \frac{4\sqrt{5}}{5}x,$$

$$\therefore GC = CN - GN = \sqrt{5}x - \frac{3\sqrt{5}}{5}x = \frac{2\sqrt{5}}{5}x.$$

$$\because AB \parallel CD, \therefore \triangle NGB \sim \triangle CGF, \therefore \frac{BN}{FC} = \frac{BG}{FG} = \frac{GN}{GC} = \frac{\frac{3\sqrt{5}}{5}x}{\frac{2\sqrt{5}}{5}x} = \frac{3}{2},$$

$$\therefore BG = \frac{3}{2}FG, \text{ 则 } BG = \frac{3}{5}BF, FC = \frac{2}{3}BN = \frac{2}{3}x.$$

$$\text{又 } BF = \sqrt{BC^2 + FC^2} = \sqrt{(2x)^2 + \left(\frac{2}{3}x\right)^2} = \frac{2\sqrt{10}}{3}x,$$

$$\therefore \sqrt{2}BG = \sqrt{2} \times \frac{3}{5} \times \frac{2\sqrt{10}}{3}x = \frac{4\sqrt{5}}{5}x, \therefore GN + EG = \sqrt{2}BG, \text{ 故⑤正确. 故选 D.}$$

3. A 【解析】经过的路径长 = $\frac{60\pi \cdot a}{180} \times 2 + \frac{60\pi \cdot \sqrt{3}a}{180} \times 2 + \frac{60\pi \cdot 2a}{180} = \frac{4+2\sqrt{3}}{3}\pi a$. 故选 A.

4. D 【解析】 $\because$ 方程 $(x-1)(x^2-2x+m)=0$ 有三根, $\therefore x_1=1, x^2-2x+m=0$ 有两实根. $\therefore \Delta=4-4m \geq 0$, 得 $m \leq 1$.

又 $\because$ 原方程有三根, 且为三角形的三边之长.

$\therefore x_2+x_3 > x_1=1, |x_2-x_3| < x_1=1$, 而 $x_2+x_3=2 > 1$ 已成立. 当 $|x_2-x_3| < 1$ 时, 两边平方得 $(x_2+x_3)^2 - 4x_2x_3 < 1$. 即 $4-4m < 1$. 解得 $m > \frac{3}{4}$.

$\therefore \frac{3}{4} < m \leq 1$. 故选 D.

5. B 【解析】作 $\odot O$ 的直径 BF , 连接 CF , 取 OB 的中点 M , 连接 EM , 易求 $BF=24\sqrt{3}, \therefore OB=12\sqrt{3}, \therefore BE \perp OD, M$ 为 OB 的中点, $\therefore EM = \frac{1}{2}OB = BM=6\sqrt{3}, \therefore$ 点 E 在以 OB 为直径的 $\odot M$ 上, 当点 D 在 B 点时, 此时 B 点和 E 点重合; 当 D 沿 $\widehat{BC}$ 运动到点 C 和 C 点重合时, 连接 OC , 则 $\angle BOC = 2\angle A = 120^\circ, \angle BOE = 60^\circ, \angle BME = 120^\circ, \therefore E$ 点经过的路径是以 OB 为直径的圆上弦 BE 所对的优弧的长, 由弧长公式得点 E 经过的路径长为 $\frac{240\pi \times 6\sqrt{3}}{180} = 8\sqrt{3}\pi$. 故选 B.

6. D 【解析】如图, 连接 $EG, \therefore EG$ 是 $\triangle ADC, \triangle BCD$ 的中位线, $\therefore GE \parallel AD, GE = \frac{1}{2}AD$, 同理 $GE \parallel BC$,

$$\frac{1}{2}BC = GE, \therefore GE \parallel AD, \therefore \frac{FG}{DF} = \frac{GE}{AD} = \frac{1}{2}, \text{ 同理}$$

$$\frac{CH}{EH} = \frac{1}{2}, \text{ 设 } S_{\triangle EFG} = x, S_{\triangle GEH} = y, \text{ 则 } S_{\triangle DFE} = 2x,$$

$$S_{\triangle ECH} = 2y, \therefore S_{\triangle DCE} = 3x + 3y = 45 \text{ cm}^2, \therefore S_{\triangle ABCD} = 45 \times 4 = 180 \text{ cm}^2. \text{ 故选 D.}$$

7. 11 【解析】 $(\sqrt{x} + \sqrt{y})^4 = x^2 + y^2 + 4x\sqrt{xy} + 4y\sqrt{xy} + 6xy, (\sqrt{x} - \sqrt{y})^4 = x^2 + y^2 - (4x + 4y)\sqrt{xy} + 6xy$.

8. $-3 < a + b < 3$ 【解析】 $x + \frac{a}{x} + b = 0$, 去分母得 $x^2 + bx + a = 0$, 设 $x^2 + bx + a = 0$ 的两个根为 x_1, x_2 , 由根与系数的关系可知, $x_1 \cdot x_2 = a, x_1 + x_2 = -b, \therefore a + b = x_1 \cdot x_2 - (x_1 + x_2)$. $\therefore x_1, x_2$ 均大于 1 且小于 2, $\therefore 1 < x_1 \cdot x_2 < 4, 1 < x_1 + x_2 < 4, \therefore -4 < -(x_1 + x_2) < -1, \therefore -3 < x_1 \cdot x_2 - (x_1 + x_2) < 3, \therefore -3 < a + b < 3$.

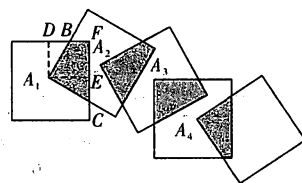
9. $1 + \sqrt{3}$ 【解析】连接 AB , 则 AB 为 $\odot M$ 的直径. 在 $\text{Rt} \triangle ABO$ 中, $\angle BAO = \angle OCB = 60^\circ, \therefore OB = \sqrt{3}OA = \sqrt{3} \times \sqrt{2} = \sqrt{6}$. 过点 B 作 $BD \perp OC$ 于点 D . 在 $\text{Rt} \triangle OBD$ 中, $\angle COB = 45^\circ$, 则 $OD = BD = \frac{\sqrt{2}}{2}OB = \sqrt{3}$. 在 $\text{Rt} \triangle BCD$ 中, $\angle OCB = 60^\circ$, 则 $CD = \frac{\sqrt{3}}{3}BD = 1, \therefore OC = CD + OD = 1 + \sqrt{3}$.

10. $\frac{n-1}{4}$ 【解析】如图, 过点 A_1 分别作正方形两边的垂线 A_1D 与 A_1E , $\therefore$ 点 A_1 是正方形的中心, $\therefore A_1D = A_1E$, 四边形 A_1EFD 是正方形, $\therefore \angle BA_1D + \angle BA_1E = 90^\circ$, $\therefore \angle BA_1D = \angle CA_1E$. 在 $\triangle A_1BD$ 和 $\triangle A_1CE$ 中, $\begin{cases} \angle BA_1D = \angle CA_1E, \\ A_1D = A_1E, \\ \angle A_1DB = \angle A_1EC = 90^\circ, \end{cases}$

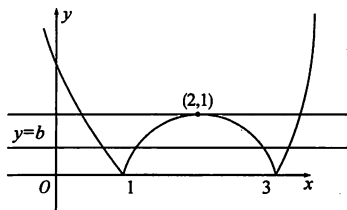
$$\therefore \triangle A_1BD \cong \triangle A_1CE (\text{ASA}), \therefore S_{\triangle A_1BD} = S_{\triangle A_1CE},$$

$$\therefore S_{\text{阴影}} = S_{\text{正方形} A_1EFD} = \frac{1}{4} \times 1^2 = \frac{1}{4} (\text{cm}^2).$$

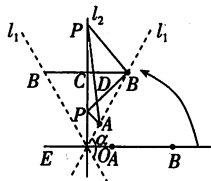
同理可求, 每一个阴影部分的面积都是正方形面积的 $\frac{1}{4}$, 为 $\frac{1}{4} \text{cm}^2$, $\therefore$ 重叠部分的面积和 = $\frac{1}{4} \times (n-1) = \frac{n-1}{4} (\text{cm}^2)$.



第10题图



第 11 题图



第 12 题图

11. $0 < b \leq 1$ 【解析】如图.

12. $45^\circ \leq \alpha < 90^\circ$ 或 $90^\circ < \alpha \leq 135^\circ$ 【解析】如图, 作 $BC \perp l_2$ 于 C 点. 在 $\triangle PBC$ 中, $BC < BP$. $\therefore BP = BA = \sqrt{2}$, $\therefore BC < \sqrt{2}$.

$\therefore \cos \angle OBC = \frac{CB}{OB} < \frac{\sqrt{2}}{2}$, $\therefore \angle OBC > 45^\circ$, 而 $\alpha = 90^\circ$ 时两直线重合,

$\therefore \angle OBC \neq 90^\circ$, $\therefore 45^\circ \leq \alpha < 90^\circ$; 同理当 l_1 旋转到 l_2 的左边时 $\angle OBC < 45^\circ$, $\therefore \alpha = 180^\circ - \angle OBC$, 而 $0^\circ < \alpha \leq 135^\circ$, $\therefore 90^\circ < \alpha \leq 135^\circ$.

13. 解: (1) $x = 1$ 代入方程, 得 $4k(1-a) = 2a^2 - b - 1$. $\therefore$ 方程任意实数 k 都成立, $\therefore 1-a = 0$ 且 $2a^2 - b - 1 = 0$, $\therefore a = 1, b = 1$.

(2) 设 $x_1 = 1$, 则 $x_2 = \frac{k^2 + 4k + b}{k^2 + 1} = \frac{k^2 + 4k + 1}{k^2 + 1} = 1 + \frac{4k}{k^2 + 1} = 1 + \frac{4}{k + \frac{1}{k}}$ ($k \neq 0$).

当 $k > 0$ 时, $1 + \frac{4}{k + \frac{1}{k}} \leq 3$; 当 $k < 0$ 时, $1 + \frac{4}{k + \frac{1}{k}} \geq -1$. $\therefore$ 方程另一根的取值范围是 $-1 \leq x_2 \leq 3$.

14. 解: (1) 如图, 由 $\frac{1}{m} = \frac{2}{m+2}$ 得 $m = 2$, 连接 AD, BD , $\therefore AB$ 是 $\odot O$ 的直径, $\therefore \angle ACB = 90^\circ$, $\angle ADB = 90^\circ$, 又 $\therefore \angle BCD = 2\angle ACD$, $\angle ACB = \angle BCD + \angle ACD$, $\therefore \angle ACD = 30^\circ$, $\angle BCD = 60^\circ$.

(2) 如图, 连接 AD, BD , 则 $\angle ABD = \angle ACD = 30^\circ$, $AB = 4$.

$\therefore AD = 2, BD = 2\sqrt{3}$, $\therefore \frac{AP}{PB} = \frac{1}{2}$, $\therefore AP = \frac{4}{3}, BP = \frac{8}{3}$. $\therefore \angle APC = \angle DPB$,

$\angle ACD = \angle ABD$. $\therefore \triangle APC \sim \triangle DPB$. $\therefore \frac{AC}{DB} = \frac{AP}{DP} = \frac{PC}{BP}$. $\therefore AC \cdot DP = AP \cdot$

$DB = \frac{4}{3} \times 2\sqrt{3} = \frac{8\sqrt{3}}{3}$ ①, $PC \cdot DP = AP \cdot BP = \frac{4}{3} \times \frac{8}{3} = \frac{32}{9}$ ②; 同理

$\triangle CPB \sim \triangle APD$, $\therefore \frac{BP}{DP} = \frac{BC}{AD}$, $\therefore BC \cdot DP = BP \cdot AD = \frac{8}{3} \times 2 = \frac{16}{3}$ ③, 由

①得 $AC = \frac{8\sqrt{3}}{3DP}$, 由③得 $BC = \frac{16}{3DP}$, $AC:BC = \frac{8\sqrt{3}}{3} : \frac{16}{3} = \frac{\sqrt{3}}{2}$, 在 $\text{Rt} \triangle ABC$

中, $\left(\frac{8\sqrt{3}}{3DP}\right)^2 + \left(\frac{16}{3DP}\right)^2 = 4^2$, $\therefore DP = \frac{2\sqrt{7}}{3}$, 由② $PC \cdot DP = PC \cdot \frac{2\sqrt{7}}{3} = \frac{32}{9}$,

解得 $PC = \frac{16\sqrt{7}}{21}$, $\therefore DC = CP + PD = \frac{16\sqrt{7}}{21} + \frac{2\sqrt{7}}{3} = \frac{10\sqrt{7}}{7}$.

(3) 如图, 连接 OD , 由 $\frac{AP}{PB} = \frac{2-\sqrt{3}}{2+\sqrt{3}}$, $AB = 4$, 则 $\frac{AP}{4-AP} = \frac{2-\sqrt{3}}{2+\sqrt{3}}$, 解得

$AP = 2 - \sqrt{3}$, $OP = 2 - AP = \sqrt{3}$, 要使 CD 最短, 则 $CD \perp AB$ 于点 P , 于是

$\cos \angle POD = \frac{OP}{OD} = \frac{\sqrt{3}}{2}$, $\therefore \angle POD = 30^\circ$, $\angle ACD = 15^\circ$, $\angle BCD = 75^\circ$,

$\therefore m = 5$, 故存在这样的 m 值, 且 $m = 5$.

15. 解: (1) $y = \begin{cases} 8+c & (0 < x \leq a), \\ 8+b(x-a)+c & (x > a). \end{cases}$ ① ②

(2) 由题意, $0 < c \leq 5$, $8+c \leq 13$, 从表中知, 2月, 3月的费用均大于13元, 故用水量 $15 \text{ m}^3, 22 \text{ m}^3$ 均大于最低限量 $a \text{ m}^3$, 将 $x = 15, x = 22$ 分别

代入②可得 $\begin{cases} 19 = 8 + b(15-a) + c, \\ 33 = 8 + b(22-a) + c, \end{cases}$ 解得 $b = 2, 2a = c + 19$ ③. 若1月

份用水量 9 m^3 超过最低限量 $a \text{ m}^3$, 则 $a < 9$, 将 $x = 9$ 代入②式中, 得 $9 = 8 + 2(9-a) + c$, 得 $2a = c + 17$, 这与③式矛盾. $\therefore a \geq 9$, 因此1月份的付款应为 $8+c=9$, $\therefore c = 1$, 故 $a = 10, b = 2, c = 1$.

16. (1) ①解: $\therefore \angle BAC = 30^\circ, AD = AE$, $\therefore \angle AED = \angle ADE = 75^\circ, \angle CEH = 75^\circ$. 又 $\therefore \angle ACB = \angle ACH = 90^\circ$, $\therefore \angle BHD = 15^\circ$;

②证明: $\therefore CD$ 是 $\odot A$ 的切线, 则 $\angle CDA = 90^\circ, \angle CDH + \angle ADE = 90^\circ$. 又 $\therefore \angle CHD + \angle CEH = 90^\circ, \angle CEH = \angle AED = \angle ADE$, $\therefore \angle CDH = \angle CHD$, $\therefore CD = CH$.

(2) 解: ①如图②, 作 $AM \perp FG$ 于点 $M, FG = 2MG, AB = \sqrt{3^2 + 4^2} = 5$, 易证 $\triangle CAD \sim \triangle CCF$, 则 $\frac{AD}{FG} = \frac{AC}{CG}$, 设 $AD = AG = 5x$, 在 $\text{Rt} \triangle AMG$ 中, $MG = AG \cdot \cos$

$\angle FGA = 5x \cdot \cos \angle BAC = 5x \times \frac{4}{5} = 4x$, $\therefore FG = 8x$, 即 $\frac{5x}{8x} = \frac{4}{4+5x}$, 解得 $x =$

$\frac{12}{25}$, $\therefore \odot A$ 的半径长为 $\frac{12}{5}$;

②过点 A 分别作 $AP \perp DE$ 于点 $P, AQ \perp DF$ 于点 Q , 过点 C 作 $CR \perp AB$ 于点 R , 易证 $\triangle ECH \sim \triangle EPA$, $\therefore PE \cdot EH = CE \cdot AE$, 即 $\frac{1}{2} DE \cdot EH = CE \cdot AE$, 即

$DE \cdot EH = 2CE \cdot AE$, 设 $CE = x$, 则 $AE = 4 - x$, 则 $DE \cdot EH = 2x(4 - x) =$

$-2x(x-4)$, $\therefore$ 当 $x = \frac{1}{2} \times (0+4) = 2$ 时, $DE \cdot EH$ 取得最大值, 即 $CE =$

$2 = AE = AD$, 在 $\triangle ACR$ 中, $AR = AC \cdot \cos \angle CAB = 4 \times \frac{AC}{AB} = \frac{4}{5} \times 4 = \frac{16}{5}$, 则

$DR = AR - AD = \frac{16}{5} - 2 = \frac{6}{5}$, 在 $\text{Rt} \triangle RCD$ 和 $\text{Rt} \triangle AQD$ 中, $\cos \angle QDA =$

$\frac{QD}{AD} = \cos \angle RDC = \frac{RD}{CD}$, 即 $AD \cdot DR = QD \cdot CD$, 而 $CD \cdot DF = CD \cdot 2DQ =$

$2CD \cdot DQ = 2AD \cdot DR = 2 \times 2 \times \frac{6}{5} = \frac{24}{5}$.

17. (1) 证明: $\Delta = (-2m)^2 - 4(n^2 - mn + \frac{5}{4}m^2) = -(m-2n)^2 \geq 0$, $\therefore (m-2n)^2 \leq 0$, $\therefore m-2n = 0$, $\therefore \Delta = 0$, $\therefore$ 一元二次方程 $x^2 - 2mx + n^2 - mn + \frac{5}{4}m^2 = 0$ 有两个相等实根, $\therefore AM = AN$.

(2) 解: $\therefore \angle BAC = 90^\circ, AD \perp BC$, $\therefore \angle ADC = \angle ADB = 90^\circ, \angle DAC = \angle DBA$, $\therefore \triangle ADC \sim \triangle BDA$, $\therefore \frac{AD}{BD} = \frac{DC}{AD}$. $\therefore AD^2 = BD \cdot DC$. 同理可证

$\triangle EBD \sim \triangle CND$, $\therefore \frac{ED}{CD} = \frac{BD}{DN}$, $\therefore BD \cdot DC = ED \cdot DN$. $\therefore AD^2 = ED \cdot DN$,

$\therefore AN = \frac{15}{8}, DN = \frac{9}{8}$,

$\therefore AD = DN + AN = 3$. $\therefore 3^2 = \frac{9}{8} DE$, $\therefore DE = 8$.

(3) 解: 由 (1) 知 $AM = AN$, 易证 $\angle BCM = \angle FBM$, 由 (2) 可知 $\angle E = \angle FCB$, $\therefore \angle ABE = \angle E$. $\therefore AB = AE$, 过点 M 作 $MG \perp AN$ 于点 G . 由 $MG \parallel$

BD 得 $\frac{MG}{BD} = \frac{AM}{AB}$. $\therefore \frac{S_{\triangle AMN}}{S_{\triangle ABE}} = \frac{\frac{1}{2} AN \cdot MG}{\frac{1}{2} AE \cdot BD} = \frac{AM^2}{AB^2} = \frac{9}{64}$, $\therefore \frac{AM}{AB} = \frac{3}{8}$,

$\therefore \frac{AN}{AE} = \frac{AM}{AB} = \frac{3}{8}$, 过点 A 作 $AH \perp EF$ 于点 H , 由 $AH \parallel FN$ 得 $\frac{EH}{HF} = \frac{AE}{AN} =$

$\frac{8}{3}$, 设 $EH = 8a$, 则 $FH = 3a$. $\therefore AE = AB$, $\therefore BH = HE = 8a$, $\therefore BF = 5a$,

$EF = 11a$, 由根与系数关系得 $\begin{cases} BF + EF = 16a = \frac{16}{5}k, \\ BF \cdot EF = 55a^2 = 2k^2 + 1, \end{cases}$ 解得 $a = \pm \frac{\sqrt{5}}{5}$.

$\therefore a > 0, a = \frac{\sqrt{5}}{5}$, $\therefore BF = \sqrt{5}$. 由 $\angle ACM = \angle MCB, \angle DAC = \angle DBA$ 可知

$\triangle ACN \sim \triangle BCM$, $\therefore \frac{AC}{BC} = \frac{AN}{BM} = \frac{3}{5}$. 设 $AC = 3b$, 则 $BC = 5b$, 在 $\text{Rt} \triangle ABC$ 中,

有 $AB = 4b$. $\therefore AM = \frac{3}{2}b$. 在 $\text{Rt} \triangle ACM$ 中, 有 $MC = \frac{3\sqrt{5}}{2}b$. 由 $\triangle ACM \sim \triangle FCB$

得 $\frac{BC}{BF} = \frac{CM}{AM}$, $\therefore \frac{BC}{\sqrt{5}} = \frac{\frac{3\sqrt{5}}{2}b}{\frac{3}{2}b}$, $\therefore BC = 5$.

全国重点高中提前招生 考试全真试卷(十四)

1. B 【解析】令 $x = \sqrt{3 + \sqrt{5+2\sqrt{3}}} + \sqrt{3 - \sqrt{5+2\sqrt{3}}}$, 则 $x > 0$,

则 $x^2 = (\sqrt{3 + \sqrt{5+2\sqrt{3}}} + \sqrt{3 - \sqrt{5+2\sqrt{3}}})^2 = (3 + \sqrt{5+2\sqrt{3}}) + 2 \times$

$\sqrt{3 + \sqrt{5+2\sqrt{3}}} \times \sqrt{3 - \sqrt{5+2\sqrt{3}}} + (3 - \sqrt{5+2\sqrt{3}}) = 6 + 2\sqrt{9-5-2\sqrt{3}} =$

$6 + 2\sqrt{4-2\sqrt{3}} = 6 + 2\sqrt{2} \times \sqrt{2-\sqrt{3}}$,

$\therefore 2 - \sqrt{3} = \frac{4-2\sqrt{3}}{2} = \frac{3-2\sqrt{3}+1}{2} = \frac{(\sqrt{3}-1)^2}{2} = \left(\frac{\sqrt{3}-1}{\sqrt{2}}\right)^2$,

$\therefore \sqrt{2-\sqrt{3}} = \frac{\sqrt{3}-1}{\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{2}$,

$\therefore x^2 = 6 + 2\sqrt{2} \times \frac{\sqrt{6}-\sqrt{2}}{2} = 6 + \sqrt{12} - 2 = 4 + 2\sqrt{3} = (3 + 2\sqrt{3} + 1) =$

$(\sqrt{3})^2 + 2\sqrt{3} + 1 = (\sqrt{3}+1)^2$, $\therefore x = \sqrt{3}+1$. 故选 B.

2. C 【解析】解不等式 $-11x - 5 \leq 6$ 得 $x \geq -1$, 解不等式 $\frac{m+x}{3} > x - m$ 得

$x < 2m$,

$\therefore$ 关于 x 的不等式组 $\begin{cases} -11x - 5 \leq 6, \\ \frac{m+x}{3} > x - m \end{cases}$ 至少有一个非负整数解,

$\therefore 2m > -1$, 解得 $m > -\frac{1}{2}$.

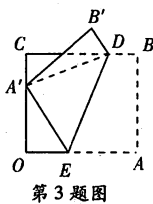
解分式方程 $\frac{1}{x-2} - 3 = \frac{m-x}{2-x}$ 得 $x = \frac{m+7}{4}$, 且 $x \neq 2$,

$\therefore$ 关于 x 的分式方程 $\frac{1}{x-2} - 3 = \frac{m-x}{2-x}$ 有不大于 5 的整数解,

$\therefore \frac{m+7}{4} \leq 5$ 且 $\frac{m+7}{4} \neq 2$, 解得 $m \leq 13$ 且 $m \neq 1$, 则符合要求的 m 的值为 5,

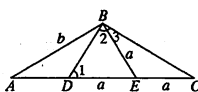
9,13,共3个.故选C.

3. B 【解析】如图,连接 $A'D$, $\because$ 四边形 $OABC$ 是矩形,
 $\therefore BC=OA=4, OC=AB=3. \therefore CD=3DB,$
 $\therefore CD=3, DB=1, \therefore A'B'=CD$. 根据折叠的性质可得
 $\angle C=\angle B'=90^\circ, B'D=DB=1$, 在 $Rt\triangle A'DC$ 和
 $Rt\triangle A'B'D$ 中, $\begin{cases} A'D=A'D \\ A'B'=CD \end{cases}, \therefore Rt\triangle A'DC \cong Rt\triangle A'B'D$,
 $\therefore A'C=B'D=1, \therefore A'O=3-1=2$, 设 $OE=x$, 则
 $A'E=4-x$, 根据勾股定理可得 $2^2+x^2=(4-x)^2$, 解
得 $x=\frac{3}{2}$, 即 OE 的长为 $\frac{3}{2}$. 故选 B.



第3题图

4. A 【解析】设 $DN=x, BM=y$, 则 $NC=1-x, MC=1-y, C_{\triangle NCM}=NC+CM+NM=2, \therefore NM=x+y$. 将 $\triangle DNA$ 绕点 A 顺时针旋转 90° 至 $\triangle ABF$, 则
 $\triangle ANM \cong \triangle AFM, \therefore \angle NAM=\angle FAM=45^\circ, \angle DNA=\angle AFB$. 过点 A 作 $AE \perp NM$ 于
点 E , 则 $\triangle DAN \cong \triangle EAM, \therefore AE=AD=1$. 在 $Rt\triangle CNM$ 中, $\therefore (1-x)^2+(1-y)^2=(x+y)^2$, 整理得 $xy+x+y-1=0, \textcircled{1}$. $S_{\triangle ANM}=\frac{1}{2}(x+y) \textcircled{2}$,
 $\therefore (x-y)^2 \geq 0, \therefore (x+y)^2 \geq 4xy, \therefore xy \leq \frac{(x+y)^2}{4} \textcircled{3}$, $\therefore$ 将 $\textcircled{2}\textcircled{3}$ 代入 $\textcircled{1}$ 并整
理得 $S^2+2S-1 \geq 0, \therefore (S+1)^2 \geq 2, \therefore S > 0, \therefore S \geq \sqrt{2}-1, \therefore \triangle MAN$ 的面
积的最小值为 $\sqrt{2}-1$. 故选 A.



第5题图

5. A 【解析】如图, $\because BE=DE, \therefore \angle 1=\angle 2$, 在
 $Rt\triangle BCD$ 中, $\angle 1+\angle C=90^\circ, \angle 2+\angle 3=90^\circ$,
 $\therefore \angle 3=\angle C, \therefore BE=CE=a, \therefore AD+AC=(c-a)+(a+c)=2c, AD \cdot AC=(c-a)(c+a)=c^2-a^2=b^2, \therefore$ 以 AD, AC 的长为根一元二次方
程是 $x^2-2cx+b^2=0$. 故选 A.

6. A 【解析】由 $a^2+4a+1=0$ 知 $a+\frac{1}{a}=-4, a^2+\frac{1}{a^2}=14$, 原式化为

$$\frac{a^2+\frac{1}{a^2}-m}{2\left(a+\frac{1}{a}\right)+m}=3, \frac{14-m}{-8+m}=3, m=\frac{19}{2}. \text{故选 A.}$$

7. $-\frac{2}{3}$ 【解析】由 $a+b+c=2$, 两边平方, 得 $a^2+b^2+c^2+2ab+2bc+2ac=4$,

$$\text{将已知代入, 得 } ab+bc+ac=\frac{1}{2}; \text{ 由 } a+b+c=2 \text{ 得 } c-1=1-a-b, \therefore ab+c-1=ab+1-a-b=(a-1)(b-1), \text{同理得 } bc+a-1=(b-1)(c-1), ca+b-1=(c-1)(a-1), \therefore \text{原式}=\frac{1}{(a-1)(b-1)}+\frac{1}{(b-1)(c-1)}+\frac{1}{(c-1)(a-1)}=\frac{c-1+a-1+b-1}{(a-1)(b-1)(c-1)}=\frac{-1}{(ab-a-b+1)(c-1)}=\frac{-1}{abc-ac-bc+c-ab+a+b-1}=\frac{-1}{1-\frac{1}{2}+2-1}=-\frac{2}{3}.$$

8. 5 【解析】两边平方得 $x_1=1, x_2=5, x_1=1$ (舍).

9. $x=\frac{a+3}{2}$ 或 $x=\frac{3a+1}{2a}$ 【解析】两边同时乘以 2, 得 $2x+\frac{1}{2x-3}=\frac{a^2+3a+1}{a}$,
 $2x-3+\frac{1}{2x-3}=\frac{a^2+3a+1}{a}-3, 2x-3+\frac{1}{2x-3}=a+\frac{1}{a}, \therefore 2x-3=a$ 或 $2x-3=\frac{1}{a}$, 解得 $x=\frac{a+3}{2}$ 或 $2ax-3a=1, x=\frac{3a+1}{2a}$.

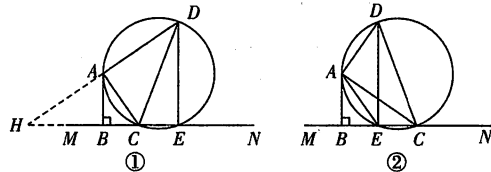
10. $\sqrt{2}-1$ 【解析】由已知可得: $\triangle ADE$ 和 $\triangle CDE$ 均为等腰直角三角形, 计算得
 $BD=\sqrt{2}-1$, 在 $Rt\triangle BCD$ 中, $\tan \angle BCD=\frac{BD}{CD}=\sqrt{2}-1$.

11. $\frac{2035}{48}$ 【解析】设折叠后所成圆形覆盖桌面的面积为 S , 则 $S=S_{\triangle ABC}+S_{\triangle ADC}-S_{\triangle AEC}=S_{\text{矩形}ABCD}-S_{\triangle AEC}, S_{\triangle AEC}=\frac{1}{2}AB \cdot EC=\frac{5}{2}EC$, 由
 $Rt\triangle ABE \cong Rt\triangle CD_1E$ 知 $EC=AE$, 设 $EC=x$, 则 $AB^2+BE^2=x^2$, 即 $5^2+(12-x)^2=x^2$, 解得 $x=\frac{169}{24}, S_{\triangle AEC}=\frac{5}{2} \times \frac{169}{24}=\frac{845}{48}, \therefore S=5 \times 12-\frac{845}{48}=\frac{2035}{48}$.

12. $\frac{\sqrt{13}}{4} \leq x \leq \frac{3\sqrt{13}}{2}$ 【解析】当点 P 落在 BE 上时, 延长 GF 交 DC 于点 H ,
作 $PM \perp AB$ 于点 $M, PN \perp AD$ 于点 N . 易求 $EC=\frac{10}{3}$,
 $\therefore \tan \angle CEB=\tan \angle PBM, \therefore \frac{BC}{EC}=\frac{PM}{BM}=\frac{3}{2}$, 设 $PM=3k$, 则 $BM=2k$,
 $PM=AN=3k, PN=AM=\frac{16}{3}-2k$, 在 $Rt\triangle PDN$ 中,
 $\therefore PD=AD=5, DN=5-3k, PN=\frac{16}{3}-2k, \therefore 25=(5-3k)^2+(\frac{16}{3}-2k)^2$, 整理得 $117k^2-462k+256=0$, 解得 $k=\frac{2}{3}$ 或 $\frac{128}{39}$ (舍).
 $\therefore PM=2, BM=\frac{4}{3}, AM=4$, 设 $AG=GP=m$, 在 $Rt\triangle PGM$ 中, $m^2=(4-$

$$m)^2+2^2, \text{解得 } m=\frac{5}{2}, \therefore AH=AG=\frac{5}{2}, \therefore EH=\frac{1}{2}, \therefore \frac{HF}{EH}=\frac{BC}{CE}=\tan \angle CEB=\frac{3}{2}, \therefore HF=\frac{3}{4}, \therefore EF=\frac{\sqrt{13}}{4}. \text{当点 } P \text{ 落在 } DC \text{ 上时, } EP=3, \tan \angle CEB=\frac{PF}{PE}=\frac{3}{2}, \therefore PF=\frac{9}{2}, \therefore EF=\sqrt{3^2+(\frac{9}{2})^2}=\frac{3\sqrt{13}}{2}, \therefore \frac{\sqrt{13}}{4} \leq x \leq \frac{3\sqrt{13}}{2}.$$

13. 解: (1) DE 的长不会发生变化. 理由如下:
延长 DA 交直线 MN 于点 H , 如图①. $\because CD$ 为直径,
 $\therefore \angle CAH=\angle DAC=90^\circ$.
又 $\because \angle ACD=\angle ACB$,
 $\therefore \angle ADC=\angle AHC, \therefore DC=HC. \therefore AC \perp DH, \therefore AD=AH$ (三线合一).
 $\therefore AB \perp MN, DE \perp MN, \therefore AB \parallel DE, \therefore \triangle HAB \sim \triangle HDE, \therefore \frac{AB}{DE}=\frac{AH}{DH}=\frac{1}{2}$,
 $\therefore DE=2AB=6$, 故点 C 运动过程中, DE 的弦长不发生变化.
(2) $\because \angle ABC=\angle DAC=90^\circ, \angle ACB=\angle ACD, \therefore \triangle ACD \sim \triangle BCA$, 要使
 $\triangle ACD$ 与 $\triangle CDE$ 相似, 则 $\triangle CDE$ 必与 $\triangle BCA$ 相似, 又 $\because \angle ABC=\angle DEC=90^\circ$, 故分两种情况讨论:



第13题图

- ①当 $\triangle ADC \sim \triangle DEC$, 如图①, $\angle ACB=\angle ACD=\angle DCE=60^\circ$, 在
 $Rt\triangle ABC$ 中, $\frac{AB}{BC}=\tan 60^\circ$, 即 $\frac{3}{x}=\sqrt{3}$, 故 $x=\sqrt{3}$;
②当 $\triangle ABC \sim \triangle CED$, 如图②, $\angle ACB=\angle ACD=\angle DCE=30^\circ$, 在
 $Rt\triangle ABC$ 中, $\frac{AB}{BC}=\tan 30^\circ$, 即 $\frac{3}{x}=\frac{\sqrt{3}}{3}$, 故 $x=3\sqrt{3}$. 综上所述, 当 $x=\sqrt{3}$ 或
 $3\sqrt{3}$ 时, $\triangle ACD$ 和 $\triangle CDE$ 相似.

14. 解: (1) $\begin{cases} kx^2-x-y+\frac{1}{2}=0 & \textcircled{1} \\ y=k(2x-1) & \textcircled{2} \end{cases}$, 将 $\textcircled{2}$ 代入 $\textcircled{1}$, 得 $kx^2-x-k(2x-1)+\frac{1}{2}=0$. 整理得 $kx^2-(2k+1)x+k+\frac{1}{2}=0$ ③, 依题意有两个不同实
数解 $x_1, x_2, \therefore k \neq 0$ 且 $\Delta > 0$, 即 $k \neq 0$, 且 $[-(2k+1)]^2-4k(k+\frac{1}{2}) > 0$, $\therefore$ 解得 $k > -\frac{1}{2}$ 且 $k \neq 0$.

- (2) 由 $\textcircled{3}$ 得 $x_1+x_2=\frac{2k+1}{k}, x_1x_2=\frac{k+\frac{1}{2}}{k}=\frac{2k+1}{2k}$. 由 $y_1=k(2x_1-1), y_2=k(2x_2-1), \therefore y_1y_2=k^2(2x_1-1)(2x_2-1)=k^2[4x_1x_2-2(x_1+x_2)+1]=k^2[4 \times \frac{2k+1}{2k}-2 \times \frac{2k+1}{k}+1]=k^2$. 由 $y_1y_2+\frac{1}{x_1}+\frac{1}{x_2}=3$, 得 $k^2+\frac{x_1+x_2}{x_1x_2}=\frac{2k+1}{2k}+k^2=k^2+2=3, \therefore k^2=1, \therefore k=\pm 1$. 经验证: 当 $k=-1$ 时, 方程
 $\textcircled{3}$ 为 $x^2-x+\frac{1}{2}=0, \Delta=(-1)^2-4 \times \frac{1}{2}<0, \therefore k=-1$ 应舍去. $\therefore k=1$.

15. 解: (1) $\because AB \perp x$ 轴, $\therefore \angle ABP+\angle APB=90^\circ$. 又 $BP \perp PD, \therefore \angle BPD=90^\circ, \therefore \angle DPQ+\angle APB=90^\circ, \therefore \angle DPQ=\angle ABP. \therefore AP=OQ=t, B(-4, 4), \therefore PQ=AO=4, \therefore PQ=AB. \therefore DQ \perp x$ 轴, $\angle DQP=\angle BAP=90^\circ, \therefore \triangle BAP \cong \triangle PQD, \therefore DQ=AP=t, \therefore \angle DOQ=45^\circ, \therefore \angle PDO=\angle DOQ-\angle DPQ=45^\circ-15^\circ=30^\circ. \therefore \cos \angle PDO=\frac{\sqrt{3}}{2}$.
(2) 过点 O 作 $OG \perp PD$ 于点 $G, \therefore P$ 为 AO 的中点, $\therefore PO=OQ=DQ=2$, 在
 $Rt\triangle ODQ$ 中, $OD=\sqrt{2^2+2^2}=2\sqrt{2}$, 在 $Rt\triangle PDQ$ 中, $PD=\sqrt{4^2+2^2}=2\sqrt{5}, \therefore OG \perp PD, DQ \perp PQ, \therefore \frac{1}{2}PO \cdot DQ=\frac{1}{2}PD \cdot OG, \therefore OG=\frac{PO \cdot DQ}{PD}=\frac{2 \times 2}{2\sqrt{5}}=\frac{2\sqrt{5}}{5}$, 在 $Rt\triangle ODG$ 中, $\sin \angle PDO=\frac{OG}{OD}=\frac{\sqrt{10}}{10}$.
(3) 过 D 点作 $DN \perp AB$, 分别交 y 轴, AB 于点 $M, N, \therefore \triangle DEM \sim \triangle DBN, \therefore \frac{EM}{BN}=\frac{DM}{DN}, \therefore EM=\frac{BN \cdot DM}{DN}=\frac{t(4-t)}{4+t}, \therefore OE=t+\frac{t(4-t)}{4+t}=\frac{8t}{4+t}$, 设
 $S_{\triangle POF}=S_{\triangle DEF}$, 则 $S_{\triangle POD}=S_{\triangle EDO}, \therefore \frac{1}{2}EO \cdot DM=\frac{1}{2}PO \cdot DQ$, 即 $\frac{1}{2}EO \cdot$
 $t=\frac{1}{2}PO \cdot t, \therefore EO=PO, \therefore \frac{8t}{4+t}=4-t, \therefore t^2+8t-16=0, \therefore t_1=4\sqrt{2}-4 < 4,$
 $t_2=-4-4\sqrt{2}$ (舍). 因此, 当 $t=4\sqrt{2}-4$ 时, $\triangle POF$ 与 $\triangle DEF$ 的面积相等.

16. 解: (1) 连接 $EF, FA, \therefore FH$ 为圆的切线又和 EC 垂直, $\therefore CE \parallel AF$,

$\therefore \angle CEF = \angle AFE$. 又 $\because \angle AFE = \angle FEB$, $\therefore \angle CEF = \angle BEF$. $\therefore EF$ 为 $\angle BEC$ 的平分线. $\because \angle EFB = 90^\circ$, $\therefore EF \perp BC$. $\therefore BE = CE$. $\therefore \triangle BEC$ 为等腰三角形. $\therefore BF$ 为 BC 的一半.

$\because EA \parallel CF$, $\therefore$ 四边形 $CEAF$ 为平行四边形, 即 $AE = CF = \frac{5}{2}$.

(2) $\because FH \parallel BE, FH \perp CE, \therefore BE \perp CE$. $\therefore \angle AEB + \angle DEC = 90^\circ$. $\because \angle ABE + \angle AEB = 90^\circ, \therefore \angle ABE = \angle DEC$. $\because \angle A = \angle D = 90^\circ$, $\therefore \triangle ABE \sim \triangle DEC$. $\therefore \frac{AB}{DE} = \frac{AE}{CD}$. $\because AB = 2, AD = 5, \therefore CD = AB = 2$.

$\therefore \frac{2}{5-AE} = \frac{AE}{2}, \therefore AE = 1$ 或 $AE = 4$.

(3) 连接 EF, OF, OG , 则 $\angle BFE = 90^\circ$. 设 $AE = x$, 则 $EF = AB = 2, BF = AE = x, CF = DE = 5 - x$. 若 $\triangle OFG$ 是等腰直角三角形, 则 $\angle FOG = 90^\circ$. 连接 BG, EG , 设 BG, EF 交于点 K . $\therefore \triangle BFK$ 和 $\triangle EGK$ 都是等腰直角三角形. $\therefore BF = KF = x, BK = \sqrt{2}x, EK = 2 - KF = 2 - x, \therefore GK = EG = \frac{\sqrt{2}}{2}(2 - x), BG = GK + BK = \frac{\sqrt{2}}{2}(2 + x)$. 又 $\because \angle EBG = \angle EFG = \angle FCH$,

$\therefore \triangle BEG \sim \triangle CEF$. $\therefore \frac{BG}{EG} = \frac{FC}{EF}$, 即 $\frac{\frac{\sqrt{2}}{2}(2+x)}{\frac{\sqrt{2}}{2}(2-x)} = \frac{5-x}{2}$, 解得 $x = \frac{9-\sqrt{57}}{2}$.

或 $x = \frac{9+\sqrt{57}}{2}$ (舍). $\therefore AE$ 的长度是 $\frac{9-\sqrt{57}}{2}$.

17. 解: (1) 由直线 $y = \frac{3}{4}x + 8$ 与 x 轴, y 轴分别交于 A, B 两点, 则 $A(-\frac{32}{3}, 0), B(0, 8)$. $\therefore OA = \frac{32}{3}, OB = 8$. 又 $OB \perp AC, AB$ 切 $\odot C$ 于 B 点, $\therefore \triangle ABO \sim \triangle BCO$. $\therefore \frac{OB}{OA} = \frac{OC}{OB}$. $\therefore \frac{8}{\frac{32}{3}} = \frac{OC}{8}$. $\therefore OC = 6$. $\therefore C(6, 0)$, 半径 $CB = 10$.

(2) ① 设 $\odot C$ 的运动时间为 t s, 则 $C(6 - \frac{10}{3}t, 0), R = 10 - \frac{3}{2}t$, 直线 $l: 3x - 4y + 32 = 0$, 点 C 到直线 l 的距离 $h = |10 - 2t|$, $\therefore 10 - \frac{3}{2}t \geq |10 - 2t|, \therefore 10 - \frac{3}{2}t \geq 10 - 2t \geq -(10 - \frac{3}{2}t)$, 解得 $0 \leq t \leq \frac{40}{7}$; ② $\because (\frac{a}{2})^2 = R^2 - h^2 = (10 - \frac{3}{2}t)^2 - (10 - 2t)^2 = (-\frac{32}{9}) \cdot (t - \frac{15}{4})^2 + 50, (\frac{a}{2})^2 = 50, \therefore a_{\max} = 10\sqrt{2}$.

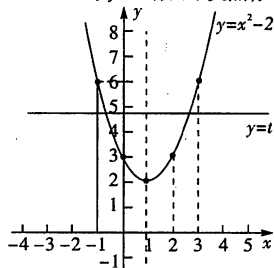
(3) $\because D(-4, 5), \therefore$ 直线 $MN: y = kx + 4k + 5$, 联立抛物线得 $x^2 - 4kx - 16k - 20 = 0, \therefore x_M + x_N = 4k, x_M x_N = -16k - 20$. 由 $\triangle MEP \sim \triangle PFN$ 得

$$\left| \frac{y_M - y_P}{x_P - x_M} \right| = \left| \frac{x_N - x_P}{y_N - y_P} \right|, \therefore \frac{\frac{1}{4}x_M^2 - \frac{1}{4}t^2}{t - x_M} = \frac{x_N - t}{\frac{1}{4}x_N^2 - \frac{1}{4}t^2}, \therefore (x_M^2 - t^2) \cdot$$

$(x_N^2 - t^2) = 16(t - x_M)(x_N - t), \therefore (x_M + t)(x_N + t) = -16, \therefore t^2 + 4kt - 16k - 4 = 0, \therefore t = 2k \pm \sqrt{k^2 - 4k + 1}, \therefore k$ 变, t 也随之变, $\therefore$ 不存在.

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1. D 【解析】: 抛物线 $y = x^2 + bx + 3$ 的顶点坐标为 $(1, 2)$, $\therefore 2 = 1 + b + 3$, 解得 $b = -2$. $\therefore$ 抛物线解析式为 $y = x^2 - 2x + 3$. $\because x = 5$ 比 $x = -1$ 离函数的对称轴远, $\therefore$ 当 $x = 5$ 时, $y = x^2 - 2x + 3 = 25 - 10 + 3 = 18$, 关于 x 的一元二次方程 $x^2 - 2x + 3 - t = 0$ (t 为实数) 在 $-1 \leq x \leq 5$ 范围内有两个不同的实数根, 则 $y = x^2 - 2x + 3$ 与 $y = t$ 有两个交点, 如图所示,

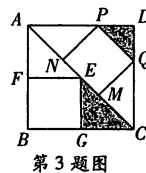


第1题图

由图象可得实数 t 的取值范围是 $2 < t \leq 6$. 故选 D.

2. C 【解析】: 函数 $y = (x+a)(x+b)$ 的图象与 x 轴有 2 个交点, $\therefore M = 2$. $\because$ 函数 $y = (ax+1)(bx+1) = abx^2 + (a+b)x + 1$, $\therefore$ 当 $ab \neq 0$ 时, $\Delta = (a+b)^2 - 4ab = (a-b)^2 > 0$, 函数 $y = (ax+1)(bx+1)$ 的图象与 x 轴有 2 个交点, 即 $N = 2$, 此时 $M = N$; 当 $ab = 0$ 时, 不妨令 $a = 0, \therefore a \neq b, \therefore b \neq 0$, 函数 $y = (ax+1)(bx+1) = bx + 1$ 为一次函数, 与 x 轴有一个交点, 即 $N = 1$, 此时 $M = N + 1$. 故选 C.

3. A 【解析】如图, $\because$ 四边形 $ABCD$ 是正方形, $\therefore AD = CD, \angle D = \angle B = 90^\circ$, $\therefore \angle DAC = \angle DCA = 45^\circ$, 同理 $\angle BAC = \angle BCA = 45^\circ$. $\therefore$ 四边形 $PQMN$ 是正方形, $\therefore PN = QM = MN = PQ, \angle PNM = \angle QMN = 90^\circ$, $\therefore \angle PNA = \angle QMC = 90^\circ$, $\therefore \angle NPA = \angle NAP = 45^\circ, \angle MQC = \angle MCQ = 45^\circ$, $\therefore AN = PN = MN = CM = QM = PQ = \frac{1}{3}AC$.



第3题图

设正方形 $ABCD$ 的边长为 m , 则 $S_{\triangle DAC} = \frac{1}{2}m^2$,

$\because PQ \parallel AC, \therefore \triangle DPQ \sim \triangle DAC$,

$\therefore \frac{S_{\triangle DPQ}}{S_{\triangle DAC}} = (\frac{PQ}{AC})^2 = (\frac{1}{3})^2 = \frac{1}{9}, \therefore S_1 = \frac{1}{9} \times \frac{1}{2}m^2 = \frac{1}{18}m^2$.

$\because$ 四边形 $BFEQ$ 是正方形, $\therefore EF = EG = BF = BG, EF \parallel BC, EG \parallel AB$, $\therefore \angle AFE = \angle B = 90^\circ, \angle CGE = \angle B = 90^\circ, \therefore \angle FEA = \angle FAE = 45^\circ, \angle GEC = \angle GCE = 45^\circ$,

$\therefore AF = EF = BF = CG = EG = BG = \frac{1}{2}BC = \frac{1}{2}m$,

$\therefore S_2 = S_{\triangle GCE} = \frac{1}{2} \times (\frac{1}{2}m)^2 = \frac{1}{8}m^2$,

$\therefore \frac{S_1}{S_2} = \frac{\frac{1}{18}m^2}{\frac{1}{8}m^2} = \frac{4}{9}$. 故选 A.

4. C 【解析】由题意知二次函数 $y = x^2 + 2x + c$ 有两个相异的不动点 $x_1, x_2, \therefore x_1, x_2$ 是方程 $x^2 + 2x + c = x$ 的两个不相等的实数根, 整理得 $x^2 + x + c = 0, \therefore \Delta = 1 - 4c > 0$, 又 $x^2 + x + c = 0$ 的两个不相等实数根为 $x_1, x_2, x_1 < 2 < x_2, \therefore (x_1 - 2)(x_2 - 2) < 0, \therefore x_1x_2 - 2(x_1 + x_2) + 4 < 0$, 由根与系数的关系可得 $x_1 + x_2 = -1, x_1x_2 = c$, 综上 $\begin{cases} 1 - 4c > 0, \\ c + 2 + 4 < 0, \end{cases}$ 解得 $c < -6$. 故选 C.

5. A 【解析】点 D 是点 B 关于直线 AC 的对称点, 连接 DE 交 AC 于点 P , 则 $PB = PD$, 此时 y 取得最小值, $y = PE + PB = PD + PE = DE$ 为最小, $ED = 3\sqrt{5}$, 设正方形的边长为 x , 则 $AE = \frac{1}{2}x$, 则 $x^2 + (\frac{1}{2}x)^2 = (3\sqrt{5})^2$, 解得 $x = 6$ (舍负). 故选 A.

6. B 【解析】由已知 $1 \leq x \leq 11$, 得 $(x-1)(x-11) \leq 0, x^2 - 12x + 11 \leq 0, x^2 \leq 12x - 11, \therefore x \geq 1, \therefore x^2 > 0, 12x - 11 > 0, \therefore \sqrt{x^2} \leq \sqrt{12x - 11}$, 即 $x \leq \sqrt{12x - 11}, \therefore y = \frac{\sqrt{12x - 11}}{x} \geq 1$, y 最小值是 1. 故选 B.

7. 2 020 【解析】 $\because x = \frac{3+\sqrt{5}}{2}, \therefore 2x = 3 + \sqrt{5}, \therefore (2x-3)^2 = 5, \therefore x^2 - 3x = -1, \therefore 2x^3 - 3x^2 - 7x + 2 = 2x(x^2 - 3x) + 3x^2 - 7x + 2 = -2x + 3x^2 - 7x + 2 = 3x^2 - 9x + 2 = 3(x^2 - 3x) + 2 = -3 + 2 = -1$.

8. $\frac{4\sqrt{10}}{5}$ 【解析】连接 $OE, \therefore \angle PEF = 90^\circ - \angle OEB = 90^\circ - \angle OBE = \angle OFB = \angle EFP, \therefore PF = PE = 4, \therefore OP = \sqrt{3^2 + 4^2} = 5$ cm. $OF = OP - PF = 5 - 4 = 1$ cm, $DF = OD - OF = 2$ cm, $CF = OF + OC = 4$ cm. $\therefore FB = \sqrt{3^2 + 1^2} = \sqrt{10}$ cm. 又 $BF \cdot EF = CF \cdot DF, \therefore EF = \frac{4 \times 2}{\sqrt{10}} = \frac{4\sqrt{10}}{5}$.

9. 5 750 【解析】设甲产品的成本价格为 b 元, 则 $\frac{72-b}{b} = 20\%, \therefore b = 60$, $\therefore$ 甲产品的成本价格为 60 元, $\therefore 1.5$ kg A 原料与 1.5 kg B 原料的成本和为 60 元, $\therefore A$ 原料与 B 原料的成本和为 40 元. 设 A 种原料成本价格为 x 元, B 种原料成本价格为 $(40-x)$ 元, 生产甲产品 m 袋, 乙产品 n 袋, 则有 $\begin{cases} m+n \leq 100, \\ 60m + (2x+40-x)n + 500 = 60m + n(80-2x+x), \end{cases}$ $\therefore xn = 20n - 250$. 设生产甲、乙产品的实际成本为 W 元, 则有 $W = 60m + 40n + xn = 60m + 40n + 20n - 250 = 60(m+n) - 250$, $\therefore m+n \leq 100, \therefore W \leq 5750, \therefore$ 生产甲、乙产品的实际成本最多为 5 750 元.

10. $4\sqrt{5}$ 【解析】设 $CE = 3k$, 则 $CF = 4k, EF = DE = 5k, AB = CD = 8k$, $\therefore \angle EFC = \angle BAF, \therefore BF = 6k$. 在 $\text{Rt} \triangle ADE$ 中, $(10k)^2 + (5k)^2 = (5\sqrt{5})^2, \therefore k = 1$. 由勾股定理得 $DF = 4\sqrt{5}, \frac{DF}{2} = 2\sqrt{5}$, 故 A 到 DF 的距离为 $\sqrt{AD^2 - (\frac{DF}{2})^2} = \sqrt{10^2 - (2\sqrt{5})^2} = 4\sqrt{5}$.

11. $-1 \leq a < -\frac{1}{2}$ 或 $a = 3 - 2\sqrt{3}$ 【解析】(1) 若图象的顶点在 AB 上, 则有 $\begin{cases} 1 \leq -\frac{a-3}{2} \leq 2, \\ \Delta = (a-3)^2 - 12 = 0, \end{cases}$ 解得 $a = 3 - 2\sqrt{3}$. (2) 若图象的顶点在 x 轴下方, 则有 $\begin{cases} f(1) = 1 + a - 3 + 3 \leq 0, \\ f(2) = 4 + 2(a-3) + 3 > 0 \end{cases}$

或 $\begin{cases} f(1)=1+a-3+3 \geq 0, \\ f(2)=4+2(a-3)+3 < 0, \end{cases}$ 解得 $-1 \leq a < -\frac{1}{2}$.

12.4 【解析】令 $f(x)=0$, 则 $ax^2+bx+c=0$, 解得 $x_1 = \frac{-b+\sqrt{b^2-4ac}}{2a}$,

$x_2 = \frac{-b-\sqrt{b^2-4ac}}{2a}$.

$\therefore$ 抛物线 $f(x)=ax^2+bx+c$ 与 x 轴两个交点的距离为 $|x_1-x_2| = \frac{\sqrt{b^2-4ac}}{|a|}$.

即抛物线的顶点与 x 轴的两交点围成的等腰直角三角形斜边为 $\frac{\sqrt{b^2-4ac}}{|a|}$.

$\therefore f(x)=ax^2+bx+c=a(x+\frac{b}{2a})^2+\frac{4ac-b^2}{4a}$, $\therefore$ 该抛物线的顶点坐标为 $(-\frac{b}{2a}, \frac{4ac-b^2}{4a})$.

$\therefore$ 抛物线的顶点与 x 轴的两交点围成的等腰直角三角形斜边上的高为 $|\frac{4ac-b^2}{4a}|$.

$\therefore$ 等腰直角三角形斜边上的高与斜边上的中线互相重合, 直角三角形斜边上的中线等于斜边的一半, $\therefore |\frac{4ac-b^2}{4a}| = \frac{1}{2} \times \frac{\sqrt{b^2-4ac}}{|a|}$. 两边平方得 $\frac{(4ac-b^2)^2}{16a^2} = \frac{b^2-4ac}{4a^2}$,

$\therefore (b^2-4ac)^2 = 4(b^2-4ac)$.

$\therefore$ 抛物线 $f(x)=ax^2+bx+c$ 与 x 轴有两个交点, $\therefore b^2-4ac > 0$, $\therefore b^2-4ac=4$.

13. 解: 去分母, 得 $2kx-x=(kx+1)(x-1)$, $kx^2+(2-3k)x-1=0$ ①. 当 $k=0$ 时, 原方程有唯一的解 $x=\frac{1}{2}$; 当 $k \neq 0$ 时, 由于 $\Delta=(2-3k)^2+4k=5k^2+4(k-1)^2 > 0$, $\therefore$ 方程①恒有两个实数根, 要使原方程只有一解, 则方程①必有一根为 0 或 1. 当 $x=0$ 时, 代入①得 $-1=0$, k 无解. 当 $x=1$ 时, 代入①, 得 $k=\frac{1}{2}$, 这时另一根为 $x=-2$. $\therefore$ 当 $k=0$ 时, 方程的解为 $x=\frac{1}{2}$. 当 $k=\frac{1}{2}$ 时, 方程的解为 $x=-2$.

14. (1) 证明: $\because y=x^2-(m^2+4)x-2m^2-12=-(x+2)(x-6)$, $\therefore$ 当 $x=-2$ 时, $y=0$, $\therefore$ 不论 m 为何值, 函数都过定点 $(-2, 0)$.

(2) 解: 由 $\begin{cases} y=x+2, \\ y=x^2-(m^2+4)x-2m^2-12, \end{cases}$ 消去 y , 得 $(x+2)(x-m^2-7)=0$, $x_1=-2, x_2=m^2+7, AB=(m^2+9)\sqrt{2}=13\sqrt{2}$, $\therefore m=\pm 2$.

(3) 解: 设该函数图象的顶点为 (x, y) ,

则 $\begin{cases} x=\frac{m^2+4}{2} & ①, \\ y=\frac{4(-2m^2-12)-(m^2+4)^2}{4} & ②, \end{cases}$

由①得 $m^2=2x-4$, 代入②, 得 $y=-(x+2)^2$. $\therefore m^2=2x-4 \geq 0, \therefore x \geq 2$, $\therefore$ 所求的曲线解析式为 $y=-(x+2)^2 (x \geq 2)$.

15. 解: (1) $\because OA=3, OB=4, \therefore B(4, 0), C(4, 3)$. $\therefore F$ 是 BC 的中点, $\therefore F(4, \frac{3}{2})$. $\therefore$ 点 F 在反比例函数 $y=\frac{k}{x}$ 的图象上,

$\therefore k=4 \times \frac{3}{2}=6, \therefore$ 反比例函数的解析式为 $y=\frac{6}{x}$.

$\therefore E$ 点的纵坐标为 3, $\therefore E(2, 3)$.

(2) 依题意有 $E(\frac{k}{3}, 3), F(4, \frac{k}{4})$, $\therefore CF=BC-BF=3-\frac{k}{4}=\frac{12-k}{4}$, $CE=AC-AE=4-\frac{k}{3}=\frac{12-k}{3}$, 在 $Rt\triangle CEF$ 中, $\tan \angle EFC=\frac{CE}{CF}=\frac{4}{3}$.

(3) 由(2)知 $\frac{CE}{CF}=\frac{4}{3}$, 过点 E 作 $EH \perp OB$ 于点 H , 则可证 $\triangle EHG \sim \triangle GBF$, $\therefore \frac{EH}{BG}=\frac{EG}{FG}=\frac{CE}{CF}$, $\therefore \frac{3}{BG}=\frac{4}{3}$, $\therefore BG=\frac{9}{4}$.

在 $Rt\triangle FBG$ 中, $\therefore (\frac{12-k}{4})^2 - (\frac{k}{4})^2 = (\frac{9}{4})^2$, $\therefore k=\frac{21}{8}$. $\therefore$ 反比例函数解析式为 $y=\frac{21}{8x}$.

16. 解: (1) 连接 PC , 易证 $\triangle OPC \sim \triangle OAB$, $\therefore \frac{PC}{AB}=\frac{OP}{OA}$, $\therefore A(-8, 0), B(0, \frac{32}{3})$, $\therefore OA=8, OB=\frac{32}{3}, \therefore AB=\frac{40}{3}, \therefore PC=5, \therefore \odot P$ 的半径为 5.

(2) 证明: 连接 CP , $\therefore AP=CP, \therefore \angle PAC=\angle PCA$. $\because AC$ 平分 $\angle OAB$, $\therefore \angle PAC=\angle EAC, \therefore \angle PCA=\angle EAC, \therefore PC \parallel AE$. $\therefore CE \perp AB, \therefore CP \perp EF$, 即 EF 是 $\odot P$ 的切线.

(3) $\frac{OH}{FH}=\frac{3}{5}$ 是定值. 连接 PH , 由(1)得 $AP=PC=PH=5$,

$\therefore OA=8, \therefore OP=OA-AP=3$. 在 $Rt\triangle POC$ 中, $OC=\sqrt{5^2-3^2}=4$, 由射影定理可得 $OC^2=OP \cdot OF, \therefore OF=\frac{16}{3}, \therefore PF=PO+OF=\frac{25}{3}, \therefore \frac{OP}{PH}=\frac{3}{5}$,

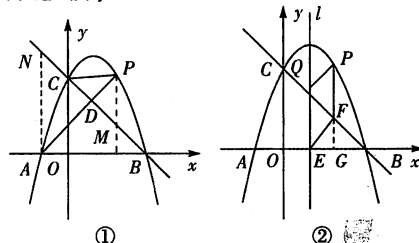
$\frac{PH}{PF}=\frac{5}{25}=\frac{3}{5}, \therefore \frac{OP}{PH}=\frac{PH}{PF}$. 又 $\because \angle HPO=\angle FPH, \therefore \triangle POH \sim \triangle PHF$.

$\therefore \frac{OH}{FH}=\frac{OP}{PH}=\frac{3}{5}$. 当点 H 与点 D 重合时, $\frac{OH}{FH}=\frac{OD}{DF}=\frac{3}{5}$.

17. 解: (1) $\because BO=3AO=3, \therefore AO=1, \therefore A(-1, 0), B(3, 0)$.

把 $A(-1, 0), B(3, 0)$ 代入 $y=-x^2+bx+c$ 得 $\begin{cases} 0=-(1)^2-b+c, \\ 0=-3^2+3b+c, \end{cases}$ 解得 $\begin{cases} b=2, \\ c=3, \end{cases}$

$\therefore$ 抛物线的表达式为 $y=-x^2+2x+3$.



第 17 题图

(2) ① $\because y=-x^2+2x+3, \therefore$ 点 C 坐标为 $(0, 3)$, 把 $B(3, 0), C(0, 3)$ 代入 $y=kx+n$ 得 $\begin{cases} 0=3k+n, \\ 3=n, \end{cases}$ 解得 $\begin{cases} k=-1, \\ n=3, \end{cases}$

$\therefore$ 直线 BC 的表达式为 $y=-x+3$.

过 P 作 $PM \perp x$ 轴交 BC 于 M , 过 A 作 $AN \perp x$ 轴交 BC 于 N , 如图①,

$\therefore AN \parallel PM, \therefore \triangle PMD \sim \triangle AND$,

$\therefore \frac{PD}{AD}=\frac{PM}{AN}, \therefore S_1:S_2=PD:AD=PM:AN=1:2$.

设 $P(m, -m^2+2m+3)$, 则 $M(m, -m+3)$, $\therefore PM=-m^2+2m+3-(-m+3)=-m^2+3m$,

$\therefore A(-1, 0), \therefore N(-1, 4), \therefore AN=4$, $\therefore \frac{-m^2+3m}{4}=\frac{1}{2}, \therefore m=1$ 或 2 ,

$\therefore$ 点 P 的坐标为 $(1, 4)$ 或 $B(2, 3)$.

② 存在. 理由如下: 过点 F 作 $FG \perp OB$ 于 G , 如图②.

$\because y=-x^2+2x+3$ 的对称轴为 $x=1, \therefore OE=1$.

$\therefore B(3, 0), C(0, 3), \therefore OC=OB=3$.

又 $\because \angle COB=90^\circ, \therefore \triangle OCB$ 是等腰直角三角形.

$\therefore \angle EFB=90^\circ, BE=OB-OE=2, \therefore \triangle EFB$ 是等腰直角三角形, $\therefore FG=GB=EG=1, \therefore$ 点 F 的坐标为 $(2, 1)$,

I. 当 EF 为边时, $\therefore$ 四边形 $EFQP$ 为平行四边形,

$\therefore QE=PF, QE \parallel PF \parallel y$ 轴, $\therefore$ 点 P 的横坐标与点 F 的横坐标同为 2.

当 $x=2$ 时, $y=-2^2+2 \times 2+3=3, \therefore$ 点 P 的坐标为 $(2, 3)$,

$\therefore QE=PF=3-1=2$, 点 Q 的坐标为 $(1, 2)$.

根据对称性当 $P(0, 3), Q(1, 4)$ 时, 四边形 $EFQP$ 也是平行四边形.

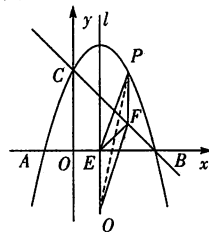
II. 当 EF 为对角线时, 如图③. $\therefore$ 四边形 $PEQF$ 为平行四边形,

$\therefore QE=PF, QE \parallel PF \parallel y$ 轴, 同理求得点 P 的坐标为 $(2, 3), \therefore QE=PF=3-1=2$, 点 Q 的坐标为 $(1, -2)$.

综上, 点 P 的坐标为 $(2, 3)$ 时, 点 Q 的坐标为 $(1, 2)$

或 $(1, -2)$, 点 P 的坐标为 $(0, 3)$ 时, 点 Q 的坐标为

$Q(1, 4)$, 以点 E, F, P, Q 为顶点的四边形是平行四边形.



第 17 题图③

全国重点高中提前招生 考试全真试卷(十六)

1. A 【解析】当 $x=4$ 时, $\frac{\sqrt{x-2\sqrt{3}}}{\sqrt{x^2-4\sqrt{3}x+12}}=\frac{\sqrt{4-2\sqrt{3}}}{\sqrt{4^2-16\sqrt{3}+12}}=$

$\frac{\sqrt{(\sqrt{3}-1)^2}}{\sqrt{28-16\sqrt{3}}}=\frac{\sqrt{(\sqrt{3}-1)^2}}{\sqrt{(4-2\sqrt{3})^2}}=\frac{\sqrt{3}-1}{4-2\sqrt{3}}=\frac{\sqrt{3}-1}{(\sqrt{3}-1)^2}=\frac{1}{\sqrt{3}-1}$, 同理

$\frac{\sqrt{x+2\sqrt{3}}}{\sqrt{x^2+4\sqrt{3}x+12}}=\frac{1}{\sqrt{3}+1}$, $\therefore$ 原式 $=\frac{1}{\sqrt{3}-1}-\frac{1}{\sqrt{3}+1}=\frac{\sqrt{3}+1}{2}-\frac{\sqrt{3}-1}{2}=1$.

故选 A.

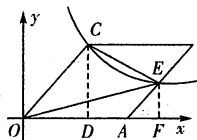
2. D 【解析】过点 C 作 $CD \perp x$ 轴于点 D , 过点 E 作 $EF \perp x$ 轴于点 F , 如图,

$\therefore$ 四边形 $OABC$ 为平行四边形,

$\therefore OC=AB, OC \parallel AB, \therefore \angle EAF=\angle AOC=60^\circ$.

在 $Rt\triangle COD$ 中,

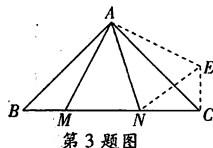
$\therefore \angle DOC = 60^\circ, \therefore \angle OCD = 30^\circ$.
 设 $OD = t$, 则 $CD = \sqrt{3}t, OC = AB = 2t$,
 在 $\text{Rt}\triangle EAF$ 中,
 $\therefore \angle EAF = 60^\circ, AE = \frac{1}{2}AB = t$.



第2题图

$\therefore AF = \frac{1}{2}t, EF = \sqrt{3}AF = \frac{\sqrt{3}}{2}t$.
 $\therefore$ 点 C 与点 E 都在反比例函数的图象上, $\therefore OD \times CD = OF \times EF$,
 $\therefore OF = 2t, \therefore OA = 2t - \frac{1}{2}t = \frac{3}{2}t$,
 $\therefore S_{\text{四边形}OABC} = 2S_{\triangle OCE}, \therefore \frac{3}{2}t \times \sqrt{3}t = 2 \times 4\sqrt{3}$,
 $\therefore$ 解得 $t = \frac{4\sqrt{3}}{3}$ (舍负), $\therefore OC = \frac{8\sqrt{3}}{3}$. 故选 D.

3. C 【解析】如图, 过点 C 作 $CE \perp BC$, 垂足为点 E , 截取 $CE = BM$, 连接 AE, EN . 则易证 $\triangle ABM \cong \triangle ACE$ (SAS), $\triangle MAN \cong \triangle EAN$ (SAS), $\therefore MN = EN$. 在 $\text{Rt}\triangle ENC$ 中, $EN^2 = EC^2 + NC^2, \therefore MN^2 = BM^2 + NC^2$.



第3题图

$\therefore AB = AC = 6\sqrt{2}$,
 $\therefore BC = \sqrt{(6\sqrt{2})^2 + (6\sqrt{2})^2} = 12$,
 $\therefore MN = BC - BM - NC = 9 - NC$.
 $\therefore BM = 3, \therefore (9 - NC)^2 = 3^2 + NC^2$, 解得 $NC = 4$. 故选 C.

4. D 【解析】 $\therefore \angle CAP + \angle CBP = 180^\circ, \therefore A, P, B, C$ 四点共圆,
 $\therefore \angle CPA = \angle CBA, \angle CPB = \angle CAB, \therefore CA = CB, \therefore \angle CBA = \angle CAB$,
 $\therefore \angle CPA = \angle CPB$, 故①正确;

$\therefore S = \frac{1}{2}AB \cdot CP \cdot \sin \angle APC$, 当 $\angle APC = 90^\circ$ 时, $\sin \angle APC = 1$ 取得最大值,

$\therefore S \leq \frac{1}{2}AB \cdot CP$, 故②正确;

当 $\alpha = 30^\circ$ 时, $\angle CPB = 30^\circ, \therefore BC = 4$, 当 PC 为直径时, PC 最大, 此时 $\angle PBC = 90^\circ$,

$\therefore PC = 2BC = 8, \therefore PC$ 的最大值为 8, 故③正确;
 过点 B 作 $BH \perp AP$ 交 AP 的延长线于点 H , 过点 D 作 $DE \perp AP$ 于点 E , 过点 D 作 $DF \perp PB$ 于点 F , 如图所示,

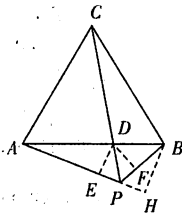
当 $\alpha = 60^\circ$ 时, $\angle APC = \angle BPC = 60^\circ$,

$\therefore DE = DF = PD \cdot \sin 60^\circ = \frac{\sqrt{3}}{2}PD, \angle APB = 120^\circ$,

$\therefore \angle BPH = 60^\circ, \therefore BH = PB \cdot \sin 60^\circ = \frac{\sqrt{3}}{2}PB$.

$\therefore PA = 4, PB = 2, \therefore S_{\triangle APB} = \frac{1}{2}AP \cdot DE + \frac{1}{2}BP \cdot DF = \frac{1}{2}AP \cdot BH$,

$\therefore 4 \times \frac{\sqrt{3}}{2}PD + 2 \times \frac{\sqrt{3}}{2}PD = 4\sqrt{3}, \therefore PD = \frac{4}{3}$, 故④正确. 故选 D.



第4题图

5. C

6. B 【解析】 $\therefore$ 当点 M 运动到 AB 的中点时, 此时点 N 为 AD 的中点, 显然 $S_{\triangle AMN} = S_{\triangle MON}$, 故 A 说法不正确, 不符合题意;

$\therefore \angle ABD = 60^\circ$, 点 O 为 BD 的中点, $\therefore AB = AO = BO$.

设 $AB = 1$, 则 $AD = \sqrt{3}, BD = 2$.

当点 M 与点 B 重合时, 点 N 是 BD 的垂直平分线与 AD 的交点,

令 $AN = x$, 则 $BN = DN = \sqrt{3} - x$,

$\therefore 1^2 + x^2 = (\sqrt{3} - x)^2$, 解得 $x = \frac{\sqrt{3}}{3}, \therefore AN = \frac{\sqrt{3}}{3}$; 当点 M 运动到 AB 的中点

时, 此时点 N 为 AD 的中点, 则点 M 的速度 $V_M = \frac{1}{2}$, 点 N 的速度 $V_N =$

$AN - \frac{\sqrt{3}}{3} = \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{3} = \frac{\sqrt{3}}{6}$;

从而点 M, N 的运动速度不同, 故 B 说法正确, 符合题意;

由 B 选项知, AM 减小的速度比 AN 增大的速度快, 则 $S_{\triangle AMN}$ 逐渐减小, 故 C 说法不正确, 不符合题意;

只有当点 M 在 AB 的中点时, 才满足 $MN^2 = BM^2 + DN^2$, 故 D 说法错误, 不符合题意;

故选 B.

7. 5 041 【解析】设有 p 个 x 取 1, q 个 x 取 -2, 则有 $\begin{cases} p - 2q = -17, \\ p + 4q = 37, \end{cases}$ 解得

$\begin{cases} p = 1, \\ q = 9, \end{cases} \therefore$ 原式 $= [1 \times 1^3 + 9 \times (-2)^3]^2 = (-71)^2 = 5 041$.

8. $\frac{24}{5}$ 【解析】由 $\triangle AEF \sim \triangle ABC$, 则 $\frac{EF}{BC} = \frac{AE}{AB}, \frac{3}{5} = \frac{AE}{6}, AE = \frac{18}{5}$, 则 $BE =$

$\sqrt{6^2 - (\frac{18}{5})^2} = \frac{24}{5}$.

9. $-3 < a \leq 2$

10. 7 【解析】由 $\frac{1}{x^4} - \frac{1}{x^2} = 3$ 得 $(\frac{1}{x^2})^2 - \frac{1}{x^2} - 3 = 0$, 可以把方程看作关于 $\frac{1}{x^2}$ 的一元二次方程, $\therefore \frac{1}{x^2} = \frac{1 + \sqrt{13}}{2}$ (舍负). 又 $2y^4 + y^2 = \frac{3}{2}$,

$\therefore (2y^2)^2 + 2y^2 - 3 = 0$, 可以把方程看作关于 $2y^2$ 的一元二次方程,

$\therefore 2y^2 = \frac{-1 + \sqrt{13}}{2}$ (舍负), $\therefore$ 原式 $= (\frac{1 + \sqrt{13}}{2})^2 + (\frac{-1 + \sqrt{13}}{2})^2 = 7$.

11. $(\frac{\sqrt{3}}{2}, 2)$ 【解析】过点 C 作 $CD \perp AB$ 于点 D . 在 $\text{Rt}\triangle ACB$ 中, $BC = AB \cdot$

$\cos \angle ABC = 3\sqrt{3}$, 在 $\text{Rt}\triangle BCD$ 中, $CD = BC \cdot \sin B = \frac{3\sqrt{3}}{2}, BD = BC \cdot \cos B =$

$\frac{9}{2}, \therefore AD = AB - BD = \frac{3}{2}$. 设 $C(\frac{\sqrt{3}}{m}, \frac{\sqrt{3}}{m}), A(n, \frac{\sqrt{3}}{n})$, 依题意知 $n > m > 0$,

故 $CD = n - m, AD = \frac{\sqrt{3}}{m} - \frac{\sqrt{3}}{n}$, 于是 $\begin{cases} n - m = \frac{3\sqrt{3}}{2}, \\ \frac{\sqrt{3}}{m} - \frac{\sqrt{3}}{n} = \frac{3}{2}, \end{cases}$ 解得 $\begin{cases} m = \frac{\sqrt{3}}{2}, \\ n = 2\sqrt{3}, \end{cases}$ 故点

C 的坐标为 $(\frac{\sqrt{3}}{2}, 2)$.

12. 36 【解析】设正方形③与⑥边长为 a , 则正方形④边长为 $a + 1$, 正方形⑤边长为 $a + 2$, 正方形②边长为 $a - 1$. 故有 $a + 1 + a + 2 = a + a - 1 + a$, $\therefore a = 4, \therefore$ 正方形⑤的面积为 $6^2 = 36$.

13. $k = -\frac{33}{8}$ 或 $k = -4$ 或 $k \geq -3$

【解析】原方程可化为 $2x^2 - 3x - (k + 3) = 0$ ①,

(1) 当 $\Delta = 0$ 时, $k = -\frac{33}{8}, x_1 = x_2 = \frac{3}{4}$ 满足条件;

(2) 若 $x = 1$ 是方程①的根, 得 $2 \times 1^2 - 3 \times 1 - (k + 3) = 0, k = -4$;

此时方程①的另一个根为 $\frac{1}{2}$, 故原方程也只有一个根 $x = \frac{1}{2}$;

(3) 当方程①有异号实根时, $x_1 x_2 = \frac{-k-3}{2} < 0$ 且 $x \neq 1$ 即 $k \neq -4$, 得 $k > -3$, 此时原方程也只有一个正实数根;

(4) 当方程①有一个根为 0 时, $k = -3$, 另一个根为 $x = \frac{3}{2}$, 此时原方程也只有一个正实根.

综上所述, 满足条件的 k 的取值范围是 $k = -\frac{33}{8}$ 或 $k = -4$ 或 $k \geq -3$.

14. 解: 由题意, 得 $ax + by + cz = \frac{x+y+z}{2}, ax + x = \frac{x+y+z}{2}, x(1+a) = \frac{x+y+z}{2}, 1+a = \frac{x+y+z}{2x}$, 同理可得 $by + y = \frac{x+y+z}{2}, 1+b = \frac{x+y+z}{2y}, 1+c = \frac{x+y+z}{2z}$, $\therefore$ 原式 $= \frac{2-(1+a)}{1+a} + \frac{2-(1+b)}{1+b} + \frac{2-(1+c)}{1+c} = \frac{2}{1+a} + \frac{2}{1+b} + \frac{2}{1+c} - 3 = \frac{4x}{x+y+z} + \frac{4y}{x+y+z} + \frac{4z}{x+y+z} - 3 = 1$.

15. 解: (1) $\therefore t = 1, y = kx + 2, \therefore A(1, 0), B(0, 2)$, 则有 $\begin{cases} -1 + b + c = 0, \\ c = 2, \end{cases}$ 解得

$\begin{cases} b = -1, \\ c = 2, \end{cases} \therefore$ 所求抛物线的解析式为 $y = -x^2 - x + 2$.

(2) 过点 C 作 $CH \perp x$ 轴于点 H , 易证 $\triangle AOB \sim \triangle CHA, \therefore \frac{OA}{CH} = \frac{OB}{AH} = \frac{AB}{AC}$.

$\therefore \tan \angle ACB = \frac{AB}{AC} = \frac{1}{2}, \therefore \frac{OA}{CH} = \frac{OB}{AH} = \frac{1}{2}, \therefore OA = t, OB = 2, \therefore CH = 2t, AH = 4, \therefore$ 点 C 的坐标为 $(t - 4, -2t), \therefore x = t - 4, y = -2t$.

(3) 由点 $A(t, 0)$, 点 $C(t - 4, -2t)$ 在第三象限内得 $0 < t < 4, \therefore Z = xy = (t - 4)(-2t) = -2t^2 + 8t = -2(t - 2)^2 + 8 (0 < t < 4),$

$\therefore$ 当 $t = 2$ 时, Z 的最大值为 8.

16. 解: (1) 连接 AD , 如图①. 设 $\angle BDC = \gamma, \angle CAD = \beta$,

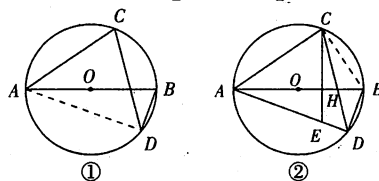
则 $\angle CAB = \angle BDC = \gamma, \therefore$ 点 C 为 $\widehat{ABD}$ 中点, $\therefore \widehat{AC} = \widehat{CD}$,

$\therefore \angle ADC = \angle CAD = \beta, \therefore \angle DAB = \beta - \gamma, \therefore AB$ 为 $\odot O$ 直径,

$\therefore \angle ADB = 90^\circ, \therefore \gamma + \beta = 90^\circ, \therefore \beta = 90^\circ - \gamma$,

$\therefore \angle ABD = 90^\circ - \angle DAB = 90^\circ - (\beta - \gamma) = 90^\circ - 90^\circ + \gamma + \gamma = 2\gamma$,

$\therefore \angle ABD = 2\angle BDC, \therefore \angle BDC = \frac{1}{2}\angle ABD = \frac{1}{2}\alpha$.

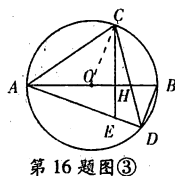


第16题图

(2) 连接 BC , 如图②. $\therefore AB$ 为 $\odot O$ 直径, $\therefore \angle ACB = 90^\circ$, 即 $\angle BAC + \angle ABC = 90^\circ, \therefore \angle ACE + \angle BAC = 90^\circ, \therefore \angle ACE = \angle ABC$,

$\therefore$ 点 C 为 $\widehat{ABD}$ 中点, $\therefore \widehat{AC} = \widehat{CD}$,

$\therefore \angle ADC = \angle CAD = \angle ABC = \beta$,
 $\therefore \angle ACE = \beta$.
 (3) 连接 OC , 如图③. $\therefore \angle COB = 2\angle CAB$,
 $\therefore \angle ABD = 2\angle BDC$. $\therefore \angle BDC = \angle CAB$,
 $\therefore \angle COB = \angle ABD$. $\therefore \angle OHC = \angle ADB = 90^\circ$,
 $\therefore \triangle COH \sim \triangle ABD$, $\therefore \frac{OH}{BD} = \frac{OC}{AB} = \frac{1}{2}$,
 $\therefore BD = 2OH = 10$,
 $\therefore AB = \sqrt{AD^2 + BD^2} = \sqrt{24^2 + 10^2} = 26$,
 $\therefore AO = 13$,
 $\therefore AH = AO + OH = 13 + 5 = 18$.
 $\therefore \angle EAH = \angle BAD$, $\angle AHE = \angle ADB = 90^\circ$, $\therefore \triangle AHE \sim \triangle ADB$,
 $\therefore \frac{AH}{AD} = \frac{AE}{AB}$, 即 $\frac{18}{24} = \frac{AE}{26}$, $\therefore AE = \frac{39}{2}$,
 $\therefore DE = AD - AE = 24 - \frac{39}{2} = \frac{9}{2}$.



第 16 题图③

17. 解: (1) 易求 $B(4, -\frac{1}{2})$, 将 $A(3, 2)$, $B(4, -\frac{1}{2})$ 代入 $y = -\frac{1}{2}x^2 + bx + c$ 中得

$$\begin{cases} -\frac{1}{2} \times 3^2 + 3b + c = 2, \\ -\frac{1}{2} \times 4^2 + 4b + c = -\frac{1}{2}, \end{cases}$$

$$\text{解得 } \begin{cases} b = 1, \\ c = \frac{7}{2}, \end{cases} \therefore \text{抛物线的解析式为 } y = -\frac{1}{2}x^2 + x + \frac{7}{2}.$$

(2) 设 $D(m, -\frac{1}{2}m^2 + m + \frac{7}{2})$, 则 $E(m, -m + \frac{7}{2})$, $DE = (-\frac{1}{2}m^2 + m + \frac{7}{2}) - (-m + \frac{7}{2}) = -\frac{1}{2}m^2 + 2m = -\frac{1}{2}(m-2)^2 + 2$,

$\therefore$ 当 $m=2$ 时, DE 有最大值为 2, 此时 $D(2, \frac{7}{2})$, 作点 A 关于对称轴的对称点 A' , 连接 $A'D$, 与对称轴交于点 P , 连接 PA . $PD + PA = PD + PA' = A'D$, 此时 $PD + PA$ 最小. $\therefore A(3, 2)$, $\therefore A'(-1, 2)$, $A'D = \sqrt{(-1-2)^2 + (2-\frac{7}{2})^2} = \frac{3\sqrt{5}}{2}$, 即 $PD + PA$ 的最小值为 $\frac{3\sqrt{5}}{2}$.

(3) 作 $AH \perp$ 对称轴于点 H , 连接 AM, AQ, MQ, HQ , 则 $M(1, 4)$, $H(1, 2)$.

$\therefore \angle AQM = 45^\circ$, $\angle AHM = 90^\circ$, $\therefore \angle AQM = \frac{1}{2} \angle AHM$, 可知 $\triangle AQM$ 外接圆的圆心为 H , $\therefore QH = HA = HM = 2$. 设 $Q(0, t)$, 则 $(0-1)^2 + (t-2)^2 = 2^2$, 解得 $t = 2 + \sqrt{3}$ 或 $2 - \sqrt{3}$, $\therefore$ 符合条件的点 Q 的坐标 $Q_1(0, 2 + \sqrt{3})$, $Q_2(0, 2 - \sqrt{3})$.

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1. D 【解析】由题意可得 $a+b=-20$, $ab=1$, $c+d=17$, $cd=1$,

$$\begin{aligned} & \therefore (a+c)(b+c)(a-d)(b-d) \\ &= [ab + (a+b)c + c^2][ab - (a+b)d + d^2] \\ &= (1 - 20c + c^2)(1 + 20d + d^2) \\ &= 1 + 20d + d^2 - 20c - 400 - 20d + c^2 + 20c + 1 \\ &= d^2 + c^2 + 2 - 400 \\ &= (c+d)^2 - 400 \\ &= 17^2 - 400 \\ &= -111. \text{ 故选 D.} \end{aligned}$$

2. C 【解析】 $\therefore [\frac{4x+a}{3}] = 2$, $\therefore 2 \leq \frac{4x+a}{3} < 3$, $\therefore \frac{6-a}{4} \leq x < \frac{9-a}{4}$.

$$\therefore [\frac{4x+a}{3}] = 2 \text{ 有正整数解, } a \text{ 是正数, } \therefore x < \frac{9}{4}, \text{ 即 } x \text{ 可取 } 1, 2,$$

当 $x=1$ 时, $6 \leq 4x+a < 9$, 即 $2 \leq a < 5$;

当 $x=2$ 时, $6 \leq 4x+a < 9$, 即 $-2 \leq a < 1$. $\therefore a > 0$, $\therefore 0 < a < 1$.

综上, a 的取值范围是 $2 \leq a < 5$ 或 $0 < a < 1$. 故选 C.

3. D 【解析】设 OB 与抛物线的交点为 M , 作 $MN \perp x$ 轴于点 N , 则 $S_{\triangle AOB} =$

$$4 \cdot \frac{OM}{OB} = \frac{1}{2}, \triangle MON \sim \triangle BOA, \therefore \frac{S_{\triangle MON}}{S_{\triangle AOB}} = (\frac{OM}{OB})^2 = \frac{1}{4},$$

$$\therefore S_{\triangle MON} = \frac{1}{2} \cdot k = 1, \therefore k = 2, \therefore y = \frac{2}{x}. \therefore \text{反比例函数与直线 } y = x \text{ 的交点}$$

为 P 时, OP 最小, 联立 $\begin{cases} y = \frac{2}{x}, \\ y = x, \end{cases}$ 解得 $\begin{cases} x = \sqrt{2}, \\ y = \sqrt{2} \end{cases}$ 或 $\begin{cases} x = -\sqrt{2}, \\ y = -\sqrt{2} \end{cases}$. $\therefore P(\sqrt{2}, \sqrt{2})$, $\therefore OP = 2$. 故选 D.

4. B 【解析】 $\therefore a^2(b+c) = b^2(a+c)$, $\therefore a^2b + a^2c - ab^2 - cb^2 = 0$,
 $\therefore ab(a-b) + c(a+b)(a-b) = 0$, $\therefore (a-b)(ab+ac+bc) = 0$. $\therefore a, b, c$
 互不相等, $\therefore ab+ac+bc = 0$, $\therefore c^2(a+b) - 2 \cdot 021 = c^2(a+b) - [a^2(b+c) - 1] = ac^2 + bc^2 - a^2b - a^2c + 1 = ac(c-a) + b(a+c)(c-a) + 1 = (c-a)(ac+ab+bc) + 1 = (c-a) \cdot 0 + 1 = 1$. 故选 B.

5. A 【解析】①延长 AD 交 EF 于点 H , 延长 FG 与 BA 的延长线交于点 M .

当 $0 \leq x \leq 4$ 时, $y = 6 \times 4 - \frac{1}{2} \times 2 \cdot x - \frac{1}{2}(6-x) \cdot x - \frac{1}{2} \times (4-x +$

$2) \times 6 = \frac{1}{2}(x-1)^2 + \frac{11}{2}$, 此时的函数图象是开口向上的抛物线的一部分

分, 且顶点坐标是 $(1, \frac{11}{2})$. 故 C, D 选项错误; ②当点 Q 在 GF 上时, $4 < x \leq 6$, $BP = x$, $MQ = 6 + 4 - x = 10 - x$, 则 $y = S_{\text{梯形} MBPQ} - S_{\triangle BPA} - S_{\triangle AMQ} =$

$$\frac{1}{2}(x+10-x) \times 4 - \frac{1}{2} \times 2 \cdot x - \frac{1}{2}(10-x) \cdot 2 = 10. \text{ 故选 A.}$$

6. B 【解析】如图, 设 $PE = x$, $PF = y$, $PQ = z$, 连接 AP, BP, CP , 则 $S_{\triangle ABC} = S_{\triangle ABP} + S_{\triangle ACP} +$

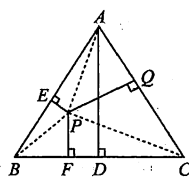
$$S_{\triangle BCP}, \therefore \frac{1}{2}BC \cdot h = \frac{1}{2}AB \cdot x + \frac{1}{2}BC \cdot y +$$

$$\frac{1}{2}AC \cdot z, \therefore AC = BC = AB, \therefore x + y + z = h,$$

又 $\therefore$ 以 x, y, z 为边可以组成三角形, $\therefore x + y >$

$$z, x + y + z > 2z, \text{ 即 } h > 2z, z < \frac{1}{2}h, \text{ 又 } x \leq y \leq$$

$$z, \therefore z \geq \frac{1}{3}(x + y + z), \text{ 即 } z \geq \frac{1}{3}h, \therefore \frac{1}{3}h \leq z < \frac{1}{2}h. \text{ 故选 B.}$$



第 6 题图

7. $2\sqrt{-b}$ 【解析】 $\therefore \sqrt{2\sqrt{ab}-a-b} =$

$$\sqrt{(\sqrt{-a})^2 + 2\sqrt{-a} \cdot \sqrt{-b} + (\sqrt{-b})^2} =$$

$$\sqrt{(\sqrt{-a} + \sqrt{-b})^2} = \sqrt{-a} + \sqrt{-b},$$

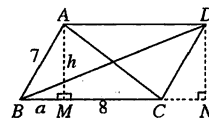
$$\text{原式} = \frac{(\sqrt{-b} + \sqrt{-a})(\sqrt{-b} - \sqrt{-a})}{\sqrt{-a} + \sqrt{-b}} + \sqrt{-a} + \sqrt{-b} =$$

$$\sqrt{-b} - \sqrt{-a} + \sqrt{-a} + \sqrt{-b} = 2\sqrt{-b}.$$

8. 226 【解析】如图, 过点 A 作 $AM \perp BC$, 过点 D 作 $DN \perp BC$, 设 $AM = h$,

$BM = a$, 由勾股定理可得 $AC^2 = h^2 + (8-a)^2$ ①, $BD^2 = (8+a)^2 + h^2$

②, 在 $\text{Rt} \triangle ABM$ 中, $a^2 + h^2 = 7^2$ ③, ①+②得 $m^2 + n^2 = 8^2 \times 2 + (2a^2 + 2h^2) = 128 + 98 = 226$.



第 8 题图

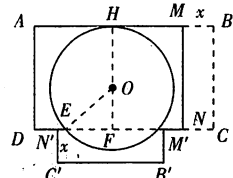
9. $\sqrt{5}$

10. $\sqrt{10}$ 【解析】作点 C 关于直线 AB 的对称点 D , 连接 AD, BD , 延长 DA 到点 H , 使得 $AH = AD$, 连接 EH, BH, DE , 可证四边形 $ACBD$ 是矩形, 四边形 $ACBD$ 是正方形, $\triangle ACF \cong \triangle DAE$ (SAS), $\therefore AF = DE$, $\therefore AF + BE = ED + EB$. $\therefore CA$ 垂直平分线段 DH , $\therefore ED = EH$. $\therefore AF + BE = EB + EH$. $\therefore EB + EH \geq BH$, $\therefore AF + BE$ 的最小值为线段 BH 的长, $BH = \sqrt{(\sqrt{2})^2 + (2\sqrt{2})^2} = \sqrt{10}$. $\therefore AF + BE$ 的最小值为 $\sqrt{10}$.

11. $2 \leq x \leq 3$ 25π 【解析】如图, 设 $\odot O$ 与 AB 相切于点 H , 交 CD 于 E , 连接 OH , 延长 HO 交 CD 于 F , 设 $\odot O$ 的半径为 r . 在 $\text{Rt} \triangle OEF$ 中, 当点 E 与 N' 重合时, $\odot O$ 的面积最大, 此时 $EF = 4$, 则有 $r^2 = (8-r)^2 + 4^2$, $\therefore r = 5$. $\therefore \odot O$ 的最大面积为 25π , 由题意

$$\begin{cases} 3+x \geq 5, \\ 13-x \geq 10, \end{cases}$$

$$\therefore 2 \leq x \leq 3.$$



第 11 题图

12. $\sqrt{3}$ 【解析】连接 BI, DI, OB, OD , 作 $OE \perp BD$ 于点 E , 设 $\angle A = x^\circ$, 则

$$\angle ABD + \angle ADB = (180 - x)^\circ, \text{ 由题意知 } \angle DBI = \frac{1}{2} \angle ABD, \angle BDI =$$

$$\frac{1}{2} \angle ADB, \therefore \angle DBI + \angle BDI = \frac{1}{2}(180 - x)^\circ = (90 - \frac{1}{2}x)^\circ,$$

$$\therefore \angle BID = 180^\circ - (90 - \frac{1}{2}x)^\circ = (90 + \frac{1}{2}x)^\circ, \text{ 又点 } C, \text{ 点 } I \text{ 关于 } BD \text{ 对称,}$$

$$\therefore \angle BCD = \angle BID = (90 + \frac{1}{2}x)^\circ. \text{ 又四边形 } ABCD \text{ 为圆内接四边形,}$$

$$\therefore x + 90 + \frac{1}{2}x = 180, \therefore x = 60, \text{ 即 } \angle A = 60^\circ. \therefore \angle BOD = 120^\circ, \angle OBE =$$

$$\frac{1}{2}(180^\circ - \angle BOD) = 30^\circ. \text{ 又 } OE \perp BD, \therefore BE = \frac{1}{2}BD = \frac{3}{2}, \therefore OB =$$

$$\frac{BE}{\cos \angle OBE} = \sqrt{3}.$$

13. 解: (1) 方程有两个实数根, 则 $m^2 - 1 \neq 0$, 解方程得 $x_1 = \frac{6}{m+1}, x_2 = \frac{3}{m-1}$. 由

题意, 得 $\begin{cases} m+1=1, 2, 3, 6, \\ m-1=1, 3, \end{cases}$ 即 $\begin{cases} m=0, 1, 2, 5, \\ m=2, 4. \end{cases}$ 故 $m=2$.

(2) 把 $m=2$ 代入两等式, 化简得 $a^2 - 4a + 2 = 0, b^2 - 4b + 2 = 0$. 当 $a=b$ 时, $a=b=2 \pm \sqrt{2}$. 当 $a \neq b$ 时, a, b 是方程 $x^2 - 4x + 2 = 0$ 的两根, 而 $\Delta > 0$, 由韦达定理得 $a+b=4 > 0, ab=2 > 0$, 则 $a > 0, b > 0$.

① $a \neq b, c = 2\sqrt{3}$ 时, 由于 $a^2 + b^2 = (a+b)^2 - 2ab = 16 - 4 = 12 = c^2$, 故

$\triangle ABC$ 为直角三角形,且 $\angle C=90^\circ$, $S_{\triangle ABC}=\frac{1}{2}ab=1$.

② $a=b=2-\sqrt{2}$, $c=2\sqrt{3}$ 时,因 $2(2-\sqrt{2})<2\sqrt{3}$,故不能构成三角形,不合题意,舍去.

③ $a=b=2+\sqrt{2}$, $c=2\sqrt{3}$ 时,因 $2(2+\sqrt{2})>2\sqrt{3}$,故能构成三角形.

$S_{\triangle ABC}=\frac{1}{2}\times 2\sqrt{3}\times \sqrt{(2+\sqrt{2})^2-(\sqrt{3})^2}=\sqrt{9+12\sqrt{2}}$. 综上, $\triangle ABC$ 的面积为 1 或 $\sqrt{9+12\sqrt{2}}$.

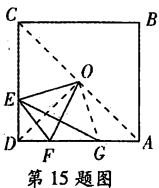
14. (1) 证明: $\because AD\perp CD$, G 为 AC 中点, 易证 A, D, F, C 四点共圆, $\therefore \angle AFC=\angle ADC=90^\circ$. 又 $\because \angle ADE=\angle ACF$, $\therefore \text{Rt}\triangle AED\sim\text{Rt}\triangle AFC$.

(2) 证明: 由 (1) 知 $\text{Rt}\triangle AED\sim\text{Rt}\triangle AFC$, $\therefore \frac{AF}{CF}=\frac{AE}{ED}$. 又易证 $\text{Rt}\triangle AED\sim\text{Rt}\triangle DEB$, $\therefore \frac{AE}{DE}=\frac{DE}{EB}$, $\therefore \frac{AF}{CF}=\frac{DE}{EB}$, $\therefore \left(\frac{AF}{CF}\right)^2=\frac{AE}{ED}\cdot\frac{DE}{EB}=\frac{AE}{EB}$, 即 $AE\cdot CF^2=BE\cdot AF^2$.

(3) 解: $\because AD^2+BD^2=36$, $AD^2+(7-BD)^2=64$, $\therefore AD=\frac{3\sqrt{15}}{2}$, $BD=\frac{3}{2}$, $CD=\frac{11}{2}$. 在 $\text{Rt}\triangle ADB$ 中, $DE=\frac{AD\cdot BD}{AB}=\frac{3\sqrt{15}}{8}$, $AE=\frac{AD^2}{AB}=\frac{45}{8}$,

$\therefore S_{\triangle ADE}=\frac{135\sqrt{15}}{128}$. 而 $S_{\triangle ACF}=\left(\frac{AC}{AD}\right)^2\cdot S_{\triangle ADE}$, $\therefore S_{\triangle ACF}=2\sqrt{15}$. 同理利用 $\text{Rt}\triangle AEF\sim\text{Rt}\triangle ADC$ 得 $S_{\triangle AEF}=S_{\triangle ADC}\cdot\left(\frac{AE}{AD}\right)^2=\frac{33\times 15\sqrt{15}}{128}$, $\therefore S_{\text{四边形}ACFD}=S_{\triangle AEF}+S_{\triangle ACF}-S_{\triangle AED}=\frac{77\sqrt{15}}{16}$.

15. (1) 解: $\triangle EDF$ 的周长与 AD 的长相等. 理由如下: 如图, 连接 OD, OG, CA , 则 CA 必过点 O , $\therefore$ 点 O 为正方形 $ABCD$ 的中心, $\therefore OD=OA$, $\angle OAG=\angle ODE$. 在 $\triangle OED$ 和 $\triangle OGA$ 中, $\begin{cases} OD=OA, \\ \angle OAG=\angle ODE, \\ DE=AG, \end{cases} \therefore \triangle OED\cong\triangle OGA(\text{SAS})$, $\therefore OE=OG$.



第 15 题图

$\therefore OF\perp EG$, $\therefore OF$ 是 EG 的垂直平分线, $\therefore FE=FG$, $\therefore \triangle EDF$ 的周长 $=DF+EF+ED=DF+FG+AG=AD$.

(2) 解: $\because OD\perp OA$, $\therefore \angle DOA=90^\circ$. 由 (1) 可得 $\triangle OED\cong\triangle OGA$, $\therefore \angle EOD=\angle GOA$, $\therefore \angle EOG=\angle EOD+\angle DOG=\angle AOG+\angle DOG=90^\circ$.

$\therefore \triangle OEG$ 为等腰直角三角形, $OF\perp EG$, $\therefore \angle EOF=\frac{1}{2}\angle EOG=45^\circ$.

(3) 证明: $\because \angle EOF=45^\circ$, $\therefore \angle COE+\angle AOF=135^\circ$. $\because \angle OAF=45^\circ$, $\therefore \angle AFO+\angle AOF=135^\circ$,

$\therefore \angle COE=\angle AFO$, $\therefore \triangle AOF\sim\triangle CEO$, $\therefore \frac{S_{\triangle AOF}}{S_{\triangle CEO}}=\left(\frac{OF}{OE}\right)^2=n^2$. $\therefore$ 点 O 到 AF 与 CE 的距离相等,

$\therefore S_{\triangle AOF}=S_{\triangle CEO}=AF\cdot CE=m$, $\therefore m=n^2$.

16. 解: (1) 证明: 连接 AD, BD, BH , 则 $DA\perp AC, DB\perp BC$, 又 H 为 $\triangle ABC$ 垂心, $BH\perp AC, AH\perp BC$, $\therefore AD\parallel BH, BD\parallel AH$, $\therefore$ 四边形 $ADBH$ 为平行四边形, $\therefore DB=AH=2OM$. 又 O 为 CD 中点, $\therefore M$ 为 BC 中点.

(2) 证明: 过点 E 作 $EG\perp BC$, 连接 GH, GF, CH 由 (1) 可知四边形 $EGHF$ 为平行四边形, 四边形 $EGFA$ 为平行四边形. $\therefore CH\perp AB, AB\parallel GF$, $\therefore CH\perp GF$, $\therefore H$ 为 $\triangle FGC$ 垂心, $\therefore GH\perp CF$, 而 $GH\parallel EF$, $\therefore EF\perp FC$.

(3) 证明: 设 AM 与 OF 交点为 I , 由 (1) 可知四边形 $OMFA$ 为平行四边形, $\therefore I$ 为直径 AM 中点而 $\odot I$ 与 $\odot O$ 相交弦为 AP , $\therefore OF\perp AP$. 而 $MH\parallel OF$. 设 MC 与 AP 交于点 Q , 则 H 为 $\triangle AMQ$ 垂心, $\therefore QH\perp AM$, $\therefore AP, BC, OH$ 三线共点, $\therefore O, H, Q$ 三点共线, $\therefore OH\perp AN$, $\therefore AH=HN$.

17. 解: (1) 易求直线 l 的解析式为 $y=\frac{3}{4}x-\frac{3}{2}$. $\therefore \frac{9}{4}=\frac{3}{4}n-\frac{3}{2}$. 解得 $n=5$. $\therefore$ 抛物线 $y=\frac{3}{4}x^2+bx+c$ 经过点 $C\left(n, \frac{9}{4}\right)$ 和点 $B\left(0, -\frac{3}{2}\right)$,

$\therefore \begin{cases} c=-\frac{3}{2}, \\ \frac{9}{4}=\frac{3}{4}\times 5^2+5b+c, \end{cases}$ 解得 $\begin{cases} b=-3, \\ c=-\frac{3}{2}. \end{cases}$ $\therefore$ 抛物线的解析式为 $y=\frac{3}{4}x^2-3x-\frac{3}{2}$.

(2) 易得 $A(2, 0)$, $\therefore OA=2$. 在 $\text{Rt}\triangle OAB$ 中, $AB=\sqrt{2^2+\left(\frac{3}{2}\right)^2}=\frac{5}{2}$, 易证 $\triangle OAB\sim\triangle FDE$, $\therefore \frac{OA}{FD}=\frac{OB}{FE}=\frac{AB}{DE}$, $\therefore FD=\frac{OA}{AB}\cdot DE=\frac{4}{5}DE$, $FE=\frac{OB}{AB}\cdot DE=\frac{3}{5}DE$, $\therefore p=2(FD+FE)=2\times\left(\frac{4}{5}+\frac{3}{5}\right)DE=\frac{14}{5}DE$.

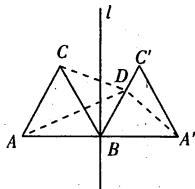
$\therefore D\left(t, \frac{3}{4}t^2-3t-\frac{3}{2}\right)$, $E\left(t, \frac{3}{4}t-\frac{3}{2}\right)$, 且 $0<t<5$, $\therefore DE=\left(\frac{3}{4}t-\frac{3}{2}\right)-\left(\frac{3}{4}t^2-3t-\frac{3}{2}\right)=-\frac{3}{4}t^2+\frac{15}{4}t$, $\therefore p=\frac{14}{5}\times\left(-\frac{3}{4}t^2+\frac{15}{4}t\right)=-\frac{21}{10}t^2+\frac{21}{2}t=-\frac{21}{10}\left(t-\frac{5}{2}\right)^2+\frac{105}{8}$,

且 $-\frac{21}{10}<0$, $\therefore$ 当 $t=\frac{5}{2}$ 时, p 有最大值 $\frac{105}{8}$.

(3) 依题意知 O_1B_1 与 x 轴平行, O_1A_1 与 y 轴平行. ① 当 O_1, B_1 在抛物线上时, 设 $O_1(t, y_1)$, $B_1\left(t+\frac{3}{2}, y_1\right)$, 则 $\frac{3}{4}t^2-3t-\frac{3}{2}=\frac{3}{4}\left(t+\frac{3}{2}\right)^2-3\left(t+\frac{3}{2}\right)-\frac{3}{2}$, 解得 $t=\frac{5}{4}$; ② 当 A_1, B_1 在抛物线上时, 设 $A_1(t, y_1)$, $B_1\left(t+\frac{3}{2}, y_1-2\right)$, 则 $\frac{3}{4}t^2-3t-\frac{3}{2}=\frac{3}{4}\left(t+\frac{3}{2}\right)^2-3\left(t+\frac{3}{2}\right)-2$, 解得 $t=\frac{13}{36}$. 综上所述, 点 A_1 的横坐标为 $\frac{5}{4}$ 或 $\frac{13}{36}$.

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1. C 【解析】如图, 连接 $A'D$, $\because$ 正三角形 ABC 的边长为 4, $\therefore AB=BC=4$, $\angle ABC=60^\circ$. $\therefore \triangle ABC$ 与 $\triangle A'BC'$ 关于直线 l 对称, $\therefore \triangle A'BC'$ 也是边长为 4 的等边三角形, $\therefore A'B=4$, $\angle A'BC'=60^\circ$, $\therefore \angle CBD=180^\circ-\angle ABC-\angle A'BC'=60^\circ$. 在



第 1 题图

$\triangle BCD$ 和 $\triangle BA'D$ 中, $\begin{cases} BC=BA'=4, \\ \angle CBD=\angle A'BD=60^\circ, \\ BD=BD, \end{cases}$

$\therefore \triangle BCD\cong\triangle BA'D(\text{SAS})$, $\therefore CD=A'D$, $\therefore AD+CD=AD+A'D$. 由三角形的三边关系定理、两点

之间线段最短可知, 当点 D 与点 B 重合, 即点 A, D, A' 共线时, $AD+A'D$ 取得最小值, 最小值为 $AA'=AB+A'B=4+4=8$, 即 $AD+CD$ 的最小值为 8. 故选 C.

2. C 【解析】观察数轴有 $a<-1<0<b<c<1$, 则 ① $a^2-a-2=(a-2)(a+1)>0$; ② $|a-b|+|b-c|=-a+b-b+c=-a+c$, $|a-c|=-a+c$, $\therefore |a-b|+|b-c|=|a-c|$; ③ $a+b<0$, $b+c>0$, $c+a<0$, $\therefore (a+b)\cdot(b+c)(c+a)>0$; ④ $|a|>1$, $1-bc<1$, $\therefore |a|>1-bc$, 故正确的结论有 ②③. 故选 C.

3. A 【解析】 $\because$ 抛物线开口向下, 与 x 轴正半轴交于 A, B 两点, 与 y 轴负半轴交于点 C , $\therefore a<0$, $c<0$, a, b 异号, $\therefore b>0$. $\therefore abc>0$, $\therefore$ ① 正确;

$\because c<0$, 抛物线过点 $(4, 0)$, $\therefore 16a+4b+c=0$, $\therefore 4a+b=-\frac{1}{4}c>0$, $\therefore$ ②

正确;

$\because$ 抛物线开口向下, 当 $M(x_1, y_1)$ 与 $N(x_2, y_2)$ 两点在对称轴左侧时, y 随 x 的增大而增大, 若 $x_1<x_2<0$, 则 $y_1<y_2<0$; $\therefore$ ③ 正确;

$\because$ 抛物线开口向下, 对称轴为 $x=3$, $\therefore$ 当 $x=3$ 时, 函数有最大值, $\therefore$ 对任意实数 k , 其对应的函数值不大于 $x=3$ 对应的函数值,

$\therefore ak^2+bk+c\leq 9a+3b+c$, $\therefore ak^2-9a\leq 3b-bk$,

$\therefore a(k-3)(k+3)\leq b(3-k)$. $\therefore$ ④ 正确;

当 $AB\geq 3$ 时, $x=1$ 时, $y\geq 0$, $\therefore a+b+c\geq 0$,

$\therefore$ 抛物线过点 $(4, 0)$, $\therefore 16a+4b+c=0$,

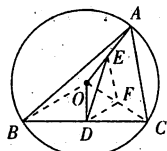
消去 a 得 $4b+5c\geq 0$, $\therefore 4b+3c\geq -2c>0$. $\therefore$ ⑤ 正确. 故选 A.

4. D 【解析】连接 AF , 设 $S_{\triangle AEF}=y$, $S_{\triangle ADF}=x$, 则 $\begin{cases} \frac{x}{y+16}=\frac{10}{20}, \\ \frac{y}{x+10}=\frac{16}{20}, \end{cases}$ 解得

$\begin{cases} x=20, \\ y=24, \end{cases}$ 故 $S_{\text{四边形}ADFE}=44$. 故选 D.

5. A 【解析】作 $P_1C\perp y$ 轴于点 C , $P_2D\perp x$ 轴于点 D , $P_3E\perp x$ 轴于点 E , $P_3F\perp P_2D$ 于点 F , 设 $P_1\left(a, \frac{2}{a}\right)$, 则 $CP_1=a$, $OC=\frac{2}{a}$, 易证 $\triangle P_1B_1C\cong\triangle B_1A_1O$, $\triangle B_1A_1O\cong\triangle A_1P_2D$, $\therefore OB_1=P_1C=A_1D=a$, $\therefore OA_1=B_1C=P_2D=\frac{2}{a}-a$, $\therefore OD=a+\frac{2}{a}-a=\frac{2}{a}$. $\therefore P_2\left(\frac{2}{a}, \frac{2}{a}-a\right)$. 于是有 $\left(\frac{2}{a}-a\right)\cdot\frac{2}{a}=2$, 解得 $a=-1$ (舍) 或 $a=1$, $\therefore P_2(2, 1)$. 设 $P_3\left(b, \frac{2}{b}\right)$, 同理 $P_3E=P_3F=DE$, $\therefore OE=OD+DE=2+\frac{2}{b}$, $\therefore 2+\frac{2}{b}=b$, 解得 $b=1-\sqrt{3}$ (舍), $b=1+\sqrt{3}$, $\therefore \frac{2}{b}=\frac{2}{1+\sqrt{3}}=\sqrt{3}-1$, $\therefore$ 点 $P_3(\sqrt{3}+1, \sqrt{3}-1)$. 故选 A.

6. C 【解析】如图, 连接 OC , 取 OC 中点 F , 连接 EF, DF ,



第 6 题图

$\therefore \angle AOC=2\angle ABC=80^\circ$, $OE=OF$, $\therefore \angle OEF=\angle OFE=\frac{1}{2}\times(180^\circ-80^\circ)=50^\circ$, 连接 OB ,

$\therefore D$ 为 BC 中点,

$\therefore BD=CD$, $OD\perp BC$, $\therefore \angle DOC=\frac{1}{2}\angle BOC$.

$\therefore \angle BAC=\frac{1}{2}\angle BOC$, $\therefore \angle DOC=\angle BAC$, $\therefore \angle DOC=\angle BAC=180^\circ-$

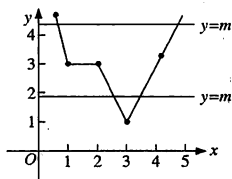
$$40^\circ - 80^\circ = 60^\circ.$$

$\therefore F$ 为 OC 中点, $\therefore OF = FD$, $\therefore \triangle OFD$ 为等边三角形,

$\therefore OD = OF = OE$, $\therefore O, E, F, D$ 四点共圆,

$\therefore \angle FED = \frac{1}{2} \angle FOD = 30^\circ$, $\therefore \angle OED = 50^\circ - 30^\circ = 20^\circ$. 故选 C.

7. $-\frac{1}{2}$ 【解析】 $\begin{cases} a+2b+3c=0, \\ 2a+3b+4c=0, \end{cases} \therefore a+b+c=0, \therefore (a+b+c)^2=0,$
 $\therefore a^2+b^2+c^2+2ab+2bc+2ac=0, \therefore a^2+b^2+c^2=-2(ab+bc+ac),$
 $\therefore \text{原式} = \frac{ab+bc+ca}{-2(ab+bc+ca)} = -\frac{1}{2}.$



第 8 题图

8. $1 < m < 3$ 或 $m > 3$ 【解析】化简可得 $m = |x-1| - |x-2| + 2|x-3|$, 作出 $y = |x-1| - |x-2| + 2|x-3|$ 与 $y = m$ 的函数图象, 由图象可得, 当 $1 < m < 3$ 或 $m > 3$ 时有两个解.

9. -23 【解析】 a, b 可看作是 $x^2 + 5x + 1 = 0$ 的两个不等实根, 所以 $a + b = -5$, $ab = 1$, a, b 均为负数. 原式 $= -\frac{b}{a} \sqrt{ab} - \frac{a}{b} \sqrt{ab}$.

10. $\frac{8\sqrt{5}}{5}$ 【解析】连接 OE . 依题意知 A, O, E, D 四点共圆, $\therefore \angle EOC = \angle EAD$ 是定值, $\therefore$ 点 E 在射线 OE 上运动, $\angle EOC$ 是定值.

$\therefore \tan \angle EOD = \tan \angle OAB = \frac{1}{2}$, 设 $E(-2m, m)$, 当点 D 与 C 重合时, $AC = \sqrt{5^2 + 2^2} = \sqrt{29}$, $\therefore AE = 2EC$, $\therefore EC = \frac{\sqrt{29}}{3} = \frac{\sqrt{145}}{5}$,

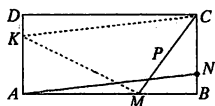
$\therefore (-2m + 5)^2 + m^2 = \left(\frac{\sqrt{145}}{5}\right)^2$, 解得 $m = \frac{8}{5}$ 或 $\frac{12}{5}$ (舍),

$\therefore E\left(-\frac{16}{5}, \frac{8}{5}\right)$, $\therefore$ 点 E 的运动轨迹 OE 的长 $= \frac{8\sqrt{5}}{5}$.

11. $\sqrt{3}$

12. $\sqrt{6}$ 【解析】在 $\text{Rt} \triangle COE$ 中, $OE = CO \cdot \cos \angle BOC = 4 \times \frac{1}{4} = 1$, 于是有 $BE = 3$, $AE = 5$, $\therefore CE = \sqrt{4^2 - 1^2} = \sqrt{15}$, $AC = \sqrt{(\sqrt{15})^2 + 5^2} = 2\sqrt{10}$, $BC = \sqrt{(\sqrt{15})^2 + 3^2} = 2\sqrt{6}$, 由垂径定理得 $CD = 2CE = 2\sqrt{15}$, 又 $CM = \frac{1}{2}AO = \frac{1}{2} \times 4 = 2$, $\therefore \triangle ACM \sim \triangle DCN$, $\therefore \frac{CM}{CN} = \frac{AC}{CD}$, $\therefore CN = \frac{CM \cdot CD}{AC} = \frac{2 \times 2\sqrt{15}}{2\sqrt{10}} = \sqrt{6}$, $\therefore BN = BC - CN = 2\sqrt{6} - \sqrt{6} = \sqrt{6}$.

13. 解: 如图, 在 AD 上取 $AK = CN = MB$, 连接 KM, KC , 则 $\triangle KAM \cong \triangle MBC$, $\therefore KM = CM$, $\angle AMK = \angle MCB$. 又 $\angle CMB + \angle MCB = 90^\circ$, $\therefore \angle CMB + \angle AMK = 90^\circ$, $\therefore \angle KMC = 90^\circ$, $\therefore \triangle KMC$ 是等腰直角三角形, $\therefore \angle KCM = 45^\circ$. $\therefore AK \parallel CN$, $\therefore$ 四边形 $AKCN$ 是平行四边形, $\therefore AN \parallel CK$, $\therefore \angle APM = \angle KCM = 45^\circ$.



第 13 题图

14. 解: (1) 若 $q = 0$, 则方程 $x^2 + 2px - 3p^2 + 5 = 0$, $\therefore$ 方程有两个不相等的实数根, $\therefore \Delta = 16p^2 - 20 > 0$, $p^2 > \frac{5}{4}$, $\therefore x_1 + x_2 = -2p$, $x_1 x_2 = 5 - 3p^2$,

$\therefore \frac{x_1 + x_2}{x_1 x_2} = \frac{-2p}{5 - 3p^2} = \frac{1}{7}$, $p_1 = 5$, $p_2 = -\frac{1}{3}$, 经检验, 只有 $p = 5$ 符合题意, $\therefore p = 5$.

(2) 显然 $q > 0$, 方程化为 $x^2 + 2px - 3p^2 + 5 = \pm q$, $\therefore$ 方程有三个不同实根, 即函数 $y_1 = x^2 + 2px - 3p^2 + 5$ 与 $y_2 = \pm q$ 的图象有三个不同交点, $\therefore x_1 = -p$, $-q = \frac{4(5-3p^2)-4p^2}{4} = 5-4p^2$, 即 $q = 4p^2 - 5$. $\therefore x_1, x_2$ 是方程 $x^2 + 2px - 3p^2 + 5 = q$ 的两根, $\therefore x^2 + 2px - 7p^2 + 10 = 0$. 则 $x_1 + x_2 = -2p$, $x_1 x_2 = 10 - 7p^2$, $x_3 = -p$. $\Delta = (2p)^2 - 4(-7p^2 + 10) = 32p^2 - 40 > 0$, 解得 $p^2 > \frac{5}{4}$. 由 $\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} = 0$, 得 $\frac{x_2 + x_1}{x_1 x_2} + \frac{1}{x_3} = \frac{-2p}{10 - 7p^2} + \frac{1}{-p} = \frac{10 - 5p^2}{(7p^2 - 10)p} = 0$, 解得 $p^2 = 2 > \frac{5}{4}$, $\therefore p = \pm\sqrt{2}$, $q = 4p^2 - 5 = 3$.

15. (1) 证明: $\because AB = AC$, $\angle BAC = 120^\circ$, $\therefore \angle B = \angle C = 30^\circ$. 又 $\angle DAE = 30^\circ$, $\therefore \angle B = \angle C = \angle DAE$. $\therefore \angle ADC = \angle B + \angle BAD$, $\angle BAF = \angle DAE + \angle BAD$, $\therefore \angle BAF = \angle ADC$, $\therefore \triangle ABF \sim \triangle DCA$.

(2) 解: 过点 A 作 $AH \perp BC$ 于点 H , 则 $BC = 2BH$, $AH = \frac{1}{2}AB = \sqrt{3}$, $BH = \sqrt{(2\sqrt{3})^2 - (\sqrt{3})^2} = 3$, $\therefore BC = 6$. $\because \triangle ABF \sim \triangle DCA$, $\therefore \frac{AB}{CD} = \frac{BF}{AC}$, 即 $CD \cdot BF = AB \cdot AC$, 设 $BD = x$, 则 $CD = 6 - x$, $BF = x + 1$, $\therefore (6 - x) \cdot (x + 1) = 2\sqrt{3} \times 2\sqrt{3}$, 解得 $x = 2$ 或 $x = 3$, $\therefore BD = 2$ 或 3 . 当 $BD = 2$ 时, $BF = 3$, 即 F 为 BC 中点, 则可得 BC 垂直平分 AE , $\therefore BE = BA = 2\sqrt{3}$; 当 $BD = 3$ 时, D 为 BC 中点, $AD \perp BC$, $\angle BAD = \frac{1}{2} \angle BAC = 60^\circ$, $\angle BAE = \angle BAD + \angle DAE = 90^\circ$, 过点 D 作 $DG \perp AE$ 于点 G , 则 $AG = AD \cdot \cos 30^\circ = \frac{3}{2}$,

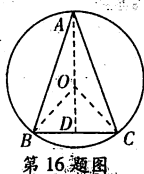
$AE = 2AG = 3$, $\therefore BE = \sqrt{(2\sqrt{3})^2 + 3^2} = \sqrt{21}$. 综上所述, BE 的长为 $2\sqrt{3}$ 或 $\sqrt{21}$.

16. 解: (1) $\because x^2 - 6x + m = 0$ 有两个不相等的实数根, 假设为 a, b 且 $0 < a < b$, $\therefore \Delta = (-6)^2 - 4m > 0$, 解得 $m < 9$. $\therefore$ 两根作底和腰的等腰三角形有且只有一个, $\therefore 2a \leq b$,

$\therefore \frac{6 - \sqrt{36 - 4m}}{2} \times 2 \leq \frac{6 + \sqrt{36 - 4m}}{2}$ 且 $6 - \sqrt{36 - 4m} > 0$, 解得 $0 < m \leq 8$, 综上, m 的取值范围是 $0 < m \leq 8$.

(2) $\because 0 < m \leq 8$, $\therefore m$ 取最大值时, $m = 8$, 解得两根为 2 和 4, 则该等腰三角形三边分别是 4, 4, 2, 如图, $AB = AC = 4$, $BC = 2$, 作其外接圆 $\odot O$, 连接 AO 并延长交 BC 于点 D .

$\therefore AB = AC = 4$, $\therefore AD \perp BC$, $BD = CD = 1$, $\therefore AD = \sqrt{15}$. 设 $AO = BO = CO = r$, $OD = \sqrt{15} - r$, $\therefore BO^2 = BD^2 + DO^2$, $\therefore r^2 = 1^2 + (\sqrt{15} - r)^2$, 解得 $r = \frac{8\sqrt{15}}{15}$.



第 16 题图

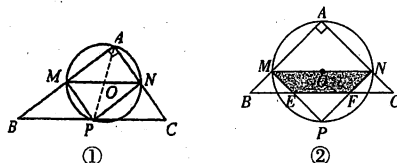
17. 解: (1) $\because MN \parallel BC$, $\therefore \angle AMN = \angle B$, $\angle ANM = \angle C$,

$\therefore \triangle AMN \sim \triangle ABC$, $\therefore \frac{AM}{AB} = \frac{AN}{AC}$, 即 $\frac{x}{4} = \frac{AN}{3}$,

$\therefore AN = \frac{3}{4}x$. $\therefore S = S_{\triangle MNP} = S_{\triangle AMN} = \frac{1}{2} \cdot \frac{3}{4}x \cdot x = \frac{3}{8}x^2$ ($0 < x < 4$).

(2) 随着点 M 的运动, 当点 P 落在直线 BC 上时, 如图①, 连接 AP , 则点 O 为 AP 的中点. $\because MN \parallel BC$, $\therefore \angle AMN = \angle B$, $\angle AOM = \angle APB$,

$\therefore \triangle AMO \sim \triangle APB$, $\therefore \frac{AM}{AB} = \frac{AO}{AP} = \frac{1}{2}$, $AM = MB = 2$. 故分以下两种情况讨论: ①当 $0 < x \leq 2$ 时, $y = S_{\triangle PMN} = \frac{3}{8}x^2$. 当 $x = 2$ 时, $y_{\text{最大}} = \frac{3}{8} \times 2^2 = \frac{3}{2}$.



第 17 题图

②当 $2 < x < 4$ 时, 如图②, 设 PM, PN 分别交 BC 于点 E, F .

$\because$ 四边形 $AMPN$ 是矩形, $\therefore PN \parallel AM$. 又 $\because MN \parallel BC$, $\therefore$ 四边形 $MBFN$ 是平行四边形, $\therefore FN = BM = 4 - x$, $\therefore PF = x - (4 - x) = 2x - 4$. 又 $\triangle PEF \sim \triangle ACB$, $\therefore \left(\frac{PF}{AB}\right)^2 = \frac{S_{\triangle PEF}}{S_{\triangle ABC}}$, $\therefore S_{\triangle PEF} = \frac{3}{2}(x - 2)^2$, $y = S_{\triangle MNP} - S_{\triangle PEF} =$

$\frac{3}{8}x^2 - \frac{3}{2}(x - 2)^2 = -\frac{9}{8}x^2 + 6x - 6$. 即当 $2 < x < 4$ 时, $y = -\frac{9}{8}x^2 + 6x - 6 = -\frac{9}{8}\left(x - \frac{8}{3}\right)^2 + 2$. 当 $x = \frac{8}{3}$ 时, 满足 $2 < x < 4$, $y_{\text{最大}} = 2$.

综上所述, $y = \begin{cases} \frac{3}{8}x^2 & (0 < x \leq 2), \\ -\frac{9}{8}x^2 + 6x - 6 & (2 < x < 4), \end{cases}$

当 $x = \frac{8}{3}$ 时, y 取最大值 2.

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1. C 【解析】 $\because a < b$, $\therefore a - b < 0$. $\frac{-(b-a)^2}{a} > 0$, $\therefore a < 0$, 即 $\frac{a}{a-b} > 0$,

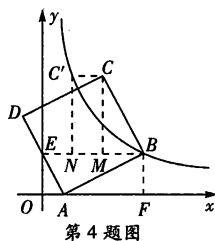
$\sqrt{\frac{(b-a)^2}{-a}} = \frac{b-a}{\sqrt{-a}} = \frac{(b-a)\sqrt{-a}}{-a}$, $\therefore$ 原式 $= \frac{a}{a-b} \cdot \frac{(b-a)\sqrt{-a}}{-a} = \frac{\sqrt{-a}}{a-b}$.

2. D 【解析】 $\because a = \sqrt{3+2\sqrt{2}} + \sqrt{3-2\sqrt{2}} = \sqrt{(\sqrt{2}+1)^2} + \sqrt{(\sqrt{2}-1)^2} = \sqrt{2} + 1 + \sqrt{2} - 1 = 2\sqrt{2}$. $\therefore a + \frac{1}{a} = 2\sqrt{2} + \frac{\sqrt{2}}{4} = \frac{9\sqrt{2}}{4} \approx 3.18$. 故选 C.

3. D 【解析】 $\frac{b+c}{a}-1=\frac{c+a}{b}-1=\frac{a+b}{c}-1$, 设 $\frac{b+c}{a}=\frac{c+a}{b}=\frac{a+b}{c}=k$,

$\therefore 2(a+b+c)=k(a+b+c)$, $\therefore k=2$ 或 -1 , $\frac{abc}{(a+b)(b+c)(c+a)}=\frac{1}{k^3}=\frac{1}{8}$ 或 -1 . 故选 D.

4. C 【解析】如图, 过点 C 作 x 轴的平行线, 交反比例函数的图象于点 C', 过点 B 作 $BF \perp x$ 轴, 垂足为 F, 分别过点 C, C' 分别作 y 轴的平行线, 交过点 B 作 x 轴的平行线交 BE 于点 M, N, 则 $CC'=MN$.



$\therefore ABCD$ 是正方形,

$\therefore AB=BC$, $\angle DAB=\angle ABC=90^\circ$.

又 $\angle OAE+\angle BAF=180^\circ-90^\circ=90^\circ$,

而 $\angle BAF+\angle ABF=90^\circ$, $\therefore \angle OAE=\angle ABF$.

$\angle AOE=\angle BFA=90^\circ$, $\therefore \triangle AOE \sim \triangle BFA$.

又 $AE=DE=\frac{1}{2}AD$, $\therefore \frac{AE}{AB}=\frac{OA}{BF}=\frac{1}{2}$, $\therefore OA=1$, $\therefore BF=2$, 同理可证

$\angle ABF=\angle CBM$. 又 $AB=BC$, $\angle AFB=\angle CMB$, $\therefore \triangle ABF \cong \triangle CBM$ (AAS),

$BF=BM=2$. 又点 B 在反比例函数的图象上, $\therefore$ 点 $B(\frac{k}{2}, 2)$, $\therefore AF=\frac{k}{2}-1=CM$, $\therefore C'(\frac{k}{2}-2-\frac{4}{3}, \frac{k}{2}-1+2)$, 即 $C'(\frac{k}{2}-\frac{10}{3}, \frac{k}{2}+1)$. 又

点 C 在反比例函数的图象上, $\therefore (\frac{k}{2}-\frac{10}{3})(\frac{k}{2}+1)=k$. 即 $3k^2-26k-40=0$, 解得 $k=10$ 或 $k=-\frac{4}{3}$ (舍去). 故选 C.

5. D 【解析】连接 AB, OP, OQ,

$AC \perp BC$, $\therefore \angle ACB=90^\circ$, $\therefore AB$ 为直径.

P 为 BE 的中点, Q 为 AD 的中点,

$\therefore OP \parallel AC$, $OP=\frac{1}{2}AE$, $OQ \parallel BD$, $OQ=\frac{1}{2}BD$,

$\therefore OP \perp OQ$, $\therefore \angle POQ=90^\circ$.

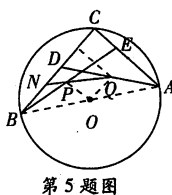
$\therefore BD=AE$, $\therefore OP=OQ$,

$\therefore \angle OPQ=\angle OQP=45^\circ$. $\therefore \angle A=30^\circ$,

$\therefore \angle CDA=60^\circ$,

$\therefore \angle NDQ=120^\circ$, $\therefore \angle OQA=120^\circ$, $\therefore \angle NQD=15^\circ$, $\therefore \angle DNQ=45^\circ$, 过点 Q 作 $QM \perp BC$ 交 BC 于 M, 则 $\triangle NQM$ 为等腰直角三角形. $\therefore NQ=2\sqrt{6}$,

$\therefore MQ=2\sqrt{3}$, 在 $Rt\triangle DMQ$ 中, $\angle MDQ=60^\circ$, $\therefore DQ=\frac{MQ}{\sin 60^\circ}=4$. 故选 D.



第 5 题图

6. 42 cm 【解析】连接 OA, OB, 则有 $\angle OAB=\angle OPB=\angle OBA=45^\circ$, $\therefore O, P, A, B$ 四点共圆. $\therefore \angle APB=\angle AOB=180^\circ-45^\circ-45^\circ=90^\circ$. 在 $Rt\triangle PAB$ 中, 设 $PA=5x$, $PB=14x$, 则 $(5x)^2+(14x)^2=1989$. 解得 $x=3$, $\therefore PB=14x=42$.

7. $\frac{11}{3}$ 【解析】 $\frac{a^2}{1+2a^2}=\frac{b}{4+3b^2}=\frac{c}{12+4+6c^2}=\frac{a}{2}$, $\therefore \frac{4}{b}=\frac{1}{a^2}+2$, $\frac{12}{c}=\frac{4}{b^2}+3$, $\frac{2}{a}=\frac{4}{c^2}+6$, $\therefore \frac{4}{b}+\frac{12}{c}+\frac{2}{a}=\frac{1}{a^2}+\frac{4}{b^2}+\frac{4}{c^2}+11$,

$\therefore (\frac{1}{a^2}-\frac{2}{a}+1)+(\frac{4}{b^2}-\frac{4}{b}+1)+(\frac{4}{c^2}-\frac{12}{c}+9)=0$, 即 $(\frac{1}{a}-1)^2+$

$(\frac{2}{b}-1)^2+(\frac{2}{c}-3)^2=0$, $\therefore \frac{1}{a}=1$, 即 $a=1$, $\frac{2}{b}=1$, 即 $b=2$, $\frac{2}{c}=3$, 即 $c=\frac{2}{3}$, $\therefore a+b+c=\frac{11}{3}$.

8. $y=\frac{3}{4}x+3$ 或 $y=\frac{3}{4}x-12$ 【解析】分两种情况 $C(-4,0), D(0,3)$ 或 $C(16,0), D(0,-12)$.

9. $\frac{9}{2}$ 【解析】设 $x+2y=t$, 则 $(t-2y)^2-2(t-2y)+4y=5$, $4y^2+(8-4t)y+(t^2-2t-5)=0$, 由 $\Delta \geq 0$ 知 $t \leq \frac{9}{2}$.

10. 25 【解析】连接 AF, BD, 则四边形 ABDF 是矩形, $\therefore AB=DF$. 取 AB 的中点 O, 作 $OG \perp CE$ 交 CE 于点 G, 连接 OC, 设 $DF=10k$, $CE=6k$, 则 $CG=\frac{1}{2}CE=3k$, $OC=OA=5k$, $OG=4k$, $AF=BD=4k$, $CF=DE=2k$,

$\therefore AC=\sqrt{CF^2+AF^2}=2\sqrt{5}k$, $BC=\sqrt{CD^2+BD^2}=4\sqrt{5}k$, $\therefore 2\sqrt{5}k+4\sqrt{5}k=15$, $\therefore k=\frac{\sqrt{5}}{2}$. $\therefore AC=5$, $BC=10$. $\therefore S_{阴影}$ 是直径为 AC 的半圆的面积 + 直径为 BC 的半圆的面积 + $S_{\triangle ABC}$ - 直径为 AB 的半圆的面积 = $\frac{1}{2}\pi \cdot$

$(\frac{AC}{2})^2 + \frac{1}{2}\pi(\frac{BC}{2})^2 + \frac{1}{2}AC \times BC - \frac{1}{2}\pi(\frac{AB}{2})^2 = \frac{1}{8}\pi(AC^2+BC^2-AB^2) + \frac{1}{2}AC \times BC = \frac{1}{2}AC \times BC = \frac{1}{2} \times 5 \times 10 = 25$.

11. $\frac{2}{9}$ 【解析】解不等式 $\frac{1}{3}(2x+7) \geq 3$, 得 $x \geq 1$, 解不等式 $x-m < 0$, 得 $x <$

m , $\therefore$ 不等式组无解, $\therefore m \leq 1$, $\therefore$ 符合此条件的有 $-3, -2, -1, -\frac{1}{2}, 0, \frac{1}{2}$,

1 这 7 个数, 解分式方程得 $x=\frac{-m-1}{2}$, $\therefore$ 方程有非负整数解, $\therefore$ 在以上 7 个数中, 符合此条件的有 $-3, -1$ 这 2 个, $\therefore$ 从这 9 个数中抽到满足条件的 m 的概率是 $\frac{2}{9}$.

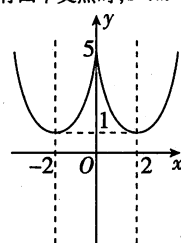
12. $\frac{5}{2}$ 【解析】连接 DB, $\therefore S_{\triangle ABE}=S_{\triangle DBE}, S_{\triangle DBF}=S_{\triangle DBF}$ 都是等底等高的三

角形, 连接 CA, $S_{\triangle DEF}=\frac{1}{4}S_{\triangle ACD}$, $\therefore S_{\text{四边形}ABCD}=6$, $\therefore S_{\triangle ABE}+S_{\triangle BCF}=3$,

$\therefore S_{\triangle ABC}=\frac{2\sqrt{2} \times 2\sqrt{2}}{2}=4$, $\therefore S_{\triangle ACD}=6-4=2$,

$\therefore S_{\triangle DEF}=\frac{1}{4} \times 2=\frac{1}{2}$, $S_{\triangle BEF}=6-3-\frac{1}{2}=\frac{5}{2}$.

13. 解: 当 $x > 0$ 时, $y=x^2-4x+5$, 当 $x < 0$ 时, $y=x^2+4x+5$, 画图象如图所示, 当直线 $y=m$ 与图象有四个交点时, $1 < m < 5$.



第 13 题图

14. 解: $\because c^2=a^2+b^2$, $\therefore c^2=[\frac{1}{3}ab-(a+b)]^2=\frac{1}{9}a^2b^2-\frac{2}{3}ab(a+b)+$

$a^2+b^2+2ab=a^2+b^2$, 化简, 得 $(ab)^2-6ab(a+b)+18ab=0$, $\therefore ab > 0$,

$\therefore ab-6(a+b)+18=0$, $(a-6)(b-6)=18$. $\therefore a, b$ 均为正整数, 不妨

设 $a < b$, 则 $\begin{cases} a-6=1, \\ b-6=18; \end{cases} \begin{cases} a-6=2, \\ b-6=9; \end{cases} \begin{cases} a-6=3, \\ b-6=6; \end{cases} \therefore (a, b, c)=(7, 24, 25),$

$(8, 15, 17), (9, 12, 15)$, $\therefore$ 满足条件的直角三角形有三个, 周长分别为 56 或 40 或 36.

15. (1) 证明: 取 BE 中点 M, 连接 CM,

$\therefore M$ 为 BE 的中点, $\therefore ME=\frac{1}{2}BE$, $\therefore EF=\frac{1}{2}BE$, $\therefore EF=ME$, $\therefore E$ 为 AC 的中点,

$\therefore AC=2AE=2CE$, 在 $\triangle AEF$ 和 $\triangle CEM$ 中,

$\begin{cases} AE=CE \\ \angle AEF=\angle CEM, \\ EF=EM \end{cases}$

$\therefore \triangle AEF \cong \triangle CEM$ (SAS),

$\therefore \angle AFE=\angle CME$, $\therefore AC=2BC$, $\therefore BC=CE$, $\therefore M$ 为 BE 的中点, $\therefore CM \perp$

BE, $\therefore \angle CME=90^\circ$, $\therefore \angle AFE=90^\circ$, $\therefore D$ 是 AB 的中点, $\therefore DF=\frac{1}{2}AB$,

$DB=\frac{1}{2}AB$, $\therefore DF=DB$.

(2) 证明: 延长 CB 到 P, 使 $BP=CB$, 连接 EP, 如图②.

$\therefore CP=2BC$, $\therefore AC=2BC$, $\therefore AC=CP$, 在

$\triangle ACB$ 和 $\triangle PCE$ 中, $\begin{cases} AC=CP, \\ \angle C=\angle C, \\ BC=EC, \end{cases}$

$\therefore \triangle ACB \cong \triangle PCE$ (SAS), $\therefore AB=PE$,

$\therefore BP=CB$, H 是 CE 的中点,

$\therefore BH \parallel PE$, $BH=\frac{1}{2}PE$,

由 (1) 知, $DF=\frac{1}{2}AB$, $\therefore DF=BH$.

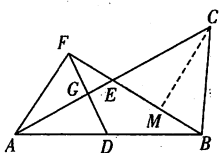
(3) 解: 连接 AF, FH, $\therefore HE=\frac{1}{2}AE$, $EF=\frac{1}{2}BE$, $\therefore \frac{HE}{AE}=\frac{EF}{BE}=\frac{1}{2}$,

$\therefore \angle HEF=\angle AEB$, $\therefore \triangle FHE \sim \triangle BAE$, $\therefore \angle FHE=\angle BAE$, $\frac{FH}{AB}=\frac{EF}{BE}=\frac{1}{2}$, $\therefore FH \parallel AB$, $FH=\frac{1}{2}AB$, $\therefore BD=\frac{1}{2}AB$, $\therefore FH=BD$, $\therefore$ 四边形

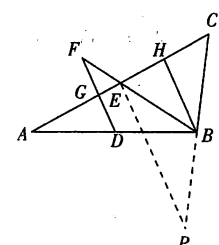
BDFH 是平行四边形, 当四边形 BDFH 是正方形时, $FD \perp AB$, $\therefore \angle ABH=90^\circ$, $HB=FD$, $\therefore \angle AFE=90^\circ$, $AD=BD$, $\therefore DF=\frac{1}{2}AB$, $\therefore HB=FD=\frac{1}{2}AB$, $\therefore AH=AE+EH=\frac{1}{2}AC+\frac{1}{4}AC=\frac{3}{4}AC=\frac{3}{2}BC$, $AB^2+HB^2=$

$\frac{5}{4}AC^2$, $\therefore AB^2+(\frac{1}{2}AB)^2=(\frac{3}{2}BC)^2$, $\therefore \frac{BC^2}{AB^2}=\frac{4}{9}=\frac{5}{9}$, $\therefore \frac{BC}{AB}=\frac{\sqrt{5}}{3}$.

16. 解: (1) 连接 OD, 过点 D 作 $DH \perp BC$ 于点 H, 设 OC 交 DF 于点 G, 则



第 15 题图①



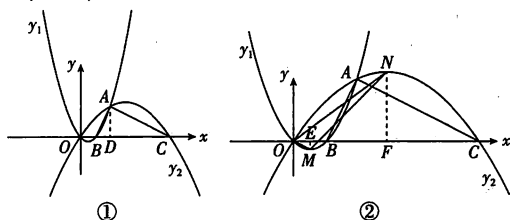
第 15 题图②

$OC = OD = DC = 2\sqrt{3}$, $\triangle CDE \sim \triangle CAD$, $\therefore \frac{CD}{CA} = \frac{CE}{CD}$, $\therefore \frac{2\sqrt{3}}{CA} = \frac{\sqrt{7}}{2\sqrt{3}}$, $\therefore CA = \frac{12\sqrt{7}}{7}$, $\therefore AE = AC - CE = \frac{12\sqrt{7}}{7} - \sqrt{7} = \frac{5\sqrt{7}}{7}$. 在 $\text{Rt}\triangle CDG$ 中, $CG = \sqrt{3}$, $DG = 3$, 在 $\text{Rt}\triangle CGE$ 中, $GE = \sqrt{CE^2 - CG^2} = 2$, $\therefore DE = DG + GE = 5$, $\therefore DE \parallel BC$, $\therefore \triangle ADE \sim \triangle ABC$, $\therefore \frac{DE}{BC} = \frac{AE}{AC}$, $\therefore \frac{5}{BC} = \frac{5}{12}$, $\therefore BC = 12$. $\therefore DE \parallel BC$, $\therefore \angle DCH = \angle CDF = 30^\circ$. 在 $\text{Rt}\triangle CHD$ 中, $DH = DC \cdot \sin 30^\circ = \sqrt{3}$, $CH = 3$, $\therefore BH = BC - CH = 9$, $\therefore \tan B = \frac{DH}{BH} = \frac{\sqrt{3}}{9}$.

(2) 连接 CF , 设 $AD = 4k$, 则 $AF = 3k$. 易证 BC 为 $\odot O$ 的切线, $\triangle CDB \sim \triangle AFC$, $\therefore \frac{DB}{CF} = \frac{DC}{FA}$, $\therefore \frac{DB}{2\sqrt{3}} = \frac{2\sqrt{3}}{3k}$, $\therefore DB = \frac{4}{k}$, $\therefore BC + BD = 8$, $\therefore BC = 8 - \frac{4}{k}$, 又可证 $\triangle BCD \sim \triangle BAC$, $\therefore \frac{DB}{BC} = \frac{BC}{AB}$, 即 $BC^2 = BD \cdot AB$, $\therefore \left(8 - \frac{4}{k}\right)^2 = \left(\frac{4}{k}\right) \cdot \left(\frac{4}{k} + 4k\right)$, 解得 $k = \frac{4}{3}$, $\therefore BD = 3$.

17. (1) 解: 联立 $\begin{cases} y = mx^2 + (1-m)x \\ y = nx^2 + (1-n)x \end{cases}$, 得 $(m-n)x^2 + (n-m)x = 0$, $\therefore (m-n)(x^2 - x) = 0$. $\because m > 0, n < 0, \therefore m \neq n, \therefore x^2 - x = 0$, 解得 $x_1 = 0, x_2 = 1$. 当 $x = 1$ 时, $y_1 = y_2 = 1$, $\therefore A(1, 1)$.

(2) ① 证明: 令 $y_1 = 0$, 得 $mx^2 + (1-m)x = 0$, 解得 $x_1 = 0, x_2 = \frac{m-1}{m}$, $\therefore B\left(\frac{m-1}{m}, 0\right)$, 同理得 $C\left(\frac{n-1}{n}, 0\right)$. 如图①, 过点 A 作 $AD \perp BC$ 于点 D , 易证 $\triangle ABD \sim \triangle CAD$, $\therefore \frac{AD}{CD} = \frac{BD}{AD}$, $\therefore AD^2 = BD \cdot CD$, $\therefore BD = 1 - \frac{m-1}{m} = \frac{1}{m}$, $CD = \frac{n-1}{n} - 1 = -\frac{1}{n}$, $\therefore \frac{1}{m} \cdot \left(-\frac{1}{n}\right) = 1$, $\therefore mn = -1$;



第 17 题图

② 解: $\because y_1 = mx^2 + (1-m)x = m\left(x - \frac{m-1}{2m}\right)^2 - \frac{(1-m)^2}{4m}$, $y_2 = nx^2 + (1-n)x = n\left(x - \frac{n-1}{2n}\right)^2 - \frac{(1-n)^2}{4n}$, $\therefore$ 函数 y_1, y_2 图象的顶点分别为 $M\left[\frac{m-1}{2m}, -\frac{(1-m)^2}{4m}\right], N\left[\frac{n-1}{2n}, -\frac{(1-n)^2}{4n}\right]$, 如图②, 过点 M 作 $ME \perp x$ 轴于点 E , 过点 N 作 $NF \perp x$ 轴于点 F , 则 $\tan \angle MOE = \frac{(1-m)^2}{4m} = \frac{m-1}{2}$, $\tan \angle NOF = \frac{(1-n)^2}{4n} = \frac{1-n}{2}$. $\because \angle BAC = 90^\circ$, $\therefore \text{Rt}\triangle ABC$ 外心 P 在 x 轴上, $\therefore$ 当 O, P, Q 三点共线时, Q 也在 x 轴上, 此时, $\angle NOF = \angle MOE$, $\therefore \frac{1-n}{2} = \frac{m-1}{2}$, $\therefore m+n=2$, 联立 $\begin{cases} mn = -1, \\ m+n = 2, \end{cases}$ 解得 $\begin{cases} m_1 = 1+\sqrt{2}, \\ n_1 = 1-\sqrt{2}, \end{cases} \begin{cases} m_2 = 1-\sqrt{2}, \\ n_2 = 1+\sqrt{2}, \end{cases}$ (舍).

$\therefore$ 存在 $m = 1+\sqrt{2}, n = 1-\sqrt{2}$, 使 O, P, Q 三点共线.

**全国重点高中提前招生
考试全真试卷(二十)**

1. D 【解析】显然, a, b 是方程 $x^2 + 2x - 2 = 0$ 的两根. 若 $a = b$, 则 $a = b = -1 \pm \sqrt{3}$, 此时, $\frac{1}{a} + \frac{1}{b} = \sqrt{3} + 1$ 或 $-\sqrt{3} + 1$; 若 $a \neq b$, 则 $a + b = -2, ab = -2$, 则 $\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab} = 1$. 综上, $\frac{1}{a} + \frac{1}{b}$ 的值是 $\sqrt{3} + 1$ 或 $-\sqrt{3} + 1$ 或 1. 故选 D.

2. D 【解析】设第一季度销售额为 m , 则有 $56\%m(1+23\%) + 44\%m(1-a\%) = 1.12m$, $\therefore a = 2$. 故选 D.

3. C 【解析】 $\because$ 过点 $(2, 3)$ 的直线 $y = ax + b (a \neq 0)$ 不经过第四象限, $\therefore a > 0, b \geq 0$. 又 $3 = 2a + b$, 即 $b = 3 - 2a$, $\therefore a > 0, 3 - 2a \geq 0$. 解得 $0 < a \leq \frac{3}{2}$,

$\therefore s = a - 2b = a - 2(3 - 2a) = 5a - 6$, 当 $a = 0$ 时, $s = -6$, 当 $a = \frac{3}{2}$ 时, $s = \frac{3}{2}$, 故 $-6 < s \leq \frac{3}{2}$. 故选 C.

4. A 【解析】设 $\angle B = x$, 则 $\angle APE = 45^\circ - x$. 又可求 $\angle ANC = 90^\circ - 2x$. 故知 $NP = NQ$. 故选 A.

5. A 【解析】点 B' 的运动轨迹是以 E 为圆心, 以 AE 的长为半径的圆, 则当 B' 点落在 DE 上时, $B'D$ 取得最小值. 此时 $\triangle EBF \sim \triangle EB'F, EB' \perp B'F, EB' = EB, AB = 4, \therefore AE = EB' = 2, \therefore DE = \sqrt{6^2 + 2^2} = 2\sqrt{10}, \therefore DB' = 2\sqrt{10} - 2$. 故选 A.

6. B 【解析】 $\because$ 将射线 OB 绕着点 O 逆时针旋转到经过甲或乙的旋转角记为自变量 x , 成员乙所在的位置为 Q , $\therefore$ 根据图 20-2③所示, 实线部分图象中距离先保持不变, 再下降至 0, 然后再上升, 可判断实线部分为乙的图象. (由于点 Q 在以 A 为圆心, AB 为半径的 $\widehat{BC}$ 上, 则 AQ 的距离保持不变). $\therefore$ 当 Q 点从点 B 开始逆时针运动时, 检测器应该在 A 点. 故选项 D 说法正确, 不符合题意;

$\therefore$ 点 Q 从 B 点运动到 A 点, 运动的角度为 $\frac{2}{3}$ 个圆周,

$\therefore \alpha = 360^\circ \times \frac{2}{3} = 240^\circ$. 故 B 选项说法不正确, 此选项

符合题意; 由图 20-2③得 $AB = 6\sqrt{3}$, 可得 $\angle AOB = \frac{1}{3} \times 360^\circ = 120^\circ$, 连接 OA, AB , 过点 O 作 $OD \perp AB$ 于点 D ,

如图, $\therefore OD \perp AB, \therefore \angle BOD = 60^\circ, BD = \frac{1}{2}AB = 3\sqrt{3}$. 在 $\text{Rt}\triangle BOD$ 中,

$\sin \angle BOD = \frac{BD}{OB} = \frac{3\sqrt{3}}{\frac{\sqrt{3}}{2}} = 6, \therefore OB = 6$. 即 $\odot O$ 的半径为 6m. 故 A 选项说法

正确, 不符合题意; 由图 20-2②可知, 当射线 OB 旋转至 $\widehat{BC}$ 的中点时, 此时点 P 在直线 OA 上, y_1 取最大值, 长度为 $\odot O$ 的直径 12m, 此时射线 OB 转过的角度为 60° , 即 $x = 60$. 故 C 选项说法正确, 不符合题意. 故选 B.

7. $\frac{\sqrt{74}}{5}$ 【解析】连接 AC, AG, AC' , 易证 $\triangle ABB' \sim \triangle ACC'$, $\therefore \frac{CC'}{BB'} = \frac{AC}{AB}$, $\triangle AB'G$ 是等腰直角三角形, $\therefore AG = \sqrt{2}AB'$, 设 $AB = AB' = x$, 则 $AG = \sqrt{2}x$, $DG = x - 4$, 在 $\text{Rt}\triangle ADG$ 中, $7^2 + (x - 4)^2 = (\sqrt{2}x)^2$, 解得 $x_1 = 5, x_2 = -13$ (舍), $\therefore AB = 5$, 在 $\text{Rt}\triangle ABC$ 中, $AC = \sqrt{5^2 + 7^2} = \sqrt{74}, \therefore \frac{CC'}{BB'} = \frac{AC}{AB} = \frac{\sqrt{74}}{5}$.

8. 18 【解析】 $x\sqrt{y} + y\sqrt{x} - \sqrt{2022x} - \sqrt{2022y} + \sqrt{2022xy} = 2022$, 变形为 $\sqrt{xy}(\sqrt{x} + \sqrt{y}) - \sqrt{2022}(\sqrt{x} + \sqrt{y}) + \sqrt{2022xy} - \sqrt{2022^2} = 0$,

$\therefore (\sqrt{x} + \sqrt{y} + \sqrt{2022})(\sqrt{xy} - \sqrt{2022}) = 0$,

$\therefore \sqrt{xy} - \sqrt{2022} = 0, xy = 2022 = 2 \times 3 \times 337$.

$\because x, y$ 均为整数, $x - y - 7 > 0$,

$\therefore \sqrt{x - y - 7}$ 最小值时 $x = 337, y = 6$,

$\therefore \sqrt{x - y - 7}$ 最小值为 $\sqrt{337 - 6 - 7} = \sqrt{324} = 18$.

9. $\frac{\sqrt{5}+1}{2}$ 【解析】设原式 $= x$, 则 $x = 1 + \frac{1}{x}, \therefore x > 0, \therefore x = \frac{\sqrt{5}+1}{2}$.

10. 100 【解析】连接 BC, AD , 则 $\angle C = \angle D = 90^\circ$, 由相交弦定理得 $AE \cdot CE = DE \cdot EB, \therefore AE \cdot AC + BE \cdot BD = AC^2 - AC \cdot CE + BD^2 - BD \cdot DE = 100 - BC^2 + 100 - AD^2 - AC \cdot CE - BD \cdot DE = 200 - BE^2 + CE^2 - AE^2 + DE^2 - AC \cdot CE - BD \cdot DE = 200 + (DE + BE)(DE - BE) + (CE + AE) \cdot (CE - AE) - AC \cdot CE - BD \cdot DE = 200 + BD(DE - BE) + AC(CE - AE) - AC \cdot CE - BD \cdot DE = 200 - AE \cdot AC - BE \cdot BD, \therefore AE \cdot AC + BE \cdot BD = 100$.

11. -29 【解析】设有 x 个 1, y 个 -2, 则 $\begin{cases} x - 2y = -5, \\ x + 4y = 19, \end{cases}$

故 $\begin{cases} x = 3, \\ y = 4. \end{cases}$ 原式 $= 3 \times 1^3 + 4 \times (-2)^3 = -29$.

12. $\frac{48}{13}$ 【解析】连接 CM , 则 $AC = 2HM = 2 \times \frac{3}{2} = 3, BC = 4, OC = OB = 2$, 连接 OA, CN, ON , OA 与 CN 交于点 F , 则 $AO = \sqrt{AC^2 + CO^2} = \sqrt{13}$.

$\therefore S_{\triangle AOC} = \frac{1}{2}AC \cdot OC = \frac{1}{2}AO \cdot CF$, 即 $3 \times 2 = \sqrt{13}CF, \therefore CF = \frac{6\sqrt{13}}{13}$,

$\therefore CN = 2CF = \frac{12\sqrt{13}}{13}$. 设 $OE = x$, 则 $CE = 2 + x, ON = OC = 2$, 在 $\text{Rt}\triangle CNE$

和 $\text{Rt}\triangle ONE$ 中, $\left(\frac{12\sqrt{13}}{13}\right)^2 - (2 + x)^2 = 4 - x^2$, 解得 $x = \frac{10}{13}, \therefore OE =$

$\frac{10}{13}, \therefore EN = \sqrt{2^2 - \left(\frac{10}{13}\right)^2} = \frac{24}{13}$, 由垂径定理得 $NQ = 2EN = \frac{48}{13}$.

13. 解: 令 $a + \frac{3}{b} = b + \frac{3}{c} = c + \frac{3}{a} = k$, 则 $ab + 3 = bk, bc + 3 = ck, ac + 3 = ak$, 由 $ab + 3 = bk$, 可得 $abc + 3c = kbc = k(ck - 3)$, 即 $abc + 3k = (k^2 - 3)c$, 同理

可得 $abc+3k=(k^2-3)a, abc+3k=(k^2-3)b, \therefore (k^2-3)c=(k^2-3)a=(k^2-3)b, \therefore a, b, c$ 为互不相等的非零实数, $\therefore k^2-3=0$, 即 $k^2=3$, 则 $(a+\frac{3}{b})^2+(b+\frac{3}{c})^2+(c+\frac{3}{a})^2=3k^2=9$.

$$\therefore \sqrt{(a+\frac{3}{b})^2+(b+\frac{3}{c})^2+(c+\frac{3}{a})^2}=\sqrt{9}=3.$$

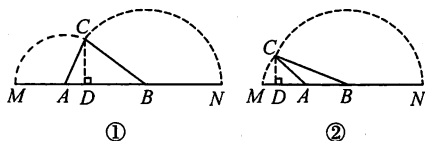
14. 解: (1) 在 $\triangle ABC$ 中, $AC=1, AB=x, BC=3-x$,

$$\therefore \begin{cases} 1+x>3-x, \\ 1+3-x>x, \end{cases} \text{解得 } 1<x<2.$$

(2) 若 AC 为斜边, 则 $1^2=x^2+(3-x)^2$, 方程无解; 若 AB 为斜边, 则 $x^2=(3-x)^2+1, x=\frac{5}{3}$; 若 BC 为斜边, 则 $(3-x)^2=1^2+x^2, x=\frac{4}{3}, x=\frac{5}{3}$ 或 $x=\frac{4}{3}$ 均满足 $1<x<2, \therefore x=\frac{4}{3}$ 或 $x=\frac{5}{3}$.

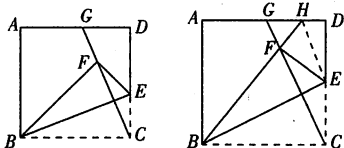
(3) 如图①所示, 若点 D 在线段 AB 上, 设 $CD=h$, 则 $AD+BD=x, \sqrt{1^2-h^2}+\sqrt{(3-x)^2-h^2}=x$, 化简得 $x^2h^2=-8x^2+24x-16, \therefore S^2=(\frac{xh}{2})^2=\frac{1}{4}x^2h^2=-2x^2+6x-4=-2(x-\frac{3}{2})^2+\frac{1}{2}(\frac{4}{3}\leq x\leq 2)$,

$\therefore$ 当 $x=\frac{3}{2}$ 时, S^2 取最大值 $\frac{1}{2}, \therefore S$ 最大值为 $\frac{\sqrt{2}}{2}$. 如图②所示, 若点 D 在线段 MA 上, 则 $\sqrt{(3-x)^2-h^2}-\sqrt{1^2-h^2}=x$, 同理可得 $S^2=\frac{1}{4}x^2h^2=-2x^2+6x-4=-2(x-\frac{3}{2})^2+\frac{1}{2}(1<x\leq \frac{4}{3})$, 易知此时 $S<\frac{\sqrt{2}}{2}$. 综上所述, $\triangle ABC$ 面积的最大值为 $\frac{\sqrt{2}}{2}$.



第 14 题图

15. (1) 证明: 如图①, $\therefore \triangle BFE$ 是由 $\triangle BCE$ 折叠得到, $\therefore BE \perp CF, \therefore \angle ECF + \angle BEC = 90^\circ, \therefore$ 四边形 $ABCD$ 是正方形, $\therefore \angle D = \angle BCE = 90^\circ, \therefore \angle ECF + \angle CGD = 90^\circ, \therefore \angle BEC = \angle CGD$. 在 $\triangle BCE$ 和 $\triangle CDG$ 中, $\begin{cases} \angle BEC = \angle CGD, \\ \angle BCE = \angle CDG, \\ BC = CD, \end{cases} \therefore \triangle BCE \cong \triangle CDG (AAS).$



第 15 题图①

第 15 题图②

(2) 解: 如图②, 连接 $EH, \therefore \triangle BCE \cong \triangle CDG, \therefore CE = DG = 7$, 由折叠可知 $BC = BF, CE = FE = 7, \therefore \angle BCF = \angle BFC, \therefore$ 四边形 $ABCD$ 是正方形, $\therefore AD \parallel BC, \therefore \angle BCG = \angle HGF, \therefore \angle BFC = \angle HFG, \therefore \angle HFG = \angle HGF, \therefore HF = HG, \therefore \frac{HD}{HF} = \frac{3}{4}, DG = 7, \therefore HD = 3, HF = HG = 4, \therefore \angle D = \angle HFE = 90^\circ, \therefore HF^2 + FE^2 = DH^2 + DE^2, \therefore 4^2 + 7^2 = 3^2 + DE^2, \therefore DE = 2\sqrt{14} \text{ 或 } -2\sqrt{14} \text{ (舍去)}, \therefore DE = 2\sqrt{14}.$

(3) 解: 如图③, 连接 HE . 由题意 $\frac{HD}{HF} = \frac{3}{4}$, 可以假设 $DH=3m, HG=4m, \text{设 } \frac{DE}{CE} = x$.

① 当点 H 在点 D 的左侧时, 如图③. $\therefore HF = HG, \therefore DG = 7m$, 由折叠可知 $BE \perp CF, \therefore \angle ECF + \angle BEC = 90^\circ, \therefore \angle D = 90^\circ, \therefore \angle ECF + \angle CGD = 90^\circ, \therefore \angle BEC = \angle CGD$.

$\therefore \angle BCE = \angle D = 90^\circ, \therefore \triangle CDG \sim \triangle BCE, \therefore \frac{DG}{CE} = \frac{CD}{BC}.$

$$\therefore \frac{CD}{BC} = \frac{AB}{BC} = k, \therefore \frac{7m}{CE} = \frac{k}{1}, \therefore CE = \frac{7m}{k}, \therefore DE = \frac{7mx}{k}.$$

$\therefore \angle D = \angle HFE = 90^\circ, \therefore HF^2 + FE^2 = DH^2 + DE^2,$

$$\therefore (4m)^2 + (\frac{7m}{k})^2 = (3m)^2 + (\frac{7mx}{k})^2,$$

$$\therefore x = \frac{\sqrt{7k^2+49}}{7} \text{ 或 } -\frac{\sqrt{7k^2+49}}{7} \text{ (舍)}, \therefore \frac{DE}{CE} = \frac{\sqrt{7k^2+49}}{7}.$$

② 当点 H 在点 D 的右侧时, 如图④, 同理 $HG = HF, \triangle BCE \sim \triangle CDG,$

$$\therefore DG = m, CE = \frac{m}{k} = FE,$$

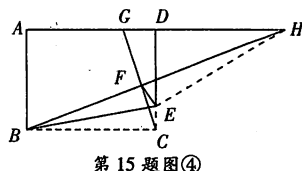
$$\therefore DE = \frac{mx}{k},$$

$$\therefore HF^2 + FE^2 = DH^2 + DE^2,$$

$$\therefore (4m)^2 + (\frac{m}{k})^2 = (3m)^2 + (\frac{mx}{k})^2,$$

$$\therefore x = \sqrt{7k^2+1} \text{ 或 } -\sqrt{7k^2+1} \text{ (舍)}, \therefore \frac{DE}{CE} = \sqrt{7k^2+1}.$$

综上所述, $\frac{DE}{CE}$ 的值为 $\frac{\sqrt{7k^2+49}}{7}$ 或 $\sqrt{7k^2+1}$.



第 15 题图④

16. (1) 解: 由于 AD, BE, CF 交于点 $O, \therefore \frac{OD}{AD} = \frac{S_{\triangle OBC}}{S_{\triangle ABC}}, \frac{OE}{BE} = \frac{S_{\triangle OAC}}{S_{\triangle ABC}}, \frac{OF}{CF} = \frac{S_{\triangle OAB}}{S_{\triangle ABC}}, \therefore \frac{OD}{AD} + \frac{OE}{BE} + \frac{OF}{CF} = 1.$

(2) 证明: 如图, 延长 AD 交 $\odot O$ 于 M , 设 R 为 $\triangle ABC$ 的外接圆半径, AD, BE, CF 交于点 O .

$$\therefore \frac{OD}{AD} = \frac{R-DM}{2R-DM} = 1 - \frac{R}{2R-DM} = 1 - \frac{R}{AD},$$

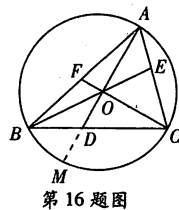
$$\text{同理有 } \frac{OE}{BE} = 1 - \frac{R}{BE}, \frac{OF}{CF} = 1 - \frac{R}{CF},$$

$$\text{代入 } \frac{OD}{AD} + \frac{OE}{BE} + \frac{OF}{CF} = 1,$$

$$\text{得 } (1 - \frac{R}{AD}) + (1 - \frac{R}{BE}) + (1 - \frac{R}{CF}) = 1,$$

$$\therefore \frac{R}{AD} + \frac{R}{BE} + \frac{R}{CF} = 2,$$

$$\therefore \frac{1}{AD} + \frac{1}{BE} + \frac{1}{CF} = \frac{2}{R} = \frac{4}{d}.$$



第 16 题图

17. 解: (1) 连接 NM , 过点 N 作 $NH \perp x$ 轴于 H , 如图①.

$\therefore PM \perp x$ 轴, $\therefore NH \parallel PM$.

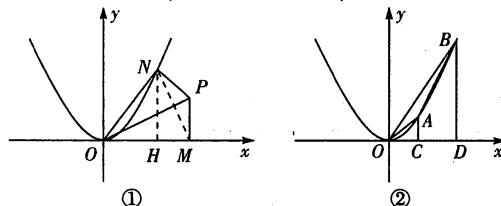
$\therefore$ 将 $\triangle OPM$ 沿 OP 翻折, 得到 $\triangle OPN$,

$\therefore OP$ 垂直平分 $MN, ON = OM$,

$\therefore \angle MOP + \angle NMH = 90^\circ, \angle HNM + \angle NMH = 90^\circ,$

$\therefore \angle MOP = \angle HNM$.

$\therefore \angle OMP = \angle NHM = 90^\circ, \therefore \triangle OMP \sim \triangle NHM,$



第 17 题图

$$\therefore \frac{NH}{HM} = \frac{OM}{MP}. \text{ 设 } P, N \text{ 的横坐标分别为 } a, b,$$

$$\therefore P(a, \frac{1}{2}a), M(a, 0), N(b, b^2), H(b, 0).$$

$$\therefore NH = b^2, HM = a - b, OM = a, MP = \frac{1}{2}a, \therefore \frac{b^2}{a-b} = \frac{a}{\frac{1}{2}a} = 2,$$

$$\therefore b^2 = 2(a-b) \text{ ①. 又 } \because ON = OM, OH^2 + NH^2 = OM^2, \therefore b^2 + b^4 = a^2 \text{ ②, 联立 ①② 得 } b = \frac{4}{3}, \therefore \text{点 } N \text{ 的坐标为 } (\frac{4}{3}, \frac{16}{9}).$$

(2) 如图②, $\therefore$ 抛物线 $y=x^2, A(x_1, y_1), B(x_2, y_2)$ 为抛物线在第一象限图象上的两个动点, $\therefore y_1 = x_1^2, y_2 = x_2^2$, 由题意得 $OC = x_1, AC = y_1 = x_1^2, OD = x_2, BD = y_2 = x_2^2, CD = x_2 - x_1 = 2,$

$$\therefore \text{梯形 } ABDC \text{ 的面积为 } S_2 = \frac{1}{2}CD(AC + BD) = \frac{1}{2} \times 2(x_1^2 + x_2^2) = x_1^2 + x_2^2, \triangle OAB \text{ 的面积为 } S_1 = S_{\triangle OBD} - S_{\triangle OAC} - S_2 = \frac{1}{2}OD \cdot BD -$$

$$\frac{1}{2}OC \cdot AC - S_2 = \frac{1}{2}x_2 \cdot x_2^2 - \frac{1}{2}x_1 \cdot x_1^2 - S_2 = \frac{1}{2}(x_2^3 - x_1^3) - S_2,$$

$$\therefore 5S_1 = 2S_2, \therefore 5 \times [\frac{1}{2}(x_2^3 - x_1^3) - S_2] = 2S_2, \therefore \frac{5}{2}(x_2^3 - x_1^3) - 5S_2 = 2S_2, 5(x_2^3 - x_1^3) = 14S_2.$$

$$\therefore x_2^3 - x_1^3 = (x_2 - x_1)(x_2^2 + x_1x_2 + x_1^2) = 2(x_2^2 + x_1x_2 + x_1^2),$$

$$\therefore 10(x_2^2 + x_1x_2 + x_1^2) = 14(x_1^2 + x_2^2), \therefore 2x_2^2 - 5x_1x_2 + 2x_1^2 = 0. \text{ 设直线 } AB \text{ 的解析式为 } y = kx + m, \therefore A(x_1, y_1), B(x_2, y_2) \text{ 为抛物线 } y = x^2 \text{ 在第一象限图象上的两个动点,}$$

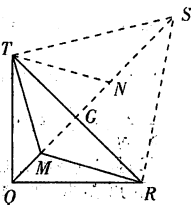
$$\therefore \begin{cases} y = kx + m, \\ y = x^2, \end{cases} \therefore x^2 - kx - m = 0,$$

$$\therefore x_1 + x_2 = k, x_1x_2 = -m, \therefore x_2 - x_1 = 2, \therefore (x_2 - x_1)^2 = 4, \text{ 即 } (x_2 + x_1)^2 - 4x_1x_2 = 4,$$

$$\therefore k^2 + 4m = 4, \therefore 2x_2^2 - 5x_1x_2 + 2x_1^2 = 2k^2 + 9m = 0, \therefore m = -8, k = 6,$$

$$\therefore \text{直线 } AB \text{ 的解析式为 } y = 6x - 8.$$

∴ $\triangle ADE \cong \triangle ABP$ (SAS),
 ∴ $PB = DE$, ∴ $PA + PB = DE + EP$.
 ②解: 将 $\triangle APC$ 绕点 C 顺时针旋转 60° 得到 $\triangle A'P'C'$
 于是 $AP = A'P'$, 连接 PP' , 如图①.
 ∵ $\angle PCP' = 60^\circ$, $PC = P'C'$, ∴ $\triangle PCP'$ 是等边三角形,
 ∴ $PC = PP'$, ∴ $PA + PB + PC = BP + PP' + P'A'$, 当 B, P, P', A' 四点共线时, $PA + PB + PC$ 有最小值.
 由旋转的性质知 $\angle APP' = 60^\circ$.
 又 ∵ $\triangle CPP'$ 为等边三角形,
 ∴ $\angle CPP' = 60^\circ$,
 ∴ $\angle APC = \angle APP' + \angle CPP' = 120^\circ$,
 由旋转的性质知 $A'C = AC = BC$,
 ∴ $\triangle BCA'$ 是等腰三角形.
 ∴ $\angle P'CA' + \angle BCP = \angle ACP + \angle BCP = 60^\circ$,
 ∴ $\angle BCA' = \angle P'CA' + \angle BCP + \angle P'CP = 120^\circ$,
 ∴ $\angle PBC = \angle A' = 30^\circ$,
 ∴ $\angle BPC = 120^\circ$, ∴ $\angle APB = 360^\circ - 120^\circ - 120^\circ = 120^\circ$.
 ∴ $PA + PB + PC$ 最小时, $\angle APB = \angle APC = 120^\circ$.
 (2)解: 如图②所示, 将 $\triangle TMR$ 绕点 T 逆时针旋转 60° 至 $\triangle TNS$, 连接 MN , 交 RT 于点 G , 连接 RS , 则 $\triangle TMR \cong \triangle TNS$, ∴ $TM = TN$, $NS = MR$, 显然, $\triangle TMN$ 为等边三角形,
 ∴ $MT = MN$, ∴ $MR + MQ + MT = MQ + MN + NS$, 显然, Q, M, N, S 四点共线时, $MQ + MN + NS$ 最小, 此时, $MQ + MN + NS = QS$.
 ∴ $\triangle TMR \cong \triangle TNS$, ∴ $\angle NTS = \angle MTR$.
 又 ∵ $\angle MTR + \angle MTQ = 45^\circ$,
 ∴ $\angle NTS + \angle MTQ = 45^\circ$,
 ∴ $\angle QTS = (\angle NTS + \angle MTQ) + \angle MTN = 105^\circ$.
 由旋转的性质易知 $\triangle RTS$ 为等边三角形,
 ∴ $ST = SR$, 又 ∵ $QT = QR$,
 ∴ $QS \perp RT$, 即 $\angle QGT = 90^\circ$, 则 $\triangle TQG$ 为等腰直角三角形,

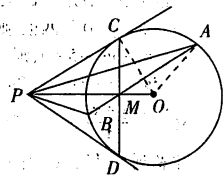


第 15 题图②

∴ $QG = QT \sin 45^\circ = 2 \times \frac{\sqrt{2}}{2} = \sqrt{2}$, $SG = TS \sin 60^\circ = 2\sqrt{2} \times \frac{\sqrt{3}}{2} = \sqrt{6}$,
 ∴ $QS = QG + SG = \sqrt{2} + \sqrt{6}$,
 ∴ $MR + MQ + MT$ 的最小值为 $\sqrt{2} + \sqrt{6}$.

16. 证明: 连接 OA, OC , 如图, ∵ PC, PD 为 $\odot O$ 的切线, ∴ $PC = PD$, $PO \perp CD$, ∴ $CM = MD$.

∵ PC 为 $\odot O$ 的切线,
 ∴ $OC \perp PC$, ∴ $CM \perp OP$, ∴ $\triangle OCM \sim \triangle CPM$,
 $\triangle OCM \sim \triangle OPC$,
 ∴ $\frac{OM}{CM} = \frac{CM}{OP} = \frac{OC}{OP}$,
 ∴ $CM^2 = OM \cdot PM$, $OC^2 = OM \cdot OP$.
 ∵ OP 与 CD 交于点 M , 弦 AB 经过点 M ,
 ∴ $BM \cdot MA = CM \cdot DM = CM^2$, ∴ $BM \cdot MA = OM \cdot MP$,
 ∴ $\frac{OM}{MA} = \frac{MB}{MP}$.

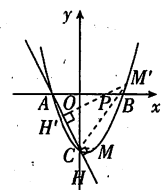


第 16 题图

∴ $\angle AMO = \angle PMB$, ∴ $\triangle AMO \sim \triangle PMB$, ∴ $\angle MAO = \angle MPB$.
 ∵ $OA = OC$, ∴ $OA^2 = OM \cdot OP$, ∴ $\frac{OA}{OM} = \frac{OP}{OA}$.
 ∴ $\angle AOM = \angle POA$, ∴ $\triangle AOM \sim \triangle POA$, ∴ $\angle MAO = \angle OPA$,
 ∴ $\angle OPA = \angle BPM$, ∴ PO 平分 $\angle APB$.

17. 解: (1)依题意设该二次函数的解析式为 $y = a(x+1)(x-2)$, 将 $C(0, -2)$ 代入得 $-2 = a(0+1)(0-2)$, 解得 $a = 1$, ∴ 抛物线的解析式为 $y = (x+1)(x-2)$, 即 $y = x^2 - x - 2$.
 (2)若 OC 为平行四边形的边, 设 $P(p, p^2 - p - 2)$, $Q(p, p)$, 则 $PQ = |p^2 - 2p - 2|$, 以 P, Q, O, C 为顶点的四边形为平行四边形, 则 $|p^2 - 2p - 2| = 2$, 解得 $p_1 = 0$ (舍去), $p_2 = 2$, $p_3 = 1 + \sqrt{5}$, $p_4 = 1 - \sqrt{5}$, 则 $Q_1(2, 2)$, $Q_2(1 + \sqrt{5}, 1 + \sqrt{5})$, $Q_3(1 - \sqrt{5}, 1 - \sqrt{5})$; 若 OC 为平行四边形的对角线, 则 $Q_4(-2, -2)$.

(3) ∵ $\triangle CHM \sim \triangle AOC$, 点 C 与点 A 对应,
 ∴ $\angle MCH = \angle CAO$. ①如图, 当点 H 在点 C 下方时,
 ∴ $\angle MCH = \angle CAO$, ∴ $CM \parallel x$ 轴,
 ∴ $y_M = -2$, 点 M 在二次函数图象上,
 ∴ $x^2 - x - 2 = -2$, 解得 $x = 0$ (舍) 或 $x = 1$,
 ∴ $M(1, -2)$; ②如图, 当点 H 在点 C 上方时,
 ∴ $\angle M'CH = \angle CAO$, 设 CM' 交 x 轴于点 P , 设 $OP = x$, 则 $PC = PA = x + 1$, 在 $Rt\triangle POQ$ 中, $x^2 + 2^2 = (x + 1)^2$, 解得 $x = \frac{3}{2}$, 即 $OP = \frac{3}{2}$, 点 M' 为直线 CP 与



第 17 题图

抛物线的另一交点, 设直线 CM' 的解析式为 $y = kx - 2$, 把 $P(\frac{3}{2}, 0)$ 代入得 $\frac{3}{2}k - 2 = 0$, 解得 $k = \frac{4}{3}$, ∴ $y = \frac{4}{3}x - 2$, 由 $\frac{4}{3}x - 2 = x^2 - x - 2$, 解得

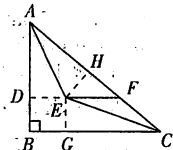
$x = 0$ (舍去) 或 $x = \frac{7}{3}$, 此时 $y = \frac{10}{9}$, ∴ $M'(\frac{7}{3}, \frac{10}{9})$, ∴ 点 M 的坐标为 $(1, -2)$ 或 $(\frac{7}{3}, \frac{10}{9})$.

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1. C 【解析】 $b^2 - ab + \frac{1}{2}a + 2 = 0$, $\Delta = (-a)^2 - 4(\frac{1}{2}a + 2) \geq 0$, $a^2 - 2a - 8 \geq 0$, $(a-4)(a+2) \geq 0$, ∴ $a \geq 4$ 或 $a \leq -2$. 故选 C.
 2. C 【解析】∵ $a + b = -2$, ∴ $a = -b - 2$, $b = -2 - a$. 又 ∵ $a \geq 2b$, ∴ $-b - 2 \geq 2b$, $a \geq -4 - 2a$. 则 $-3b \geq 2, 3a \geq -4$, 解得 $b \leq -\frac{2}{3} < 0$, $a \geq -\frac{4}{3}$, 由 $a \geq 2b$, 得 $\frac{a}{b} \leq 2$; A. 当 $a > 0$ 时, $\frac{b}{a} < 0$, 即 $\frac{b}{a}$ 的最小值不是 $-\frac{1}{2}$; B. 当 $-\frac{4}{3} \leq a < 0$ 时, $\frac{b}{a} \geq \frac{1}{2}$, $\frac{b}{a}$ 有最小值 $-\frac{1}{2}$, 无最大值; C. $\frac{a}{b}$ 有最大值 2; D. $\frac{a}{b}$ 无最小值. 故选 C.

3. D 【解析】 $x^2 + 2bx - a^2 = 0$ 的根为 $x = \frac{-2b \pm \sqrt{4b^2 + 4a^2}}{2} = -b \pm \sqrt{b^2 + a^2}$, ∴ $\sqrt{b^2 + a^2} = AB$, ∴ 正根 $x = -b + AB = -AD + AB = BD$. 故选 D.

4. C 【解析】如图, 延长 FE 交 AB 于点 D , 作 $EG \perp BC$ 于点 G , 作 $EH \perp AC$ 于点 H , ∴ $EF \parallel BC$, $\angle ABC = 90^\circ$, ∴ $FD \perp AB$.
 ∴ $EG \perp BC$, ∴ 四边形 $BDEC$ 是矩形.
 ∴ AE 平分 $\angle BAC$, CE 平分 $\angle ACB$,
 ∴ $ED = EH = EG$, $\angle DAE = \angle HAE$,
 ∴ 四边形 $BDEC$ 是正方形, ∴ $BD = ED = EH$.



第 4 题图

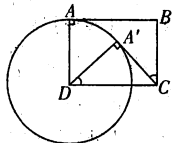
在 $\triangle DAE$ 和 $\triangle HAE$ 中, $\begin{cases} \angle DAE = \angle HAE, \\ \angle ADE = \angle AHE, \\ AE = AE, \end{cases}$
 ∴ $\triangle DAE \cong \triangle HAE$ (AAS), ∴ $AD = AH$. 同理可得 $CG = CH$.

设 $BD = BG = x$, 则 $AD = AH = 6 - x$, $CG = CH = 8 - x$,
 在 $Rt\triangle ABC$ 中, $AC = \sqrt{AB^2 + BC^2} = \sqrt{6^2 + 8^2} = 10$,
 ∴ $6 - x + 8 - x = 10$, 解得 $x = 2$, ∴ $BD = ED = 2$, $AD = 4$.
 ∴ $DF \parallel BC$, ∴ $\triangle ADF \sim \triangle ABC$,
 ∴ $\frac{AD}{AB} = \frac{DF}{BC}$, ∴ $DF = \frac{AD \cdot BC}{AB} = \frac{4 \times 8}{6} = \frac{16}{3}$,
 ∴ $EF = DF - ED = \frac{16}{3} - 2 = \frac{10}{3}$. 故选 C.

5. B 【解析】连接 EF , 设 $AB = 3a = CD$, $AD = BC = 5a$, 则 $BE = 2a$, $DF = a$,
 ∴ $CF = 2a = BE$, $EC = AB = 3a$. 可证 $\triangle ABE \cong \triangle ECF$,
 ∴ $AE = EF$, $\angle 1 = \angle FEC$, ∴ $\angle AEF = 90^\circ$ 且 $AE = EF$, ∴ $\angle EAF = 45^\circ$,
 ∴ $\angle 1 + \angle 2 = 45^\circ$. ∴ $\cos(\angle 1 + \angle 2) = \frac{\sqrt{2}}{2}$. 故选 B.

6. D 【解析】图象过 $(-2, 0)$, $(6, 0)$. ∴ $m = -2 \times 6 \times 2 = -24$. 故选 D.
 7. -6 【解析】 $\begin{cases} x^2 + 2y = 5 & ① \\ y^2 + 2x = 5 & ② \end{cases}$ ① + ②, 得 $(x+y)^2 - 2xy + 2(x+y) = 10$ ③, ① - ②, 得 $(x+y)(x-y) - 2(x-y) = 0$,
 ∴ $x \neq y$, ∴ $x + y = 2$, 代入 ③ 得 $xy = -1$.

8. $\frac{\sqrt{5}}{3}$ 【解析】以 D 为圆心, 4 为半径作圆,
 ∴ $DA' = DA = 4$,
 ∴ A' 在 $\odot D$ 上, 当 CA' 与 $\odot D$ 相切时, $\angle BCA'$ 最小,
 此时 $\angle BCA' + \angle DCA' = 90^\circ$,
 又 ∵ $\angle A'DC + \angle DCA' = 90^\circ$,
 ∴ $\angle BCA' = \angle A'DC$.



第 8 题图

在 $Rt\triangle DCA'$ 中, $CA' = \sqrt{CD^2 - A'D^2} = \sqrt{6^2 - 4^2} = 2\sqrt{5}$,
 ∴ $\sin \angle BCA' = \sin \angle A'DC = \frac{CA'}{CD} = \frac{2\sqrt{5}}{6} = \frac{\sqrt{5}}{3}$.

9. 21 【解析】由题意得 $(a+3)(b-1) = 60$, 即 $(a+3)(2b-2) = 120$,
 ∴ $(a+2b+1)^2 = [(a+3) + (2b-2)]^2 = [(a+3) - (2b-2)]^2 + 4 \times 120$, 故当 $a+3 > 0$, $2b-2 \geq 0$ 时, 要使 $a+2b-1$ 取最小值, 必须 $|(a+3) - (2b-2)|$ 取最小值.
 ∴ a, b 为正整数, $(a+3)(2b-2) = 120 = 12 \times 10$,
 ∴ 当且仅当 $\begin{cases} a+3=12, \\ 2b-2=10 \end{cases}$ 或 $\begin{cases} a+3=10, \\ 2b-2=12 \end{cases}$ 时, $|(a+3) - (2b-2)|$ 取最小值.
 ∴ 当 $\begin{cases} a=9, \\ b=6 \end{cases}$ 或 $\begin{cases} a=7, \\ b=7 \end{cases}$ 时, $a+2b$ 取最小值为 21.

10. $\frac{10}{3}$ 【解析】过点 A 作 $AG \perp BC$ 交 CB 的延长线于点 G , 连接 BD, AC , 则 $AG = \sin 60^\circ \times AB = \frac{3}{2}$. ∴ $S_{\triangle ABC} = \frac{1}{2}AG \cdot BC = 3$. ∴ $S_{\triangle ACD} = S_{\text{四边形}ABCD} -$

$$S_{\triangle ABC} = 9 - 3 = 6. \because \frac{DE}{AD} = \frac{DF}{DC} = \frac{2}{3}, \angle ADC = \angle ADC, \therefore \triangle DEF \sim \triangle DAC,$$

$$\therefore \frac{S_{\triangle DEF}}{S_{\triangle DAC}} = \left(\frac{2}{3}\right)^2 = \frac{4}{9}, \therefore S_{\triangle DEF} = \frac{8}{3}. \text{又} \because \frac{AE}{AD} = \frac{1}{3}, \therefore S_{\triangle ABE} = \frac{1}{3} S_{\triangle ABD},$$

$$\text{同理 } S_{\triangle BFC} = \frac{1}{3} S_{\triangle BDC}, \therefore S_{\triangle ABE} + S_{\triangle BFC} = \frac{1}{3} (S_{\triangle ABD} + S_{\triangle BDC}) = \frac{1}{3} \cdot$$

$$S_{\text{四边形}ABCD} = \frac{1}{3} \times 9 = 3, \therefore S_{\triangle BEF} = S_{\text{四边形}ABCD} - S_{\triangle DEF} - (S_{\triangle ABE} + S_{\triangle BFC}) =$$

$$9 - \frac{8}{3} - 3 = \frac{10}{3}.$$

11. $\frac{19}{7}$ 【解析】第一次用“调日法”后得 $\frac{27}{10}$, 即 $\frac{27}{10} < e < \frac{14}{5}$, 第二次用“调日法”后得 $\frac{41}{15}$, 即 $\frac{27}{10} < e < \frac{41}{15}$, 第三次用“调日法”后得 $\frac{68}{25}$, 即 $\frac{27}{10} < e < \frac{68}{25}$, 第四次用“调日法”后得 $\frac{95}{35} = \frac{19}{7}$.

12. $\frac{16}{3}$ 【解析】如图, 过点 C 作 $CF \perp y$ 轴于点 F , 作 $CG \perp x$ 轴于点 G , 过点 D 作 $DE \perp x$ 轴于点 E , 则四边形 $OFCG$ 是矩形, 设 $D(a, \frac{4}{a}), C(b, \frac{4}{b})$,

$$\therefore \text{点 } B(\frac{am}{4}, \frac{4}{a}), A(\frac{bm}{4}, \frac{4}{b}),$$

$$\therefore BD = \frac{am}{4} - a, AC = \frac{bm}{4} - b, DE = \frac{4}{a},$$

$$CG = \frac{4}{b}, EG = a - b. \therefore BD = 3AC,$$

$$\therefore \frac{am}{4} - a = 3(\frac{bm}{4} - b), \text{化简得 } a = 3b,$$

$$\therefore S_{\text{五边形}OFCDE} = S_{\triangle OCF} + S_{\triangle ODE} + S_{\triangle OCD} = S_{\text{矩形}OFCG} + S_{\text{梯形}CDEG}.$$

$$\because \text{点 } C \text{ 和点 } D \text{ 在反比例函数 } y_1 = \frac{4}{x} \text{ 上},$$

$$\therefore S_{\text{矩形}OFCG} = 4, S_{\triangle OCF} = S_{\triangle ODE} = 2, \therefore 2 + 2 + S_{\triangle OCD} = 4 + S_{\text{梯形}CDEG},$$

$$\therefore S_{\text{梯形}CDEG} = S_{\triangle OCD}, \therefore S_{\text{梯形}CDEG} = \frac{1}{2} \cdot (CG + DE) \cdot EG = \frac{1}{2} \cdot (\frac{4}{a} + \frac{4}{b}) \cdot (a - b) = \frac{2(a^2 - b^2)}{ab} = \frac{16}{3},$$

$$\therefore S_{\triangle OCD} = \frac{16}{3}.$$

13. 证明: $\because 1 = (a^2 + b^2)(c^2 + d^2) = (ac)^2 + a^2 \times d^2 + b^2 \times c^2 + (bd)^2$, 又 $ad = bc, \therefore 1 = (ac)^2 + a^2 \times d^2 + b^2 \times c^2 + (bd)^2 = (ac)^2 + 2 \times a^2 \times d^2 + (bd)^2 = (ac)^2 + 2acbd + (bd)^2, \therefore 1 = (ac + bd)^2, \therefore a, b, c, d > 0, \therefore ac + bd > 0, \therefore ac + bd = 1$.

14. 解: (1) $\because$ 一次函数 $y = 2x - 4$ 的图象与 x 轴, y 轴分别相交于点 A, B , $\therefore A(2, 0), B(0, -4). \therefore P$ 为线段 AB 的中点, $\therefore P(1, -2), \therefore d_1 + d_2 = 1 + 2 = 3$.

(2) $d_1 + d_2 \geq 2. \therefore$ 设 $P(m, 2m - 4), \therefore d_1 + d_2 = |m| + |2m - 4|$. 当 $m < 0$ 时, $d_1 + d_2 = |m| + |2m - 4| = -m + 4 - 2m = -3m + 4$, 由 $-3m + 4 = 3$ 解得 $m = \frac{1}{3}$, 与 $m < 0$ 不符, 舍去. 当 $0 \leq m < 2$ 时, $d_1 + d_2 = |m| + |2m - 4| = m + 4 - 2m = -m + 4$, 由 $-m + 4 = 3$ 解得 $m = 1$, 此时 $P(1, -2)$.

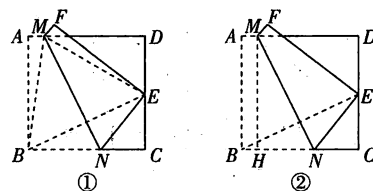
当 $m \geq 2$ 时, $d_1 + d_2 = |m| + |2m - 4| = m + 2m - 4 = 3m - 4$, 由 $3m - 4 = 3$ 解得 $m = \frac{7}{3}$, 此时 $P(\frac{7}{3}, \frac{2}{3})$. 综上所述, 当 $d_1 + d_2 = 3$ 时, 点 P 的坐标为 $(1, -2)$ 或 $(\frac{7}{3}, \frac{2}{3})$.

(3) 设 $P(m, 2m - 4), \therefore d_1 = |2m - 4|, d_2 = |m|$. $\because$ 点 P 在线段 AB 上, $\therefore 0 \leq m \leq 2, \therefore d_1 = 4 - 2m, d_2 = m. \therefore d_1 + ad_2 = 4, \therefore 4 - 2m + am = 4. (a - 2)m = 0, \therefore$ 存在无数个 P 点, $\therefore a = 2$.

15. 解: (1) 如图①, 连接 BM, EM, BE , 则 MN 垂直平分 $BE, \therefore BM = EM, BN = EN$. 设 $AB = BC = CD = DA = 2$, 则 $CE = DE = 1$. 设 $BN = x$, 则 $NE = x, NC = 2 - x$. 在 $\text{Rt} \triangle CNE$ 中, $x^2 = (2 - x)^2 + 1^2$, 解得 $x = \frac{5}{4}$, 即 $BN = \frac{5}{4}$. 在 $\text{Rt} \triangle ABM$ 和在 $\text{Rt} \triangle DEM$ 中, $AM^2 + AB^2 = DM^2 + DE^2$. 设 $AM = y$, 则 $DM = 2 - y, \therefore y^2 + 2^2 = (2 - y)^2 + 1^2$, 解得 $y = \frac{1}{4}$, 即 $AM = \frac{1}{4}, \therefore \frac{AM}{BN} = \frac{1}{5}$.

(2) 当四边形 $ABCD$ 为正方形时, 连接 BE , 令 $CD = CB = n$, 则 $CE = 1$, 设 $BN = x$, 则 $EN = x, NC = n - x, x^2 = (n - x)^2 + 1^2$, 解得 $x = \frac{n^2 + 1}{2n}$. 如图②, 作 $MH \perp BC$ 于点 H , 则易证 $\triangle EBC \cong \triangle NMH, \therefore NH = EC = 1, AM = BH =$

$$BN - NH = \frac{n^2 + 1}{2n} - 1 = \frac{n^2 - 2n + 1}{2n}, \text{则 } \frac{AM}{BN} = \frac{\frac{n^2 - 2n + 1}{2n}}{\frac{n^2 + 1}{2n}} = \frac{n^2 - 2n + 1}{n^2 + 1}.$$



第 15 题图

16. 解: (1) $\because AB = AC, \angle A = 50^\circ, \therefore \angle ABC = \angle ACB = 65^\circ. \therefore \angle A = \angle EBC = 50^\circ, \therefore \angle DBE = 15^\circ, \therefore \angle DOE = 2 \angle DBE = 30^\circ$.

(2) $\because \angle A = \angle EBC, \angle ACB = \angle BCE, \therefore \triangle ABC \sim \triangle BEC, \therefore \frac{BC}{AB} = \frac{CE}{BC} = \frac{2}{3}$, 则 $\frac{S_{\triangle BCE}}{S_{\triangle ABC}} = (\frac{BC}{AB})^2 = \frac{4}{9}, \therefore S_{\triangle BEC} = \frac{4}{9} S_{\triangle ABC}$. 设 $BC = 2m$, 则 $AC = AB = 3m, CE = \frac{4}{3}m$.

$$\therefore AE = AC - CE = \frac{5}{3}m.$$

$\because$ 四边形 $BCED$ 是圆内接四边形, $\therefore \angle AED = \angle ABC = \angle ACB, \therefore DE \parallel BC, \therefore \triangle ADE \sim \triangle ABC$,

$$\therefore \frac{S_{\triangle ADE}}{S_{\triangle ABC}} = (\frac{AE}{AC})^2 = \frac{25}{81}, \therefore S_{\triangle ADE} = \frac{25}{81} S_{\triangle ABC},$$

$$\therefore S_{\triangle BDE} = S_{\triangle ABC} - S_{\triangle ADE} - S_{\triangle BEC} = \frac{20}{81} S_{\triangle ABC}, \therefore \frac{S_{\triangle BDE}}{S_{\triangle ABC}} = \frac{20}{81}.$$

(3) 由(2)可得 $\triangle ABC \sim \triangle BCE, \therefore \frac{BC}{AB} = \frac{CE}{BC} = k, \frac{c_2}{c} = \frac{BC}{AB} = k$,

$$\therefore BC = kAB, CE = kBC = k^2 \cdot AB,$$

$$\therefore AE = AC - EC = (1 - k^2) \cdot AB, \text{由(2)可得 } \triangle ADE \sim \triangle ABC,$$

$$\therefore \frac{c_1}{c} = \frac{AE}{AC} = 1 - k^2,$$

$$\therefore \frac{c_1}{c} + \frac{c_2}{c} = k + 1 - k^2 = -(k - \frac{1}{2})^2 + \frac{5}{4} \leq \frac{5}{4},$$

$$\therefore 0 < k < 1, \therefore 1 < \frac{c_1 + c_2}{c} \leq \frac{5}{4}.$$

17. 解: (1) $A(-3, 0), B(0, 4). \therefore$ 当 $y = 2$ 时, $\frac{4}{3}x + 4 = 2, x = -\frac{3}{2}$, 则直线

$$AB \text{ 与 } CD \text{ 交点的坐标为 } (-\frac{3}{2}, 2);$$

- (2) ① 当 $0 < t < \frac{3}{2}$ 时, $\triangle MPH$ 与矩形 $AOCD$ 重合部分的面积即 $\triangle MPH$ 的面积. 过点 M 作 $MN \perp OA$ 于点 N . 则 $AB = \sqrt{3^2 + 4^2} = 5$, 由 $\triangle AMN \sim$

$$\triangle ABO \text{ 得 } \frac{AN}{AO} = \frac{AM}{AB}, \therefore \frac{AN}{3} = \frac{\frac{5}{3}t}{5}, \therefore AN = t.$$

$$\therefore S_{\triangle MPH} = \frac{1}{2} \times 2(3 - t - t) = 3 - 2t. \text{当 } 3 - 2t = 1 \text{ 时, } t = 1;$$

当 $\frac{3}{2} < t \leq 3$ 时, 设 MH 与 CD 相交于点 E , $\triangle MPH$ 与矩形 $AOCD$ 重合部分的面积即 $\triangle PEH$ 的面积. 过点 M 作 $MG \perp AO$ 于点 $G, MF \perp HP$ 交 HP 的延长线于点 F . 则 $FM = AG - AH = AM \times \cos \angle BAO - (AO - HO) = \frac{5}{3}t \times \frac{3}{5} - (3 - t) = 2t - 3, HF = GM = AM \times \sin \angle BAO = \frac{5}{3}t \times \frac{4}{5} = \frac{4}{3}t$. 由 $\triangle HPE \sim \triangle HFM$ 得 $\frac{PE}{FM} = \frac{HP}{HF}, \therefore \frac{PE}{2t - 3} = \frac{2}{\frac{4}{3}t}, \therefore PE = \frac{6t - 9}{2t},$

$$\therefore S_{\triangle PEH} = \frac{1}{2} \times 2 \times \frac{6t - 9}{2t} = \frac{6t - 9}{2t}. \text{当 } \frac{6t - 9}{2t} = 1 \text{ 时, } t = \frac{9}{4}. \text{经检验, } t = \frac{9}{4} \text{ 是原方程的解. 综上所述, 若 } \triangle MPH \text{ 与矩形 } AOCD \text{ 重合部分的面积为 } 1, t \text{ 为 } 1 \text{ 或 } \frac{9}{4};$$

- ② $BP + PH + HQ$ 有最小值. 连接 PB, CH , 则四边形 $PHCB$ 是平行四边形, $\therefore BP = CH$, 得 $BP + PH + HQ = CH + HQ + 2$. 当点 C, H, Q 在同一直线上时, $CH + HQ$ 的值最小. $\therefore$ 点 C, Q 的坐标分别为 $(0, 2), (-6, -4), \therefore$ 直线 CQ 的解析式为 $y = x + 2, \therefore H(-2, 0)$, 此时 $P(-2, 2)$.